BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 4

DCA2207

DATA MINING AND VISUALISATION

Unit – 1

Introduction to Data Mining

TABLE OF CONTENTS

1. INTRODUCTION

This unit introduces the fundamental concepts of data mining, its evolution, step-by-step processes, and real-world applications. By the end of this unit, we will have a clear understanding of how data mining techniques such as clustering, classification, and association rule mining help uncover valuable patterns in large datasets across various industries.

In today's digital era, vast amounts of data are generated every second. Extracting meaningful insights from this data is essential for businesses, researchers, and industries. This is where data mining comes into play. In this unit, we will explore the basics of data mining, including its definition, importance, and role in transforming raw data into useful knowledge. We will also understand how data mining is connected to fields like machine learning, artificial intelligence, and big data analytics.

To appreciate the significance of data mining, we will look at its history and evolution. Data mining has grown from traditional statistical techniques to advanced machine learning algorithms, making it a powerful tool for decision-making. We will learn how the development of computing power, storage systems, and sophisticated algorithms have contributed to the rise of data mining as a critical discipline.

Understanding the data mining process is essential to know how data is collected, cleaned, analysed, and interpreted. We will break down the entire process into key stages, including data collection, pre-processing, pattern discovery, evaluation, and decision-making. This will give us a structured approach to handling real-world data challenges.

Finally, we will explore the applications of data mining in various industries. From predicting customer behaviour in marketing to detecting fraud in banking, data mining has revolutionized how businesses operate. We will also discuss its role in healthcare, cybersecurity, e-commerce, and social media analytics. Additionally, ethical considerations and privacy concerns surrounding data mining will be addressed to highlight responsible data usage.

By the end of this unit, we will have a solid understanding of how data mining works, its historical development, and its impact across different fields. This knowledge will provide a strong foundation for more advanced topics in data analytics and artificial intelligence. This unit enables learners to discover patterns in data through systematic mining techniques.

1.1. Learning Objectives

- **Define** data mining
- **Identify** the core functionalities of data mining
- **Explain the basic structure of a data mining system**
- **Explore the** various applications of data mining across industries

2. DEFINITION OF DATA MINING

Data mining refers to the practice of utilising a variety of statistical and computational methods to discover patterns or insights within massive datasets. Databases, data warehouses, and data lakes are just a few examples of the many possible storage formats for organised, semi-structured, and unstructured data.

Finding previously unseen connections and patterns in data for the purpose of making better predictions and judgements is data mining's main objective. Data mining, clustering, classification, regression, association rule mining, and anomaly detection are all part of this process.

Marketing, banking, healthcare, and telecommunications are just a few of the many industries that might benefit from data mining. In healthcare, data mining can help discover risk factors for diseases and generate personalised treatment regimens; in marketing, it can help identify client categories and target marketing campaigns.

But, when dealing with sensitive or personally identifiable information, data mining can bring up ethical and privacy issues. Data mining must be done in an ethical manner with proper protections to avoid exploitation of personal information and to preserve people's privacy.

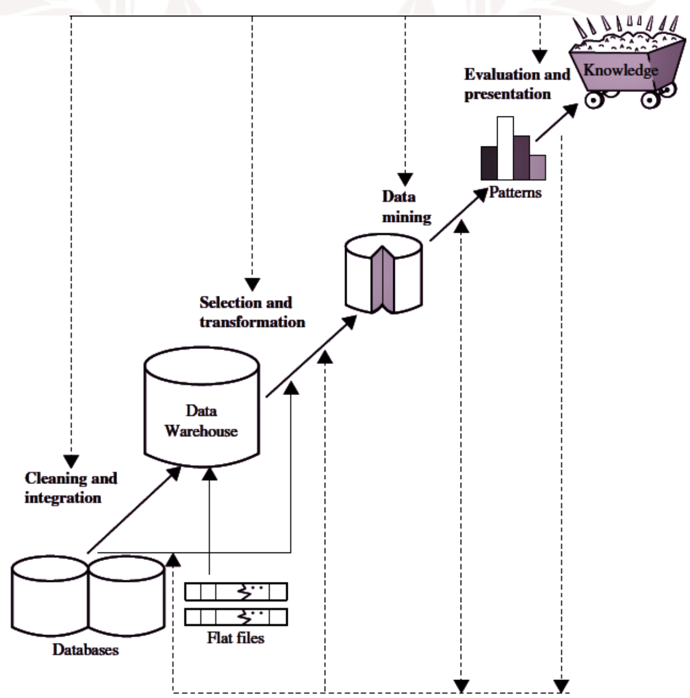

Figure 1: Data Mining – Knowledge Data Discovery

Data mining is the practice of discovering valuable insights within massive datasets by means of systematic search and analysis.

To better understand their clientele, businesses employ data mining technologies. They can use it to improve their marketing strategy, which in turn can boost sales and cut expenses. Efficient data gathering, storage, and processing are the backbone of data mining.

Based on this, it can be defined as "Data mining is the process of extracting meaningful patterns and insights from large datasets using various techniques and algorithms.

It is a critical step within the broader Knowledge Discovery in Databases (KDD) process, which also includes data cleaning, integration, transformation, pattern evaluation, and knowledge presentation, as shown in figure 1.

The aim of data mining is to uncover hidden relationships, trends, and anomalies that support decision-making and forecasting. Common data mining tasks include classification (e.g., decision trees), clustering (e.g., K-means), and association rule mining (e.g., Apriori algorithm).

Data mining is widely applied in domains such as business, healthcare, finance, and scientific research."

Figure 1 represents the structured process of extracting useful knowledge from raw data. It highlights the sequential stages through which data travels before becoming actionable insights.

The process begins with databases and flat files, which store raw, unorganised data. Since this raw data is often inconsistent and redundant, the first step is cleaning and integration. At this stage, errors, duplicates, and irrelevant entries are removed, while multiple data sources are integrated to ensure consistency and accuracy.

Next, the cleaned data is stored in a data warehouse, a central repository that allows efficient storage and management. From here, the step of selection and transformation occurs, where relevant data is selected based on the research objective and transformed into suitable formats for mining. Transformation may involve normalisation, aggregation, or scaling to prepare the dataset.

Once the data is ready, the data mining stage is carried out. This is the core of the process, where advanced algorithms and statistical methods are applied to identify patterns, correlations, or associations hidden within the data. These patterns are not yet actionable until they are properly interpreted.

The next stage is evaluation and presentation, where the mined patterns are assessed for their usefulness, relevance, and validity. Here, results are visualised using charts, graphs, or reports, making them understandable for decision-makers. Only the patterns that provide meaningful insights are considered.

Finally, the evaluated patterns are converted into knowledge, which organisations can use to guide strategies, optimise operations, or predict future trends. This knowledge is the ultimate goal of KDD and is vital for informed decision-making.

3. WHAT IS DATA

For the purposes of analysis, decision-making, and problem-solving, data refers to a compilation of facts, figures, observations, or measurements. It stands for unprocessed data that, when examined, reveals important patterns.

That "data is separate bits of information, typically formatted in a special way" is what the Oxford English Dictionary says. Measuring, collecting, reporting, and analysing data typically leads to its visualisation through graphs, pictures, or other analytical tools. Before being "cleaned" and rectified by researchers, raw data, also known as "unprocessed data," might be a collection of numbers or characters. In order to eliminate outliers, instruments, or data entering mistakes, it needs to be fixed. It is reasonable to think of the "processed data" from one stage as the "raw data" for the next stage as data processing usually happens in stages. "In situ" data refers to information gathered from a natural setting rather than a controlled laboratory. Information gathered through scientific experiments is known as experimental data.

Data can be generated by:

- Humans
- Machines
- Human-Machine combines.

Typically, it can be produced in any location where data is created and saved, whether in a structured or unstructured format.

What makes data so crucial?

- Information aids in decision-making.
- Problems can be solved with the use of data by identifying the cause of underperformance.
- One can assess the performance with the use of data.
- Facts allow for the enhancement of procedures.
- Information on customers and the market can be found in data.

Classes of data

The two primary components of data categorisation are -

a. Structured Data: Information that has been organised into a precise format for the purpose of searching, analysing, and processing is known as structured data. Numbers, dates, and categories are all examples of the structured data that may be found in relational databases.

b. Unstructured Data: Data that is not structured does not adhere to any particular format or structure. It could contain data that is difficult to organise or analyse without further processing, such as photographs, videos, and text documents.

3.1. Types of Data

In most cases, there are two main categories into which data falls:

1. Categorical Data: Data that has been assigned a specific category is known as categorical data.

Examples of this type of data include:

- Family Status
- Party Affiliation
- Eye color

2. Numerical Data: Two additional types of numerical data exist:

- Discrete Data: This type of data includes items with specific numerical values, such as the number of children or the number of defects per hour.
- Data with continuous numerical values, such as weight, voltage, etc., are known as continuous data.

3. Nominal Scale: Nominal scales divide data into multiple groups without implying any particular order of importance. For instance, marital status and gender.

4. Ordinary Scale: The use of an ordinal scale implies a ranking system when data is classified into various groups. As an illustration:

  - Professor, Associate Professor, and Assistant Professor are the ranks of the faculty.
  - A, B, C, and D.E.F. are the student grades.

5. Interval scale: In the absence of a real zero point, the difference between measurements becomes a meaningful number on an interval scale, which is an ordered scale. As an illustration:

  - Celsius and Fahrenheit temperatures.

- Decades

6. Ratio scale: A ratio scale is an ordered set of measurements where the difference between each measurement is a meaningful number, giving the measurements on the scale a true zero point. This allows us to do calculations on very huge datasets. Things to think about include: weight, age, salary, etc.

3.2. Data Processing Cycle

The data processing cycle is the repeated series of operations performed on raw data in order to get useful insights. Think of it like a pipeline with different steps:

1. Data Acquisition: At this point, we have gathered all of the raw data from all of our sources. Data collection methods may include readings from sensors, web scraping, questionnaires, and application logs.
2. Data Preparation: Because of its inherent disorder, raw data must undergo cleaning and pre-processing before to analysis. Finding and dealing with missing values, fixing discrepancies, organising data into a consistent structure, and maybe eliminating outliers are all part of this stage's responsibilities.
3. Data Input: The pre-processed data is imported into the right system to make room for more processing and analysis. Making data machine-readable is a common first step in getting it ready for a data warehouse or database.
4. Data Processing: At this stage, the data undergoes several transformations and processing steps in order to extract valuable insights. One way to use machine learning to find patterns and correlations is to aggregate, filter, sort, or "feature engineered" (i.e., created from existing features) the data.
5. Data Output: We further evaluate the updated data using a number of approaches to get insights and information. Tools for visualisation, statistical analysis, or the creation of models for prediction may be required for this.
6. Data Storage: The processed data and resultant outputs are preserved in a secure and accessible manner for future use, reference, or input into further analysis cycles.

Iterativeness refers to the fact that the data processing cycle can leverage input from earlier steps. Consequently, the raw data can be continuously improved, studied more extensively, and ever-more-advanced insights can be developed.

3.3. How Do We Analyze Data

The data cycle is primarily concerned with data analysis, which is where knowledge and useful information are extracted from raw data.

A summary of the main points is as follows:

1. Establish Purpose and Enquiries:

Before you even think about collecting data, you need to figure out what you want to achieve. Is your goal to generate predictions, analyse consumer behaviour, or compile seasonal lineups? To stay on track, it's important to have well-defined objectives and use practical analysis tools.

2. Strive for Appropriate Methods

It can be difficult to know which data analysis techniques to use because there are so many. Some typical methods are as follows: Some typical methods are as follows:

In statistical analysis, you can find tools for data summarisation and preparation, such as the mean, median, standard deviation, and hypothesis testing. It lays bare these connections, among other ways to probe causal elements. Algorithms in machine learning rely on pre-existing data to learn new behaviours and make predictions. Categorisation and regression, which involve identifying data points and predicting a continuous value, are well-suited to these tasks.

- Data mining: moreover, it denotes the investigation of unknown occurrences and behaviours in massive data clusters. Finding hidden relationships is the speciality of methods like clustering and association rule learning.
- Data Visualisation: Dashboards, charts, and graphs are all tools for visualising data, which helps to reveal trends, patterns, and disclosures that can be hard to detect in raw numbers.

3. Explore and Clean the Data

Understanding the data's characteristics is crucial prior to conducting any kind of in-depth research. EDA analyses distribution graphs, finds missing values, and builds profiles to understand the data as a whole. By removing duplicates, incorrect entries, and missing values, data cleansing enables you to build a solid foundation on top-notch information.

4. Conduct the Evaluation

Data processing can begin immediately after all methods have been selected and data cleansing has been completed. Some methods that fall under this category include creating well-designed data visualisations and running specific tests using sophisticated regression or machine learning algorithms.

5. Decipher the Findings

Because they are unique to your goals, you should take your time interpreting the analytics. Instead of merely building the model, demonstrate what it represents, argue with the limitations of your study, and draw conclusions from your initial queries.

6. Share Your Discoveries

In order to make better decisions, data analysis is often performed. Whether it's through papers, presentations, or interactive charts, make sure to communicate your findings honestly to all parties involved.

SELF-ASSESSMENT QUESTIONS – 1

Fill in the blanks:	
1	Which of the following best defines data? A) Raw facts and figures without meaning B) Processed information used for decision-making C) A collection of related records D) A structured database
2	Which of the following is NOT a type of data? A) Structured Data B) Unstructured Data C) Semi-structured Data D) Mixed Data
3	Why is data important in modern organizations? A) It helps in making informed decisions B) It replaces human effort completely C) It eliminates the need for market research D) It only benefits IT companies

4	What is the correct sequence of the Data Processing Cycle? A) Input → Processing → Storage → Output → Feedback B) Storage → Input → Processing → Output → Feedback C) Input → Storage → Processing → Output → Feedback D) Input → Processing → Output → Storage → Feedback
5	Which of the following is an example of unstructured data? A) An Excel spreadsheet containing customer details B) A relational database table C) A handwritten letter or a scanned document D) An SQL query result
6	Which of the following is an example of structured data? A) Video files stored on a server B) A relational database table C) Email conversations D) Social media posts
7	What is the primary goal of the data processing cycle? A) To store data permanently B) To convert raw data into meaningful information C) To eliminate the need for human intervention D) To secure data from cyber threats

4. IMPORTANCE OF DATA MINING

In today's environment, when businesses produce and store huge volumes of data every day, data mining is quite important. It is sometimes not enough to use traditional approaches to look at such huge and complicated datasets. Data mining is a great tool for making decisions because it gives you the skills and methods to automatically find hidden patterns, relationships, and important information.

1. Extracting Knowledge from Large Data

- Companies get massive volumes of unstructured data from things like web traffic, social media, transactions, and sensors.
- This data goes unused if data mining isn't done.
- By discovering patterns, correlations, and trends, data mining transforms unstructured data into useful information.

2. Improving Decision Making

- When formulating plans, managers and decision-makers need reliable information.
- Data mining helps find facts that might not be immediately apparent, which supports decision making based on evidence.
- Data mining is useful for many industries, including banking and retail, where it helps find credit concerns and forecasts consumer behaviour.

3. Competitive Advantage

- There is a lot of rivalry in the corporate world nowadays.
- Businesses may learn about consumer preferences, industry tendencies, and rival strategies by utilising data mining.
- Organisations can use this intelligence to keep ahead of the competition, optimise resources, and provide personalised services.

4. Detection of Anomalies and Risks

- Data mining is useful for spotting suspicious activity, fraud, and hazards.
- It is possible, for instance, to identify questionable credit card transactions in the banking industry.
- Early diagnosis is possible in healthcare by identifying anomalous patient health records.

5. Automation of Analysis

- Domain knowledge and human intervention are prerequisites for conventional analysis.
- By automating the process of locating essential information, data mining saves both time and effort.
- Online recommendation systems (like Amazon's and Netflix's) rely on this automation in real time.

6. Wide Range of Applications

- Data mining isn't just for businesses
- It’s also employed extensively in healthcare, education, telecommunications, scientific research, e-commerce, and the financial sector.
- Data mining is useful in education for analysing student learning patterns and developing better teaching methods.

7. Enhancing Productivity and Efficiency

- Data mining enhances efficiency by revealing inefficiencies and finding ways to optimise operations.
- One use case is in manufacturing, where it helps with downtime prediction and maintenance scheduling.

8. Supporting Innovation

- Businesses can find untapped markets and create ground-breaking new offerings with the help of data mining.
- Better goods that live up to user expectations can be created, for instance, by analysing consumer feedback.
- Data mining's strength is in the insights it can glean from raw data, which in turn promote innovation, risk management, and decision-making. Organisations can improve their services in a variety of areas, get an advantage over competitors, and boost efficiency by using data mining.

5. DATA MINING FUNCTIONALITIES

Functionalities in data mining stand in for the patterns that must be discovered during data mining operations. Come with me as I show you nine features of data mining.

1. Classification

The term "classification" refers to the process of sorting a set of data items into distinct groups based on their attributes and capabilities. This method can be useful for organising and categorising fresh data according to its unidentified classification.

The classifier in data mining is both the method of categorisation and the instances—the findings—that are produced by applying the classifier. When dealing with qualitative variables, classification techniques might be utilised.

Decision trees, logistic regression, Naive Bayes, and random forest are some of the methods that could be used in this categorisation process. Future data can be identified by retrieving these approaches.

Marketers that wish to divide their audience into different segments often use classification. Using this data mining technique, they segment their target client into multiple groups to develop more targeted and effective marketing campaigns.

Classification is a kind of supervised learning that sorts information according to labels or predetermined categories.

Example:

age(*X*, "youth") *AND income*(*X*, "high") → *class*(*X*, "A")
age(*X*, "youth") *AND income*(*X*, "low") → *class*(*X*, "B")
age(*X*, "middle_aged") → *class*(*X*, "C")
age(*X*, "senior") → *class*(*X*, "C")

Figure 2: Example for classification

Figure 2 shows a few **if-then rules** used to classify people into groups (A, B, or C) based on their **age** and **income**.

- If someone is **youth** and has **high income**, they belong to **Class A**.

- If someone is **youth** and has **low income**, they belong to **Class B**.
- If someone is **middle-aged** or **senior**, they belong to **Class C**—regardless of income.

These rules are like conditions used to sort people into different categories.

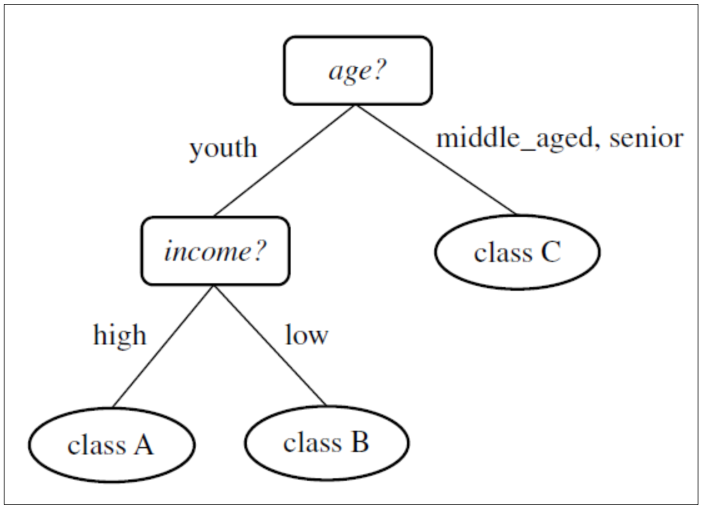

Figure 3: Visualisation of classification

Figure 3, shows a **decision tree diagram** that visually represents how we make classification decisions:

1. Start with the **age** of a person.
   a. If the person is **middle-aged** or **senior**, directly assign them to **Class C**.
   b. If the person is **youth**, we check their **income**.
      i. If income is **high**, they go to **Class A**.
      ii. If income is **low**, they go to **Class B**.

This tree helps us understand the classification process in a step-by-step manner.

2. Association analysis

Association analysis is a useful tool for discovering connections between frequently occurring objects. One of the more common data mining features in sales, it is also known as market basket analysis. It is made up of criteria and rules that define how data is grouped in different contexts.

These one-of-a-kind connections can be described using either association rules or frequent item sets.

- **Frequent item sets** symbolise a set of things that are often seen together, as the ingredients for a certain dish (for example, rice, turmeric powder, onions, and soy milk).

- **Association rules** the purchase of bread is likely to be accompanied by the purchase of butter, indicating a strong correlation between the two goods.

It is more likely to find a consequent (then) in data collecting if an antecedent (if) is present. This two-part rule indicates their link and is essential to association analysis.

When it comes to sales, this method is useful for predicting customer behaviour. As an example, it is quite probable that moviegoers will also purchase a refreshing drink if they buy a bucket of popcorn.

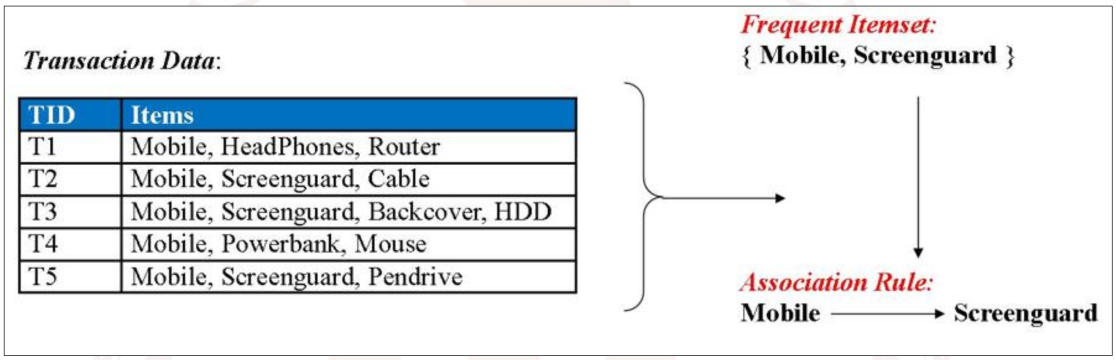

Figure 4: Example for Association

Figure 4 shows how we can find patterns in shopping data using association rules. Each row in the table represents a transaction where a customer buys multiple items, like a mobile, headphones, or screenguard. By looking at all the transactions, we notice that "Mobile" and "Screenguard" are often purchased together. This pair is called a frequent itemset. From this, we create an association rule: **if a customer buys a Mobile, they are likely to also buy a Screenguard**. This kind of rule helps businesses suggest related products and improve sales.

3. Cluster analysis

Cluster analysis involves assembling sets of related data into larger sets and then assigning each set a newly-defined class label. In clustering techniques, data is grouped according to similarities; within each data group, there is a higher degree of similarity than there is between any two data groups. Cluster analysis is utilised in deep learning, picture processing, pattern recognition, and NLP. Cluster analysis is quite similar to classification; however, the key difference is that elements in classification are already established. In contrast, clustering is often known as an unsupervised classification or an unsupervised learning method because it produces the same results as a classification but does not require any predefined classifications.

Example1:

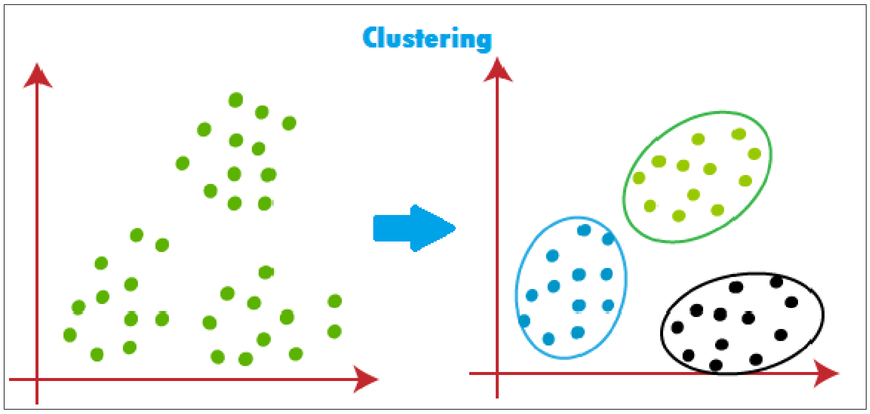

Figure 5: Example1 for cluster analysis

Figure 5 shows how **clustering** works in data analysis. On the left side, we see many data points scattered without any clear grouping. After applying clustering, as shown on the right side, these data points are grouped into three clear clusters—each marked with a different color and boundary. This means that the algorithm has found patterns and similarities among the points and grouped them accordingly. Clustering helps to organize data into meaningful groups so that similar data points fall into the same category without any prior labeling.

Example2:

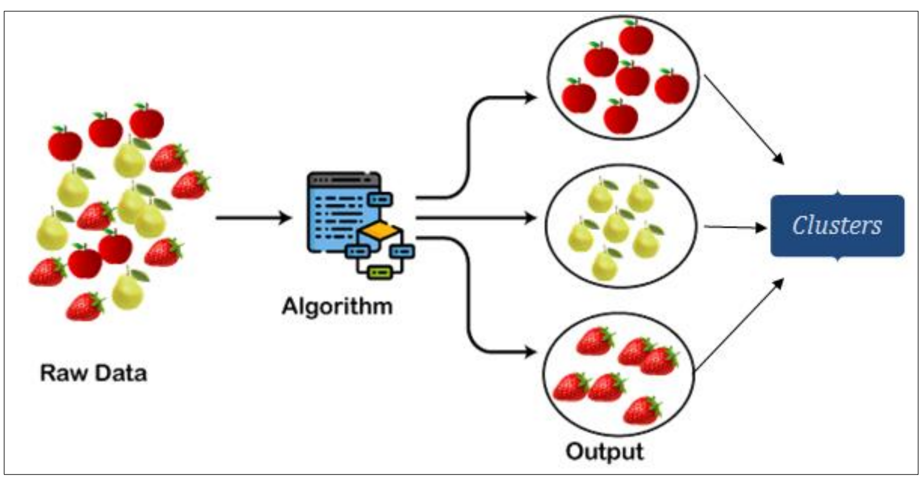

Figure 6: Example 2 for Cluster analysis

Figure 6 shows how clustering can organize different types of data into meaningful groups. On the left, we have **raw data** consisting of a mix of apples, pears, and strawberries. This data is fed into a **clustering algorithm**, which processes and identifies similarities among the items. On the right side, the algorithm groups similar fruits into separate **clusters—**apples in one cluster, pears in another, and strawberries in a third. This shows how clustering helps in automatically sorting unlabelled data into well-defined groups based on their characteristics.

4. Data characterisation

The purpose of data characterisation is to define a target class by summarising its general properties or elements. You can characterise the data with minimal user intervention using an attribute-oriented induction technique. Graphs, pie charts, bar graphs, and table formats are just a few examples of how the "characteristics rule of the target class" can be used to illustrate the connection between the described data and its visual representation.

5. Prediction

If you have a data collection with some missing or unclear pieces, you can utilise prediction, a data mining capability, to find them. Businesses can use linear regression models that are based on previous data to generate numerical projections for any event's outcome, good or bad. The following two methods are available for making predictions:

- Data prediction: Use prediction analysis to fill in any gaps in a dataset that may contain unknown or missing values.
- Class predictions: Predicting the class label using an existing class model

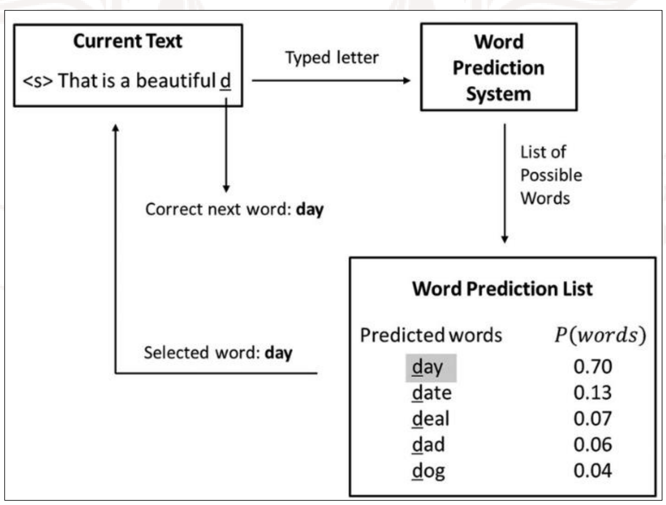

Figure 7: Example for prediction

Figure 7 shows how a **word prediction system** works. A user is typing the sentence: "That is a beautiful d...", and the system tries to predict the next word. Based on what has been typed and past data, the system generates a list of possible words like **day**, **date**, **deal**, **dad**, and **dog**, each with a probability. The word **"day"** has the highest probability (0.70), so the system suggests it first. If the user selects "day", it becomes the next word in the sentence. This is how predictive text systems, like those on smartphones or search engines, help complete our sentences quickly and accurately.

6. Data discrimination

When one set of data is intentionally or unintentionally handled differently from another set of data, this is called data discrimination. Akin to data characterisation, this data mining feature facilitates the separation of anomalous data sets. Data from two classes can be compared and a target class can be mapped to an existing one with its support.

7. Evolution analysis

The term "evolution analysis" describes a method for studying datasets that might have changed or transformed over time. It helps with trend identification using characteristics like periodicity and time-series data as well as trend similarity and time-related data clustering.

In addition to helping with multivariate time series data categorisation, characterisation, discrimination, and grouping, the evolution analysis model also depicts data trends as they change over time.

8. Outlier analysis

Data quality can be better understood by examining outliers. Data anomalies are called outliers. A poorer level of data quality is indicated by a larger number of outliers in a dataset. In order to discover patterns or draw conclusions, it is not a good idea to use a data collection that has a lot of outliers. When dealing with data that defies algorithmic classification due to unique qualities that do not fit into any pre-existing class or general model, this analytical tool comes in handy. Nevertheless, it is essential to record unusual data so that this data mining method may detect outliers promptly and predict how they may affect the company.

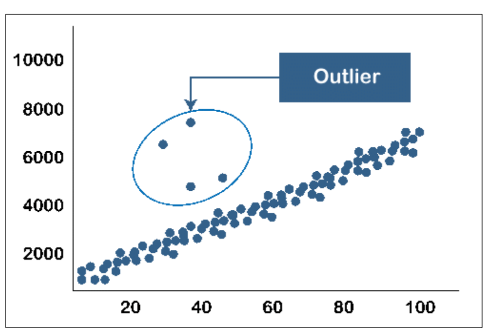

Figure 8: Example for Outlier analysis

Figure 8 illustrates the concept of **outlier detection** in data analysis. Most of the data points in the graph follow a consistent upward pattern or trend. These are shown as tightly packed dots along a diagonal line, indicating a typical behavior or expected values. However, a few data points—highlighted within a blue circle—fall far outside this normal pattern. These are called **outliers**. Outliers are unusual or unexpected values in a dataset that do not fit the general trend. Detecting them is important because they might indicate errors, fraud, or rare but significant events. For example, in a sales dataset, an extremely high purchase amount might be an outlier worth investigating.

9. Correlation analysis

The mathematical concept of correlation can also be used to determine the existence or strength of a relationship between two variables. You may find out how closely two numerical variables (like height and weight) that are being tracked all the time connect to one another by looking at their correlation. This kind of analysis can help researchers find potential relationships between their study variables, such the fact that taller people tend to be heavier.

6. BASIC DATA MINING STRUCTURE

An important part of any data mining system is the data source, the engine that processes the data, the server that stores the data, the module that evaluates patterns, the GUI, and the knowledge base.

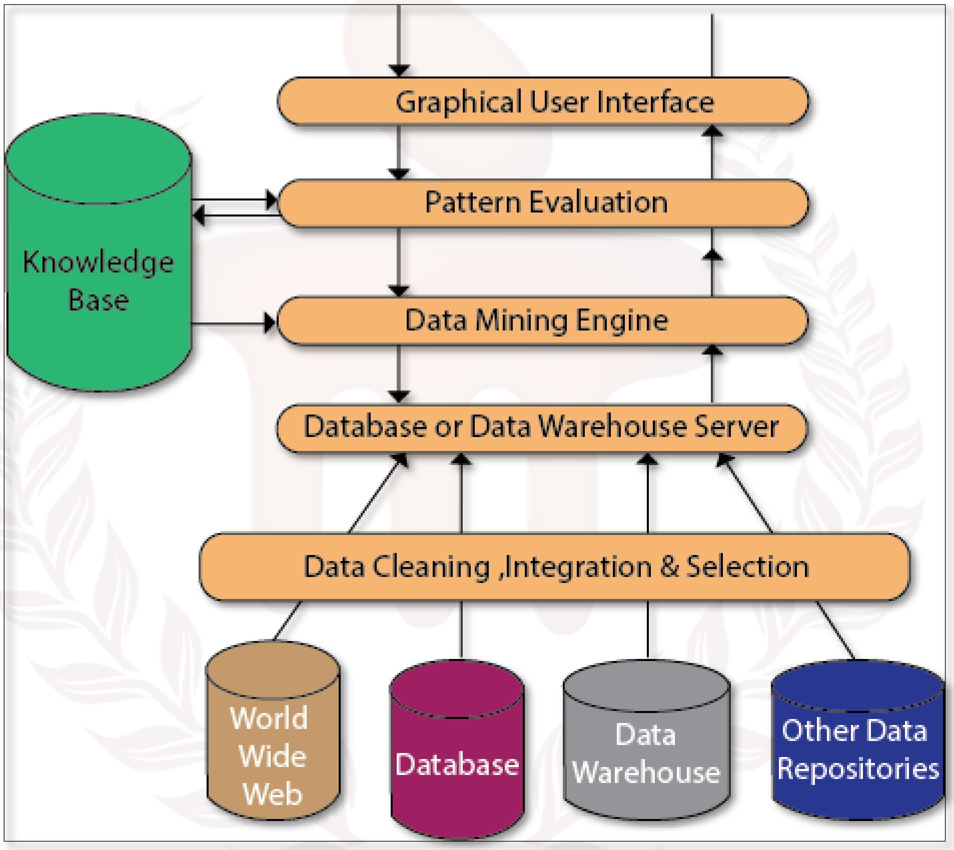

Figure 9: Data Mining Structure

Data Source:

Databases, data warehouses, the Internet, text files, and other publications are the true origins of data. For data mining to work, you need a mountain of historical records. Databases and data warehouses are common places for organisations to keep their data. Data warehouses can be made up of various types of repositories, including databases, text files, spreadsheets, and more. It is possible for spreadsheets and plain text files to contain information on occasion. The internet, or World Wide Web, is another main data source.

Different processes:

Data must be cleansed, integrated, and chosen before it can be passed to the server running the database or data warehouse. Due to the data's heterogeneity in source and format, it cannot be simply fed into the data mining process without first ensuring its accuracy and completeness. Therefore, cleaning and unifying the first data set is essential. We will gather more data than is necessary from several sources; then, we can provide only the relevant data to the server. We underestimated how difficult these procedures would be. Data may undergo a number of procedures throughout the selection, integration, and cleansing processes.

Database or Data Warehouse Server:

The raw, processed data is stored on the server of the database or data warehouse. Based on the user's request, the server then retrieves relevant data retrieved from data mining.

Data Mining Engine:

A data mining system would be incomplete without the data mining engine. Association, characterisation, classification, clustering, prediction, time-series analysis, and other data mining activities are all part of its functionality. Put simply, our data mining architecture is built upon data mining. It consists of tools and programs that are used to get understanding and insights from data that has been saved in the data warehouse from a variety of sources.

Pattern Evaluation Module:

Determining the level of pattern research using a threshold value is mainly the responsibility of the Pattern evaluation module. In order to zero in on interesting patterns, it works in tandem with the data mining engine.

This part of the process often makes use of stake measures, which work in tandem with the data mining modules, to zero in on interesting trends. To exclude previously identified patterns, it may use a stake threshold. The data mining methods utilised to deploy them will determine whether the pattern evaluation and mining modules are coordinated. It is highly recommended to incorporate pattern stake evaluation into the mining operation to limit the search to only interesting patterns for efficient data mining.

Graphical User Interface:

One way in which the data mining system and the user can communicate is through the GUI module. Without understanding the intricacies of the process, this module enables the user to utilise the system efficiently and with ease. When the user enters a query or task and wants to see the results, this module works with the data mining system to do just that.

Knowledge Base:

Throughout data mining, the knowledge base is a great asset. To direct the search or assess the significance of the pattern results might be useful. Users' perspectives and experience data might be stored in the knowledge base, which could be used for data mining. Data mining engines often use inputs from knowledge bases to improve the accuracy and reliability of their results. On a regular basis, the pattern assessment module updates and retrieves inputs from the knowledge base through interactions.

SELF-ASSESSMENT QUESTIONS – 2

Fill in the blanks:	
8	The _________ layer in data mining architecture is responsible for extracting, transforming, and loading data from various sources into the data warehouse.
9	In a typical data mining architecture, the _________ serves as the core component that applies algorithms to analyze patterns and relationships within the data.

7. TYPES OF DATA MINING

The several subfields that make up data mining fall into one of three main categories: descriptive, predictive, or prescriptive

- Descriptive data mining requires describing and summarising the features of a dataset. Data exploration, data understanding, pattern and trend identification, and meaningful data summarisation are common goals of this subset of data mining.

The goal of this subfield of data mining is to discover hidden structures inside datasets by identifying links and trends. Question such, "What are the most common patterns or relationships in the data?" can be answered with descriptive data mining, which is commonly used for data summarisation and exploration. Is the data organised into any meaningful groups or clusters? How do we identify and understand the data's outliers?

Descriptive data mining often makes use of the following methods:
Grouping data into clusters:
Cluster analysis: Finding clusters of data points with shared attributes is the goal of this method. The three main applications of clustering are summary generation, anomaly detection, and segmentation.
Association Rule: One method for discovering correlations in datasets is association rule mining. Use it to find patterns in your transaction data or to find events that happen at the same time.
Visualization: When data is represented visually, consumers may be better able to spot trends and patterns than when looking at the raw data alone. This method is known as visualisation.

- Predictive data mining is the process of constructing models with the purpose of making predictions or forecasts regarding future occurrences or results utilising data. Predicting future occurrences or outcomes by identifying and modelling correlations between distinct variables is a common use case for this data mining technique. Some common techniques used in predictive data mining include:

Decision trees: With this method, one can build a model that, given the values of a number of input variables, can forecast the value of a target variable. Classification problems frequently employ decision trees.

Neural networks: Using this method, one can build a model with pattern recognition capabilities. Natural language processing, picture identification, and audio recognition are common applications of neural networks.

Regression analysis: With this method, one can build a model that, given the values of a number of input variables, can forecast the value of a target variable. Prediction jobs frequently employ regression analysis.

- **Prescriptive data mining** proposes courses of action or choices based on analysis of data and models. Process optimisation, resource allocation, and other decision-making aides are common uses of this subset of data mining.

In general, these three subfields of data mining are utilised for data exploration, modelling, and decision-making. They have many uses because they are effective at finding patterns and relationships in large data sets.

8. BENEFITS OF DATA MINING

"Data mining" refers to the practice of discovering new patterns and insights in massive datasets. Among its many advantages, this instrument is both powerful and versatile:

1. **Improved decision-making –** Data mining's ability to improve decision-making is one of its primary advantages. Data mining is the process of analysing large amounts of data in search of previously unseen patterns and trends. This process can yield useful insights and information that can enhance decision-making.

2. **Increased efficiency and productivity –** Data mining may also aid businesses in being more productive and efficient. Organisations can save time and resources through data mining, which automates and streamlines the data analysis process.

3. **Reduced costs –** Organisations might also benefit from data mining in terms of cost reduction. Data mining helps businesses save money and increase profits by finding and fixing wasteful and inefficient processes.

4. **Increased customer satisfaction** Improving customer happiness is another possible use of data mining. Data mining is a powerful tool that businesses can use to gain a deeper understanding of their customers, allowing them to create products and services that are more tailored to their needs.

5. **Improved risk management** Risk management is another area that can benefit from data mining. Data mining is a powerful tool that organisations may use to make better strategic decisions, find and fix vulnerabilities, and detect dangers.

In conclusion, data mining is an effective technology with numerous advantages for businesses. Organisations can benefit from data mining in many ways, including enhanced decision-making, increased efficiency and production, decreased costs, higher customer satisfaction, and improved risk management.

9. LIMITATIONS OF DATA MINING

Data mining allows for the extraction of valuable insights and information from massive data sets in a powerful and versatile manner. Data mining does have its uses, but it also has its limitations and difficulties. Here are a few of data mining's most significant drawbacks:

1. Data quality – The data's quality is a major constraint of data mining. If the data used for data mining is of low quality, the results may be incorrect or misleading because data mining is only as good as the data used for it.

2. Model bias – The possibility of bias in the resulting models is another drawback of data mining. The models constructed from data run the risk of being biassed and failing to effectively portray the underlying relationships in the data if the data is not representative of the population or if the data collection or analysis processes are biassed.

3. Ethical considerations – There are moral questions that data mining brings up as well. It is imperative that organisations carefully and in accordance with applicable rules and regulations handle any personally identifiable information (PII) that may be gathered and processed.

4. Technical challenges Data mining, particularly when working with complicated and massive datasets, can be extremely demanding. In addition to requiring time and resources, specialised knowledge and expertise may be necessary to extract valuable information and insights from data.

While data mining offers many advantages, it is not without its flaws and restrictions. Businesses need to be cognisant of these constraints and work to overcome them if they want their data mining to be trustworthy, accurate, and ethical.

10. APPLICATIONS OF DATA MINING

Data mining has been instrumental in numerous measurable advancements across various domains. Now, let's talk about some of the many uses of data mining:

Scientific Analysis: Every day, massive amounts of data are being produced by scientific models. This encompasses information gathered from nuclear power plants, statistics on human behaviour, etc. These data can be analysed using data mining techniques. The rate at which we can collect and store new data has surpassed the rate at which we can evaluate previously collected data. Example are - Analysis based on scientific principles, Bioinformatics sequence analysis, Classification of astronomical objects and Decision support on healthcare.

Intrusion Detection: Any unlawful action on a digital network is called a network intrusion. Theft of important network resources is a common consequence of network intrusions. When looking for abnormalities, intrusion detection, or network attacks, data mining techniques are crucial. When applied to big data sets, these methods aid in the extraction and refinement of relevant and usable information. Intrusion Detection System data can be better organised with the use of data mining techniques. Network traffic can be used by an Intrusion Detection system to trigger alarms in the event that the system is invaded from outside. As an example, we can find instances of misuse • or find security breaches

Business Transactions: Remembered forever is every part of the economy. Examples of such interactions include time-based business talks with both internal and external parties. The ability to make competitive judgements in real time based on correct data is a critical concern for businesses in today's fiercely competitive industry. Data mining can shed light on these business deals, as well as marketing tactics and decision-making procedures. To name a few examples: • Personalised advertising • Investing

One of the most common applications of big data in business is churn prediction, which involves consumer segmentation.

Market Basket Analysis: Market Basket Analysis is one way to look at a customer's food spending in detail. We can find out which products are most popular by using this technique. Businesses can use data mining technologies to their advantage by advertising promotions, deals, and discounts, as shown in this study.

Example:

- In order to improve customer service, increase the effectiveness of cross-selling opportunities, and increase the response rate to direct mail, the marketing and sales teams are implementing data mining concepts.

- Through the discovery of trends and the prediction of customer churn, data mining can aid in customer retention efforts.

- A number of domains make use of the data-mining concept, including risk assessment and fraud detection, to identify instances of suspicious or improper activity.

Education: The area of education is investigated through data mining using the Educational Data Mining (EDM) method. The patterns that this technique generates are useful for both students and teachers. The following educational objectives can be achieved by the utilisation of data mining EDM:

- Acknowledging prospective college students
- Anticipating the profiling of students
- Developing curricula
- Forecasting student placement opportunities
- Predicting student performance
- Teachers' teaching performance

Research: For academic purposes, data mining allows for faultless prediction, classification, clustering, association, and grouping of data. Data mining produces one-of-a-kind rules for discovering outcomes. It is common practice in data mining research to develop both a training and testing model. A method for gauging the suggested model's accuracy is the training/testing model. The reason we divide the dataset into a training set and a testing set is why it's referred to as Train/Test. When building a training model, one uses a training data set; when testing, one uses a testing data set.

Healthcare and Insurance: By analysing the results of its new deals force efforts, the pharmaceutical industry can better target valuable doctors and choose the most effective promotional strategies for the next months. In contrast, data mining can aid the insurance industry in predicting which consumers will purchase new policies, identifying patterns of behaviour among high-risk clients, and detecting instances of customer fraud. Analysing the claims, or the medical treatments that are associated with each other. Determine which medical treatments have been effective in treating certain diseases. Uses patient behaviour modelling to forecast the likelihood of office visits.

Transportation: Use data mining to find the ideal customers for your varied transportation company's services if you have a big direct sales team. Using information mining, a big consumer goods company can enhance its business cycle with retailers. Analyse loading patterns. • Get the distribution schedules for each outlet

Financial/Banking Sector: By mining its massive database of consumer spending, a credit card issuer can zero in on the demographic most likely to sign up for a new line of credit.

- Credit card fraud detection is one of our strong suits.
- Discover who your "loyal" consumers are.
- Obtaining data pertaining to clients.
- Categorise purchases made with credit cards by kind of consumer.

SELF-ASSESSMENT QUESTIONS – 3

Fill in the blanks:	
10	Which of the following is a key benefit of data mining? a) Reduces data redundancy b) Helps in discovering hidden patterns and trends c) Eliminates the need for databases d) Increases data storage costs
11	Which of the following is a limitation of data mining? a) It is always 100% accurate b) It requires complex algorithms and high computational power c) It does not require any data preprocessing d) It replaces human decision-making completely
12	Which of the following industries widely use data mining for fraud detection? a) Healthcare b) Banking and Finance c) Education d) Entertainment

13	One of the main concerns of data mining is: a) Data security and privacy issues b) Easy implementation without skilled professionals c) Lack of data storage options d) Limited usage in real-world applications
14	Which of the following is NOT an application of data mining? a) Customer relationship management (CRM) b) Market basket analysis c) Handwritten letter typing d) Healthcare diagnosis

11. SUMMARY

- Data mining is the process of extracting meaningful patterns, trends, and insights from large datasets using statistical, machine learning, and artificial intelligence techniques. It plays a crucial role in transforming raw data into useful information that helps in decision-making across various industries.
- Data itself refers to raw facts and figures, which when processed and analyzed, become information that is meaningful and useful. There are different types of data, including structured, semi-structured, and unstructured data, each serving unique purposes in analytical processes.
- The history of data mining can be traced back to the evolution of database systems, machine learning, and statistical analysis, with its development accelerating due to advancements in computing power and big data technologies.
- The advantages of data mining include improved decision-making, fraud detection, market trend analysis, and personalized recommendations. However, it also comes with disadvantages such as data privacy concerns, high computational requirements, and potential biases in models.
- The limitations of data mining include dependency on high-quality data, complex implementation, and challenges in handling real-time data.
- There are different types of data mining, such as predictive data mining, descriptive data mining, classification, clustering, and association rule mining, each catering to different analytical needs.
- The architecture of data mining consists of components like data sources, data preprocessing, data mining engines, pattern evaluation, and visualization tools, all working together to derive valuable insights.
- The applications of data mining span various fields, including healthcare, finance, retail, education, and cybersecurity, where it is used for predictive analytics, risk assessment, customer segmentation, and fraud detection.
- Overall, data mining has become an essential technology in the modern data-driven world, helping organizations leverage vast amounts of data to gain a competitive edge while addressing the challenges associated with its implementation and ethical considerations.

12. GLOSSARY

Data Mining	-	The process of discovering patterns, trends, and useful information from large datasets using statistical, machine learning, and artificial intelligence techniques.
Data	-	Raw facts, figures, and observations that can be processed to generate meaningful insights.
Information	-	Processed data that has been organized and structured to be meaningful and useful for decision-making.
Structured Data	-	Data that is organized in a predefined format, such as rows and columns in databases.
Unstructured Data	-	Data that does not have a predefined structure, such as images, videos, and social media posts.
Semi-Structured Data	-	A mix of structured and unstructured data, such as XML or JSON files.
Big Data	-	Extremely large and complex datasets that require advanced tools and techniques to store, process, and analyze.
Classification	-	A data mining technique that categorizes data into predefined groups based on patterns.
Clustering	-	A technique that groups similar data points together without predefined categories.
Association Rule Mining	-	A data mining technique used to discover relationships between variables in large datasets, commonly used in market basket analysis.
Predictive Analytics	-	The use of data mining techniques to forecast future trends and behaviors based on historical data.
Descriptive Analytics	-	Analyzing past data to identify patterns and trends without making future predictions.
Data Preprocessing	-	The process of cleaning, transforming, and organizing raw data before applying data mining techniques.

Data Warehouse	-	A centralized repository that stores and manages large volumes of structured data for analysis and reporting.
Data Mining Architecture	-	The framework and components that define how data mining processes are structured, including data sources, preprocessing, mining engines, and visualization tools.
Machine Learning	-	A subset of artificial intelligence that enables computers to learn patterns from data and make decisions without explicit programming.
Privacy and Security in Data Mining	-	The ethical considerations and legal aspects of ensuring that data mining does not compromise sensitive or personal information.
Applications of Data Mining	-	The various fields where data mining is applied, including healthcare, finance, retail, education, and cybersecurity, to improve decision-making and operational efficiency.

13. TERMINAL QUESTIONS

1. What is data mining?
2. What are the main types of data?
3. Why is data mining important?
4. What is the difference between data and information?
5. What are some advantages of data mining?
6. What are the key limitations of data mining?
7. What are the main types of data mining techniques?
8. What is a data warehouse, and how does it relate to data mining?
9. What is the role of machine learning in data mining?
10. How is data mining applied in real-world scenarios?

14. ANSWERS

14.1. Self-Assessment Answers:

1. Answer: A) Raw facts and figures without meaning
2. Answer: D) Mixed Data
3. Answer: A) It helps in making informed decisions
4. Answer: A) Input → Processing → Storage → Output → Feedback
5. Answer: C) A handwritten letter or a scanned document
6. Answer: B) A relational database table
7. Answer: B) To convert raw data into meaningful information
8. Answer: Data integration
9. Answer: Data mining engine
10. Answer: b) Helps in discovering hidden patterns and trends
11. Answer: b) It requires complex algorithms and high computational power
12. Answer: b) Banking and Finance
13. Answer: a) Data security and privacy issues
14. Answer: c) Handwritten letter typing

14.2. Terminal Question Answers

Answer 1: Data mining is the process of analyzing large datasets to discover patterns, relationships, and useful insights using machine learning, statistical methods, and artificial intelligence.(For more information Refer Section- 1.1)

Answer 2: The main types of data include structured data (organized in rows and columns), unstructured data (text, images, videos), and semi-structured data (JSON, XML). (For more information Refer Section-1.3.1)

Answer 3: Data mining helps businesses and organizations make informed decisions by identifying trends, predicting future events, and improving efficiency in various applications such as marketing, healthcare, and fraud detection. (For more information Refer Section-1.3)

Answer 4: The difference between data and information is as follows: Data refers to raw, unprocessed facts and figures that have no meaningful context. It can be in the form of numbers, text, images, or symbols. Examples include a list of sales transactions, temperature readings, or survey responses. Information is processed, organized, and structured data that has been analyzed to provide meaning and context. It is useful for decision-making. For example, analyzing sales data to determine which product is selling the most in a specific region transforms raw data into valuable information.

Answer 5: Some advantages of data mining include improved decision-making, fraud detection, personalized recommendations, cost savings, and better customer insights. (For more information Refer Section-1.8)

Answer 6: Limitations include data privacy concerns, high computational costs, data quality issues, potential biases in data, and ethical considerations. (For more information Refer Section-1.9)

Answer 7: The main types include classification, clustering, association rule mining, regression, and anomaly detection. (For more information Refer Section-1.6)

Answer 8: A data warehouse is a centralized storage system that collects, integrates, and manages structured data from multiple sources for analysis and reporting. It is designed to support business intelligence and decision-making by providing a consolidated view of historical and current data. Unlike operational databases, which handle day-to-day transactions, a data warehouse is optimized for analytical queries and large-scale data processing. Data mining and data warehousing are closely related, as data mining techniques are applied to warehouse data to discover patterns, relationships, and trends. The structured and historical nature of a data warehouse makes it an ideal source for data mining, enabling businesses to perform predictive analysis, customer segmentation, fraud detection, and market trend analysis. By leveraging data mining on warehouse data, organizations can gain valuable insights that drive strategic decisions and improve overall efficiency.

Answer 9: Machine learning enables data mining systems to learn from data, recognize patterns, and make predictions without explicit programming. (For more information Refer Section-1.10)

Answer 10: Data mining is applied in various industries such as healthcare (disease prediction), finance (fraud detection), retail (customer segmentation), education (personalized learning), and cybersecurity (threat detection). (For more information Refer Section-1.10)

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 4

DCA2207

DATA MINING AND VISUALISATION

Unit – 2

Operational Database and Data Warehouse

TABLE OF CONTENTS

1. INTRODUCTION

Learners must study this subject as it establishes a solid base for managing and analysing vast data quantities, essential for today's data-centric industries. These courses improve students' proficiency in data modelling, ETL processes, and advanced analytics, equipping them for positions such as data analysts, data scientists, and database administrators. They integrate with other fields of computer science, offering practical, hands-on learning while enhancing analytical reasoning and problem-solving skills. In addition, these programmes equip students with the necessary skills and knowledge to excel in advanced positions and explore research possibilities. This ensures they stay competitive and well-informed about current and emerging technology developments.

In this unit, learners will learn that Data warehousing and OLAP (Online Analytical Processing) are essential elements of modern business intelligence, enabling thorough data analysis and well-informed decision-making. A data warehouse is a centralised storage system that consolidates data from numerous sources, facilitating complex searches and analysis of historical data. It is characterised by its focus on subjects, integration, changes throughout time, and lack of volatility. Data warehouses differ from OLTP systems in that they prioritise deep analysis and strategic insights rather than transactional efficiency. Data warehouses are composed of data sources, ETL procedures, optimised storage, and access tools. The multidimensional data model enables versatile data analysis from many viewpoints. OLAP systems improve query performance and enable complicated, multidimensional inquiries through operations such as slicing, dicing, and drilling down. These technologies are utilised in diverse sectors, like healthcare and retail, to enhance operational efficiency and strategic planning, showcasing their substantial influence on enhancing business intelligence.

To study this unit learners should commence by understanding fundamental principles and essential vocabulary. Clear representations and active engagement with technological tools are essential for understanding complex subjects. To improve understanding, one can follow an organised learning route, make use of internet resources, and participate in collaborative learning activities such as study groups or peer conversations. Engaging in projects and keeping up of industry developments are essential for practical implementation and maintaining relevance.

1.1. Learning Objectives:

At the end of this topic, students should be able to:

- Define the concept of a data warehouse and its key characteristics.
- Summarise the multidimensional data model and its significance in data analysis.
- Perform basic OLAP operations such as slicing, dicing, and drill-down on sample datasets.
- Analyse the impact of data warehousing on decision-making processes in various industries.

2. OPERATIONAL DATABASE

An operational database, often called an OLTP database, is a type of database system that can handle real-time data changes. This is commonly used in industries that require instantaneous processing and storage of transactions, such as banking, retail, and telecommunications. Daily transactional operations within organisations are supported by operational databases, which are also called OLTP databases. Data integrity, concurrency, and consistency are guaranteed by these databases, which are built to efficiently manage huge quantities of real-time transactions. Transactional processing, including inserting, updating, and removing records, is the primary focus of operational databases.

Operational Database Features

- Transactional Processing: Operational databases are particularly well-suited for real-time processing of sales transactions, inventory updates, and customer interactions.
- Normalised Data Structures: Normalised data models, which divide data into numerous interconnected tables, are commonly used by operational databases to reduce data duplication and guarantee data consistency.
- Concurrency Control: To avoid conflicts that can arise when numerous users try to access or edit the same data at the same time, operational databases have systems for handling concurrent transactions.
- Indexes for Performance: Optimal query performance and efficient access to specific data subsets are made possible by indexes in operational databases, which in turn allow for speedy data retrieval.

Information needed to run a business on a daily basis can be better managed and stored in an operational database. These databases manage transactional data in real-time and guarantee the seamless operation of business processes. It is compatible with Online Transaction Processing (OLTP), a method whereby a large number of frequent, tiny activities (such as inserting, updating, and retrieving information) take place all at once. Order processing, account balance updates, aircraft reservation management, and patient data entry are all examples of such operations. These operations must be fast, reliable, and secure, as even minor errors can disrupt critical business processes.

Role in Day-to-Day Business Operations

The day-to-day operations of every organisation rely on operational databases. One of their responsibilities is managing transactions, which includes taking care of real-time sales, deposits, payments, and reservations from customers. Keeping accurate records of stock levels, data from the supply chain, or personnel schedules is what inventory and resource tracking is all about. Customer profiles, support tickets, and service requests are managed in CRM systems.

- Decision Support in Real Time: Giving staff and management quick access to real-time data to help them make quick operational decisions. Maintaining correct and trustworthy data after a transaction has been committed (such as a money transfer from one account to another) is known as data consistency.
- Integration and collaboration: letting different users or programs (such as POS systems, mobile apps, or web portals) access the same real-time data at the same time. Most contemporary organisations, including banks, airlines, e-commerce platforms, and hospitals, rely on operational databases to run smoothly in real time.

Difference Between Operational Databases and Analytical Databases (OLTP vs OLAP)

Operational databases are often confused with analytical databases, but they serve **different purposes.**

Table 1: Difference between OLTP and OLAP

Aspect	Operational Database (OLTP)	Analytical Database (OLAP)
Primary Purpose	Supports day-to-day operations (transactions, real-time updates)	Supports analysis, decision-making, and long-term planning
Data Type	Current, real-time transactional data	Historical, aggregated, summarised data
Operations	Frequent read/write (insert, update, delete) operations	Complex queries, aggregations, and reporting
Users	Operational staff (cashiers, salespeople, clerks)	Analysts, managers, executives
Speed Requirement	Must be very fast for simple queries and updates	Optimised for complex queries, not for frequent updates

Aspect	Operational Database (OLTP)	Analytical Database (OLAP)
Database Size	Relatively smaller (only operational data kept)	Very large (years of historical data stored)
Example Systems	Banking systems, airline booking, e-commerce orders	Data warehouses, BI tools, dashboards
Example Technology	Oracle, MySQL, SQL Server (OLTP mode)	Snowflake, Teradata, Amazon Redshift, OLAP cubes

The main idea is that OLTP databases allow you to "run the business" in real-time through transactional data. The "Analyse the business" (historical, decision-making) function is part of analytical databases like OLAP.

2.1. Characteristics of Operational Databases

1. Real-time Data Processing

Because of their real-time processing capabilities, operational databases automatically capture and store any data that changes throughout a transaction, whether it's a bank transfer, an internet purchase, or a hospital updating a patient record. All stakeholders, including customers, employees, and managers, are guaranteed to view the most recent information quickly thanks to real-time processing. Take online rail ticket booking as an example. The database refreshes in real-time to ensure that no other user may book the same ticket. This functionality is essential for applications that cannot handle delays or outdated data.

2. High Availability & Reliability

Since operational databases are relied upon by organisations for continuous operations, their availability is of the utmost importance. To ensure that the database keeps running in the event of a component failure, systems are designed with redundancy, such as backup servers, failover systems, and cloud replication. This is known as high availability (HA). The system stores transactions accurately, prevents data loss in the event of a breakdown, and has recovery measures (such as backups and logs) to get back up and running. Bank customers' money and confidence are at stake in the event of an outage, thus online banking databases must be available at all times.

3. Support for Concurrent Users

Multiple users can access operational databases at the same time. A large number of users (hundreds or thousands) can see and edit data simultaneously. Consider an online marketplace like Amazon, where millions of users may be perusing products, making purchases, and paying all at once. To keep transactions from interfering with one another, we use concurrency management techniques such as locking, isolation levels, and timestamps.

Without proper concurrency control, two users might accidentally withdraw the same amount or reserve the same seat.

4. Transactional Nature (ACID Properties)

Reliability of transactions is ensured by operational databases adhering to ACID properties: An atomic unit is an all-or-nothing transaction. As an example, a transfer from one account to another will only result in a debit from that account. A transaction must move the database from a legitimate state to another in a consistent manner, without breaking any constraints. As an example, if the rules prohibit it, the bank balance cannot fall below zero. Isolation: Results of concurrent transactions should remain independent of each other. For instance, two clients who are purchasing the final item should not be processed at the same time. Stability: A transaction is recorded forever once it is committed, regardless of whether the system fails right after. These characteristics guarantee the accuracy, reliability, and safety of data in real-time processes.

5. Large Volume of Short, Frequent Transactions

Rather than long analytical queries, operational databases are designed to accommodate a large number of smaller activities. Typically, very little quantities of data are involved in each action (update, insert, delete, easy retrieval). Scanning products at the checkout of a grocery store, updating a customer's order status, registering a payment, and verifying account balances are all examples. These procedures occur on a daily basis in major organisations at a rate of millions. To handle this deluge, the system needs to be highly optimised for speed and throughput.

2.2. Types of Operational Databases

The data model, design, and deployment environment of an operational database determine its type. Here are the most common types:

1. Relational Databases (RDBMS)

Relational databases categorise information using rows representing records and columns representing attributes. It is based on relational algebra and set theory, ensuring structured data retrievals utilised for data retrieval. Make sure that the data is consistent by using constraints and keys (both main and foreign). They are trustworthy for operational usage since they support transactions and have ACID features. Some examples include Microsoft SQL Server, PostgreSQL, Oracle Database, and MySQL. Financial services, online shopping, enterprise resource planning, and payroll automation. For operational databases, RDBMS is the gold standard because of its dependability, standardisation, and ease of use.

2. Object-Oriented Databases (OODBMS)

The principles of object-oriented programming and database technology are brought together in an Object-Oriented Database. Like objects in programming languages like C++ or Java, data is stored in this format, complete with properties and methods. It can handle complicated data types like scientific data, CAD/CAM designs, pictures, and multimedia.

It supports inheritance, encapsulation, and polymorphism.

er at odds with one another. db4o, ObjectDB, GemStone/S, Versant OODBMS are other examples. Real-time simulations, engineering applications, medical imaging, and multimedia databases are some of the use cases. When it comes to efficiently managing complex objects and multimedia data, OODBMS is perfect.

3. Distributed Databases

Data stored in several physical locations (servers or sites) yet accessible as one logical database is known as a distributed database. It distributes data among sites to ensure dependability and speed. Provides data fragmentation and replication to boost availability and performance. Makes sure everything is open and accessible, so users aren't obligated to know the exact location of their data. Harmonisation, concurrency control, and recovery methods that are state-of-the-art are necessary. Some examples include CockroachDB, Apache Cassandra, and Google Spanner. Global organisations, communication systems, enterprise apps hosted in the cloud, and systems for airline reservations are examples of use cases. For large-scale, globally dispersed businesses that require fault tolerance and high availability, distributed databases are a must.

4. Cloud-based Operational Databases

Third-party suppliers host and administer these operational databases in a cloud environment. By offering Database-as-a-Service (DBaaS), they save businesses from the burden of handling hardware and maintenance in-house. Ability to handle spikes in transactions quickly: scalable on demand. The reliability and resilience of cloud services are guaranteed by their providers. Infrastructure costs are decreased by adopting a pay-as-you-go pricing model. Support for operational workloads using both SQL and NoSQL databases. MongoDB Atlas, Microsoft Azure SQL Database, Amazon RDS, and Google Cloud SQL are a few examples. Examples of Use Cases: New companies, software as a service platform, online marketplaces, and mobile and web apps. Modern IT infrastructures increasingly value cloud-based databases for their cost-effectiveness, scalability, and flexibility.

2.3. Components of an Operational Database System

All the moving parts of a functional database system must be in place for real-time transactions to be processed quickly, reliably, and securely.

1. Engine for Databases

The backbone of the database system that handles data storage, retrieval, and management. It Runs SQL and NoSQL queries oversees the storage of data in memory and on disc enhances the efficiency of queries by implementing strategies such as indexing, caching, and query optimisation. Allows programs to access and modify data through specified interfaces. He database engine ensures fast and reliable data access, serving as the system's core component. Some examples of database engines are MySQL InnoDB, Oracle Database, and SQL Server Database.

2. Transaction Manager

Verifies that the ACID principles are adhered to by all database operations (atomicity, consistency, isolation, durability). Assists in initiating, overseeing, and concluding transactions. If an operation succeeds or fails, it will either be committed or rolled back.

Tracks modifications for recovery by maintaining transaction logs. Keeps data intact and keeps the system dependable in the face of crashes or mistakes. The transaction manager checks if the debit and credit operations are successful, or that they are unsuccessful, while transferring funds between accounts.

3. Concurrency Control

A system that allows numerous users to access and edit the database at the same time without causing disputes. Avoids issues such as deadlocks, unclean reads, and lost updates.

Uses methods like locking, timestamps, and isolation levels Read Uncommitted, Read Committed, Repeatable Read, Serialisable where applicable. Maintains data integrity even when faced with demanding workloads. keeps things honest and dependable even when a lot of people are looking at the same data (like when people order plane tickets online).

4. Recovery Manager

one who, in the event of a failure (such as a power loss, software fault, or hardware crash), is in charge of returning the database to its original, consistent condition. Keeps records of backups and undo/redo operations, guarantees that committed transactions will remain in effect. It can implement rollback for transactions that aren't finished and rollforward to apply modifications from logs again. Prevents data loss and guarantees rapid system recovery in the event of failure. To illustrate the point, the recovery manager will never leave a transaction in an incomplete state; for instance, if the system crashes after a money transfer, it will either be finished or undone.

5. Security Manager

Guarantees the database is safe from abuse, intrusion, and misuse. Manages authentication, which is the process of confirming the identity of a user. Oversees authorisation, which involves granting users certain permissions, to safeguard information, employs measures such as firewalls, encryption, and audit trails Makes sure everyone follows data privacy laws (including HIPAA and GDPR). Data security is of the utmost importance since it prevents unauthorised individuals from accessing sensitive information and keeps sensitive documents safe from cyber dangers.

Use Cases for Operational Databases

Industries that rely on operational databases for real-time data processing, high availability, and short reaction times find them indispensable. By facilitating consistent data, concurrent access, and quick transactions, they are the backbone of how businesses run on a daily basis. The following are examples of typical and useful applications:

1. **E-commerce**

   In order to process a large number of consumer transactions all at once, online shopping platforms depend significantly on operational databases. They handle things like product catalogues, up-to-the-minute inventory counts, information about shopping carts, processing payments, and confirming orders. As an illustration, to provide a smooth purchasing experience, the database immediately changes the stock levels, creates an order ID, and processes payment whenever a customer makes a purchase.

2. **Banking and Finance**

   Banks and other financial organisations rely on operational databases to handle millions of transactions every day. These transactions include deposits, withdrawals, money transfers, and credit card payments. These databases ensure real-time updates on account balances, fraud detection, and regulatory compliance. For instance, rapid logging of mobile banking transactions and ATM withdrawals helps prevent overdrafts and ensures accuracy.

3. **Customer Relationship Management (CRM)**

   Information about customers, including their contact details, history of interactions, buying habits, and comments, is stored and managed by customer relationship management systems through operational databases. As a result, companies can keep tabs on potential customers, manage their contacts with customer care, and tailor their marketing efforts to each individual. When interacting with customers, sales and support personnel can rest assured that they have access to the most current information thanks to real-time data.

4. **Supply Chain Management (SCM)**

   From the acquisition of raw materials to the shipment of completed items, operational databases play a crucial role in keeping tabs on shipments, their records of stock, warehouse activities, suppliers, shipments, and demand predictions are accurate and up-to-date. When a shipment is behind schedule, for instance, the database immediately changes the status, letting businesses respond swiftly to fix the problem.

SELF-ASSESSMENT QUESTIONS – 1

Choose the Right Answer:	
1	The primary role of an operational database is to: a) Support complex analytical queries b) Store historical data for decision-making c) Handle day-to-day transaction processing d) Generate business intelligence reports

2	Which of the following is a type of operational database? a) Relational Database b) Hierarchical Database c) Network Database d) All of the above
3	Which characteristic best describes operational databases? a) Data is updated in real-time b) Data is mainly historical c) Optimised for analytical processing d) Used only for batch processing
Fill in the blanks:	
4	An operational database is mainly designed for ___________ processing, where real-time transactions are managed.
5	A key characteristic of operational databases is that they ensure ___________ and ___________ during concurrent transactions.

3. DATA WAREHOUSE

A data warehouse is a specialised type of data management system designed to support business intelligence (BI) operations, with a focus on analytics. These systems are dedicated to executing complex queries and conducting research, typically housing vast amounts of historical data. Data warehouses aggregate information from various sources, including transaction systems and application log files, to provide a unified view for analysis.

A data warehouse brings together and organises a lot of data from different sources. Because it can analyse data, businesses can get useful business insights from it that help them make better decisions. It keeps track of history over time, which can be very helpful for data scientists and business experts. A data warehouse serves as the standard source of truth for an organisation by consolidating, storing, and managing vast amounts of data from various sources into a single, consistent repository.

The following things frequently appear in a standard data warehouse:
You can store and handle data in a relational database.

- A process to prepare the data for research through extraction, loading, and transformation (ELT).
- Ability to do statistical analysis, report, and data mining.
- Client research tools that help business users see and understand data.
- Additionally, by applying data science and AI methodologies, modern analytical tools produce useful insights, and these tools have graph and geographical functionalities, allowing for in-depth data analysis.

Organisations can also choose a solution combining machine learning, processing of transactions and analytics in real time across data lakes and warehouses in a single MySQL Database service. This way, organisations don't have to deal with the risk, cost, complexity, and delay of extract, transform, and load (ETL) duplication.

Characteristics of a Data Warehouse

One big and special benefit of data warehouses is that they let companies examine many different kinds of data, find useful patterns, and keep records.

This big benefit is made possible by four unique features explained by computer scientist William Inmon, known as the "father of the data warehouse." This description says that data warehouses are:

i. Subject-oriented: They can examine data about a certain topic or job, like sales data, customer, product, etc. The primary focus is on modeling and analysing data for decision-makers instead of routine operations or processing of transactions. They present a concise and simple perspective on specific topics by eliminating irrelevant data that does not contribute to the decision-making process.

ii. Integrated: Data storage is essential for ensuring consistency among diverse data types emerging from multiple sources. These stores are created by combining many data sources, including relational databases, flat files, and online transaction systems.

Cleaning and data integration strategies are used to carry out this task. These strategies help in the standardisation of naming standards, encoding structures, attribute measurements, and other features across a wide range of data sources.

Example: Consider hotel pricing which could include elements such as currency conversion, taxes, and included services like breakfast. When data is transported to a centralised warehouse, it undergoes conversion to ensure uniformity and consistency.

iii. Nonvolatile: It is steady and doesn't change once it is in a data warehouse. A physically distinct repository of information derived from the operating environment. Data warehouse environment does not undergo operational updates of data. It does not necessitate the use of transaction processing, recovery, and concurrency control techniques.

iv. Time-variant: Data warehouse analysis studies the time-based evolution of occurrences.

An efficiently built data warehouse will provide quick query response, accommodate large volumes of data, and offer users the flexibility to selectively analyse data to satisfy various requirements, whether at a macro or micro level. The data warehouse serves as the operational foundation for middleware business intelligence (BI) systems, which provide end users with reports, dashboards, and other interfaces.

OLTP Vs Data Warehousing

Databases often require an activation of real-time SQL transaction processing to serve time-sensitive applications like e-commerce. Online transaction processing (OLTP) refers to this specific sort of processing.

Queries used for Online Transaction Processing (OLTP) and Decision Support Systems (DSS) have different performance demands. OLTP queries are somewhat less complex, and the estimated response time varies from milliseconds to a few minutes. In contrast, DSS queries are characterised by their increased complexity, resulting in response times that might vary from a few minutes to several hours. Traditionally, OLTP and DSS were managed on separate systems since they have different uses and require specific performance capabilities. Businesses face higher complexity demands and are under pressure to reduce overall costs. As a result, data servers are being forced to accept a mixed workload environment, in which the online transaction processing (OLTP) system also undertakes the duties of a decision support system. These environments that combine several tasks are called operational decision support systems.

OLTP System:

An OLTP system is specifically built to optimise transaction throughput, however it is not suitable for storing actual information and historical data for business analytical purposes.

There is a difficulty in promptly responding to spontaneous queries. The ability to retrieve information quickly is nearly impossible. The data is inconsistent and constantly changing, with duplicate entries and possible missing entries. The raw data provided by OLTP is vast but not easily understandable. A typical OLTP query involves a basic aggregation, such as determining the current account balance for a specific customer.

OLTP and Data Warehousing

Table 2: Difference between OLTP and Data warehousing

Characteristics	OLTP	Data Warehousing
Role	Execute daily operational tasks	the process of retrieving and analysing information.
Structure	RDBMS	RDBMS
Data Model	Normalised	A data warehouse employs a multidimensional data model that systematically gathers and stores data from various sources in an organised format. The data is organised in a multi-dimensional format,

		allowing for efficient analysis and reporting.
Accessibility	Online Transaction Processing (OLTP) is a database system that uses Structured Query Language (SQL) for managing and processing transactions.	An expansion of SQL that incorporates data analytic capabilities.
Type of Data	refers to the data that is essential for the functioning of an organisation.	refers to the process of collecting and storing data that is used to analyse business operations.
Aggregation of data	Changing incomplete	Data warehousing involves the storage and organisation of historical and descriptive data.

3.1. Data Warehouse Architecture

A data warehouse can be built using three approaches:

- A top-down approach
- A bottom-up approach
- A combination of both approaches

The top-down approach starts with developing a full design and strategic planning. It is beneficial when the technology is well-established and widely recognised and when the business challenges that need to be addressed are evident and thoroughly understood.

The bottom-up approach starts by performing experiments and creating prototypes. This is advantageous during the initial phases of business modelling and technology development. It allows an organisation to develop at a considerably reduced expense and evaluate the benefits of the technology before making substantial investments.

The mixed method enables an organisation to benefit from the strategic and intentional nature of the top-down approach while also preserving the fast implementation and adaptable application of the bottom-up approach.

The warehouse design process generally includes the following stages:

- Select a particular business process, such as orders, invoices, shipments, inventories, account administration, sales, or the accounting system, for which you will develop a model. If the business process is organisational and contains multiple complex sets of items, it is recommended to stick to a data warehouse method. However, if the procedure is exclusive to a given department and focuses on analysing a single business process, it is recommended to choose a data mart model.
- Choose the exact business process workflow. The grain represents the fundamental data element that is stored in the fact table for this particular operation. The data can consist of individual transactions, daily snapshots, or other granular atomic-level information.
- Define the specific dimensions that will be associated with each record in the fact table. The dimensions that are considered standard are time, item, customer, supplier, warehouse, transaction type, and status.
- Choose the metrics to be incorporated into each fact table entry. Typical metrics include numerical quantities that can be added together, such as the overall revenue and the total number of units sold. After designing and building a data warehouse, the first step in deploying it involves installing it, organising the rollout, and providing training and orientation.

It is important to consider the updates and maintenance of the platform. Data warehouse administration includes tasks such as updating data, synchronising data sources, preparing for emergencies, controlling access and ensuring security, managing data expansion, optimising database performance, and improving and expanding the data warehouse.

A Three-tier Data Warehouse Architecture:

Data Warehouses typically employ a three-level architecture comprising:

The bottom tier, which encompasses the Data Warehouse server, typically implemented as a Relational Database Management System (RDBMS). The system may consist of multiple specialist data marts and a metadata repository.

The middle tier consists of an OLAP server, facilitating effective querying of the data warehouse. The OLAP server is typically implemented using either a Relational OLAP (ROLAP) model, which is an expanded version of a relational DBMS that translates operations on multidimensional data into standard relational operations, or a Multidimensional OLAP (MOLAP) model, which is a specialised server that directly supports multidimensional data and operations.

The top tier consists of front-end tools for showing results created by OLAP, together with supplementary tools for data mining the data produced by OLAP.

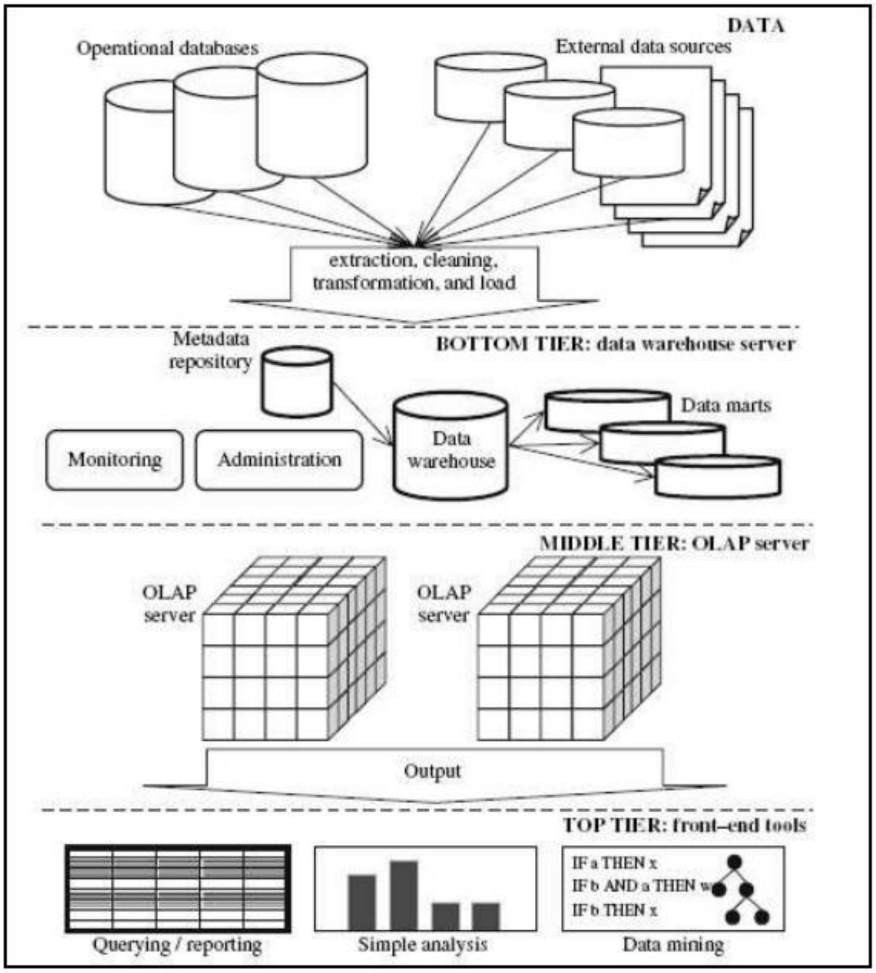

Figure 1: A three-tier Data Warehousing Architecture

SELF-ASSESSMENT QUESTIONS – 2

Choose the Right Answer:	
6	A Data Warehouse is primarily designed for: a) Real-time transaction processing b) Analytical processing and decision support c) Maintaining operational records only d) Simple data storage without analysis
7	Which layer of Data Warehouse Architecture is responsible for extracting, cleaning, and loading data? a) Data Source Layer b) ETL Layer c) Metadata Layer d) Presentation Layer

4. OLAP

OLAP, or Online Analytical Processing, is a technology that allows users to analyse their data quickly and interactively. Instead of viewing the data as flat files or spreadsheets, an OLAP application enabl es users to explore the data using several dimensions.

A flat file is a sequential collection of textual material that is stored line by line and is always read in a linear manner from the beginning to the end.
A dimension is a method for determining the value of a performance measure.
The capability to arrange material in a manner that aligns with users' cognitive understanding is refe rred to as multidimensionality.

The distinguishing factor between OLAP capability and a regular system is this.
For example, you may categorise a sale based on the occurrence of the purchase and the price of the product. Each object can be seen as a distinct dimension within the database.

There are four main factors to consider: time, product, geography, and price.
The dimensions are the essential component of the OLAP database.
By inputting different dimensions, the point where numerous dimensions intersect is referred to as a cell.

The cell holds the values that intersect across all dimensions. A cell is a discrete data point located at the intersection formed by choosing one value from each dimension in a multidimen sional array.

Think about the OLAP tasks that need to be done on data with more than one dimension. The image shows info cubes for a store's sales. Each cube has measurements, location, time, and item. The location is grouped by city values, the time is grouped by quarters, and the item is grouped by item types.

4.1. Characteristics of OLAP

The characteristics of OLAP are: (F A S M I)

- Fast
- Analysis
- Shared

- Multidimensional
- Information

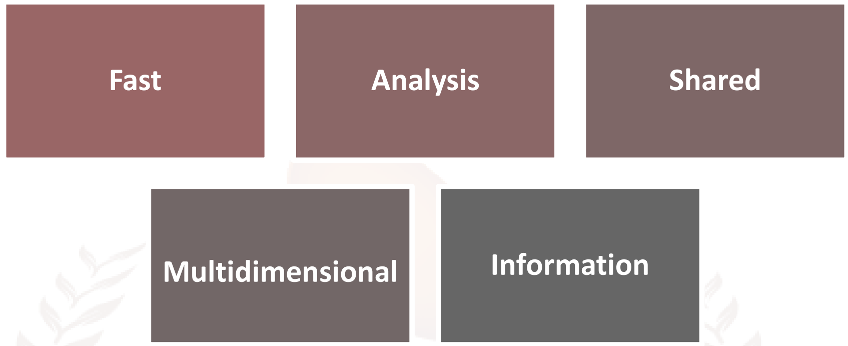

Figure 2: Characteristics of OLAP

i. Fast: OLAP systems are specifically built to deliver immediate responses to complex queries. The objective is to get query response speeds of five seconds or fewer for the majority of inquiries, enabling users to engage with the data in real time without significant delays.

ii. Analysis: OLAP systems enable complex analytical and mathematical computations. They offer robust analytical skills for doing trend analysis, forecasting, and statistical computations, which are essential to conducting thorough data analysis and making informed decisions.

iii. Shared: OLAP systems provide concurrent access and analysis of data by several users in a shared environment. This guarantees that data and analysis outcomes may be distributed throughout the organisation, encouraging collaborative decision-making and uniformity in data interpretation.

iv. Multidimensional: One of the fundamental features of OLAP is its capacity to offer a multidimensional perspective on data. Data is represented and analysed in numerous dimensions, including time, geography, product lines, and customer segments, enabling users to manipulate and examine the data from different viewpoints.

v. Information: OLAP systems provide access to both detailed and summarised data from several sources, hence providing comprehensive availability of information. The information is completely merged and logical, hence enhancing accurate and dependable data analysis and reporting.

4.2. OLAP Operations:

Five basic analytical operations can be performed on an OLAP cube:

i. Roll up

ii. Drill down

iii. Slice

iv. Dice

v. Pivot

i. Roll-up: The roll-up process, also known as the drill-up or aggregation operation, involves descending through idea hierarchies to combine data into a single collection. This process is referred to as dimension reduction.

Roll-up involves summarising data to a higher level in the dimensional hierarchy. The result of consolidation operations on the location dimension is shown in the figure. The data is aggregated along location hierarchy, such as Order Street → City → Province → Country, along with the city, province, state, or nation. The roll-up procedure involves aggregating the data by location, starting from the city level and moving up to the country level.

During a roll-up, dimensions reduction removes one or more dimensions from the cube. Take, for example, a sales data cube that covers two dimensions: time and location. To obtain a summary of sales by place rather than by location and time, you can exclude the time dimensions from the list.

For example, take a look at these cubes that show the temperatures for certain days of the week:

Temperature	64	65	68	69	70	71	72	75	80	81	83	85
Week1	1	0	1	0	1	0	0	0	0	0	1	0
Week2	0	0	0	1	0	0	1	2	0	1	0	0

Figure 3: Temperature of the week

Let's say we want to set three temperature values from the cubes above: hot (80–85), mild (70–75), and cool (64–69).

To achieve this, it is necessary to categorise the columns and aggregate the data according to the hierarchical structure. This process is called a roll-up.

In this way, we can contain the following cube:

Temperature	Cool	Mild	Hot
Week 1	2	1	1
Week 2	1	3	1

Figure 4: Temperature data in the cube

When you roll up the data, it is put into groups based on temperature levels.

The following image shows how roll-up works.

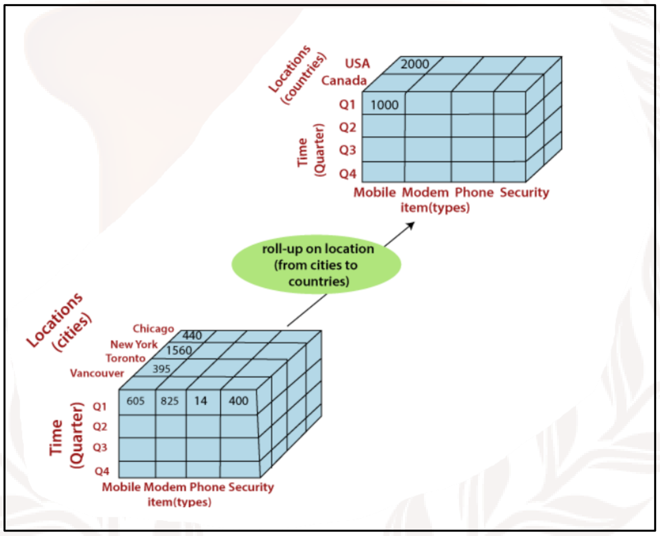

Figure 5: Demonstrate of roll up operations

ii. Drill-down: Drill-down, sometimes called "roll-down," is the opposite of roll-up.

Drilling down is like getting a closer look at the data cube. It moves from a record with less information to one with more information. You can drill down by moving down an idea hierarchy for a dimension or adding more dimensions.

The figure shows a drill-down action on the time dimension by moving down a concept hierarchy made up of day, month, quarter, and year.

Drill-down works by going down the time structure from the quarter to the month level, which is more specific.

It is possible to do a drill-down by adding a new layer to a cube because it gives you more information. As an example, adding a new dimension, like a customer group, can lead to a drill-down on the figure's centre cubes.

Example: Drill-down gives you more information about the material you're given.

Temperature	cool	mild	hot
Day 1	0	0	0
Day 2	0	0	0
Day 3	0	0	1
Day 4	0	1	0
Day 5	1	0	0
Day 6	0	0	0
Day 7	1	0	0
Day 8	0	0	0
Day 9	1	0	0
Day 10	0	1	0
Day 11	0	1	0
Day 12	0	1	0
Day 13	0	0	1
Day 14	0	0	0

Figure 6: Drill down data

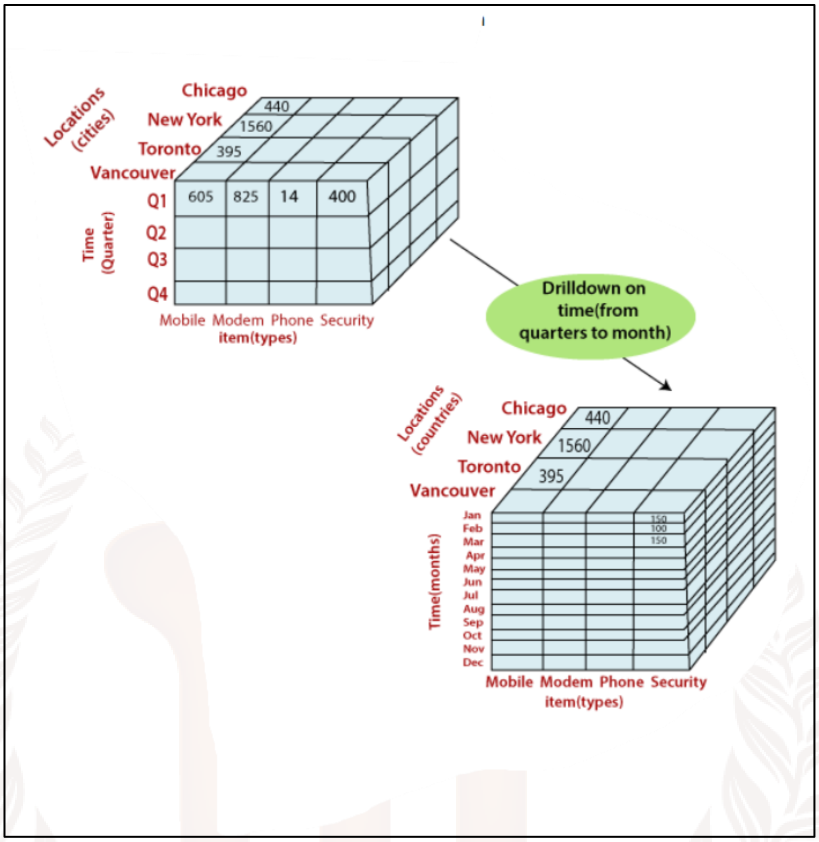

Figure 7: Visualisation of drill down operation

Slice:

A slice is a group of cubes with the same number of one or more dimension members. For example, a slice operation is used when the customer only wants to pick one side of a cube, making the site two-dimensional. In other words, the Slice processes pick out one dimension of the given cube, making a sub-cube. As an example, if we choose "temperature=cool," we'll get the following cube:

Temperature	cool
Day 1	0
Day 2	0
Day 3	0
Day 4	0
Day 5	1
Day 6	0
Day 7	1
Day 8	1
Day 9	0
Day 11	0
Day 12	0
Day 13	0
Day 14	0

Figure 8: Temperature based on day wise

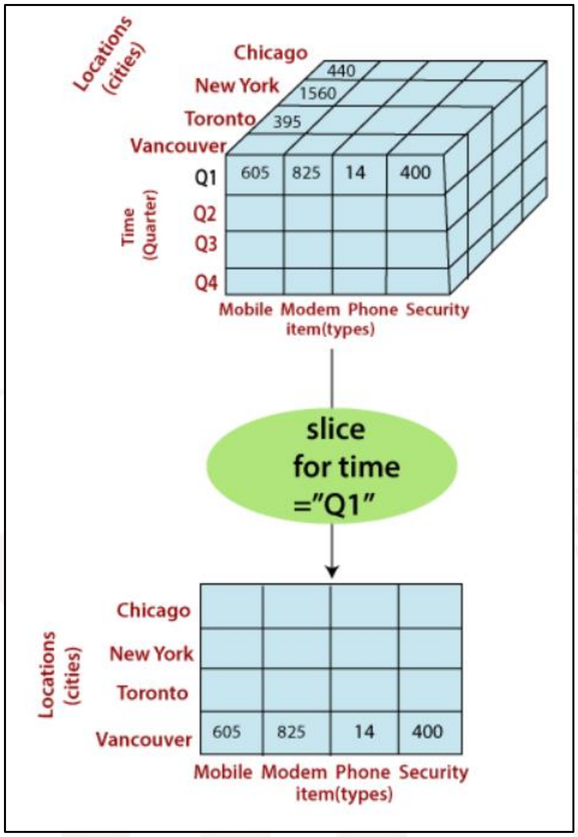

Figure 9: Slice operation Visualisation

This is how Slice works for the dimension "time" with the condition "time = "Q1". You can choose one or more elements to make a new sub-cube.

Dice

The dice procedure defines a subcube by applying a selection on two or more dimensions. As an example, if we apply the selection criteria of (time = day 3 OR time = day 4) AND (temperature = cold OR temperature = hot) to the initial cubes, we obtain a subcube that remains two-dimensional.

Time	Temperature	Value
Day 3	Cool	0
Day 3	Hot	1
Day 4	Cool	0
Day 4	Hot	0

Observe the diagram below, which illustrates the actions with dice.

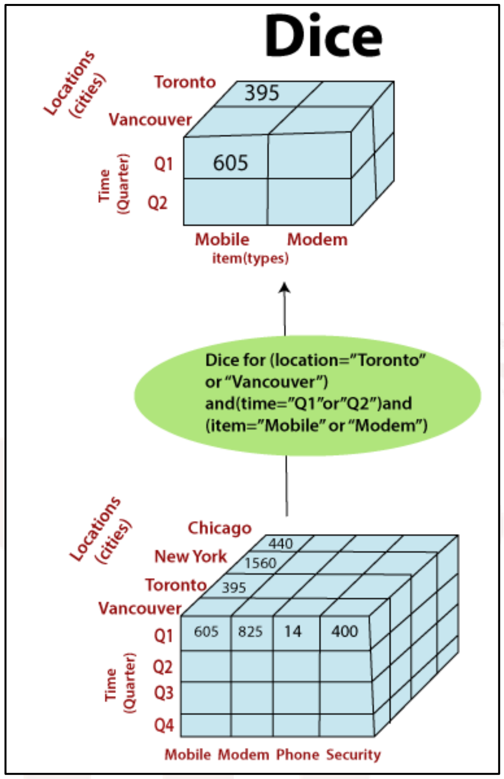

Figure 10: Dice operation Visualisation

The process of rolling the dice on the cubes is determined by three dimensions according to the selection criteria.

- (location = "Toronto" or "Vancouver")
- (time = "Q1" or "Q2")
- (item =" Mobile" or "Modem")

Pivot

The pivot operation is alternatively referred to as a rotation. Pivot is a data manipulation technique that involves rotating the axis of a visualisation to present the data in a different way. It could involve interchanging the rows and columns or transferring one of the row measurements to the column dimensions.

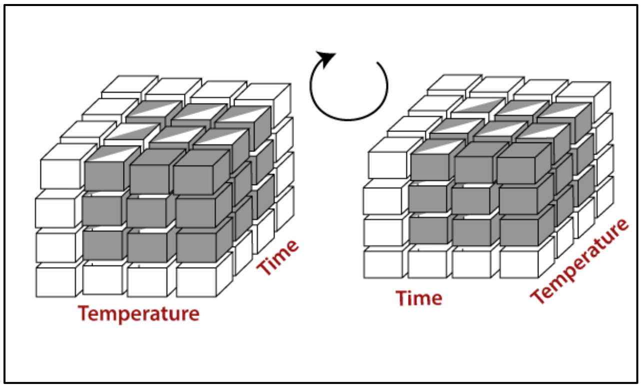

Figure 11: Visualisation of Pivot operation

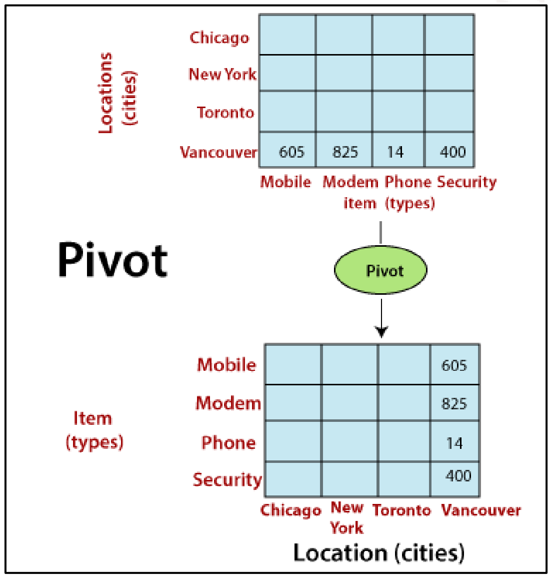

Figure 12: Pivot operation Visualisation

5. DIFFERENCE BETWEEN OPERATIONAL DATABASE AND DATA WAREHOUSE

Aspect	Operational Database (OLTP)	Data Warehouse (OLAP)
Purpose	Supports day-to-day business operations and transactions.	Supports analytical processing, reporting, and decision-making.
Data Type	Current, real-time transactional data.	Historical, consolidated, and subject-oriented data.
Data Structure	Highly normalised (to avoid redundancy).	Denormalised or multidimensional (optimised for queries).
Users	Clerks, cashiers, operational staff.	Managers, analysts, decision-makers.
Operations	Insert, update, delete, retrieve transactions.	Query, analysis, trend detection, forecasting.
Processing Type	Online Transaction Processing (OLTP).	Online Analytical Processing (OLAP).
Data Volume	Handles relatively small transactions in high numbers.	Handles large volumes of historical data.
Response Time	Very fast for small, simple queries.	Slower compared to OLTP but optimised for complex queries.
Example	Banking systems, e-commerce order management.	Sales trend analysis, business intelligence dashboards.

6. CASE STUDY

Problem Statement: HealthPlus, an established healthcare provider, manages numerous regional hospitals and clinics. HealthPlus faces significant challenges in managing and analysing patient data, which is dispersed over multiple departments and systems.

The primary concerns include:

- **Data fragmentation*:*** refers to the situation when patient data is maintained in several systems, such as Electronic Health Records (EHR), laboratory information systems, and billing systems. This fragmentation makes it challenging to obtain a complete and comprehensive overview of a patient's medical history.
- **Inadequate Data Analysis*:*** Current systems have a lack of advanced analytical skills, resulting in delays in obtaining actionable insights to enhance patient care and operational efficiency.
- **Operational inefficiencies*:*** arise when there is a lack of a comprehensive understanding of data, resulting in ineffective resource allocation management such as manpower, equipment, and beds. This often leads to either shortages or excesses.

Physicians' ability to promptly access and analyse patient data compromises the quality of care, as it limits their capacity to make well-informed judgements.

Solution:

In order to solve these problems, HealthPlus has made the decision to use a data warehousing solution that is paired with an OLAP engine. The objective of this solution is to centralise patient data, facilitate advanced data analysis, and enhance operational effectiveness and patient treatment.

In order to address the challenges encountered by HealthPlus, the following measures are implemented:

1. Implementation of Data Warehouse:

a. Data Consolidation: HealthPlus establishes a data warehouse to combine data from diverse sources, such as EHR systems, laboratory information systems, and billing systems. This repository serves as a centralised location for storing both past and present data.

b. The ETL process is implemented to assure the accurate extraction of data from source systems, its transformation into a standardised format, and its subsequent loading into the data warehouse. This technique involves the removal of inconsistencies and inaccuracies through data cleansing.

2. Integration of OLAP Engine

a. Multidimensional Analysis: The data warehouse is equipped with an OLAP engine, allowing for multidimensional analysis. This enables the examination and evaluation of data from several angles, including time, patient characteristics, types of diseases, and treatment results.

b. Data Cubes: OLAP cubes are constructed to show several dimensions of healthcare data, including patient visits, treatments, diagnosis, and financials. These cubes facilitate rapid retrieval and analysis of data.

3. Sophisticated Data Analysis and Reporting

a.Interactive dashboards are designed to offer healthcare management and doctors immediate acces s to essential key performance
indicators (KPIs), like patient admission rates, average length of stay, and treatment success rates.

b.Predictive Analytics: Predictive analytics models are constructed by utilising past data to anticipat e forthcoming patterns, such as the rates of patient admissions and the likelihood of disease outbrea ks. These predictive models enable proactive resource planning, such as scheduling staff or managing ICU beds during seasonal surges.

4. Enhancements in operations:

a. Resource Allocation: Through the analysis of patient admissions data and resource utilisation, HealthPlus can enhance staffing schedules, guarantee sufficient availability of medical equipment, and enhance bed management. This leads to a decrease in operational expenses and improves overall efficiency.

b. Clinical Decision Support: The combined data warehouse and OLAP engine offer clinicians a thorough overview of patient profiles, encompassing medical history, laboratory findings, and treatment strategies. This facilitates enhanced clinical decision-making and individualised patient care.

5. Improvement of Care Quality:

a. Patient Care Analytics: HealthPlus utilises OLAP capabilities to analyse patient outcomes by evaluating different therapies and processes. This facilitates the identification of optimal methods and regions that require enhancement, resulting in improved patient care.

b. Rapid Data Accessibility: Healthcare providers may quickly retrieve essential patient data, facilitating timely and well-informed decision-making. This decreases the duration required to diagnose and treat patients, while enhancing overall patient happiness and results.

Conclusion:

HealthPlus effectively resolves issues related to data fragmentation, ineffective data analysis, operational inefficiencies, and quality of care by installing a data warehousing system that is connected with an OLAP engine. The system offers an extensive system for consolidating data, doing advanced analytics, and generating real-time reports. This eventually results in improved patient outcomes and more efficiency in healthcare operations.

7. SUMMARY

A data warehouse is an exclusive database created to facilitate decision-making processes by serving as a centralised repository for aggregated and historical data. It is famous for its focus on certain subjects, its integration of different elements, its ability to change with time, and its ability to retain information without being lost. These attributes guarantee that data is structured based on important topics, consolidated from several origins into a cohesive format, maintained with historical context for analysing patterns, and remains consistent and unchanged after being inputted. A data warehouse architecture typically consists of multiple layers: the data source layer (which includes different data sources), the data staging layer (which involves ETL processes), the data storage layer (which houses the primary data warehouse database), the data presentation layer (which provides tools for data access and reporting), and the metadata layer (which describes the structure, operations, and contents of the data warehouse). The multidimensional data model is the fundamental structure of the data warehouse. It includes dimensions (also known as perspectives or attributes), measures (numerical data used for analysis), hierarchies (levels of detail within dimensions), and cubes (efficient storage and retrieval structures for multidimensional data).

OLAP operations are crucial for analysing data in a data warehouse. These operations include slicing, which involves selecting a single dimension to create a subset of data. Dicing is the process of selecting specific values for multiple dimensions to create a subcube. Roll-up is the aggregation of data to higher levels of hierarchy. Drill-down involves breaking down data into finer details. Pivoting refers to changing the dimensional orientation of the data cube. These two operations Ranking and Cross-tabulation are not introduced or explained in the OLAP section above. These procedures enable users to conduct intricate, interactive analyses of extensive datasets, allowing the extraction of insights and aiding decision-making in diverse real-world situations. For instance, in the retail and e-commerce industries, data warehouses facilitate the examination of sales patterns and the categorisation of customers. Within the realm of financial services, their role involves providing assistance in risk management and the detection of fraudulent activities. Within the healthcare field, these tools facilitate the study of clinical data and the allocation of resources. Telecommunications corporations utilise data warehouses to conduct network performance monitoring and customer churn analysis, while manufacturing enterprises employ them for supply chain management and quality control. Data warehouses are employed by government organisations to analyse tax revenue and examine the impact of policies. Similarly, educational institutions utilise data warehouses to analyse student performance and efficiently manage resources. These applications showcase the crucial significance of data warehousing and OLAP in improving operational efficiency and strategic planning in several sectors.

8. GLOSSARY

Data Warehouse	-	A centralised storage facility designed to hold huge quantities of data from multiple sources. The data is organised in a way that facilitates easy retrieval and analysis, rather than being optimised for transaction processing.
ETL (Extract, Transform, Load)	-	refers to the procedure of retrieving data from various sources, converting it into a compatible format, and then storing it in a data warehouse.
Metadata	-	refers to the information that provides a description of other data present in the data warehouse. This includes details on its organisation, definitions, and connections with other data.
Data Consistency	-	refers to the process of ensuring that data obtained from different sources is uniform and standardised, hence establishing a singular and accurate representation of reality.
Data integration	-	is the process of merging data from several sources into a unified repository, facilitating the analysis and reporting tasks.
A cube	-	is a multidimensional array of data used in OLAP (Online Analytical Processing) to efficiently search and analyse data.
A slice	-	is a process that retrieves a portion of a data cube by choosing a single value for one dimension.
Dice	-	A computational process that generates a smaller cube by choosing particular values for many dimensions.
Drill-Down	-	The procedure of breaking data into increasingly detailed layers within a hierarchy.
Roll-Up	-	refers to the procedure of condensing data by combining it into broader categories within a hierarchical structure.
Pivot (Rotate)	-	A transformative action that adjusts the multidimensional viewpoint of data to offer alternative viewpoints.

9. TERMINAL QUESTIONS

1. Define Datawarehouse and explain its benefits.
2. Explain the Multidimensional Data Model.
3. Elucidate the features of Multidimensional data models.
4. Write a short note on Multidimensional data transformation.
5. Explicate the OLAP Operations and mention the basic analytical Operations.
6. Explicate how Roll-up and Drill down Operations are different. Explain with an example.
7. Explain Slice Operations with an example.
8. Explain Dice Operations with an example.
9. Explain Pivot Operations with an example.
10. Write a short note on the OLAP engine Application of Data Warehousing in real-world scenarios.

10. ANSWERS

10.1. Self-Assessment Answers

1. Answer: c) Handle day-to-day transaction processing

2. Answer: d) All of the above

3. Answer: a) Data is updated in real-time

4. **Answer:** Online Transaction Processing (OLTP)

5. Answer: Consistency, Integrity

6. Answer: b) Analytical processing and decision support

7. **Answer:** b) ETL Layer

10.2. Terminal Answers

Answer 1: A Data Warehouse is a centralised repository that integrates data from multiple sources for analysis and reporting. Its benefits include improved decision-making, historical data storage, and support for business intelligence tools. (Refer to Section 3)

Answer 2: The Multidimensional Data Model organises data into cubes with dimensions (like time, product, location) and facts (like sales, profit), enabling fast and intuitive analysis.

Answer 3: Key features include data representation in cubes, support for hierarchical dimensions, fast OLAP queries, and easy Visualisation for decision-making.

Answer 4: Multidimensional data transformation involves converting raw transactional data into structured cubes and dimensions to support analysis and OLAP operations.

Answer 5: OLAP operations allow interactive data analysis. The main operations are Roll-up, Drill-down, Slice, Dice, and Pivot.

Answer 6: Roll-up summarises data (e.g., daily sales → monthly sales), while Drill-down provides finer details (e.g., monthly sales → daily sales).

Answer 7: Slice selects a single dimension value, creating a sub-cube. Example: Viewing sales only for “2024” across products and regions.

Answer 8: Dice selects a sub-cube by applying filters on multiple dimensions. Example: Sales for “Laptops” in “India” during “2024.”

Answer 9: Pivot rotates the data cube for a new perspective. Example: Swapping rows and columns to view "Products by Region" instead of "Region by Products."

Answer 10: An OLAP engine powers fast multidimensional analysis for applications like sales forecasting, financial planning, customer analytics, and supply chain optimisation.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 4

DCA2207

DATA MINING AND VISUALISATION

Unit – 3

Multidimensional Data model

TABLE OF CONTENTS

1. INTRODUCTION

In the previous unit, learners learnt about operational databases and their role in handling day-to-day transactions such as sales, banking activities, and inventory updates. They also explored the concept of a data warehouse, which serves as a centralized repository for integrating data from multiple sources. Learners understood how warehouses support historical data storage, business intelligence, and decision-making processes, and also examined the differences between operational databases and data warehouses in terms of purpose, structure, and usage.

In this current unit, learners will go a step further by studying how data stored in warehouses can be modelled and analysed using multidimensional approaches. They will explore the multidimensional data model, which allows information to be represented across different dimensions like time, product, and geography. Learners will learn about data cubes, which enable data summarization, aggregation, and analysis at different levels of detail. The unit also covers various schema designs such as the Star Schema, Snowflake Schema, and Fact Constellation (Galaxy Schema), each providing a unique way of structuring warehouse data. Furthermore, learners will study measures, including additive, semi-additive, and non-additive measures, to understand how quantitative values are analysed across dimensions. Finally, the unit introduces data pre-processing techniques, which are essential for cleaning, transforming, and preparing raw data before it is used for OLAP, reporting, and advanced analytics.

By the end of this unit, learners will gain a strong understanding of how to design, organize, and analyse multidimensional data, laying the groundwork for more advanced topics like OLAP operations and data mining.

1.1. Learning Objectives:

At the end of this topic, students should be able to:

- Define the concept of a multidimensional data model and its key characteristics.
- Summarize the multidimensional data model and its significance in data analysis.
- Perform basic OLAP operations such as slicing, dicing, and drill-down on sample datasets.
- Analyse the impact of data warehousing on decision-making processes in various industries.

2. MULTIDIMENSIONAL DATA MODEL

The ability to organise and view data in multiple dimensions, such as product, time, and location, is what is meant by a multidimensional data model. Multidimensional data refers to data that is organized and represented across multiple dimensions, rather than just rows and columns. Each dimension is a perspective or attribute used to analyze the data, such as time, location, product, or customer. The actual measurable values (like sales, revenue, or quantity) are stored as facts, and dimensions describe those facts.

Features of Multi-dimensional data model

- It lets us discover market or company trends by letting consumers ask analytical enquiries linked to various aspects.
- Data warehousing and online analytical processing (OLAP) both make use of multi-dimensional databases.
- Data cubes serve as its representational unit. With data cubes, you can model and examine your data from all angles.
- It is depicted by a fact table and defined by facts and dimensions.
- Tables of facts include numerical measurements of the facts themselves as well as measures of the related dimensional tables.

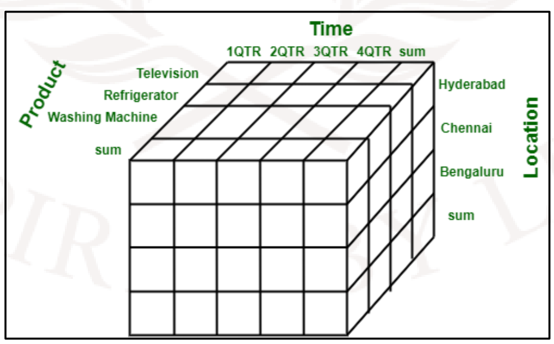

Figure 1: Multidimensional Data Representation

2.1. Working on a Multidimensional Data Model

In order to construct a Multi-Dimensional Data Model, each project must adhere to the following steps:

Stage 1: Assembling data from the client

The first step is for a Multi-Dimensional Data Model to gather accurate customer data. Experts in software development typically explain to clients in layman's terms the breadth of data that can be obtained with the chosen technology and then gather all of that data in great detail.

Stage 2: Grouping different segments of the system

Step two involves the Multi-Dimensional Data Model identifying and categorising all data into its proper part and then building it in a way that is easy to apply in stages.

Stage 3: Identifying the different dimensions: This step forms the backbone of the model's design. At this phase, the user's perspective has been used to identify the key factors. "Dimensions" is another name for these considerations.

Stage 4: Preparing the actual-time factors and their respective qualities: In the fourth stage, the linked traits are further identified by using the elements that were recognised in the previous step. The database also uses the term "attributes" to describe these characteristics.

Stage 5: Identifying the actual measurable facts and their associated attributes. **The fifth step of using a multi-dimensional data model is to identify and distinguish between the actual and gathered factors. In fact, they are crucial to the structure of a dimensional data model.**

Stage 6: Building the Schema to place the data, with respect to the information collected from the steps above: Stage six involves creating a Schema using the previously acquired data.

Example to Understand Multidimensional Data Model

1. A company can serve as an example. Various criteria can be used to determine a firm's revenue cost, including the location of the firm's workplace, the products of the firm, the ads done, the time used to flourish a product, and so forth.

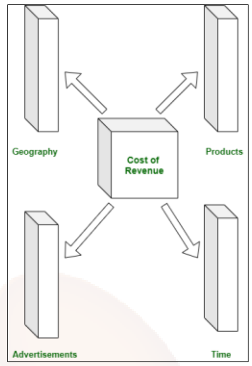

Figure 2: Example of Multidimensional Data model

2. Consider the following data from a Bangalore factory that sells products quarterly. The information is displayed in the following table:

Location = "Bangalore"				
Time (quarter)	Type of item			
	Jam	Bread	Sugar	Milk
Q1	350	389	35	50
Q2	260	528	50	90
Q3	483	256	20	60
Q4	436	396	15	40

Figure 3: 2D factory data

The factory's sales for Bangalore are shown in the following way: for the time dimension, which is divided into quarterly; and for the items dimension, which is arranged according to the type of item sold. Here, the figures are given in thousands of rupees.

Here is a diagram that shows how the sales data can be viewed in a three-dimensional table. Here, a two-dimensional table represents the sales data. Our data can be organised by item, time, and place (e.g., Kolkata, Delhi, Mumbai). See the table here the data representation as 2D and 3D

	Location="Kolkata"			Location="Delhi"			Location="Mumbai"		
	item			item			item		
Time	Milk	Egg	Bread	Milk	Egg	Bread	Milk	Egg	Bread
Q1	340	604	38	335	365	35	336	484	80
Q2	680	583	10	684	490	48	595	594	39
Q3	535	490	50	389	385	15	366	385	20

Figure 4: Data Representation as 2D

As seen in the graphic below, this data can be theoretically represented in three dimensions:

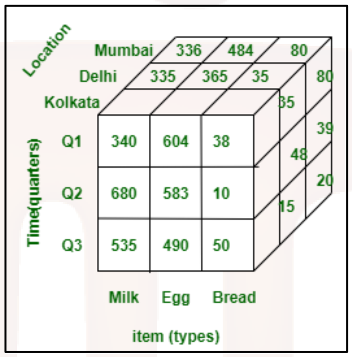

Figure 5: Data Representation as 3D

2.2. Characteristics of dimensional data sets

- Metrics: Calculable numerical values, such as sales or income, are known as measures. A multidimensional model stores them in fact tables.
- Dimensions: Time, place, or product are examples of dimensions that provide context for measurements. Dimension tables are where they are kept.
  Data can be organised into several dimensions using cubes, which allow for fast and flexible analysis by connecting measures and dimensions. Users can access data at various degrees of detail through the use of aggregation, which summarises data (e.g., total sales by month).

You can view data in more depth by drilling down, for example, from year to month. Roll-up which gets an overview of the data, for instance, from day one to quarter two. Data can be explored across layers with these. Hierarchies: These help with drill-down and roll-up by arranging dimensions into tiers (e.g., Year > Quarter > Month > Day). With the help of OLAP (Online Analytical Processing) technologies, complicated queries may be quickly executed on big data sets by utilising cubes, hierarchies, and aggregation.

Advantage	**Disadvantage**
Easy to handle	Requires skilled professionals
Simple to maintain	Complex structure
Better performance than relational databases	System performance drops if cache fails
More intuitive data representation (multi-viewed)	Dynamic and harder to design
Handles complex systems and applications well	Longer path to final output

Table 1: Advantages and Disadvantages of Data Model

3. DATA CUBES

A data cube in data mining is a multi-dimensional array that contains pre-aggregated data for efficient analysis. It provides a way to represent data in multiple dimensions, such as time, location, and product, allowing users to view data from different angles and gain insights into patterns and trends.

For example, consider a retail business that wants to analyze its sales data. The business may create a data cube with dimensions such as year/month, product, and city/country. The cube would contain pre-aggregated data, such as total sales, average sales, and the number of units sold, for each combination of the dimensions. This would allow the business to quickly analyze sales data by year, product, and city and identify trends and opportunities for improvement. For example, a data cube shown in the below figure represents sales data in multiple dimensions, i.e., product, quarter, and country.

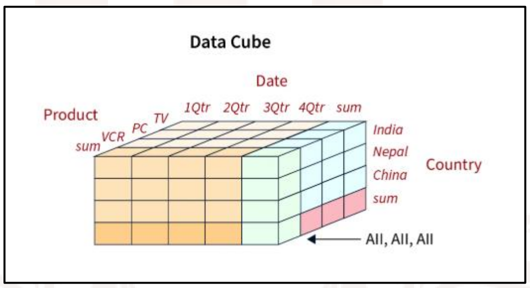

Figure 6: Data Cube

Data cube technology in data mining is commonly used in OLAP (online analytical processing) and business intelligence applications, enabling users to perform fast and efficient data analysis.

OLAP (Online Analytical Processing) is a technique used in data mining that allows users to quickly and interactively analyze large and complex datasets from multiple perspectives. OLAP is important for businesses because it enables them to analyze data in real-time and drill down into details to gain insights into business performance, identify trends, and make informed decisions. It is especially useful for decision-making and planning activities, such as budgeting, forecasting, and resource allocation.

OLAP is closely connected with data cube technology in data mining because data cubes are a popular way to represent multidimensional data for OLAP analysis. It is also called an OLAP cube. Data cube technology in data mining allows users to easily navigate through complex datasets and view data from different angles, enabling them to analyze data more efficiently and effectively.

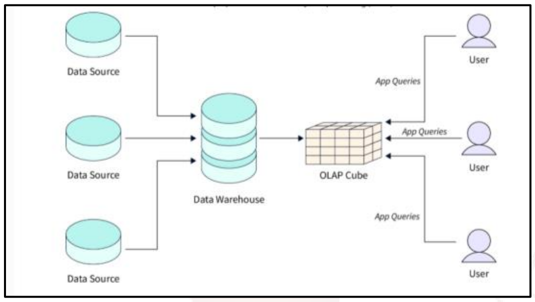

Figure 7: Data Cube Classification

Data cubes in data mining can be classified into two main categories -

- **Multidimensional data cube** - The idea of dimensions and measures forms the basis of this data cube type used in data mining. It provides a multi-dimensional representation of data that consumers may examine from several angles, including time, product, and geography. Data mining makes use of multidimensional data cubes, which are cube-shaped data structures that allow users to explore in depth and discover trends and patterns in the data by aggregating data across numerous dimensions.
- **Relational data cube -** The relational database model provides the foundation for this data cube type in data mining, which uses tabular data representations with rows and columns. The creation of this table is accomplished by using aggregate functions like sum, count, and average to data fields in many tables. When data sets are too big to store in memory alone, data miners often turn to relational data cubes for storage in databases. It lets users run complicated searches and analyses on massive datasets, but for some kinds of analysis, it might not be as efficient as multidimensional data cubes.

3.1. Operations on Data Cube

Data mining operations on a data cube allow for multi-view data analysis, revealing trends and patterns. Data cubes often undergo the following five operations:

- **Roll-up -** In this process, data is summarised along one or more data cube dimensions. A data cube with less granularity is the end outcome. As an example, we can increase the level of aggregation and decrease the number of dimensions in a sales data cube by rolling up monthly sales to quarterly sales.

- **Drill-down -** This process entails enhancing a data cube's level of detail by augmenting its current dimensions with additional features. The end product is a data cube with finer-grained findings. Adding a month dimension to an existing time dimension allows us to drill down from quarterly to monthly sales in a sales data cube.
- Slice - In this process, we fix the values of one or more dimensions in a data cube and then choose a subset of that data. It reduces the number of data points in the data cube while keeping the dimensions the same. To examine sales data for a certain area and time frame, for instance, we can slice a sales data cube.
- Dice - In this process, we fix the values of one or more dimensions and choose a range of values for another dimension in order to choose a subset of the data cube. Dice reduces the size of the data cube by applying filters across multiple dimensions, creating a more focused sub-cube To examine sales data for a specific region, time period, and product category, we can dice a sales data cube.
- Pivot - Rotating a data cube's dimensions and measures is what this procedure is all about. The end product is a data cube that looks at the data from a new angle. As an example, instead of analysing sales data by time period and product category, we can pivot a sales data cube to analyse it by product category and time period.

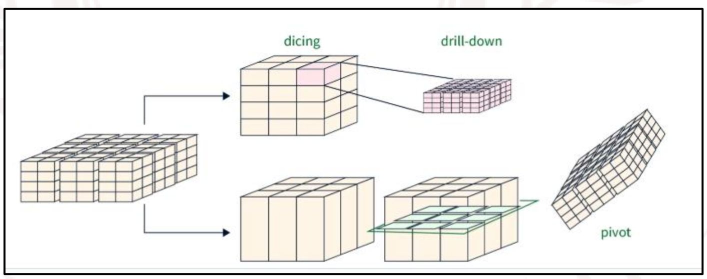

Figure 8: Data cube operations

Advantages of Data Cube

Data cube in data mining provides several advantages -

- **Multidimensional analysis -** Users are able to examine data from several angles and dimensions, including time, product, location, and customer, with the use of data cube technology in data mining. This provides a more complete picture of the data.
- **Fast query performance -** Data cubes pre-aggregate information at various granularities, which speeds up and simplifies the process of querying and retrieving results from massive databases.

- **Reduced data redundancy -** Data cubes eliminate the need to keep duplicate data in databases by storing pre-aggregated data at several degrees of granularity.
- **Data visualization -** To help users better comprehend and analyse complex data, data cubes in data mining can be visually represented using charts, graphs, and other graphical tools.
- **Improved decision-making -** Users are able to drill down, roll up, slice, and dice data using data cube technology in data mining. This empowers them to make informed decisions based on insights gleaned from the data.
- **Scalability -** Data cubes are scalable for data mining at the corporate level since they can manage big datasets and be saved in a database.

Disadvantages of Data Cube

Although data cubes offer numerous benefits in data mining, they can come with a few drawbacks as well.

- **Data cube creation -** Because of the complexity and time required to carefully evaluate the dimensions, measurements, and aggregation levels, data cube creation in data mining is sometimes a daunting task.
- **Data storage requirements -** When working with huge datasets that have numerous dimensions and metrics, data cubes may necessitate a substantial amount of storage space.
- **Limited flexibility -** If the underlying data or analysis needs change, data cubes, which are designed for multidimensional analysis, may have to be more adaptable.
- **Data quality issues -** In data mining, data cube technology is dependent on consistent and accurate underlying data, which is not always easy to attain when working with complicated datasets.
- **Complexity -** The data cube makes it easier to analyse complicated data, but understanding the dimensions, measurements, and aggregation levels employed in the cube can make the analysis itself complicated.

The data cube is a powerful structure used in data mining and OLAP, which is a multi-dimensional data representation that lets users examine massive datasets from different angles. Quick query performance, data visualisation, better decision-making, and scalability are just a few of the data analysis benefits it offers. Data mining data cube creation isn't without its drawbacks, though, including data redundancy, difficulty of analysis, and restricted adaptability. In most circumstances, the benefits of data cubes in data mining and business intelligence exceed their limits, and they continue to be a potent tool for multidimensional analysis.

SELF-ASSESSMENT QUESTIONS – 1

Choose the Right Answer:	
1	Which of the following best describes multidimensional data? a) Data stored in flat files b) Data represented as facts and dimensions for analysis c) Data stored only in relational tables d) Data stored in unstructured text format
2	In a multidimensional model, what does a fact table contain? a) Descriptive attributes b) Numerical measures like sales or revenue c) Metadata about tables d) Keys of sub-dimensions only
3	What is the role of a dimension table in a multidimensional data model? a) To store transactional data b) To describe facts with attributes like time, location, product c) To act as metadata storage d) To normalize fact tables
4	Which operation in OLAP involves moving from summarized data to more detailed levels? a) Roll-up b) Slice c) Drill-down d) Pivot
5	A sales data cube with dimensions: Time, Product, and Region is an example of: a) Two-dimensional data b) Multidimensional data c) Unstructured data d) Metadata schema
6	Why is the multidimensional model preferred in data warehousing? a) It reduces the size of the database

	b) It enables faster and more flexible data analysis c) It stores only unstructured data d) It eliminates the need for fact tables

4. DATA WAREHOUSE SCHEMA

The schema of a data warehouse is a description of the logical relationships between data objects, including tables and indexes. Three distinct warehouse schemas—Star, Galaxy, and Snowflake—describe several possible logical data groupings. An architectural plan for the storage and management of data is what a data warehouse schema is all about. Schema refers to the organisation of data storage and its relationships within a data warehouse, rather than the data itself.

Data warehouse schemas used to be enterprise-wide mandates, however with the advent of cheaper storage options, these restrictions have been relaxed in recent installations. In spite of this loosening or even complete rejection of data warehouse schemas, understanding the original schema designs can help with both the upkeep of legacy resources and the development of new data warehouse designs that draw lessons from the old. Fact and dimension tables are the building blocks of every data warehouse schema. The majority of data warehouse schema designs are based on variations on these two primary components.

4.1. Fact Table and Dimension table

A **fact table** and a **dimension table** are the two core components of a data warehouse. A fact table primarily contains the **quantitative data or measurable values** of a business process, such as sales amount, profit, or quantity sold. These values are usually numeric and represent the key performance indicators (KPIs) that organizations want to analyze. On the other hand, a dimension table provides the **descriptive context** for these facts by storing attributes like product details, customer information, time, or region. Together, fact and dimension tables enable businesses to organize data in a structured way for analysis, forming the basis of models such as the **star schema** and **snowflake schema,** which are widely used in data warehousing and business intelligence systems.

Fact Table

A fact table compiles data pertaining to business operations in the form of metrics, measurements, or facts. The schema design depicting the relationships between data in the data warehouse is formed in this example by connecting fact tables to dimension tables. As foreign keys, fact tables store the primary keys of dimension tables.

Dimension Table

Data attributes or dimensions can be stored in non-denormalized tables called dimension tables. As previously stated, the fact table uses the main key from a dimension table as a foreign key. There is no connection between the dimension tables. The primary fact table instead serves as a means of linkage, linking them together.

4.2. Types of schema used in data warehouses

One of the simplest data warehouse designs in history is the star schema. This schema adheres to specific design guidelines, such as allowing just one main table and a small number of linked one-dimensional tables. By adhering to these guidelines, a star schema can be created, which is characterised by a centre table and five joined dimension tables (thus the name). One database architecture method that Star Schema is famous for creating is denormalised dimension tables, which organise tables to introduce redundancy for enhanced speed. If denormalisation improves query performance, it will introduce duplication in more dimensions.

Characteristics of the Star Schema:

- Data warehouse schemas that use stars produce denormalised databases, which speed up replies to queries.
- A foreign key connects the fact table to the primary key in the dimension table. • In the star schema, each dimension translates to a single dimension table.
- In a star scheme, you shouldn't directly link dimension tables.
- Star schema structures use denormalised dimension tables for improved query performance.

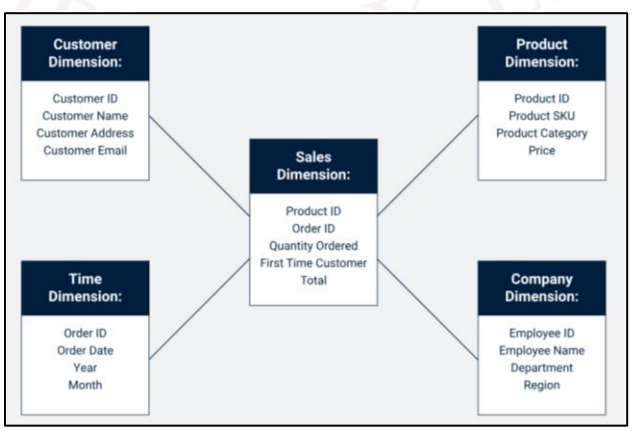

Figure 9: Example of Star Schema

Snowflake Schema

Data warehouses often employ the Snowflake Schema, which includes a specific way dimension tables are organised. Adding more sub-dimension tables that link to first-order dimension tables connected to the fact table, this data warehouse schema expands upon the star structure. The snowflake schema method mimics the relationship between fact table primary keys and dimension table foreign keys by having sub-dimension primary keys correspond to higher-order dimension table foreign keys.

One way to structure databases is the snowflake schema, which generates normalised dimension tables. These tables are organised in a way that minimises redundancy. The goal of data normalisation is to cut down on unnecessary work by removing duplicate records.

Characteristics of the Snowflake Schema:
In a Snowflake Schema, dimension tables may be further normalised into related sub-dimension tables.

- Snowflake Schema are to have one fact table only
- Snowflake Schema create normalized dimension tables
- The normalized schema reduces required disk space for running and managing this data warehouse
- Snowflake Scheme offer an easier way to implement a dimension

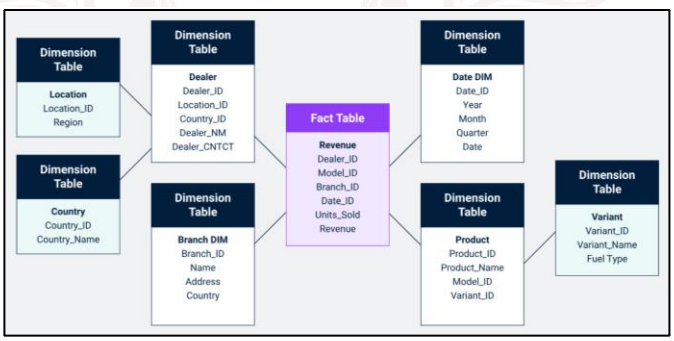

Figure 10: Example of Snowflake Schema

Galaxy Schema

Following this version of the data warehouse design is the Galaxy Data Warehouse design, which is also called a Fact Constellation Schema. The Galaxy Schema employs a network of fact tables linked by common normalised dimension tables, in contrast to the Star and Snowflake Schemas. To avoid data redundancy or inconsistency, Galaxy Schema is like a network of interconnected and fully normalised star schemas.

Characteristics of the Galaxy Schema:

- Galaxy Schema is multidimensional acting as a strong design consideration for complex database systems
- Galaxy Schema reduces redundancy to near zero redundancy as a result of normalization
- Galaxy Schema promotes high data quality and is well-suited for complex reporting and analytics

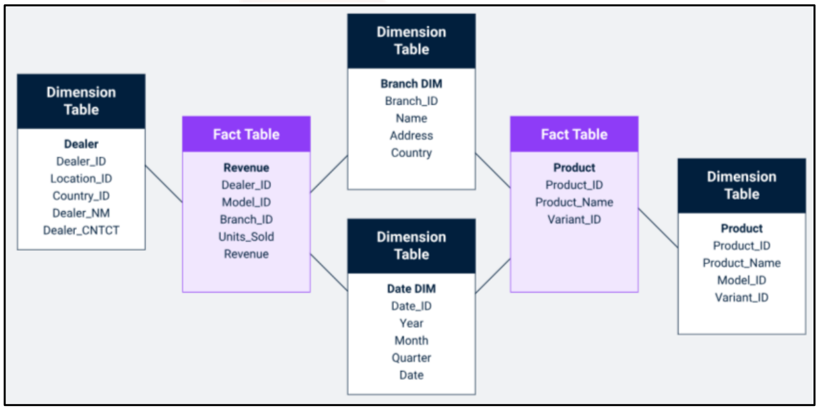

Figure 11: Example of Galaxy Schema

Key differences between Star, Snowflake and Galaxy Schema

The Star, Snowflake, and Galaxy schemas are the three most common data warehouse schema designs, each with unique strengths and trade-offs. The **Star Schema** is the simplest and most widely used, with a central fact table connected directly to dimension tables. It is easy to design and offers fast query performance, but it often results in higher data redundancy and increased storage requirements. The **Snowflake Schema** builds on the star model by normalizing dimension tables into multiple related sub-tables. This reduces redundancy and storage usage but increases design complexity and query time due to additional joins. The **Galaxy Schema,** also known as the Fact Constellation Schema, is the most advanced and complex design, involving multiple fact tables that share common dimension tables. It is ideal for handling large-scale, sophisticated business processes but requires careful management and is generally slower to query. Together, these schemas provide flexible modelling options, balancing simplicity, performance, and scalability depending on organizational needs.

Feature / Aspect	Star Schema	Snowflake Schema	Galaxy Schema
Elements	Single Fact Table connected to multiple Dimension Tables with no sub-dimension tables	Single Fact Table connects to multiple Dimension Tables that connect to multiple sub-dimension tables	Multiple Fact Tables connect to multiple Dimension Tables that connect to multiple sub-dimension tables
Normalization	Denormalized	Normalized	Normalized
Number of Dimensions	Multiple Dimension Tables map to a single Fact Table	Multiple Dimension Tables map to multiple Dimension Tables	Multiple Dimension Tables map to multiple Fact Tables
Data Redundancy	High	Low	Low
Performance	Faster queries, fewer foreign keys	Slower queries due to more foreign keys	Slower compared to Star and Snowflake, used for complex aggregations
Complexity	Simple, easy to understand	Moderate complexity, harder to understand than Star	Most complex, used for highly sophisticated data
Storage Usage	Higher storage due to redundancy	Lower storage, less redundancy	Low storage relative to complexity (due to reduced redundancy)
Design Limitations	Only one Fact Table, no sub-dimensions	One Fact Table allowed, multiple sub-dimensions permitted	Multiple Fact Tables allowed, but only first-level dimensions permitted

Table 2: Description of schema types

To gain a conceptual understanding of data warehouse schemas and their types, keep the following points in mind:

- A data warehouse schema serves as a blueprint for how data is stored, organized, and managed. It defines the structure, relationships, and arrangement of data components.

- The most common types of schemas are Star, Snowflake, and Galaxy, each differing in design and the way relationships between data are structured.
- The Star Schema is the simplest, featuring a single central fact table connected directly to dimension tables, making it easy to understand and query.
- The Snowflake Schema extends the star model by normalizing dimension tables into sub-dimensions. This reduces redundancy and storage costs but increases structural complexity.
- The Galaxy Schema, also called the Fact Constellation Schema, involves multiple fact tables sharing dimension tables. It is more sophisticated than Star and Snowflake, minimizes redundancy, and is well-suited for handling complex business processes.

Which data warehouse schema is best?

There is no single "best" data warehouse schema. The choice of schema largely depends on factors such as available resources, the nature of the data, and the intended analytical goals.

- The Star Schema is well-suited for organizations seeking simplicity and ease of use, even if it requires more storage space.
- The Galaxy Schema is preferable when dealing with complex data aggregation across multiple processes.
- The Snowflake Schema works best for organizations aiming to minimize redundancy while maintaining a more structured, normalized design.

SELF-ASSESSMENT QUESTIONS – 2

Choose the Right Answer:	
7	Which schema in data warehousing has a central fact table connected directly to multiple dimension tables? a) Snowflake Schema b) Galaxy Schema c) Star Schema d) Hierarchical Schema
8	The Snowflake Schema is an extension of which schema? a) Galaxy Schema b) Star Schema c) Network Schema d) Fact Schema
9	Which schema is also known as the Fact Constellation Schema? a) Star Schema b) Snowflake Schema c) Galaxy Schema d) Hybrid Schema
10	What is the main drawback of the Star Schema? a) Too complex to understand b) Causes data redundancy due to denormalization c) Requires multiple joins leading to slow queries d) Cannot store fact data
11	Which schema is most suitable for modeling complex business processes with multiple fact tables? a) Star Schema b) Snowflake Schema c) Galaxy Schema d) Relational Schema

5. MEASURES OF CENTRAL TENDENCY

Understanding the measure of central tendency in a dataset is essential in data analysis. Central tendency is a statistical measure that determines the central or typical value in a dataset. Standard measurements of central tendency including the mean, median, mode, and midrange. These measurements help in understanding the concentration of data points.

Important metrics for determining central tendency are:

- Mean: A data set's mean is its average value.
- Median: The median is the midpoint of a data set, the value that divides the upper and lower halves of the set.
- Mode: The data point with the highest frequency of occurrence.
- Midrange: The midrange is the average of the minimum and maximum values. These measures help understand where most data values fall within the distribution.

i. Mean: The mean $\bar{x}$ of a set of N values $x_1, x_2, \ldots, x_N$ is given by the formula:

$$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N} = \frac{x_1 + x_2 + \cdots + x_N}{N}$$

This formula sums up all the values in the dataset and divides the total by the number of values, resulting in the average value.

Mean—Example 1: Salary: Consider the given salary represented in thousands of dollars, arranged in ascending order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, and 110.

Arithmetic Mean (Average)

Most people utilise the arithmetic mean to find out where things are centrally distributed. It stands for the dataset's average or mean value. To find the average, we sum up all the values we have and split (divide) it by the total number of observations.

Using the Mean formula, we get,

$$\bar{x} = \frac{30 + 36 + 47 + 50 + 52 + 52 + 56 + 60 + 63 + 70 + 70 + 110}{12}$$
$$= \frac{696}{12} = 58.$$

This formula sums up all the values in the dataset and divides the total by the number of values, resulting in the average value.

Thus, the mean salary is $58,000.

Sometimes, within a set, each value xi can be linked to a weight w_i for i = 1,...,N.

The weights represent the relative importance or frequency of occurrence associated with their corresponding values. In this instance, we may calculate

$$\bar{x} = \frac{\sum_{i=1}^{N} w_i x_i}{\sum_{i=1}^{N} w_i} = \frac{w_1 x_1 + w_2 x_2 + \cdots + w_N x_N}{w_1 + w_2 + \cdots + w_N}$$

The term used to describe this calculation is the weighted arithmetic mean or the weighted average.

ii. Median: When data is organised in ascending order, the median—a measure of central tendency—represents the value in the middle of the collection. It splits the dataset in half, with half of the points falling below the median and half above. If your dataset is small, calculating the median should be easy. However, if your dataset is huge, you can estimate it using interval frequencies and interpolation.

Consider the above example 1 and find the median for the same example.

- When a dataset has an even number of observations, the median is not a single value but the average of the two middlemost values.

  Example: Given a dataset of 12 values sorted in increasing order, the median will be the average of the 6th and 7th values. If these values are 52 and 56, the median is calculated as

  Median=(52+56) / 2 = 54

Hence, the median in this case is 54.

- When a dataset has an odd number of observations, the median is the middlemost value.

  Example: If we have a dataset of 11 values, the median is the value at the 6th position in the sorted list. If the 6th value is 52,000, then the median is:

  Median = 52,000

Approximating the Median for Large Datasets:

- Computing the median can be computationally intensive for large datasets. An approximation can be made by grouping data into intervals and using the frequencies of these intervals.
- To approximate the median, we identify the interval containing the median frequency (the interval where the cumulative frequency reaches or exceeds half the total number of observations).
- We then use the following formula for interpolation:

$$\text{median} = L_1 + \left(\frac{N/2 - (\sum \text{freq})}{\text{freq}_{\text{median}}}\right) \text{width}$$

Where **Variables** are:

- L1 is the lower boundary of the median interval.
- N is the total count or number of observations.
- ∑freq is the cumulative frequency of all classes or intervals below the median interval.
- $\text{freq}_{\text{median}}$ is the frequency of the median class or interval.
- width is the width of the median class interval.

Example of Approximating the Median:

- Consider employees grouped by salary intervals: $10,000-$20,000, $20,000-$30,000, etc.
- If the interval $30,000-$40,000 contains the median frequency, we use the formula to approximate the median salary.

iii. Mode: In example 1, the data are bimodal, i.e., there are two most frequently occurring values: $52,000 and $70,000.

For unimodal datasets (having one peak) and moderately skewed (not symmetrical), there's a useful empirical relationship that helps estimate the mode.

$$\text{mean} - \text{mode} \approx 3 \times (\text{mean} - \text{median})$$

The above formula indicates that if you know the mean and median of a skewed dataset, you can easily approximate the mode using this relationship.

Another simple way to measure the central point of a numeric dataset is the midrange. The midrange is calculated by averaging the maximum and minimum values in the dataset:

$$\text{Midrange} = \frac{\max + \min}{2}$$

This measure can be quickly computed using SQL functions like max() and min(), making it a convenient tool for assessing central tendency.
For Example 1, the midrange is:

$$\frac{30{,}000 + 110{,}000}{2} = \$70{,}000$$

Figure 1 shows how the mean, median, and mode relate in different data distributions. All three measures coincide at the centre in symmetric data, indicating balanced data. In positively skewed data, the mean is pulled right by higher values, so it's greater than the median and mode. Conversely, in negatively skewed data, the mean is pulled left by lower values, making it less than the median and mode. These relationships help us understand the data's shape and distribution.

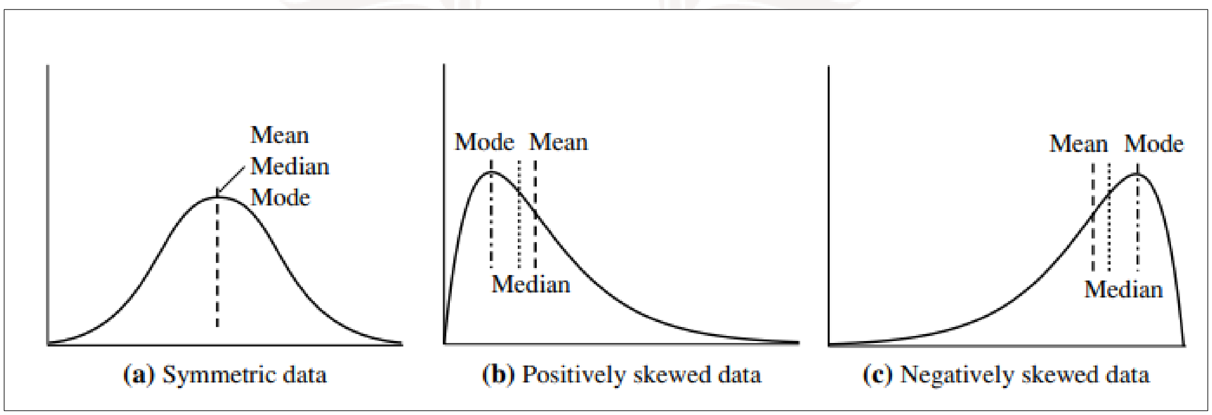

Figure 12: Symmetric, positive and negative skewed data for Mean, median and mode.

SELF-ASSESSMENT QUESTIONS – 3

Choose the Right Answer:	
12	The measure of central tendency that represents the average of all data values is called the _________.
13	The middle value of an ordered dataset is known as the _________.
14	The value that occurs most frequently in a dataset is referred to as the _________.

6. DATA PRE-PROCESSING

To get data suitable for analysis, it must first be cleaned and transformed from its raw form into a more usable shape. In data mining, preparing raw data for mining algorithms involves data cleansing, transformation, and organisation.

The objective is to enhance the data's quality.

- Facilitates data normalisation, deletion of duplicates, and management of missing values.
- Makes ensuring the dataset is accurate and consistent.

An integral component of data analysis and machine learning, data pre-processing entails cleaning, transforming, and preparing the data for further analysis. The data must be accurate, consistent, and fit for use in prediction models, and this approach guarantees it.

Steps in Data Preprocessing

Some key steps in data pre-processing are Data Cleaning, Data Integration, Data Transformation, and Data Reduction.

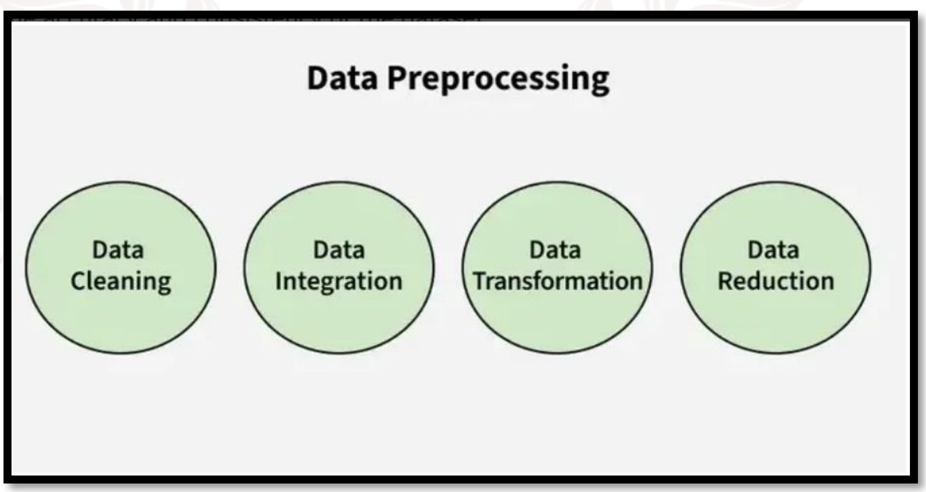

Figure 13: Steps in Data Pre-processing

a. Data cleaning: Finding and correcting inconsistencies or errors in the dataset is known as data cleansing. Fixing incorrect or outlier data, dealing with missing numbers, and removing duplicates are all steps in making a reliable dataset. If you want better results and more reliable data models, then you need clean data for your analysis.

i. Missing Values: This happens whenever a dataset is devoid of data. Rows that are missing data can be ignored or filled in manually, using the attribute mean or the most likely value. Doing so keeps the dataset complete and accurate for future research.

ii. Noisy Data: Noisy data is data that is either irrelevant or inaccurate, making it difficult for computers to process. In most cases, this is because someone made a mistake when entering or collecting the data. It has multiple possible approaches:

- Binning Method: By dividing the data into equal segments, we can smooth out each segment using the mean or boundary values instead of individual ones. This technique is called binning.
- Regression: Linear or multiple regression functions can be used to smooth out data and predict results.
- Clustering: This technique groups records that are statistically similar together, allowing outliers to either go unnoticed or be located outside of the clusters. These methods enhance data quality by eliminating noise.

iii. Removing Duplicates: In order to guarantee the accuracy and consistency of the dataset, this process entails locating and eliminating duplicate data elements. Because this procedure keeps only unique records, it avoids mistakes and guarantees accurate analysis.

b. Data Integration: It comprises merging data sets from multiple databases into a single, coherent whole. Because data may have different structures, formats, and meanings, comparing and contrasting it may be challenging. Methods like data fusion and record linking can ensure accuracy and consistency when data is merged quickly.

- Record Linkage: It involves locating and matching records from several datasets that relate to the same object, independent of their varied representations. In order to combine data from several sources, it can find matched records based on shared attributes or IDs.
- Data fusion is the process of combining datasets from different sources to make them more comprehensive and accurate. By incorporating sometimes contradictory or missing data from several sources, it streamlines and improves the dataset for analysis.

c. Data Transformation: It entails transforming data into an analytically-friendly format. Standardisation, which normalises data to a common range; discretisation, which turns continuous data into discrete categories; and normalisation, which scales data to a common range, are typical procedures. The data can be better prepared for analysis with the help of these procedures.

- Data normalisation involves bringing all variables' values into a consistent range by scaling them to a common value. To facilitate analysis, discretisation requires isolating discrete categories from continuous data.
- Data Aggregation: To make analysis easier, summarise different pieces of information by calculating their averages or totals.
- Generating a Concept Hierarchy: Organising attributes into hierarchical levels (e.g., Country → State → City) to offer a bird's-eye view for enhanced comprehension and analysis.

d. Data Reduction: It reduces the size of the dataset while preserving crucial data. There are two ways to achieve this: feature selection and feature extraction. The former includes picking out the most relevant features, while the latter requires transforming the data into a lower-dimensional space while preserving important details. There are a number of methods used for reduction, including

Dimensionality reduction refers to methods that can be used to decrease the amount of variables in a dataset while preserving all of the critical information. The principal component analysis is one such example. Numerosity Reduction use methods such as sampling to decrease the quantity of data points with the aim of simplifying the collection without losing key patterns. In order to facilitate storage and processing, data compression involves compressing data in a smaller, more compact form.

6.1. Uses of Data Pre-processing

In order to get raw data ready for analysis and decision-making, data preparation is a typical technique in numerous sectors. The following crucial domains make use of data preparation:

1. Data warehousing:

Important steps in data warehousing preprocessing include cleaning, integrating, and structuring data before storing it in a central repository. This will ensure that the data used in subsequent reports and queries is accurate and consistent.

2. Data Mining: The term "data preparation" refers to the steps used to clean and change raw data in order to make it ready for analysis. This step is essential for discovering insights and patterns in large datasets.

3. Machine Learning: In machine learning, "preprocessing" refers to the process of getting data ready to train a model. Managing missing values, standardising features, encoding categorical variables, and partitioning datasets into training and testing sets are all steps in improving model performance and accuracy.

4. Data Science: Data preprocessing, which comprises cleaning, organising, and relevance-checking the data prior to its usage for analysis or the building of prediction models, is an integral aspect of data science endeavours. The overall quality of the data insights obtained is higher.

5. Web Mining: Web usage logs can be analysed with the assistance of preprocessing in web mining to uncover valuable patterns of user behaviour. Marketing tactics and user experience can both be enhanced by incorporating this information into personalised recommendations.

6. Business Intelligence: Business intelligence (BI) relies on preprocessing, which cleans and organises data in preparation for dashboards and reports that give decision-makers useful insights.

7. Deep Learning Purpose: In order to optimise the training of models, deep learning applications, like machine learning, necessitate preprocessing to standardise or improve input data attributes.

Advantages of Data Preprocessing

- Enhanced Data Quality: Makes ensuring data is consistent, clean, and dependable enough to analyse.
- Improved Model Performance: Less data that isn't relevant or useful is discarded, resulting in more precise forecasts and insights.
- Data streamlining facilitates efficient analysis by making data processing easier and faster.
- Improved Decision-Making: Gives organised data that is easy to understand, leading to smarter business choices.

Disadvantages of Data Pre-processing

- It requires significant time and effort for cleaning, transformation, and organisation of data..
- Heavy on Resources: Complex pre-processing processes necessitate a lot of computing power and trained humans to complete.
- Possible Data Loss: Important information could be lost due to improper management.
- Difficulty: Dealing with many formats or huge datasets can be a real pain.

7. SUMMARY

This unit has provided learners with a comprehensive understanding of the fundamental building blocks of data warehousing and analytical processing. It began with the study of the **multidimensional data model**, which allows data to be structured into facts and dimensions, enabling flexible and intuitive analysis from different viewpoints such as time, location, and product. From there, students examined **data cubes**, which extend the multidimensional model by offering a visual and computational framework for performing operations like **roll-up** increasing levels of aggregation, **drill-down** moving into finer levels of detail, **slice and dice** filtering subsets of data, and **pivoting** rotating perspectives for comparative analysis. The unit then moved into **schemas in data warehouses**, where students explored the design and architecture of data storage through **Star Schema** simple and denormalised, **Snowflake Schema** normalized with reduced redundancy, and **Galaxy Schema** fact constellation for complex business processes. Each schema was studied in terms of its strengths, limitations, and use cases in real-world applications.

To strengthen analytical skills, the unit also emphasized **measures of central tendency—**namely **mean, median, and mode—**which are essential for summarizing datasets and identifying central patterns in business data. These measures serve as the foundation for statistical analysis and decision-making, helping transform raw data into meaningful insights. Finally, the unit highlighted the critical role of **data pre-processing**, which involves data cleaning, integration, transformation, reduction, and discretization. This process ensures that data used in mining and analysis is accurate, consistent, and reliable. Pre-processing not only improves data quality but also enhances the performance of mining algorithms, making the results more actionable.

Altogether, this unit bridges the gap between raw operational data and insightful analytical outcomes. By learning multidimensional modelling, cube operations, warehouse schemas, statistical measures, and pre-processing techniques, students are equipped with the knowledge to handle data systematically, transform it into valuable information, and support **business intelligence, decision-making, and predictive analytics** in modern organizations.

8. GLOSSARY

Data Warehouse	-	A centralized repository used to store, integrate, and manage large volumes of data for analysis and decision-making.
Multidimensional Data Model	-	A way of organizing data into facts (numerical measures) and dimensions (descriptive attributes) for flexible analysis.
Fact Table	-	A central table in a schema that stores quantitative data such as sales, revenue, or transactions.
Dimension Table	-	A table that contains descriptive attributes like time, location, or product, used to analyze facts from different perspectives.
Data Cube	-	A multidimensional representation of data that supports OLAP operations such as roll-up, drill-down, slice, dice, and pivot.
OLAP (Online Analytical Processing)	-	A technology that enables interactive, multidimensional analysis of data stored in warehouses.
Star Schema	-	A simple database schema with a central fact table connected to multiple dimension tables, easy to understand but may contain redundancy.
Snowflake Schema	-	A normalized version of the Star Schema where dimension tables are split into sub-dimensions, reducing redundancy but increasing complexity.
Galaxy Schema (Fact Constellation)	-	A schema design with multiple fact tables sharing dimension tables, used for complex business processes.
Roll-up	-	An OLAP operation that aggregates data to a higher level, such as moving from daily sales to monthly sales.
Drill-down	-	An OLAP operation that moves to a finer level of detail, such as going from yearly sales to quarterly sales.
Slice	-	An OLAP operation that selects a single dimension from the cube, creating a subset of data.
Dice	-	An OLAP operation that filters data on multiple dimensions, producing a smaller cube.

Pivot (Rotation)	-	An OLAP operation that rotates the cube to view data from different perspectives.
Central Tendency	-	A statistical concept that measures the central value of a dataset using mean, median, or mode.
Mean	-	The arithmetic average of a dataset, calculated by dividing the sum of all values by the number of values.
Median	-	The middle value in an ordered dataset, representing the central point.
Mode	-	The value that occurs most frequently in a dataset, useful for categorical data.
Data Preprocessing	-	The process of preparing raw data for analysis by performing tasks like cleaning, integration, transformation, and reduction.
Data Cleaning	-	Removing errors, inconsistencies, and missing values from raw data.
Data Transformation	-	Converting data into a suitable format for analysis, such as normalization or aggregation.
Data Integration	-	Combining data from multiple sources into a single unified view.
Data Reduction	-	Reducing the size of data while maintaining integrity, often using techniques like sampling or dimensionality reduction.

9. TERMINAL QUESTIONS

1. What is a multidimensional data model?
2. What is a data cube in data warehousing?
3. What is the difference between fact and dimension tables?
4. What is a Star Schema?
5. What is a Snowflake Schema?
6. What is a Galaxy Schema?
7. What are measures of central tendency?
8. Define the mean and explain its appropriate use cases in data analysis.
9. What is data preprocessing, and why is it important?
10. How does data preprocessing improve data mining?

10. ANSWERS

10.1. Self-Assessment Answers

1. **Answer:** b) Data represented as facts and dimensions for analysis

2. **Answer:** b) Numerical measures like sales or revenue

3. **Answer**: b) To describe facts with attributes like time, location, product

4 **Answer:** c) Drill-down

5. **Answer:** b) Multidimensional data

6. **Answer:** b) It enables faster and more flexible data analysis

7. Answer: c) Star Schema

8. Answer: b) Star Schema

9. **Answer:** c) Galaxy Schema

10. **Answer:** b) Causes data redundancy due to denormalization

11. **Answer:** c) Galaxy Schema

12. Answer: Mean

13. Answer: Median

14. Answer: Mode

10.2. Terminal Answers

Answer 1: The multidimensional data model organizes information into facts and dimensions, making it easier to analyze. It allows users to explore data from multiple perspectives, such as time, location, or product.

Answer 2: A data cube is a way of representing multidimensional data for analysis. It supports operations like roll-up, drill-down, slice, dice, and pivot, enabling quick and interactive insights.

Answer 3: Fact tables contain numerical or quantitative data such as sales or profit, while dimension tables store descriptive attributes like time, customer, or region. Both work together for effective analysis.

Answer 4: A Star Schema consists of a central fact table connected to multiple dimension tables. It is simple, denormalized, and easy to query, though it often causes data redundancy.

Answer 5: Snowflake Schema is an extension of the Star Schema where dimensions are normalized into sub-dimensions. This reduces redundancy and saves storage space but increases query complexity.

Answer 6: Galaxy Schema, also known as Fact Constellation, uses multiple fact tables sharing common dimensions. It is highly flexible for complex processes but is the most difficult to design and maintain.

Answer 7: Measures of central tendency are statistical methods to summarize data using a single representative value. The mean, median, and mode are the most common measures for identifying central patterns.

Answer 8: The mean is the arithmetic average of a dataset, found by dividing the sum of values by the number of items. It is best used when data is evenly distributed and not skewed by outliers.

Answer 9: Data preprocessing is the process of cleaning and preparing raw data for analysis. It ensures consistency, accuracy, and reliability, which improves the quality of insights derived from data mining.

Answer 10: Preprocessing removes noise, fills missing values, and transforms data into a suitable format. This leads to better accuracy and efficiency in mining algorithms, making the results more meaningful.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 4

DCA2207

DATA MINING AND VISUALISATION

Unit – 4

Data Mining Primitives

TABLE OF CONTENTS

1. INTRODUCTION

In the previous unit, learners gained insights into how data is stored, modelled, and analysed in a data warehouse environment. They explored the multidimensional data model, data cubes for summarization and aggregation, and different schema designs such as Star, Snowflake, and Galaxy schemas. The unit also introduced measures like additive, semi-additive, and non-additive, along with basic data pre-processing techniques that prepare raw data for OLAP and reporting. This foundation helped learners understand how multidimensional structures enable effective data organization and analysis.

In this unit, learners will explore the essential steps involved in preparing raw data for analysis. Data pre-processing is a critical phase in data mining and machine learning, ensuring that data is clean, structured, and ready for further processing. Poor-quality data can lead to inaccurate results, making pre-processing a necessary step before applying advanced analytical techniques.

By the end of this unit, you will understand the importance of data pre-processing and gain hands-on knowledge of various techniques that prepare data for meaningful analysis. Acquiring correct insights and making data-driven judgements are both made possible by this essential step.

1.1. Learning Objectives

- **Define** data cleaning
- **Describe** the various task involved in data mining task premitives
- **Identify** various techniques involved in cleaning, integrating and transforming
- **Evaluate** the challenges and the advantages of these techniques

2. DATA MINING TASK PRIMITIVES

Primitives in data mining are the fundamental elements that make up a data mining process. Data mining relies on these primitives to symbolise its most basic and prevalent operations. A more efficient, effective, and easily understood data mining process is possible with the help of data mining job primitives, which offer a modular and reusable approach.

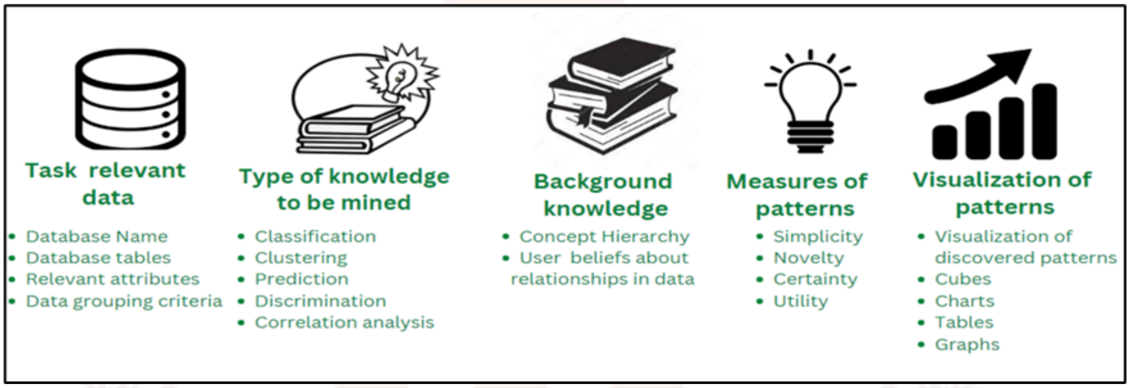

Figure 1: Data Mining Task Primitives

The Data Mining Task Primitives are as follows:

The set of task relevant data to be mined: Data that is pertinent and required for a given study or activity being performed utilising data mining techniques is what this term refers to. Customer demographics, sales figures, and website traffic statistics are all examples of the kinds of data that might be pertinent to the current endeavour. Since not all data is required or useful for the task at hand, the data chosen for mining is usually a subset of the total data available. As an example, using the input database to extract the name of the database, the tables inside the database, and any necessary attributes from the dataset.

Kind of knowledge to be mined: It describes the desired outcomes of data mining in terms of knowledge or understanding. The data mining tasks that need to be completed are outlined here. Classes, clusters, discrimination, characterisation, association, and evolution analysis are all a part of it. Data mining techniques including classification, clustering, prediction, discrimination, outlier detection, and correlation analysis are just a few examples of what it can do.

Background knowledge to be used in the discovery process: What this means is that the data mining process is guided by any prior knowledge or information. This can encompass both data-specific knowledge and domain-specific knowledge, such as jargon, trends, or best practices unique to a certain industry. Information gleaned via data mining can be more precise and applicable with the aid of prior knowledge. For instance, evaluating and performing better by drawing on prior information like idea hierarchies and user opinions around data linkages.

Interestingness measures and thresholds for pattern evaluation: Data mining quality assurance is the process of determining whether or not the patterns or insights found through data mining are useful and relevant. A pattern's perceived interest or relevance can be quantified using interestingness measures that take into account factors like its frequency, confidence, or lift. Meaningful or task-relevant patterns can be identified using these metrics. In contrast, pattern evaluation thresholds establish the bar that a pattern must cross before it is even evaluated for more study or action. For instance, determining a suitable cutoff value for the pattern evaluation and assessing the data's interest using metrics like utility, certainty, and novelty.

Representation for visualizing the discovered pattern: It describes the procedures required to easily grasp and analyse the patterns or insights found by data mining. Data visualisation tools like charts, graphs, and maps help to portray the data and reveal hidden relationships, patterns, and trends. A broader audience, including non-technical stakeholders, can access and understand the insights generated by data mining when the found pattern is visually represented. Using a variety of visualisation techniques including barplots, charts, graphs, tables, etc., to present and display the data that has been analysed for patterns.

Advantages of Data Mining Task Primitives

The use of data mining task primitives has several advantages, including:

1. **Modularity:** Data mining task primitives offer a modular approach to data mining, enabling flexibility and easy modification or replacement of certain tasks.
2. **Reusability:** To cut down on wasted time and energy, data mining task primitives can be reused across several data mining projects.
3. **Standardization:** In order to ensure that the data mining process is consistent and of high quality, data mining job primitives offer a standardised approach.
4. **Understand ability:** Primitives for data mining tasks are straightforward, which facilitates better teamwork and communication.
5. **Improved Performance:** By streamlining data processing and making data more suitable for particular data mining algorithms, data mining task primitives can boost data mining performance.

6. **Flexibility:** Because data mining job primitives can be mixed and matched in a variety of ways, the data mining process can be tailored to each project's unique requirements.
7. **Efficient use of resources:** Primitives for data mining activities enable targeted execution with appropriate tools, eliminating wasteful processes and cutting down on computational power and time requirements, ultimately leading to more efficient resource utilisation.

2.1. Role of Task Primitives

1. Bridging the Gap Between Users and Mining Systems

- **Business users** often think in terms of goals, not algorithms. Their questions are strategic:
    - "What products are often bought together?"
    - "Which customers are likely to leave?"
- **Task primitives** act as a **translation layer**, converting these high-level, domain-specific questions into **technical mining tasks** the system can execute.
- This bridge ensures that users don't need to understand the underlying data mining algorithms—they just need to express their intent.

2. Customization of Mining Tasks

- Task primitives allow users to **specify the type of knowledge** they want extracted:
    - **Association rules**: Discover relationships between items or events.
    - **Classification**: Predict categories or outcomes based on input data.
    - **Clustering**: Group similar data points without predefined labels.
    - **Sequential patterns**, **outlier detection**, and more.
- This customization ensures that mining is **goal-driven**, not generic. It aligns the mining process with the **specific business context**.

3. Flexibility and Control in Data Exploration

- Users can **fine-tune** mining tasks using parameters:
    - **Data subset**: Focus on a particular region, time frame, or product line.
    - **Interestingness thresholds**: Set minimum support/confidence levels to filter out trivial patterns.

    - **Presentation format**: Choose how results are displayed—rules, graphs, dashboards, etc.
- This empowers users to **explore data interactively**, adjusting the scope and granularity as needed.

4. Improving Relevance of Results

- Task primitives ensure that the **patterns discovered are not just statistically valid**, but **strategically useful**.
- They help align mining outputs with:
    - **Business objectives**
    - **Operational constraints**
    - **Decision-making needs**

Why These Components Matter

Together, these components ensure that data mining is:

- **Targeted**: Focused on the right data and questions
- **Customizable**: Adapted to user goals and domain knowledge
- **Relevant**: Produces insights that are meaningful and useful
- **Actionable**: Presented in a way that supports real-world decisions

SELF-ASSESSMENT QUESTIONS – 1

Multiple Choice Questions	
1	Which of the following best defines Task-Relevant Data in data mining? a) A visualization method for presenting results b) A subset of data relevant to the mining task c) A measure of pattern interestingness d) A clustering algorithm
2	Association rules such as “Customers who buy X also buy Y” fall under which type of knowledge to be mined? a) Classification b) Clustering

	c) Association d) Outlier Detection
3	Which of the following is an example of Background Knowledge? a) Support and confidence b) Product hierarchy (Electronics → Mobiles → Smartphones) c) Customer transaction log d) Scatter plot visualization
4	The role of Interestingness Measures in data mining is to: a) Collect raw data from warehouses b) Filter out trivial patterns and highlight useful ones c) Present results in a dashboard format d) Normalize input data
Fill in the blanks:	
5	__________ specifies the format in which the discovered patterns are presented to the user.
6	Support, confidence, and lift are common examples of __________.
7	__________ refers to the additional domain-specific knowledge, such as taxonomies or business rules, that enhances the mining process.

3. DATA CLEANING

It is often known that the extent and quality of your insights and analyses are directly influenced by the data quality that you utilise. In data analysis, "garbage in, garbage out" is the golden rule. Data cleaning (also spelt data cleansing or data scrubbing) is an important initial step in creating a company culture that makes good data decisions.

What is data cleaning?

Data cleaning refers to the process of repairing or removing corrupted, incorrect, poorly structured, duplicate, or incomplete data from a dataset. Integrating data from many sources often leads to issues like data duplication or mislabeling. If the data is incorrect, the algorithms and results will not be reliable, no matter how good they look. There is no agreed-upon way to describe data cleaning since every dataset is different. Nevertheless, developing a template is crucial to guarantee accurate data cleansing on a constant basis.

Data transformation and cleaning the data are not the same thing.

Data cleansing is the process of removing irrelevant information from your dataset. Transforming data means converting it from one structure or format to another. Data wrangling, data munging, and transformation operations all mean the same thing: they transform and map data from one "raw" data type into another for storage and analysis. The methods for cleaning that data are the main focus of this article.

3.1 Steps followed in cleaning the data

Data cleansing procedures may vary according to the kind of information your business stores, but generally speaking, they consist of the following:

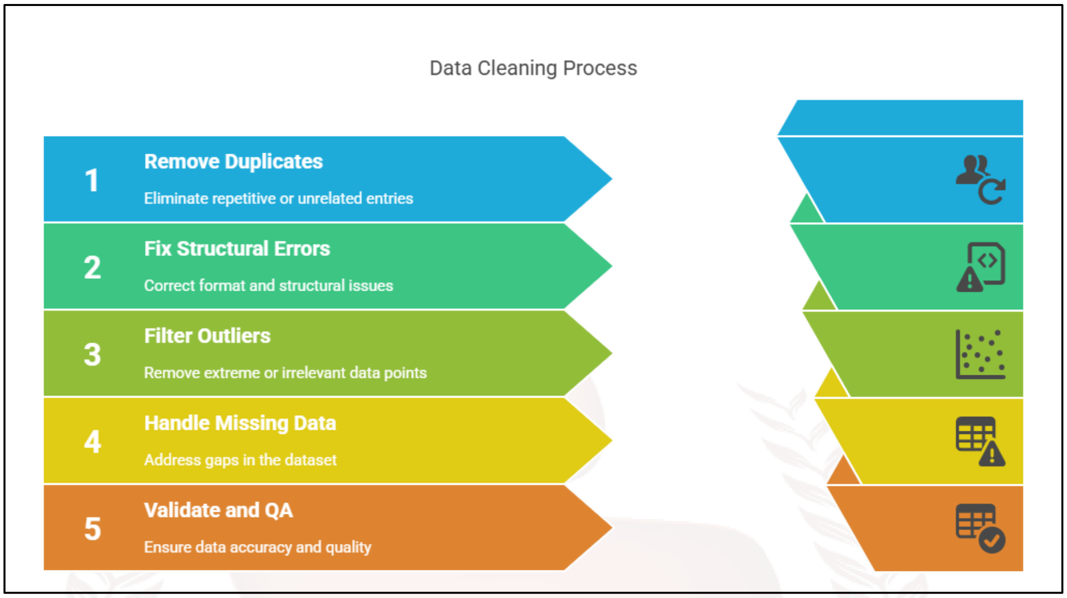

Figure 2: Data cleaning Process

1. Remove duplicate or unrelated comments

Remove duplicate or irrelevant observations and any undesirable ones from your dataset. Duplicate observations are a common occurrence when collecting data. Data duplication can occur as a result of data merging, data scraping, data acquisition from clients or other departments, or both. One of the most important things to think about in this process is de-duplication. The significance of these observations is diminished when they are unrelated to the particular topic you are attempting to investigate.

For instance, you might exclude irrelevant observations from a dataset that includes information on previous generations if your goal is to study millennial consumers. This can make the analysis go more smoothly, keep you closer to your original objective, and produce a dataset that is easier to maintain and use.

2. Fix structural errors

Odd naming conventions, typos, or incorrect capitalisation discovered while measuring or moving data are structural flaws. These discrepancies could lead mislabelled groups or classes. Though they should be examined under the same topic, "N/A" and "Not Applicable" could appear on any one sheet.

3. Filter unwanted outliers

From time to time, isolated findings that seem inappropriate for the data you are examining will surface. Your data will work better if you have a legitimate reason to exclude an outlier, such erroneous data entry.

On the other hand, it is quite unusual for an outlier to provide credence to a hypothesis you are testing. It is not necessarily incorrect simply because there is an anomaly. The credibility of the figure can only be determined by carrying out this essential step. If an outlier is found to be unfounded or unconnected to the study, it may be best to disregard it.

4. Handle missing data

Missing values are not tolerated by many algorithms, hence you cannot ignore missing data. Handling missing data has several possibilities. Although neither is perfect, both can be considered, for instance:

Though you can delete missing-value observations, doing so will cause information loss; so, be careful.

Once more, entering missing values depending about more findings allows you to operate inferring results from theoretical considerations rather than empirical facts, which may lead to faulty conclusions.

A change in data application may be necessary if null values are viewed rapidly.

5. Validate and QA

You ought to be able to respond after the data cleansing procedure is completed, then following questions as part of fundamental validation:

Can you make sense of the data?

- Are the rules that pertain to the data's specific domain being followed?
- Does it lend credence to your working theory or cast doubt on it? Can you tell me anything new from it?
- Are there any patterns in the data that you can use to back up your next theory?
- In that case, may the data quality be an issue?

Because of erroneous or noisy data, bad business strategy and decisions can be informed by false conclusions. If your data isn't robust enough to withstand more scrutiny, drawing false conclusions can put you in an embarrassing position during a reporting meeting. It is critical that you establish a data quality culture in your organisation prior to your arrival. In order to accomplish this, you need list the tools that you might use to create this plan.

3.2 Techniques for Cleaning Data

We recommend running the data via one of the several data-cleaning techniques that are now available. The steps are detailed below:

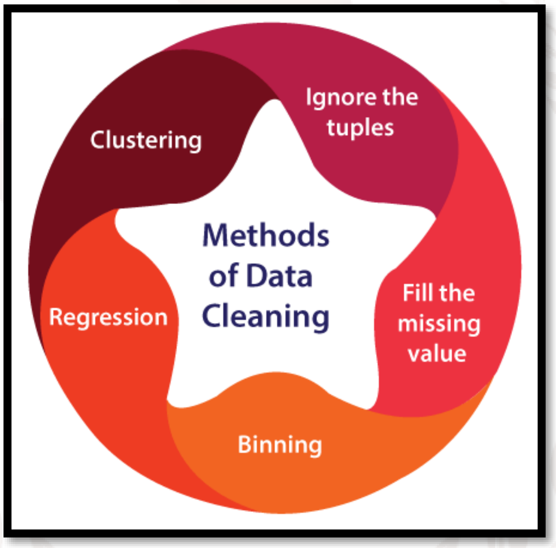

Figure 3: Data cleaning techniques

1. **Ignore the tuples:** Because it is only applicable when a tuple contains several attributes and missing values, ignoring them is not a particularly practical technique.
2. **Missing values is filled:** This approach is likewise not very effective or practical, therefore please fill in the blank. Also, the method may take a long period. A missing piece of value must be added to the strategy. Manual entry is by far the most popular way, but attribute means and the most likely value are also viable alternatives.
3. **Binning method:** The binning method is a simple and straightforward approach. The sorted data is smoothed using the adjacent values. The next step is to divide the data into multiple smaller pieces. In order to complete the task, the several methods are then employed.
4. **Regression**: The regression function is used to smooth out the data. Both linear and multivariate regression are possible. Unlike linear regressions, which only use one independent variable, multiple regressions make use of many.

5. **Clustering:** The primary emphasis of this method is on the group. Clustering is used to group data. The next step is to identify the outliers using clustering. At that point, a "group" or "cluster" of similar values is formed.

Process of Data Cleaning

What follows is an illustration of the data cleansing approach to data mining.

1. **Monitoring the errors:** Make note of the spots where mistakes tend to crop up most often. False or corrupted data can be more easily located and preserved. Data is king when it comes to integrating prospective replacements with existing management software.
2. **Standardize the mining process:** Consistent insertion points can help reduce the possibility of duplication.
3. **Validate data accuracy:** Perform data analysis and invest in data cleansing tools. Tools based on artificial intelligence were used to check for accuracy comprehensively.
4. **Scrub for duplicate data:** Locate duplicates to expedite data analysis. It is feasible to prevent retrying the same data by studying and investing in autonomous data-erasing technology that can evaluate flawed data quantitatively and automate the process.
5. **Research on data:** Please ensure that our data has been thoroughly reviewed, standardised, and double-checked before proceeding. Numerous third-party sources are available; we may provide you with data extracted directly from our databases by these verified and certified sources. They help us collect data and clean it up so it can be used for business choices in a trustworthy, accurate, and thorough way.
6. **Communicate with the team:** In addition to providing prospective clients with more targeted information, keeping the group apprised of developments will aid in developing and strengthening clients

3.3 Usage of Data Cleaning in Data Mining.

Some common applications of data cleansing in mining are as follows:

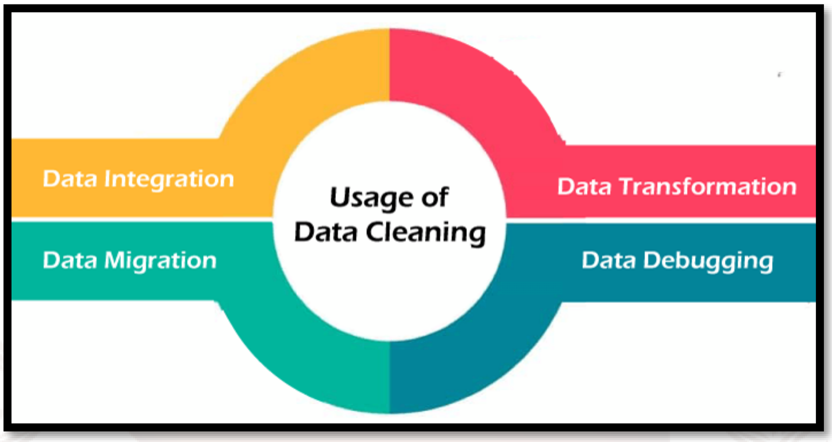

Figure 4: Steps in Data Pre-processing

Data Integration: Data integration is essential for fixing this problem because it is hard to ensure quality with poor dataThe term "data integration" describes the steps used to merge information from various databases into one. In this stage, data cleansing methods are used to standardise and format the embedded data set before transferring it to the final destination.

- **Data Migration:** Data migration refers to the process of moving files between different systems, formats, or applications. It is critical to keep the data's consistency, security, and quality while it's in transit to make sure the resultant data arrives at its destination in the right format and structure.
- **Data Transformation:** Prior to uploading to a destination, the data needs to be modified. The only way to accomplish this is through data cleansing, which considers the needs of the system in terms of formatting, organisation, etc. Data transformation methods usually include applying filters and rules before performing further analysis. Data transformation is an integral part of most data integration and management approaches. Data cleansing tools aid in cleaning the data by utilising the internal transformations of the systems.
- **Data Debugging in ETL Processes:** Cleaning data is a crucial part of the extract, transform, and load (ETL) process since it gets the data ready for analysis and reporting. Data purification ensures that only high-quality data is utilised for analysis and decision-making.

Data cleaning is a must. A retail company, for instance, may get erroneous or redundant data from various CRM or ERP systems. Finding and fixing data discrepancies would be the job of a trustworthy data debugging tool. In order to transfer the erased data to the target database, it will first be converted to a standard format.

3.4 Characteristics of cleaning the data

Data cleaning is essential for maintaining the accuracy, completeness, and safety of company records. Depending on the data's characteristics, their quality could range from average to poor. Here are the main elements of data cleansing in data mining:

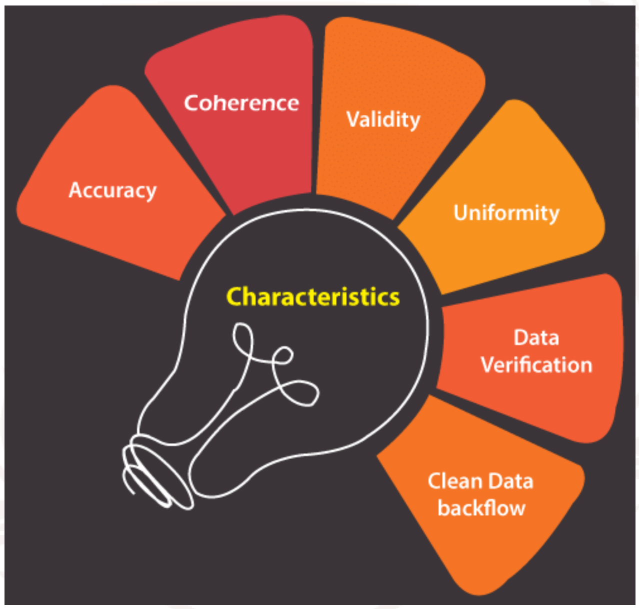

Figure 5: Steps in Data Pre-processing

- **Accuracy:** Data stored in the company's database must be absolutely precise. One way to verify their credibility is to compare them to other sources. Problems with the saved data could also arise if the source is incorrect or cannot be traced.
- **Coherence:** Data must be consistent with one another to guarantee that information about a person or body is consistent across all storage formats.
- **Validity:** Certain regulations must be put in place regarding data storage. Verification of the information is also necessary to ensure its accuracy.
- **Uniformity:** Because it simplifies things, data in a database must have consistent units or values. This is a crucial step in cleaning data.

- **Data Verification:** You must verify the procedure's appropriateness and efficacy at each stage. As a whole, verification encompasses the research, planning, and checking phases. After applying the data to a certain number of adjustments, the drawbacks often become apparent.
- **Clean Data Backflow:** For legacy applications to benefit from the cleaned data and prevent a later data-cleaning exercise from being necessary, it is necessary to replace the clean data with data that is not present in the source once quality issues have been addressed.

Tools for Data Cleaning in Data Mining

For those that lack the necessary skills or time to manually clean all of their data sets, data cleansing tools are a lifesaver. Those are tools you may have to shell out some cash for, but they're money well spent. There is an abundance of data cleansing options that can be found on the internet. Some of the best data cleansing tools are as follows:

1. Open Refine
2. Trifacta Wrangler
3. Drake
4. Data Ladder
5. Data Cleaner
6. Cloudingo
7. Reifier
8. IBM Infosphere Quality Stage
9. TIBCO Clarity
10. Winpure

Benefits of Data Cleaning

Among data cleaning's many advantages in data mining are the following: making decisions based on top-notch information; and, ultimately, increased productivity.

- Eliminating errors in the context of data from several sources. When errors are reduced, both clients and employees experience an improvement in their mood.
- The ability to lay out all the various purposes and intended uses for your data.
- By keeping an eye out for errors and making reporting better, users can fix corrupted or incorrect data for future uses. This aids in identifying where the problems are coming from.
- Decisions can be made more quickly and efficiently with the use of data purification procedures.

SELF-ASSESSMENT QUESTIONS – 2

Multiple Choice Questions	
8	What is the primary goal of data cleaning? A) To delete as much data as possible B) To ensure data is accurate, consistent, and usable C) To increase the size of the dataset D) To remove all numerical data
9	What should be done if a dataset has missing values? A) Ignore the missing values B) Fill missing values using methods like mean, median, or mode C) Delete the entire dataset D) Replace missing values with random numbers

4. DATA INTEGRATION AND TRANSFORMATION

Data integration is a record preparation technique in data mining that combines information from various disparate sources into a single, coherent whole, with the goal of preserving and presenting the data in a more cohesive light. These assets could include multiple record cubes, databases, or flat documents. The statistical integration strategy is officially stated as a triple (G, S, M) approach, where G represents the global schema, S represents the heterogeneous source of schema, and M represents the mapping between source and global schema queries. Data redundancy, inconsistency, and duplication are all aspects that must be addressed during data integration efforts.

In this unit, you will learn about Data integration in data mining and discuss its methods, issues, techniques, and tools.

What is Data Integration?

Given the variety of potential places to find the data, it has become an essential component of data operations. It is a method that makes a user's status visible in a unified display by integrating data from multiple sources. Databases, data cubes, and flat files are all examples of possible sources of information exchange between systems. Integrating data from various sources into a unified whole is known as data fusion. There can be no contradictions, inconsistencies, redundancies, or unfairness in the combined results.

The ability in order to consolidate many data sets into one whole while preserving data integrity is a key benefit of data integration. Executives and managers are able because the data-mining tool may glean valuable insights from the extracted data, allowing the organisation to make more informed strategic decisions. A formal description of the data integration methods is a triple (G, S, M), where;

A global schema (G), a heterogeneous schema source (S), and a mapping (M) between the two are all parts of a schema query.

What Makes It Crucial?

Big data, with all its advantages and disadvantages, is a boon to companies that aim to remain competitive and relevant. Gathering information about consumers and the market is a typical use case for data integration tools and services. With the help of data integration, queries can be run in these massive datasets, and real-time information delivery can be stimulated by analytics on consumer data and business intelligence. In order to facilitate enterprise reporting, analytics for prediction, and business intelligence, enterprise data integration supplies integrated data to data centres.

The healthcare industry places a special emphasis on data integration. Healthcare providers can improve their ability to diagnose illnesses and conditions by integrating data from multiple sources. This is made possible through the integration of data from different patient records and clinics. The quality of medical insurance claims processing and the consistency and accuracy of patient names and contact information are both enhanced by effective data collection and integration. The ability for various systems to exchange data with one another is known as interoperability.

4.1 Ways to Integrate Data

Data integration can be approached in two major ways. I have listed them below:

Tight Coupling

The usage of ETL allows for the centralisation of data that has been integrated from many sources. (Extraction, Transformation, and Loading).

Loose Coupling

The real source databases are the best places to store facts with loose coupling. Using this method, a user can submit a query through an interface, and the system will immediately begin transforming it into a format that the supplier database can comprehend.

Issues in Data Integration

Data integration in Data Mining can be fraught with difficulties. To name a few of the concerns:

The Challenge of Identifying Entities

You are aware that the records come from a variety of different places, therefore the question becomes how to "match the real-world entities from the data." For instance, you were provided with client data sourced from niche statistical databases. One statistics supply assigns an entity a customer identity, and another statistics supply assigns an entity a customer range. In order to avoid making mistakes when integrating schemas, it is recommended to analyse such metadata statistics.

Verifying that a character's functional dependency and referential constraints in the source machine are equal to those in the target machine is the last step in structural integration. Picture this: in one machine, the discount is applied to the whole order, but in every other machine, it's applied to each item. It is important to make note of this difference before adding the data from those assets to the goal system.

Duplicate Data and Correlation Study

Redundancy is a big problem that arises during data integration. Redundant data is meaningless information that is no longer needed. Attributes generated by the application of another property within the data set can potentially cause it to show up as well. For instance, if the patronage and separate data sets are stored in one truth set, and the purchaser's beginning date is stored in another, then the age property might be unnecessary since it can be inferred from the beginning date.

When there are discrepancies, the amount of duplication inside the attribute is even higher. Determining redundancy can be accomplished through the use of correlation analysis. By looking at how one characteristic is dependent on the others, we can find the connection between them.

Tuple Duplication

Duplicate tuples, along with redundancy, have also been dealt with using information integration. If the denormalised table was used as a deliverable for data integration, then the resulting information can also contain duplicate tuples.

Support and detection of data warfare

Data warfare, which involves merging records from many sources, is harmful. Statistics units might differ in the same manner as characteristic values. One possible explanation for the discrepancy is the differing ways in which the special data units represent them. The cost of a hotel room, for instance, may be indicated in a specific currency in small, unique cities. While integrating the data, we find and fix issues like this.

4.2 Methods for Integrating Data

The field of data mining encompasses a wide range of methods for integrating data. The following are a few of them:

Manual Integration

When integrating data, this approach stays away from automation. In order to derive useful insights from data, the data analyst must first gather, cleanse, and combine the data. For a small business with a limited amount of data, this approach works. Big, complex, and ongoing integration, nevertheless, will take a lot of time. Everything has to be done by hand, which means it takes a long time.

Middleware Integration

The data set is created by taking data from multiple sources, normalising it, and then storing it using the middleware software. Whenever a company has to transfer data from older systems to newer ones, they employ this method. The role of middleware software is to bridge the gap between older and more modern systems. You have the option to bring an adapter that enables the connection of two systems that have different interfaces. Some systems are the only ones that can use it.

Application-based integration

Data extraction, transformation, and loading from various sources is done utilising software applications. Although this approach is more involved, it saves time and effort compared to other options because developing such an app requires technical knowledge. Although this approach is more involved, it saves time and effort compared to other options because developing such an app requires technical knowledge.

Uniform Access Integration

Data from a more dispersed source is combined using this strategy. Nevertheless, in this case, the data remains in its original spot; its position is unaltered. A consolidated view of the combined data is all that this method does. Since the end-user only sees the integrated view, there's no need to save the integrated data separately.

Data Warehousing

There is a tangential relationship between this method and the uniform access integration method. Conversely, the unified view is kept in a separate place. Because of this, the data analyst is able to handle increasingly complex queries. The unified data view or copy necessitates distinct storage and maintenance expenses, despite being a promising solution and increasing storage prices.

Integration tools

In data mining, you can find a number of tools for integration. The following are a few of them:

- **Tool for integrating data on promise**
  Integrating data from local sources and connecting historical databases via middleware software is the job of an on-premise data integration solution.

- **Data integration tool available as open-source**

  An open-source data integration tool is a great substitute for costly business solutions. However, if you want to use the tool, you should know that you are fully responsible for protecting the privacy and security of the data.

- **Data integration tool hosted on the cloud**

  The term "integration platform as a service" might describe a data integration solution that is hosted in the cloud. An integral part of data mining is data transformation. Importing raw data into an analytically sound format is what this process is all about. The goal of data transformation is to improve data quality, decrease redundancy, and make data more consistent and relevant for the intended study.

SELF-ASSESSMENT QUESTIONS – 3

Multiple Choice Questions	
10	What is the main purpose of data integration? A) To separate data from different sources B) To merge data from multiple sources into a unified format C) To delete redundant data D) To visualize data in graphs
11	Which of the following is a common challenge in data integration? A) Data redundancy and inconsistency B) Increasing the number of datasets C) Reducing data storage costs D) Ignoring missing values
12	Which of the following is a common challenge in data integration? A) Data redundancy and inconsistency B) Increasing the number of datasets C) Reducing data storage costs D) Ignoring missing values

5. WHICH OF THE FOLLOWING IS A COMMON CHALLENGE IN DATA INTEGRATION?

When it comes to data mining, data transformation is crucial for a number of reasons. To begin, it may be difficult and time-consuming to analyse raw data that is either unstructured, inaccurate, or incomplete. Thus, data transformation mainly aims to clean and organise the data so it can be used for additional analysis.

Secondly, data transformation helps simplify the data. Structured data is essential for data mining algorithms to discover trends, patterns, and connections. Data transformation helps simplify data by converting it into an acceptable format and removing any unwanted or extraneous information.

Thirdly, data transformation verifies the data is accurate and relevant to the study. Varieties of scales, formats, and measuring units may be used by various data sources. When data is transformed, it becomes more standardised, which improves its comparability and analytical capabilities.

It is possible to improve the accuracy and efficiency of data mining algorithms by altering the data. Data mining algorithms find patterns and trends more accurately when they convert the data to a suitable format.

Common Techniques for Data Transformation

A number of methods exist for transforming data. As a whole, these methods can be classified into three broad categories: data reduction, data integration, and data purification.

- **Data Cleaning**

  Data cleaning is the process of finding and correcting errors, inconsistencies, and inaccurate information. There are several ways to achieve this, including as –

- Interpolation, the data's mean, median, or mode, or any number of other techniques can be used to fill in missing numbers.
- By comparing the values of each record and eliminating those that match, duplicates can be eliminated.
- Outlier management: Using statistical methods, outliers can be identified and then removed or fixed.

➢ **Data Integration**

The term "data integration" describes the action of combining data from several sources. A number of methods exist for this, such as −

- Merging involves combining datasets that contain similar variables.
- "Joining" refers to the process of merging datasets that include comparable observations.
- Appending refers to the process of adding new observations or variables to an existing dataset.

➢ **Data Reduction**

Data reduction is the process of simplifying and reducing the amount of data. This can be accomplished in a number of ways, such as

- Sampling: The word "sampling" describes taking a small portion of a dataset and analysing it instead of the whole thing.
- Reducing the number of variables in a dataset while retaining the most important data is called dimensionality reduction.
- The process of aggregation

Advantages of Data Transformation in Data Mining

- **Improved data quality:** Raw data that is unstructured and missing information is often not suitable for analysis. Data transformation helps clean and organise data so it can be analysed more effectively. As a result, the data's reliability and quality for decision-making can improve.
- **Simplified process:** Data mining algorithms necessitate structured data in order to discover trends, correlations, patterns. Data transformation is useful for data simplification because it removes unnecessary information and changes the data into the right format. The data may become less complex and easier to analyse as a result of this.
- **Improved precision:** Data mining algorithms are able to more accurately and successfully identify trends and patterns after they have transformed the data into a usable format. Improved prediction and decision-making could emerge from this.
- **Standardisation:** Different data sources may have different formats, scales, and measurement units, which can lead to issues with standardisation. When data is

transformed, it becomes more standardised, which improves its comparability and analytical capabilities. The data's consistency and relevance to the proposed analysis can be enhanced in this way.

- **Efficiency Gain:** By reducing the amount of data that needs to be analysed, data transformation helps data mining algorithms perform more efficiently. Data mining algorithms improve the speed and accuracy of analysis by reducing the quantity and complexity of data.

Disadvantages of Data Transformation in Data Mining

- **Information loss:** Data reduction strategies, in particular, can lead to information loss during data transformation. Consequently, the analysis's accuracy and reliability can suffer.
- **Overfitting:** Overfitting happens when data modification causes the model to be overfit. Consequently, the model could get too reliant on the initial data set and stop working with new data.
- **Complexity:** It is possible that data transformation will make data mining more difficult. Consequently, the analysis's findings could be hard to decipher.
- **Cost:** When dealing with massive data sets, the expense of data transformation becomes apparent. For groups operating on a shoestring budget, this can pose a significant obstacle.
- **Time-consuming:** When there is a large amount of data that needs to be modified, data transformation could take some time. Because of this, the analysis and final decisions could take longer than expected.

SELF-ASSESSMENT QUESTIONS – 4

Multiple Choice Questions	
13	What is the primary purpose of data transformation in data preprocessing? A) To visualize data in charts and graphs B) To convert data into a suitable format for analysis C) To remove all missing values from the dataset D) To increase the size of the dataset
14	Which of the following is a common technique used in data transformation? A) Normalization B) Data Encryption

	C) Data Deletion D) Data Redundancy

6. DATA REDUCTION IN DATA MINING

Extracting valuable information from large databases is what data mining is all about. Data mining and analysis on massive datasets is computationally expensive and time-consuming, rendering the procedure unrealistic and unworkable.

While reducing data, data reduction strategies keep data intact. The goal of data reduction is to provide a more compact representation of the original data by decreasing its size. In order to get a much smaller representation of the dataset while keeping the original data intact, data reduction techniques are employed. Data reduction improves data mining efficiency, which in turn yields the same analytical outputs. Full screen

Data reduction has no effect on the results of data mining. Data mining results before and after data reduction are similar, according to this.

As a result of data reduction, the definition is made more concise. When dealing with smaller amounts of data, complicated and computationally expensive techniques can be more easily employed. Increasing the number of dimensions (columns) and decreasing the number of rows (records) are two methods for reducing data.

Techniques of Data Reduction

Some examples of data reduction strategies used in data mining are:

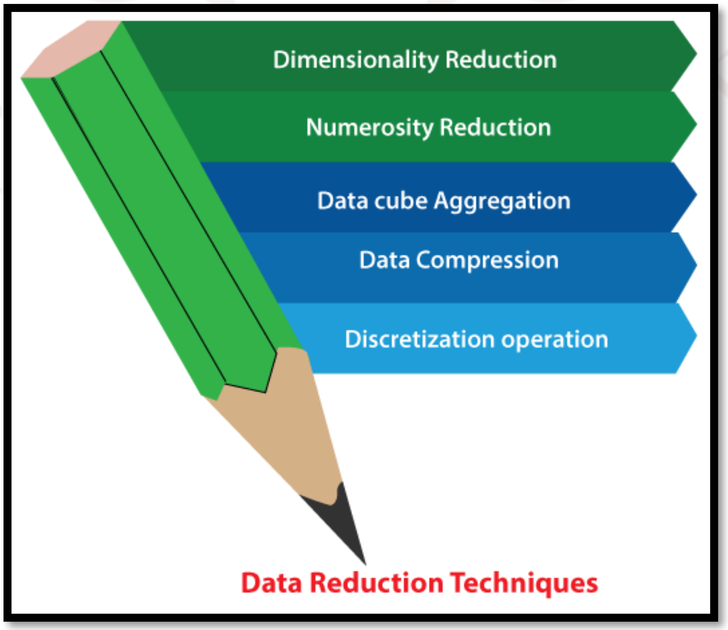

Figure 6: Steps in Data Pre-processing

1.Dimensionality Reduction

To conduct our analysis, we always make use of the necessary attribute whenever we come across data that is weakly important. The original data set's volume is reduced using dimensionality reduction, which removes the attributes from the dataset. It shrinks data by removing unnecessary or out-of-date characteristics. Presented here are three approaches to reducing dimensionality are wavelet transformation, principal component analysis and attribute subset selection

2. Numerosity Reduction

A substantially reduced representation of the original data volume is produced by the numerosity reduction. Two varieties of this method exist: parametric and non-parametric reduction of numerology.

i. Parametric: One aspect of parametric numerosity reduction is the practice of retaining solely data parameters rather than the raw data itself. The regression and log-linear technique is one way to reduce parametric numerology.

ii. Non-Parametric: The absence of an underlying model is characteristic of non-parametric numerosity reduction methods. Non-Parametric reduction is more consistent across all data sizes, although it might not be able to reduce as much data as parametric methods.

Histogram: One way to visually illustrate the distribution of values in data is via a histogram, which shows how frequently each value appears. The distribution of data for an attribute is shown by the histogram using the binning method. A separate subset known as "bin" or "buckets" is utilised. Data might be thick, thin, homogeneous, or skewed in a histogram. The histogram can be applied to more than one characteristic instead of only one.

3. Data Cube Aggregation

Data can be simplified by using this technique for aggregation. As a multidimensional aggregation, data cube aggregation reduces data by representing the original set of records by aggregate at different levels of a data cube.

4. Data Compression

In order to reduce storage requirements, data compression makes use of encoding or other transformations to the data structure. Data compression is the process of creating a smaller version of a dataset by eliminating unnecessary information and storing it in binary form. Lossless compression refers to data that can be effectively restored from its compressed form. But lossy compression is the inverse, and it's impossible to recover the original form from the compressed one. Additional methods for data compression include dimensionality and numerology reduction.

5. Discretization Operation

In order to transform continuous qualities into interval data, the data discretisation approach is employed. We substitute numerous attribute names with small interval values instead of constant values. Meaningful and easily digestible presentation of mining findings is achieved.

SELF-ASSESSMENT QUESTIONS – 5

Multiple Choice Questions	
15	What is the main objective of data reduction in data preprocessing? A) To increase the size of the dataset B) To simplify the dataset while preserving important information C) To delete irrelevant data without any analysis D) To merge multiple datasets into one
16	Which of the following is a common data reduction technique? A) Data Replication B) Dimensionality Reduction C) Data Encryption D) Data Augmentation

7. SUMMARY

- Data preprocessing is a crucial step in data mining that ensures the quality and reliability of data before analysis. It begins with data cleaning, where missing values are handled through imputation or removal, duplicates are eliminated, and noise or outliers are detected and corrected.
- Next, data integration combines data from multiple sources, resolving inconsistencies in formats and structures to create a unified dataset. Data transformation follows, where techniques like normalization (scaling values to a fixed range), standardization (adjusting data to have a mean of 0 and a standard deviation of 1), and encoding categorical data into numerical values are applied for better analysis.
- To enhance efficiency, data reduction techniques such as dimensionality reduction (e.g., PCA), feature selection (retaining only relevant attributes), and sampling (selecting a smaller representative dataset) are used.
- Data discretization further simplifies data by converting continuous attributes into categorical bins, making it easier for pattern recognition. Finally, data splitting ensures effective model training by dividing data into training, validation, and test sets.
- Proper data preprocessing enhances data quality, ensures consistency, improves model accuracy, reduces computational complexity, and prevents biases in analysis. Without it, data mining models may produce misleading or inefficient results, making preprocessing an essential step in any data-driven
- Data transformation, the process of transforming raw data into an analytically acceptable format, is an essential part of data mining. At this point, we're working to improve the data's quality and analytical significance by eliminating duplicate entries.
- In order for data mining machines to find trends and patterns more effectively, there are several methods for transforming data, including data integration, data reduction, and data purification.
- When numerous sources of data are combined, it is called data integration. Some of the challenges that data integration faces include dealing with inconsistent or duplicate data, outdated systems, and duplicate data.

8. GLOSSARY

Data Preprocessing	-	The process of cleaning, transforming, and preparing raw data for analysis to improve accuracy and efficiency in data mining.
Data Cleaning	-	The process of handling missing values, removing duplicates, correcting inconsistencies, and eliminating noise or outliers in a dataset.
Data Integration	-	Combining data from multiple sources into a single, unified dataset while resolving inconsistencies in formats and structures.
Data Transformation	-	The process of converting data into a suitable format for analysis, including normalization, standardization, and encoding categorical variables.
Normalization	-	A technique used to scale data within a fixed range (e.g., 0 to 1) to ensure uniformity in analysis.
Standardization	-	Adjusting numerical values in a dataset so that they have a mean of 0 and a standard deviation of 1, improving comparability.
Encoding Categorical Data	-	Converting non-numeric (categorical) data into numerical form, such as one-hot encoding or label encoding, for machine learning models.
Data Reduction	-	The process of simplifying large datasets by reducing the number of attributes or records while retaining important information.
Dimensionality Reduction	-	A technique used to reduce the number of features (variables) in a dataset while preserving essential information (e.g., PCA – Principal Component Analysis).
Feature Selection	-	Identifying and keeping only the most relevant attributes in a dataset to improve efficiency and accuracy in data analysis.
Sampling	-	Selecting a smaller, representative subset of data from a larger dataset to speed up processing while maintaining accuracy.
Data Discretization	-	Converting continuous numerical data into categorical bins or groups (e.g., age groups: 0–18, 19–35, etc.) to simplify analysis.

Data Splitting	-	Dividing a dataset into training, validation, and test sets to ensure proper model evaluation and performance assessment.

9. TERMINAL QUESTIONS

1. What is data preprocessing, and why is it important in data mining?
2. What are the four major steps in data preprocessing?
3. What are some common techniques used in data cleaning?
4. How does data integration help in data preprocessing?
5. What are the differences between normalization and standardization in data transformation?
6. Why is data reduction necessary in data preprocessing?
7. What are some common data reduction techniques?
8. What is the advantage of data transforming in data mining?
9. What challenges can arise during data preprocessing, and how can they be addressed?
10. How does data preprocessing improve the performance of machine learning models?

10. ANSWERS

10.1. Self-Assessment Answers:

1. Answer: B) A subset of data relevant to the mining task
2. Answer: C) Association
3. Answer: B) Product hierarchy (Electronics → Mobiles → Smartphones)
4. Answer: B) Filter out trivial patterns and highlight useful ones
5. Answer: Presentation of Discovered Patterns
6. Answer: Pattern Interestingness Measures
7. Answer: Background Knowledge
8. Answer: B) To ensure data is accurate, consistent, and usable
9. Answer: B) Fill missing values using methods like mean, median, or mode
10. Answer: B) To merge data from multiple sources into a unified format
11. Answer: A) Data redundancy and inconsistency
12. Answer: A) The process of merging different database structures into a common format
13. Answer: B) To convert data into a suitable format for analysis
14. Answer: A) Normalization
15. Answer: B) To simplify the dataset while preserving important information
16. Answer: B) Dimensionality Reduction

10.2. Terminal Question Answers

Answer 1: Data preprocessing is the process of cleaning, transforming, integrating, and reducing raw data to improve its quality and make it suitable for analysis. It is important because it ensures accuracy, consistency, and efficiency in data mining, leading to better decision-making and model performance.(For more information refer section : 2.3)

Answer 2: The four major steps in data preprocessing are:

1. Data Cleaning – Handling missing values, removing duplicates, and eliminating noise.
2. Data Integration – Combining data from multiple sources into a single dataset.
3. Data Transformation – Normalizing, standardizing, and encoding data for analysis.
4. Data Reduction – Reducing data volume or complexity while maintaining key information. .(For more information refer section : 2.3)

Answer 3: Common data cleaning techniques include:

- Filling or removing missing values (using mean, median, or mode).
- Detecting and removing duplicate records.
- Identifying and correcting outliers.
- Standardizing inconsistent formats (e.g., date formats, unit conversions).

Answer 4: Data integration combines data from multiple sources (databases, files, APIs) into a unified dataset. It helps in removing inconsistencies, resolving conflicts, and improving data completeness, ensuring better insights and analysis(for more information refer section 2.5.2)

Answer 5: Normalization: Rescales data within a specific range (e.g., 0 to 1) to maintain uniformity were as Standardization: Adjusts data so that it has a mean of 0 and a standard deviation of 1, improving comparability across different datasets

Answer 6: Normalization: Rescales data within a specific range (e.g., 0 to 1) to maintain uniformity were as Standardization: Adjusts data so that it has a mean of 0 and a standard deviation of 1, improving comparability across different datasets

Answer 7: Common data reduction techniques include (Refer section 2.7 for more information)

•Dimensionality Reduction (e.g., PCA – Principal Component Analysis) to remove less important features.
•Feature Selection to retain only the most relevant attributes.
•Sampling to select a representative subset of data.

Answer 8: Data transformation helps by converting unstructured data into a structured format using text processing, image recognition, and data encoding techniques. (Refer secition 2.6)

Answer 9: Refer Section: 2.3.1

During data preprocessing, several challenges can arise. Datasets often have missing values, which may be addressed using imputation or removal techniques. Noisy data with errors or outliers can distort results, so smoothing, normalization, or outlier detection is applied. Inconsistent formats (like date formats or categorical labels) require standardization. High dimensionality makes models complex; dimensionality reduction techniques like PCA help here. Imbalanced data can bias predictions, and resampling methods such as SMOTE can balance it. Finally, scalability issues occur with large datasets, requiring distributed processing frameworks. Proper handling ensures data quality, consistency, and reliability for accurate analysis.

Answer 10: Data preprocessing improves machine learning performance by:

- Enhancing data quality, reducing errors, and ensuring consistency.
- Removing irrelevant or redundant features, leading to better generalization.
- Ensuring that numerical data is properly scaled and formatted for accurate predictions.
- Reducing noise and handling missing values to prevent biases in model training.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 4

DCA2207

DATA MINING AND VISUALISATION

Unit – 5

Data Reduction

TABLE OF CONTENTS

1. INTRODUCTION

In the previous unit, learners explored the essential steps involved in preparing raw data for analysis. Data pre-processing is a critical phase in data mining and machine learning, ensuring that data is clean, structured, and ready for further processing. Poor-quality data can lead to inaccurate results, making pre-processing a necessary step before applying advanced analytical techniques, the learners also understood the importance of data pre-processing and gained hands-on knowledge of various techniques that prepare data for meaningful analysis. Acquiring correct insights and making data-driven judgements are both made possible by this essential step.

In this unit, learners will gain an understanding of the concept of data reduction and its importance in simplifying large datasets for efficient analysis. They will explore various techniques of data reduction, including data cube aggregation, dimensionality reduction, numerosity reduction, and data compression. The unit also introduces Discretisation methods, which transform continuous data into meaningful intervals, and the concept of hierarchy generation, which allows data to be represented at multiple levels of abstraction. Along with this, learners will examine the advantages of data reduction, such as faster processing and reduced storage requirements, as well as its disadvantages and limitations, like potential information loss or reduced accuracy. By the end of the unit, learners will be equipped to apply reduction techniques effectively while balancing efficiency and accuracy in data mining tasks.

1.1. Learning Objectives

- **Define** data reduction
- **Describe** the various techniques involved in data reduction
- **Identify** the binning techniques and its types
- **Evaluate** the challenges and the advantages of data reduction

2. DATA REDUCTION

With data reduction, it is possible to get a smaller, more concise description of the original data while preserving its quality.

In data mining, data reduction is a method for shrinking datasets while keeping all of the relevant information intact. In data mining, data reduction is a method for reducing dataset sizes while keeping key information intact. When the dataset contains a lot of unnecessary or duplicate data or is too big to be handled properly, this can be helpful. first and foremost data. When the dataset contains a lot of unnecessary or duplicate data or is too big to be handled properly, this can be helpful.

There are several different data reduction techniques that can be used in data mining, including:

1. Data Sampling: Instead of utilising the complete dataset, this method requires picking a subset to operate with. If you need to compress a dataset without losing any of its general patterns or trends, this might be a good option.
2. Dimensionality Reduction: Reducing the amount of features in the dataset is the goal of this strategy. This can be achieved by merging numerous features into one or by deleting irrelevant features.
3. Data Compression: This method reduces the size of a dataset by employing compression techniques, either lossy or lossless.
4. Data Discretisation: Separating the whole spectrum of potential values into smaller, more manageable intervals or bins is the key to this method for turning continuous data into discrete data.
5. Feature Selection: This method requires picking out the most pertinent features from the whole dataset.

Keep in mind that there may be a compromise between data accuracy and data size when data reduction is applied. The accuracy and generalisability of the model are negatively impacted by the reduction of data.

To sum up, data reduction is a crucial part of data mining since it allows for the reduction of dataset size, which in turn improves the efficiency and performance of machine learning algorithms. Before putting it into action, though, you should weigh the dangers and rewards and be cognisant of the trade-off between data size and accuracy.

Importance of Data Reduction:

- Handles large-scale datasets
- Data reduction allows organisations to manage vast amounts of information without being overwhelmed. It condenses data while keeping important insights intact.
- Reduces storage cost and computation time
- By minimising the volume of data, storage requirements drop significantly. At the same time, computations run faster, saving both time and resources.
- Improves efficiency of algorithms
- Smaller, cleaner datasets make algorithms work more smoothly. This increases accuracy and reduces the risk of errors during analysis.
- Helps focus on meaningful data
- Irrelevant or redundant details are filtered out, leaving only valuable information. This ensures that analysis highlights patterns and insights that truly matter.

Role in Data Mining:

- Data reduction as a preprocessing Step-It minimises the size of raw datasets while retaining their essential features. This ensures data is manageable and ready for mining tasks.
- Prepares data for machine learning models-By simplifying datasets, it removes redundancies and noise. This allows machine learning models to train faster and produce more accurate results.
- Helps visualisation and Interpretation-Smaller, structured data is easier to represent in charts or graphs. This improves understanding and supports decision-making.

2.1 Methods of data reduction

Data reduction methods aim to minimise the size of datasets while preserving the integrity and quality of the original information. Since large volumes of data can slow down analysis and increase storage costs, reduction techniques help in creating a compact yet meaningful representation of the data. The idea is to maintain the essential patterns, relationships, and trends while eliminating redundancy or irrelevant details.

Common approaches include dimensionality reduction (e.g., Principal Component Analysis) to reduce the number of attributes, numerosity reduction (e.g., sampling, clustering, regression) to replace the data with smaller models, and data compression to encode data more efficiently. These methods improve storage, speed up computations, and enhance the scalability of mining algorithms without significant loss of accuracy.

These are explained as following below.

1. Data Cube Aggregation:

The purpose of this method is to simplify data aggregation. As an example, let's say that your company's income was collected every three months from 2012 to 2014 and used in your study. So that we may summarise the data in a manner that shows the total sales per year instead of every quarter, they get you involved in the yearly sales instead of the quarterly average. A synopsis of the data is provided.

2. Dimension reduction:

Whenever we encounter data that is just marginally relevant, we incorporate the necessary attribute into our analysis. It shrinks data by removing unnecessary or out-of-date characteristics.

- Step-wise Forward Selection –

We start with a blank slate of traits in the selection process, and then we prioritise those attributes according to how important they are in relation to one another. In statistics, it is known as a p-value.

Imagine a data set with the following attributes, with very few of them being duplicates.

Initial attribute Set: {X1, X2, X3, X4, X5, X6}

Initial reduced attribute set: { }

Step-1: {X1}

Step-2: {X1, X2}

Step-3: {X1, X2, X5}

Final reduced attribute set: {X1, X2, X5}

- Step-wise Backward Selection -

Beginning with a set of full attributes in the source data, this selection iteratively removes the poorest remaining attribute.

Imagine a data set with the following attributes, with very few of them being duplicates.

Initial attribute Set: {X1, X2, X3, X4, X5, X6}

Initial reduced attribute set: {X1, X2, X3, X4, X5, X6 }

Step-1: {X1, X2, X3, X4, X5}

Step-2: {X1, X2, X3, X5}

Step-3: {X1, X2, X5}

Final reduced attribute set: {X1, X2, X5}

Combination of forwarding and Backward Selection - Time is saved and the process is accelerated since we are able to eliminate the worst and choose the greatest features.

3. Data Compression:

Data compression uses two encoding methods—Huffman encoding and run-length encoding—to shrink file sizes. Its methods of compression allow us to classify it into two categories.

a. Lossless Compression

Run Length Encoding is one method for encoding that enables a straightforward and minimal reduction in data size. Algorithms are employed in lossless data compression to precisely restore the original data from compressed data.

b. Lossy Compression

Compression methods include the Discrete Wavelet transform approach and PCA (principal component analysis). As an example, we can discover meaning that is identical to the original image even though the JPEG image format is a lossy compression. Even if the decompressed data could seem different from the original, it's still possible to get usable information out of them when using lossy data compression.

4. Numerosity Reduction

As the model parameter is the only piece of information that needs to be saved in this reduction strategy, mathematical models or smaller representations of the data are used in place of the actual data. Alternative non-parametric approaches include sampling, clustering, and histogram analysis.

5. Discretisation & Concept Hierarchy Operation

In order to transform continuous qualities into interval data, data discretisation techniques are employed. We substitute small interval labels for many constant attribute values. This ensures that the mining results are presented in a clear and straightforward manner.

a. Top-down Discretisation

To implement top-down discretisation, often dubbed splitting, one must first think of one or two locations (also called breakpoints or split points) to partition the entire collection of characteristics, and then continue doing so until the conclusion.

b. Bottom-up Discretisation

This method is known as bottom-up discretisation, and it involves taking all the constant values as split points and then discarding part of them using a mix of the interval's neighbourhood values.

c. Concept Hierarchies

By gathering and substituting low-level concepts (like 43 for age) with high-level concepts (like middle age or Senior) in the form of categorical variables, it minimises the data size.

For numeric data following techniques can be followed:

1. **Binning -**

   In binning, numerical variables are transformed into their category equivalents. The amount of corresponding categories is proportional to the user-specified number of bins.

2. **Histogram analysis** -

   Using the histogram is similar to using binning in that it divides the value of attribute X into separate ranges called brackets. A number of guidelines exist for partitioning:

1. Equal Frequency Partitioning: Dividing the values according to how often they appear in the dataset.
2. Partitioning with Equal Width: Dividing the values into a defined gap according to the number of bins, which is a set of numbers from 0 to 20.
3. Gathering data that is comparable together is called clustering.

Advantages and Disadvantages of Data Reduction in Data Mining:

There are a lot of potential benefits and drawbacks to data minimisation in data mining.

Advantages:

1. Enhanced efficiency: By decreasing the dataset size, data reduction can aid in making machine learning algorithms more efficient. Working with enormous datasets can become more efficient and feasible as a result of this.
2. Data reduction, by reducing unnecessary or duplicate information from a dataset, can enhance the efficiency of machine learning algorithms, which leads to improved overall performance. A more accurate and resilient model may result from this.
3. Thirdly, data reduction can assist lower storage costs by making datasets smaller, which is especially helpful for big datasets.
4. Fourth, data reduction can make the results easier to understand by reducing the amount of superfluous or unneeded information in the dataset.

Disadvantages:

1. Information loss: If crucial data is omitted during data reduction, the procedure might lead to information loss.
2. Effect on precision: Reducing the amount of the dataset could exclude crucial information required for precise predictions, which in turn might affect the precision of a model.
3. Effect on interpretability: Reducing data could make it more difficult to grasp the results because eliminating unnecessary or superfluous information also removes context.
4. Data reduction can increase the computational costs of data mining since it requires more processing time to process the reduced data.
5. Finally, there are pluses and minuses to data minimisation. By decreasing the dataset size, it can enhance the efficiency and performance of machine learning algorithms. But it can also lead to data loss, which makes it more difficult to understand the findings. Prior to adopting data minimisation, it is crucial to carefully evaluate the risks and advantages and weigh the pros and negatives.

SELF-ASSESSMENT QUESTIONS – 1

Multiple Choose Questions	
1	Data reduction in data mining mainly aims to: a) Increase the size of the dataset b) Reduce the volume of data while preserving its meaning c) Remove all attributes from data d) Make data inconsistent
2	Which of the following is a method of data reduction? a) Sampling b) PCA (Principal Component Analysis) c) Discretisation d) All of the above
3	Discretisation in data reduction converts: a) Categorical data into continuous form b) Continuous data into categorical form c) Text data into numerical values

	d) Data into compressed binary form
4	Concept hierarchy helps in data reduction by: a) Replacing detailed data with higher-level concepts b) Removing missing values c) Splitting data into training and test sets d) Increasing data precision
5	Which of the following is a limitation of data reduction? a) Increases processing speed b) Reduces storage requirements c) May lead to loss of important information d) Improves algorithm efficiency

3. DISCRETISATION

To reduce the number of values for a feature, data discretisation techniques can be used to partition the range of a continuous variable into intervals. Using interval labels, you can retrieve the true data values. By utilising a limited set of interval labels to restore multiple continuous attribute values, the original data is reduced and made simpler.

The end result is a knowledge-level representation of mining findings that is succinct and straightforward to use. Methods of discretisation can be classified according to their implementation, which might include factors like the utilisation of class data or the direction of progression (top-down vs. bottom-up). It can be said that the discretisation procedure is supervised discretisation if class data is used. This means it is not being monitored. It is called top-down discretisation or splitting if the procedure starts by finding one or a few points (split points or cut points) to divide the complete attribute range, and then iteratively continues on the resulting intervals.

As a first step in bottom-up discretisation or merging, it can look at all the continuous values as possible split-points, eliminate some by combining values from the surrounding area to create intervals, and then repeat the process with the new intervals. In order to facilitate a concept hierarchy—a hierarchical or multi-resolution split of the attribute values—Discretisation can be applied recursively on an attribute.

When mining at different levels of abstraction, concept hierarchies come in handy. One way to discretise a number attribute is to look at its idea hierarchy. Utilising concept hierarchies, one can reduce the amount of data by combining lower-level ideas (such as numerical values for the age attribute) with higher-level ones (such as youth, middle-aged, or senior). Although this kind of data generalisation hides details, the resulting data can be easier to implement and more useful. As a result, the outcomes of data mining may be described consistently across different mining jobs, meeting a common need. In addition, mining on a smaller, more specific data set is more efficient and requires fewer input/output processes than mining on a larger, more generic data set. The benefits of using discretisation techniques and idea hierarchies as pre-treatment steps before data mining make them more commonly utilised than during mining itself.

Concept hierarchies for numerical attributes can be automatically generated or refined dynamically using a number of discretisation approaches. Furthermore, many categorical attribute hierarchies are automatically represented at the schema definition level and are implicit in the database design.

3.1 Types of Discretisation Techniques

In data analysis, binning is the most common discretisation approach for transforming continuous data into discrete categories, however there are others. Some of the most popular approaches are as follows:

1. Binning in Data Mining

One way to preprocess data is by using data binning or bucketing, which helps to reduce the impact of insignificant observational mistakes. To replace the original data values, we first partition them into little intervals called bins, and then we generate a generic value for each bin. By doing so, the input data is smoothed down, and in the instance of small datasets, the likelihood of overfitting may be decreased.

Why Binning is Important?

- **Data Smoothing:** Data is smoothed down by binning, which serves to lessen the effect of small observational changes.
- **Outlier Mitigation:** Grouping values into bins helps to limit the influence of outliers.
- **Improved Analysis:** The process of discretising continuous data makes data analysis easier and allows for more accurate visualisation.
- **Feature Engineering:** For predictive modelling, binned variables might be more practical and easy to understand.

Types of Binning Techniques

Based on the definition of the bins, there are three main forms of binning:

a. Equal Width Binning

Using this method, we can partition the whole data set into intervals of uniform size. The data range is divided into n intervals to determine the width of each bin, which is equal.

Formula:

$$\text{Bin Width} = \frac{\text{Max Value} - \text{Min Value}}{n}$$

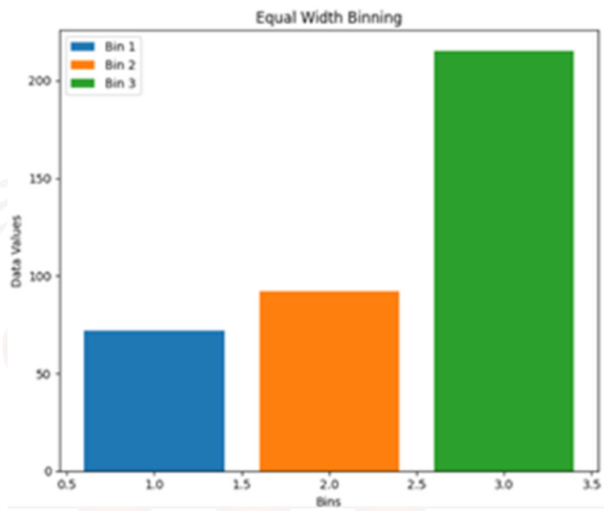

Figure:1

For example, if you have data from 1 to 100, you can divide it into 5 intervals: 1-20, 21-40, 41-60, 61-80, and 81-100.

- Benefits: Ease of use and comprehension.
- Drawbacks: Might cause extremely unequal data distribution in the bins

b. Equal Frequency Binning

By dividing the data into intervals, this strategy ensures that each interval contains an equal number of data points. Take 100 data points as an example. You may split them into 5 intervals, with each interval holding 20 points. Each bin contains approximately the same number of data points.

Benefits: Prevents sparse bins by ensuring balanced bin sizes.
Drawback: One potential drawback is that the bin width can differ greatly.

Steps in Binning:

1. Step one is to sort the data. Put the variable values in ascending order.
2. Establish Bin Boundaries: Find the intervals according to the binning strategy you've chosen.
3. The third step is to put each data point into its appropriate bin according to its value.

Implementation of Binning Technique

To make it easy to compare and contrast the two binning methods used in data processing, the code includes visualisations of the binning methods in the form of bar charts.

```
import numpy as np

import pandas as pd

# Sample dataset

data = [12, 15, 20, 21, 25, 30, 33, 35, 40, 42]

print("Original Data:", data)

# Convert to pandas series

data_series = pd.Series(data)

# --- Equal-Width Binning ---

equal_width = pd.cut(data_series, bins=3)  # 3 bins of equal width

print("\nEqual-Width Binning:")

print(pd.DataFrame({"Data": data_series, "Bin": equal_width}))

# --- Equal-Frequency Binning ---

equal_freq = pd.qcut(data_series, q=3)   # 3 bins with equal frequency

print("\nEqual-Frequency Binning:")

print(pd.DataFrame({"Data": data_series, "Bin": equal_freq}))
```

Applications of Binning

- Uses of Binning in Data Preprocessing: One common usage is to transform continuous variables into categorical ones in order to get data ready for machine learning models.
- Anomaly Detection: By analysing distributions and binning data, it helps find outliers or anomalies.
- Data visualisation: utilised in bar charts and histograms to depict data distribution by frequency.

- The goal of feature engineering is to improve the efficiency of specific ML models by developing new category features.

Challenges of Binning

- Information Loss: Granularity and precise information may be lost when continuous data is converted to discrete bins.
- Subjectivity: Making decisions about bin definitions is generally a subjective process, which could lead to biases.
- Overfitting Risk: The generalisability of the model may be reduced if custom binning causes the data to be overfit

2. K- means Cluster

K-Means clustering Data points can be helpfully grouped into clusters according to their inherent resemblance using clustering, an unsupervised machine learning process. When working with unlabelled data, K-Means can help reveal latent patterns and structures, in contrast to supervised learning, which relies on labelled data for model training. The usage of K-Means by an online retailer to classify consumers into "Budget Shoppers," "Frequent Buyers," and "Big Spenders" is just one example of how this technology might be put to use. Ko-Means Clustering Process Assume we have a vector of items in our dataset that share some characteristics, and that we are also provided with values for these attributes. Sorting those things into sets is the objective. The K-means method will be utilised to do this. The number of clusters or groups into which we wish to sort our objects is denoted by "kk".

The objects will be sorted into "kk" groups, which are clusters of similarity, by the algorithm. With the help of the Euclidean distance, we can determine how similar the two objects are. How the algorithm operates is as follows:

1. Initialization: Picking k cluster centroids at random is the first step.
2. Assignment Step: In order to create clusters, the centroid closest to each data point is used.
3. Update Step: We average the points inside each cluster and compute its centroid after the assignment is over.
4. Repeat: He keeps going until either the centroids stop moving or he reaches the maximum number of iterations.

Splitting the dataset into kk clusters where the data points are more comparable to each other than to data points from other clusters is the objective.

Use of K-Means Cluster

The simplicity, efficiency, and effectiveness of K-Means have made it useful in many different applications. The reasons for its widespread use are as follows:

- **Data Segmentation:** Separating data into different categories is a typical application of K-Means. K-Means is used by businesses to classify clients according to their actions, including how they use the website or how often they buy something.
- **Image Compression:** Through the process of clustering comparable pixels into images, K-Means can simplify images by compressing them. For processing and storing images, this is helpful.
- **Anomaly Detection:** By finding data points that do not fit into any of the clusters, K-Means can be used to locate anomalies or outliers.
- **Document Clustering:** K-Means is a tool used in natural language processing (NLP) to classify articles and texts based on their similarities. Use cases such as recommendation systems or news classification commonly make use of it.
- **Organising Large Datasets:** K-Means can improve data analysis efficiency by dividing big datasets into smaller, more manageable chunks based on similarities.

Implementation of K-Means Clustering

Using blobs datasets, we will demonstrate the Python programming language's clustering capabilities.

Step 1: Importing the necessary libraries

Before writing the program, we need to import some helpful tools (libraries).

Think of libraries as ready-made tools that make coding easier.

We will be importing the following libraries.

- Numpy: for numerical operations (e.g., distance calculation).
- Matplotlib: for plotting data and results.
- Scikit learn: to create a synthetic dataset using make_blobs

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import
make_blobs
```

Where,

- import numpy as np → Allows us to use NumPy and call it np (short and easy).
- import matplotlib.pyplot as plt → Lets us use Matplotlib to draw graphs. We call it plt.
- from sklearn.datasets import make_blobs → Brings a function called make_blobs that creates sample data points to use for clustering.

Step 2: Creating Custom Dataset

We will generate a synthetic dataset with make_blobs.

- make_blobs(n_samples=500, n_features=2, centers=3): Generates 500 data points in a 2D space, grouped into 3 clusters.
- plt.scatter(X[:, 0], X[:, 1]): Plots the dataset in 2D, showing all the points.
- plt.show(): Displays the plot

In this step 2, we will create our own dataset to use for clustering.
Since we just want to practice, we will not use real-world data.
Instead, we will generate some sample points using the function make_blobs (make_blobs() helps us create many points grouped into clusters.)

Step 3: Initializing Random Centroids

In K-Means Clustering, we need to start with some initial cluster centers.
These centers are called centroids.
At the beginning, we don't know where the correct cluster centers should be, so we start with random positions.

We will randomly initialize the centroids for K-Means clustering

- np.random.seed(23): Ensures reproducibility by fixing the random seed.
- The for loop initializes k random centroids, with values between -2 and 2, for a 2D dataset.

Where,

- np.random.seed(23) just ensures that: Everyone gets the same starting centroids when the code runs.
- for loop is used to create k random centroids. If we want k = 3 clusters, we create 3 random starting points. Each centroid is placed somewhere between -2 and 2 on the x and y axes. We do this because our dataset is 2-dimensional, so each centroid has two values (x, y).

Step 4: Plotting Random Initialized Center with Data Points

In this step, we will show the data points and the randomly chosen centroids on a graph.This helps us see where the first cluster centers are located before K-Means actually starts adjusting them.

We will now plot the data points and the initial centroids.

- plt.grid(): Plots a grid.
- plt.scatter(center[0], center[1], marker='*', c='red'): Plots the cluster center as a red star (* marker).

Where,

plt.grid() displays a light grid behind the graph to make points easier to view and compare.

- center[0] = x-coordinate of the centroid
- center[1] = y-coordinate of the centroid
- marker='*' means draw it as a star
- c='red' means color the star red

Step 5: Defining Euclidean Distance

To decide which data point belongs to which centroid, we need to measure how far each point is from each centroid.

We use a formula called Euclidean Distance (the normal straight-line distance between two points in a 2D graph).

To assign data points to the nearest centroid, we define a distance function:

- np.sqrt(): Computes the square root of a number or array element-wise.
- np.sum(): Sums all elements in an array or along a specified axis.

Step 6: Creating Assign and Update Functions

Now that we can measure distance between each data point and each centroid, we must assign every point to the nearest centroid.

After assigning, we must update the centroid position so it better represents its cluster. This is the main repeating cycle of K-Means.

The functions to assign points to the nearest centroid and update the centroids based on the average of the points assigned to each cluster.

- dist.append(dis): Appends the calculated distance to the list dist.
- curr_cluster = np.argmin(dist): Finds the index of the closest cluster by selecting the minimum distance.
- new_center = points.mean(axis=0): Calculates the new centroid by taking the mean of the points in the cluster.

Here,

dist.append(dis); We keep adding the calculated distances to a list named dist. This helps us compare distances.

curr_cluster = np.argmin(dist) : From the list of distances, we pick the cluster with the smallest distance. This means the nearest centroid.

new_center = points.mean(axis=0) : Once points are grouped in a cluster, we find the average position of all points. This average becomes the new centroid.

Step 7: Predicting the Cluster for the Data Points

Once the centroids have been adjusted and are in their final positions, we are ready to predict which cluster each data point belongs to. This step is just like Step 6, but now the centroids are final, so we won't update them anymore.

We only assign points based on the closest centroid.

We create a function to predict the cluster for each data point based on the final centroids.

- pred.append(np.argmin(dist)): Appends the index of the closest cluster (the one with the minimum distance) to pred.

Here,

Dist: A list containing distances of a point from each centroid.

np.argmin(dist): Finds the index of the smallest value in dist. This means: the closest centroid.

pred.append(...): Stores this cluster index in a list named pred.

Step 8: Assigning, Updating and Predicting the Cluster Centers

In this final step, we combine everything we learned earlier.

The K-Means algorithm works in a cycle:

1. Assign points to the nearest centroid
2. Update the centroids
3. Repeat until centroids stop moving

Once the centroids do not change much anymore, the clusters are final.

We assign points to clusters, update the centroids and predict the final cluster labels.

- assign_clusters(X, clusters): Assigns data points to the nearest centroids.
- update_clusters(X, clusters): Recalculates the centroids.
- pred_cluster(X, clusters): Predicts the final clusters for all data points.

Where,

assign_clusters(X, clusters): Finds which centroid each point is closest to, and assigns the point to that cluster.

update_clusters(X, clusters): Recalculates the centroid of each cluster by taking the average position of all points in the cluster.

pred_cluster(X, clusters): Uses the final centroids to predict the cluster label for each data point.

Step 9: Plotting Data Points with Predicted Cluster Centers

This is the final visualization step in K-Means clustering.
Now that each data point has been assigned to a cluster, we will plot all the points and show the cluster centers.
We will use different colors to show which data point belongs to which cluster.
This makes the result easy to understand by just looking at the graph.

Finally, we plot the data points, colored by their predicted clusters, along with the updated centroids.

- center = clusters[i]['center']: Retrieves the center (centroid) of the current cluster.
- plt.scatter(center[0], center[1], marker='^', c='red'): Plots the cluster center as a red triangle (^ marker).

Where,

center = clusters[j]['center']: This gets the final centroid (center point) of cluster j.

plt.scatter(center[0], center[1], marker='^', c='red'): This draws the centroid on the graph as a red triangle.

center[0] → x-coordinate of the centroid

center[1] → y-coordinate of the centroid

marker='^' → triangle shape

c='red' → color is red

The red triangle helps us clearly identify the final cluster center.

Challenges with K-Means Clustering

The K-Means algorithm comes with certain drawbacks:

- Determining the number of clusters: Selecting the appropriate value of k is often difficult and requires prior knowledge or experimentation.
- Dependence on initial centroids: The outcome of clustering may differ based on where the starting centroids are placed.
- Assumption of spherical clusters: K-Means works best when clusters are spherical and of similar size, which limits its effectiveness on datasets with irregular shapes or varying densities.
- Impact of outliers: Extreme values can shift the centroid positions, leading to inaccurate cluster formation.

Data mining is the practice of uncovering hidden patterns, useful information, and knowledge from large datasets. It applies methods from statistics, machine learning, and artificial intelligence to analyse data and extract meaningful insights. This process has applications across diverse domains such as business, finance, healthcare, and scientific research.

SELF-ASSESSMENT QUESTIONS – 2

Multiple Choose Questions	
6	What is the main purpose of data Discretisation in data mining? a) To convert categorical data into continuous data b) To convert continuous data into categorical/binned data c) To reduce the dimensionality of data d) To remove outliers
7	Which of the following is a supervised Discretisation technique? a) Equal-width binning b) Equal-frequency binning c) Entropy-based Discretisation d) Clustering-based Discretisation
8	Equal-width Discretisation works by: a) Dividing the range of values into intervals of equal size b) Dividing the dataset into intervals with an equal number of records c) Grouping values based on similarity measures

	d) Using decision tree algorithms
9	Which Discretisation method is more robust when data is unevenly distributed? a) Equal-width binning b) Equal-frequency binning c) Concept hierarchy generation d) PCA-based Discretisation
10	Concept hierarchies in data Discretisation are useful because they: a) Replace detailed values with higher-level concepts b) Eliminate noisy data c) Normalise continuous attributes d) Encode categorical attributes into binary form
11	One disadvantage of Discretisation is: a) It makes data easier to interpret b) It may result in information loss c) It helps in reducing computational complexity d) It is used for pattern discovery

4. CONCEPT HIERARCHY IN DATA MINING

The term "concept hierarchy" is used in data mining to describe a way of organising data in a hierarchical fashion, with each level standing for a more general concept than the one below. The ability to analyse data more efficiently and effectively and to drill down to more particular levels of detail when necessary are both made possible by this hierarchical organisation of data. Data is made more intelligible and easier to analyse by organising and classifying it according to the concept of hierarchy. Data can be organised hierarchically to facilitate understanding and analysis, with the fundamental premise being that the same data can have varying degrees of granularity or detail.

Example:

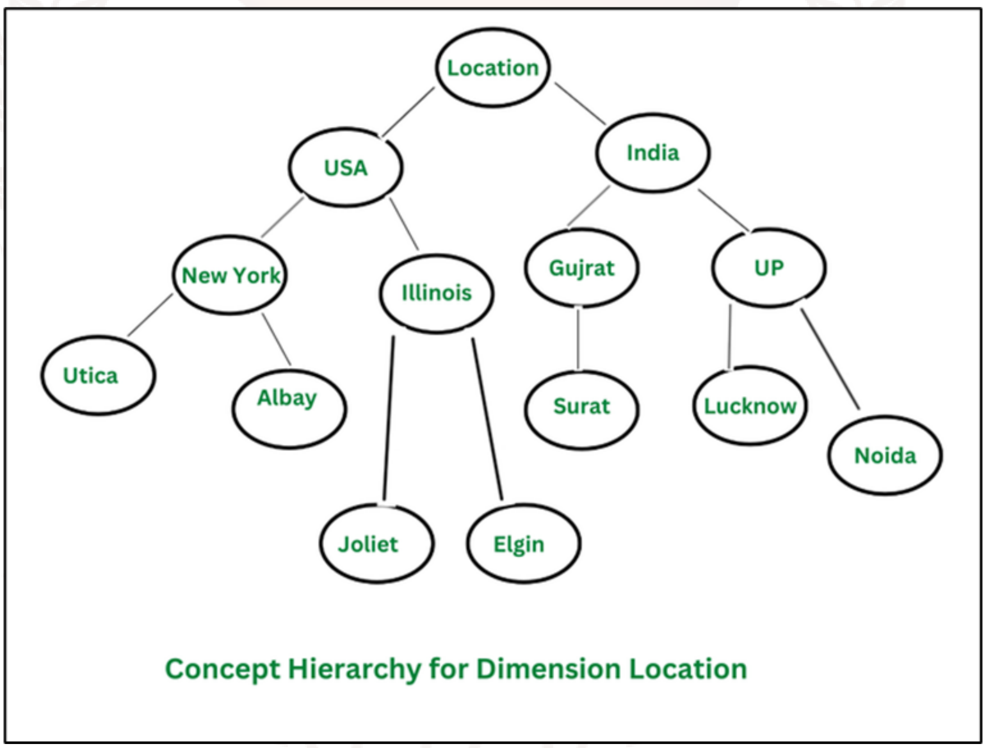

Figure:2

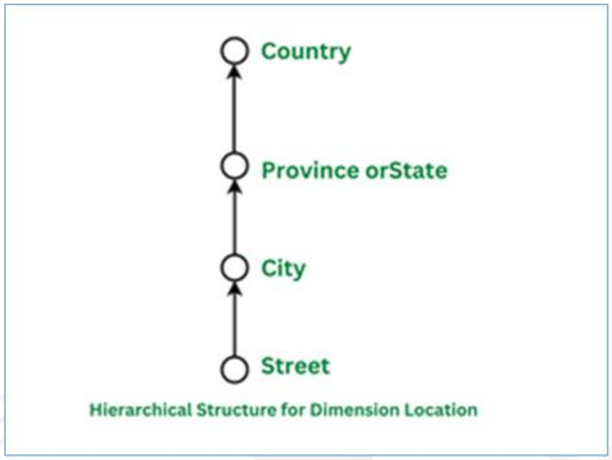

Figure:3

The user can quickly access the data from the idea hierarchy for the dimension location, as illustrated in the following diagram. The data is shown in a structure similar to a tree so that it may be easily evaluated. The main dimension location is at the very top of the tree, which then branches out into several sub-nodes. Two countries, the United States and India, make up the nodes that branch out from the root node. For example, the states of New York, Gujarat, and Uttar Pradesh are each represented by a sub-node within the larger country of New York. So, the data is organised into a tree-like structure by the idea hierarchy, which describes and expresses at a more general level than the level below it, as demonstrated in the previous example.

The hierarchical structure stands in for the location dimension's abstraction level; this level includes all of the dimension's footprints, including streets, cities, provinces, states, and countries.

4.1 Types of Concept Hierarchies

a. Schema Hierarchy: Schema Hierarchy is a concept hierarchy that groups related objects together to organise a database's schema in a sensible and logical fashion. A schema hierarchy provides a sensible and effective means of organising various data kinds, including tables, attributes, and relationships. When it comes to data warehousing, this might be helpful for integrating data from several sources into one database.

b. Set-Grouping Hierarchy: Grouping by Sets Based on set theory, hierarchies are a specific kind of notion hierarchy in which the membership of other sets is used to define each set in the hierarchy. Data cleaning, pre-processing, and integration can all benefit from set-grouping hierarchy. Data from many sources can be integrated using this kind of hierarchy, which can also be used to detect and eliminate outliers, noise, or discrepancies.

c. Operation-Derived Hierarchy: A Derivative of Operations The application of a sequence of operations or transformations to data allows for the creation of hierarchies, which are a kind of notion hierarchy. The procedures are executed in a hierarchical order, with lower levels providing a more concrete representation of the data and higher levels providing a more generic or abstract one. Data mining operations like clustering and dimensionality reduction frequently employ this hierarchical structure. Mathematics and statistics, including processes like aggregation and normalisation, can be utilised.

d. Rule-based Hierarchy: Based on predetermined rules Applying a set of conditions or rules to data allows for the creation of hierarchies, which are a kind of notion hierarchy. Data mining activities like classification, decision-making, and data exploration benefit from this hierarchical structure. It finds patterns and correlations between various data qualities and permits the assignment of a decision or class label to each data point according to its characteristics

4.2 Need of Concept Hierarchy in Data Mining

In data mining, an idea hierarchy is helpful for multiple reasons:

a. Improved Data Analysis: Data may be better organised and analysed with the use of a concept hierarchy, which makes the data more manageable. Data trends and patterns can be better seen with the use of a concept hierarchy, which groups related ideas together. Uncovering hidden or unexpected insights can greatly benefit businesses by informing decision-making and the development of new products or services.

b. Improved Data Visualisation and Exploration: By arranging data into a tree-like form, a concept hierarchy can aid in data exploration and visualisation by making it easier to browse and comprehend complicated and vast data sets. When making interactive dashboards or reports, this can be very helpful since it lets users easily go to deeper levels of detail as needed.

c. Improved Algorithm Performance: Another way to make data mining algorithms work better is to employ a concept hierarchy. Algorithms are able to handle and analyse data more efficiently and accurately when it is organised into a hierarchical structure.

d. Data Cleaning and Pre-processing: In data cleaning and pre-processing, a concept hierarchy can be employed to detect and eliminate noise and outliers.

e. Domain Knowledge: To better understand the data and the problem domain, a concept hierarchy can be utilised to describe the domain knowledge in a more structured fashion.

Limitations of Concept Hierarchies

- **May lose important details**
  When data is grouped into higher-level categories, fine-grained information is often sacrificed. For example, grouping ages into child, teen, adult, and senior simplifies analysis, but it may hide patterns that exist within smaller age differences, such as performance variations between 12-year-olds and 14-year-olds. This generalisation can reduce the accuracy of insights.
- **User-defined hierarchies can be biased**
  Since user-defined hierarchies rely on domain experts, they may unintentionally introduce bias. Experts often design groupings based on personal assumptions, business priorities, or incomplete knowledge. As a result, alternative and potentially valuable ways of organising data may be ignored, leading to a narrow view of the information.
- **Automatic hierarchies depend on data quality**
  Algorithms for automatic hierarchy generation work only as well as the underlying data. If the dataset is noisy, inconsistent, or missing values, the generated hierarchies may be unreliable. For instance, if customer data contains duplicate or incorrect entries, clustering-based hierarchies may give misleading results.
- **Not always universally applicable**
  A hierarchy designed for one dataset or domain may not translate effectively to another. For example, a product hierarchy in retail may not be applicable in healthcare or finance. This lack of universality makes concept hierarchies context-dependent, reducing their flexibility across industries.
- **Maintenance complexity**
  As data evolves, hierarchies must be updated to remain accurate and relevant. For instance, new product categories, education levels, or regional divisions require adjustments in the hierarchy. This process can be time-consuming, resource-intensive, and prone to errors, especially in large and dynamic datasets.

Applications of Concept Hierarchy

Concept hierarchy has various uses in data mining, including but not limited to:

- **Data Warehousing:** Data warehousing makes use of concept hierarchy to unify disparate data sets into a unified framework. The results of data analysis and reports can be enhanced in this w
- **Business Intelligence:** In business intelligence, concept hierarchy can be employed to arrange and evaluate data in a manner that can guide company decisions. One use case is in the analysis of consumer data for trends and patterns that might guide the creation of new offerings.
- **Online Retail:** Using concept hierarchy, online retailers can arrange products into categories, subcategories, and sub-subcategories; this makes it easier for buyers to locate what they're searching for.
- **Healthcare:** By classifying patients according to their diagnoses or treatment plans, for instance, healthcare providers can utilise concept hierarchy to better organise patient data, which in turn can aid in the discovery of trends and patterns that can guide the creation of novel treatments or the enhancement of current ones.
- **Natural Language Processing:** Natural language processing makes use of concept hierarchy to classify and analyse text data; for instance, it can aid in extracting valuable information from unstructured data by identifying themes and topics within a document.
- **Fraud Detection:** One application of concept hierarchy in fraud detection is the organisation and analysis of financial data for the purpose of spotting trends and patterns that might point to fraudulent conduct.

Data mining techniques like idea hierarchies make short work of organising and understanding complex data sets. The data cleansing and pre-processing processes, as well as algorithm performance, are enhanced. There are many potential uses for the idea hierarchy, including but not limited to data warehousing, BI, e-commerce, healthcare, NLP, and fraud detection. Data mining tasks and significant insights can be greatly enhanced by understanding and utilising idea hierarchy.

SELF-ASSESSMENT QUESTIONS – 3

Multiple Choose Questions	
12	What is the primary purpose of a concept hierarchy in data mining? a) To remove redundant data b) To provide multiple levels of abstraction for data analysis c) To Discretise continuous attributes only d) To normalize data for analysis
13	Which of the following is an example of a concept hierarchy? a) Customer ID → Customer Name b) City → State → Country c) Sales amount → Profit d) Database → Table → Record
14	Why are concept hierarchies important in OLAP and data mining? a) They improve visualisation only b) They support roll-up and drill-down operations for better insights c) They replace preprocessing methods d) They eliminate the need for dimensional modeling

5. CASE STUDY

Customer Purchase Behaviour Analysis Using Data Reduction

Background

A retail company collects millions of purchase records every month, including details such as transaction ID, product category, price, discount, customer age, and purchase date. Analysing this huge dataset directly is resource-intensive and time-consuming. To overcome this, the company applies data reduction techniques like Discretisation and concept hierarchy generation.

Methodology

1. **Discretisation**
   - The continuous attribute "Customer Age" is Discretised into intervals:
     - 18–25 (Young)
     - 26–40 (Middle-aged)
     - 41–60 (Senior)
     - 60+ (Elderly)
   - Similarly, "Purchase Amount" is divided into ranges:
     - Low (< $100)
     - Medium ($100–$500)
     - High (>$500).
2. **Concept Hierarchy Generation**
   - Product Category is simplified into a hierarchy:
     - Electronics → Mobile, Laptop, TV
     - Apparel → Men, Women, Kids
     - Groceries → Fresh Produce, Packaged Food.
   - Time attribute is generalised from date → month → quarter → year.

Results

- The original dataset of 10 million records was reduced to a condensed form of 2 million records after applying reduction techniques.
- Discretisation allowed clearer identification of high-value elderly customers who prefer electronics purchases above $500.
- Concept hierarchies made seasonal analysis easier, e.g., showing increased sales in Q4 due to holiday shopping trends.
- Reduced dataset enabled faster OLAP queries, lowering query execution time by 40%.

Output & Insights

The reduced dataset not only saved storage and computation but also provided better interpretability for managers. For example:

- Elderly customers often purchase high-value electronics during festive seasons.
- Young customers showed a preference for low-to-medium value apparel.
- Seasonal sales patterns became visible at the quarter level, supporting better inventory planning.

Illustrative Image Example:

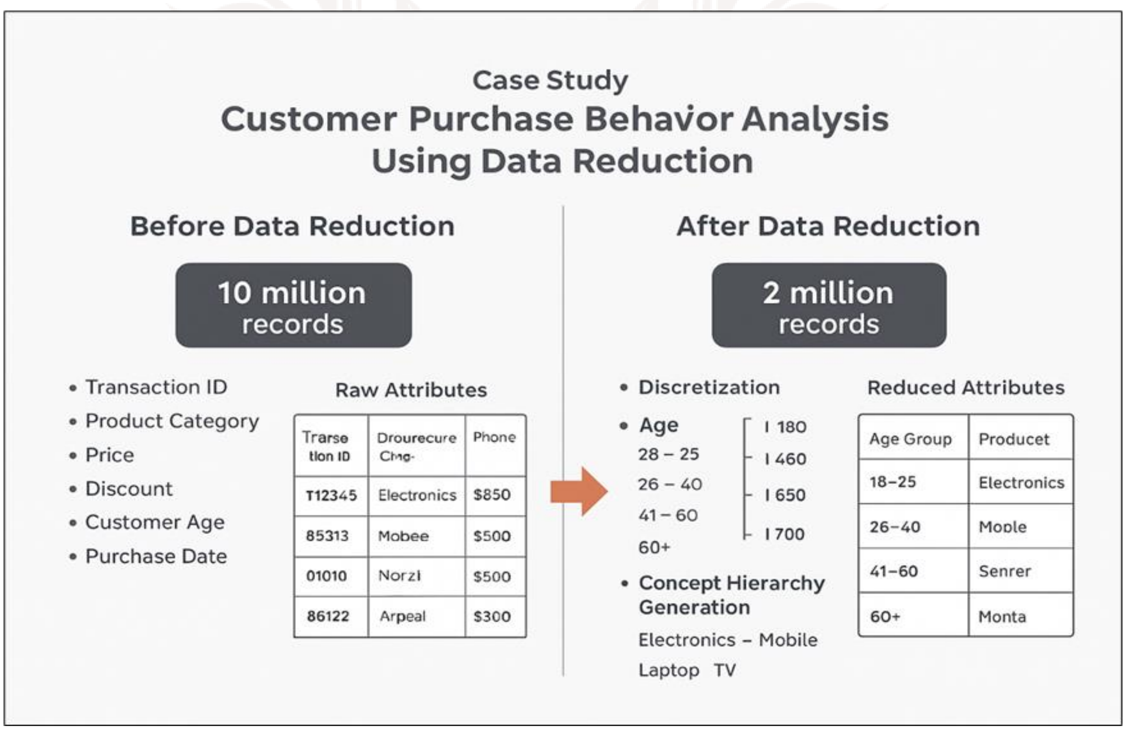

Figure 4: Impact of Data Reduction Techniques on Customer Purchase Dataset

6. SUMMARY

- Data reduction is an important step in data mining that decreases the volume of data while preserving its quality and integrity. It ensures faster query performance, reduced storage requirements, and more efficient analysis.
- Several methods of data reduction exist, including data cube aggregation (summarising data across dimensions), dimensionality reduction (e.g., PCA), numerosity reduction (e.g., clustering, regression models, sampling), and data compression (encoding data into compact forms).
- Discretisation simplifies continuous attributes by dividing them into intervals or bins, making data easier to interpret and suitable for algorithms that prefer categorical input. Techniques include binning, histogram analysis, clustering-based Discretisation, and decision tree methods.
- Concept hierarchy allows data to be represented at multiple abstraction levels (e.g., City → State → Country), enabling drill-down for detailed analysis or roll-up for summaries. Hierarchies may be schema-based, set-grouping based, or rule-based.
- The need for concept hierarchies arises from the flexibility they provide in data exploration, pattern discovery, and abstraction. They also make complex data easier to analyse across different perspectives.
- A case study on customer purchase data showed how Discretisation (e.g., age grouped into Young, Middle-aged, Senior) and hierarchy generation (e.g., Product → Category → Department) reduced millions of records into a more compact form. This improved OLAP queries and highlighted useful patterns, such as seasonal sales trends and age-wise buying behavior.
- Data reduction techniques also help in removing redundancy and irrelevant attributes, ensuring that only meaningful information is retained for analysis.
- By lowering computational costs, these methods make large-scale data mining more feasible and scalable, especially in big data environments.
- Discretisation and hierarchy generation not only simplify data but also improve interpretability for decision-makers who may not be familiar with technical details.
- Overall, data reduction, Discretisation, and concept hierarchy ensure that large datasets become manageable, interpretable, and insightful without losing essential information. These techniques enhance efficiency, maintain analytical accuracy, and form a strong foundation for further mining tasks.

7. GLOSSARY

Binning	-	A Discretisation method that divides data into equal-width or equal-frequency intervals.
Histogram Analysis	-	Discretisation technique where data is grouped into frequency intervals, creating a histogram-like distribution.
Clustering-based Discretisation	-	Grouping values based on similarity, where each cluster represents an interval/category.
Decision Tree Discretisation	-	Splitting data into intervals using decision tree algorithms (e.g., based on information gain).
Concept Hierarchy	-	A structure that organises data at multiple levels of abstraction (e.g., City → State → Country).
Schema-based Hierarchy	-	Hierarchies that are predefined in the database schema (e.g., ZIP Code → City → State)
Set-grouping Hierarchy	-	Hierarchy created by grouping values into larger sets (e.g., "Apple, Mango, Banana" → Fruit).
Rule-based Hierarchy	-	Hierarchies defined using IF-THEN rules (e.g., IF age < 18 THEN category = "Minor").
Drill-down	-	Moving from higher-level summarised data to more detailed levels in a hierarchy.
Roll-up	-	Aggregating data to a higher level of abstraction (e.g., daily → monthly sales).
Data Integrity	-	The accuracy, consistency, and reliability of data after transformation or reduction.

8. TERMINAL QUESTIONS

1. What is data reduction in data mining, and why is it important?
2. Differentiate between dimensionality reduction and numerosity reduction.
3. Explain the concept of Discretisation with an example.
4. What are the main types of Discretisation techniques?
5. Define concept hierarchy and give an example.
6. What are schema-based and rule-based hierarchies?
7. Why is Discretisation needed in data mining?
8. Explain roll-up and drill-down in the context of concept hierarchies.
9. Give a case study example of data reduction using Discretisation and hierarchy.
10. What are the benefits of applying data reduction techniques?

9. ANSWERS

9.1. Self-Assessment Answers

1. Answer: b) Reduce the volume of data while preserving its meaning
2. Answer: d) All of the above
3. Answer: b) Continuous data into categorical form
4. Answer: a) Replacing detailed data with higher-level concepts
5. Answer: c) May lead to loss of important information
6. Answer: b) To convert continuous data into categorical/binned data
7. Answer: c) Entropy-based Discretisation
8. Answer: a) Dividing the range of values into intervals of equal size
9. Answer: b) Equal-frequency binning
10. Answer: a) Replace detailed values with higher-level concepts
11. Answer: b) It may result in information loss
12. Answer: b) To provide multiple levels of abstraction for data analysis
13. Answer: b) City → State → Country
14. Answer: b) They support roll-up and drill-down operations for better insights

9.2. Terminal Question Answers

Answer 1: Data reduction is the process of minimising the volume of data while retaining essential patterns and insights. It improves efficiency by reducing storage space and computation time. It also ensures scalability when dealing with large datasets. Without data reduction, analysis may become slow and resource-heavy.

Answer 2: Dimensionality reduction reduces the number of features/attributes (e.g., PCA, feature selection). Numerosity reduction decreases the volume of data by using models or representative subsets (e.g., regression, sampling). Both aim to simplify data while maintaining integrity. Together, they make mining faster and less complex.

Answer 3: Discretisation converts continuous attributes into categorical intervals.

For example, instead of using exact ages, we can group them into bins like 0–18 (young), 19–35 (adult), 36+ (senior). This simplifies analysis and enhances pattern recognition. It is widely used in classification and rule mining.

Answer 4: The main techniques include binning, histogram analysis, clustering-based Discretisation, and decision tree-based Discretisation. Binning divides values into equal ranges, while clustering groups based on similarity. Histogram analysis organises data into frequency intervals. Decision tree Discretisation uses splitting criteria like information gain.

Answer 5: A concept hierarchy organises data at multiple levels of abstraction.

For example, "Bangalore → Karnataka → India" represents location hierarchy.

It helps in drill-down (detailed view) and roll-up (summarization). This improves flexible data analysis in OLAP and mining.

Answer 6: Schema-based hierarchies are predefined in the database, such as "ZIP → City → State".Rule-based hierarchies are user-defined using IF-THEN rules. For instance, IF income > 10 lakh THEN category = "High income". Both are used to structure data into levels of abstraction.

Answer 7: Discretisation simplifies continuous data, making it easier to interpret.

It improves the efficiency of algorithms that work better with categorical inputs.

It reduces noise by grouping nearby values. This helps in classification, association rule mining, and pattern recognition.

Answer 8: Roll-up moves data to a higher level of abstraction, e.g., daily sales → monthly sales.

Drill-down provides detailed analysis, e.g., country → state → city. Both operations are key in OLAP-based data mining. They allow flexible exploration of summarized and detailed data.

Answer 9: In customer purchase data, Discretisation can group ages into "Young, Adult, Senior". Concept hierarchy can organise products as "Shirts, Pants → Clothing → Apparel".This reduces data complexity while retaining useful insights. The result is faster mining with better interpretability.

Answer 10: Data reduction decreases computation time and storage requirements. It improves accuracy by removing redundant or irrelevant attributes. It ensures scalability for big data applications. Ultimately, it enables efficient and meaningful data mining.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 4

DCA2207

DATA MINING AND VISUALISATION

Unit – 6

Schema Design

TABLE OF CONTENTS

1. INTRODUCTION

In the previous unit, learners gained an understanding of the concept of data reduction and its importance in simplifying large datasets for efficient analysis. They had explored various techniques of data reduction, including data cube aggregation, dimensionality reduction, numerosity reduction, and data compression. The unit also introduced discretisation methods, which transform continuous data into meaningful intervals, and the concept of hierarchy generation, which allows data to be represented at multiple levels of abstraction. Along with this, learners had examined the advantages of data reduction, such as faster processing and reduced storage requirements, as well as its disadvantages and limitations, like potential information loss or reduced accuracy. By the end of the unit, learners were well equipped to apply reduction techniques effectively while balancing efficiency and accuracy in data mining tasks.

By the end of this unit, learners will understand core database-schema concepts (schema vs. instance, fact and dimension tables, grain, normalisation/denormalisation) and be able to explain, compare, and design both Star and Snowflake schemas—including their advantages star: simplicity and high query performance; snowflake: reduced redundancy and storage efficiency and limitations like star: redundancy and update overhead; snowflake: more joins and complexity. They will learn how to translate business requirements into dimensional models, choose appropriate fact table types transactional, snapshot, handle slowly changing and conformed dimensions, and address ETL considerations such as surrogate keys, loading order, unknown members, and late-arriving facts. Practically, students will build unified feature schemas, implement tables, write analytical SQL (joins, aggregations, window functions, apply indexing/partitioning and materialised views for performance, and evaluate schema trade-offs for OLAP workflows like drill-down, roll-up, slice-and-dice. The unit develops skills in schema design, ETL planning, and performance tuning through hands-on labs retail sales example: normalise → star → snowflake, mini-projects, and comparative benchmarks; assessment blends practical assignments, lab tasks, quizzes, and a short presentation defending design choices—preparing learners to recommend and implement the most suitable warehousing schema for real-world reporting needs.

1.1. Learning Objectives

- Explain core concepts of database schemas
- Compare and contrast Star and Snowflake schemas
- Design a dimensional model from a business requirement
- Implement and evaluate analytical queries and basic optimisations of schema

2. DATABASE SCHEMA

A database schema, a data model schema (such as a star or snowflake), or even a psychological schema are all examples of schemas used in data visualisation. Schemas are blueprints for how data is organised and related. Schemas for data models help to organise data so that tools like Power BI can query and analyse it efficiently, whereas schemas for databases use diagrams to outline data items and relationships. The audience's pre-existing information, or psychological schemas, impacts how they perceive and comprehend visualisations.

Forecasts indicate that the enterprise data management industry will expand worldwide from 2019 to 2030, with a CAGR of 12.1%. The significance of organisations implementing efficient data management strategies is highlighted by this growth. In order to make better data-driven business decisions, this strategy relies on the database management system (DBMS), which contains all the enterprise data needed for software applications, systems, and IT environments.

One way to formally describe the organisation and structure of data within a database is with a schema. All of the database's logical components, such as tables, fields, relationships, views, and constraints, are defined in it. Although non-relational (NoSQL) databases differ greatly from relational schemas, they both use schemas to organise data in tables and use SQL for querying.

2.1 Components of a Database Schema

Physical Database Schema: The physical database schema represents the lowest level of abstraction in a database. It deals with the way data is actually stored in the storage system. This includes specifications such as files, key-value pairs, indexes, partitions, and storage paths that determine how data is physically arranged. It is closely related to the database's hardware, operating system, and storage mechanisms. For example, in relational databases, the physical schema might define how a B-Tree index is created on a column to speed up searches or how a table is partitioned to handle large datasets efficiently. The main goal of the physical schema is to ensure performance optimisation, quick retrieval, and efficient use of storage resources.

Logical Database Schema: Defines fields, tables, relations, views, and integrity constraints; describes the logical constraints applied to the data. For the database's physical design, it gives important information. The logical database schema focuses on the logical structure and organisation of the data, without concern for how it is physically stored. It defines tables, fields (attributes), primary and foreign keys, relationships, views, and integrity constraints. The logical schema ensures that the database is properly structured using normalisation techniques, and that rules such as referential integrity are enforced. For example, in a university database, the logical schema might define tables like Students, Courses, and Enrollments, along with their relationships. Unlike the physical schema, it is platform-independent and understandable to developers, database designers, and end-users. The main purpose of the logical schema is to provide a clear model of how data is organised, related, and used in business processes.

Types of Database Schemas

Different kinds of data organisation and use cases call for different kinds of database schemas. The six most prevalent kinds are as follows:

a. Flat Model:

Organises data in a single, two-dimensional table, similar to a spreadsheet. Suitable for simple databases without complex relationship. The Flat Model organises data into a single, two-dimensional table, much like a spreadsheet. Each row represents a record, and each column represents a field or attribute of that record.

Data is stored in rows and columns; each row contains a single entry or record. Each column holds a specific type of data, such as Name, Age, or Salary. There are no relationships between tables since all data resides in one table.

Example: A flat model of an employee database might look like below figure, Here, all employee information is stored in a single table without links to other datasets.

Employee_ID	Name	Department	Salary
101	Alice Roy	HR	40,000
102	Rahul Das	IT	55,000

Table:1

The flat model of databases offers some clear advantages. It is simple and easy to design, making it suitable for beginners or small-scale applications. Since it only uses a single table, it works well for small datasets or applications with minimal complexity, such as spreadsheets. Moreover, it requires fewer resources for storage and maintenance compared to more advanced models like relational databases. However, the flat model also comes with several limitations. One major drawback is redundancy, as data repetition is common, leading to inefficiency. It also does not support relationships between entities, which makes it unsuitable for complex applications. Additionally, it faces scalability issues, meaning it cannot handle large datasets effectively. This often results in poor data integrity and a higher chance of inconsistencies. Despite these drawbacks, the flat model has useful applications. It is often used by small businesses for basic record-keeping, in spreadsheets for accounting or attendance, or for simple contact lists and flat-file storage systems.

b. Hierarchical Model of Database

The Hierarchical Model organises data in a tree-like structure, where data is represented using parent-child relationships. Each parent node can have multiple child nodes, but each child node has only one parent. Data is stored as records (nodes) connected by links (edges). A parent node represents a higher-level entity, while child nodes represent lower-level entities. Traversal is done from the root node down to its child nodes.

Example

Consider an organisational chart:

- CEO (Parent)
    - Manager 1 (Child)
        - Employee A (Sub-child)
        - Employee B (Sub-child)
    - Manager 2 (Child)
        - Employee C (Sub-child)

Here, each manager belongs to one CEO, and each employee belongs to one manager.

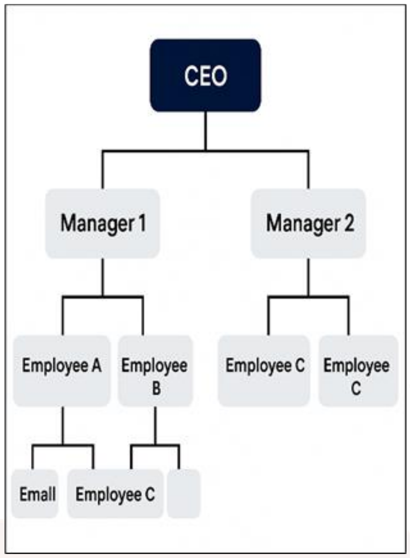

Figure:1

The hierarchical model of databases comes with several advantages. It offers a logical representation of data, making it highly effective for naturally hierarchical structures such as organisational charts or family trees. The use of clear parent-child links helps maintain data integrity by reducing redundancy, since each child belongs to only one parent. Another strength is fast access to data, as predefined paths allow for efficient retrieval of records when navigating from parent to child nodes. However, the model also has some limitations. Its structure is rigid, making it difficult to reorganise or restructure if relationships between entities change. Performing complex queries across different branches can also be challenging, as traversal is limited to defined paths. Additionally, redundancy may occur when a child needs to belong to multiple parents, leading to duplication of data. Despite these drawbacks, the hierarchical model has practical use cases. It is commonly applied in organisational charts to represent company structures, family trees to show ancestry, and computer file systems with folders and subfolders. In the past, industries such as banking and telecom also relied on hierarchical models to manage accounts and service-related data.

c. Network Model of Database

The Network Model is an extension of the hierarchical model that allows more flexible relationships among data. Instead of strictly parent-child structures, it supports many-to-many relationships, making it more powerful for representing complex real-world systems. Data is organised as records (nodes) connected through pointers or links. Unlike the hierarchical model, a child (or member) can have multiple parents (owners) Relationships are maintained through sets, where one record type is the owner and others are members.

Example

In a university database:

- A Student can enroll in multiple Courses, and each Course can have multiple Students. This many-to-many relationship is efficiently represented in the network model.

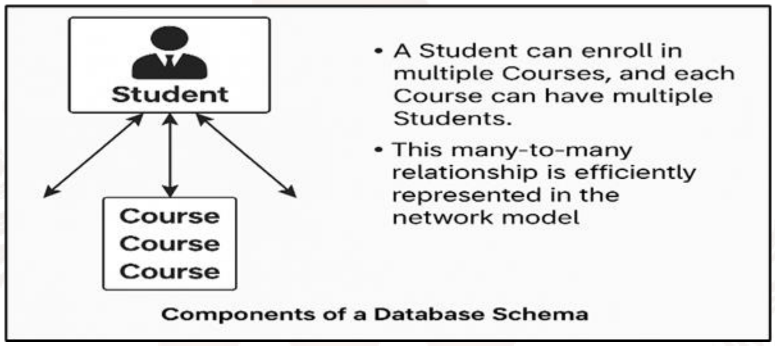

Figure:2

The network model of databases provides several advantages. It supports complex relationships, making it possible to represent many-to-many connections between entities efficiently. Its design ensures quick access to data through pointers and direct links, which allows faster retrieval compared to simple hierarchical structures. Another strength is that it reduces redundancy, since entities can be directly connected to multiple records without duplicating information. However, this model also has limitations. It comes with a complex design, making it more difficult to implement and manage compared to hierarchical or relational models. Additionally, it is navigation dependent, meaning users must know the exact access paths to retrieve data, which can be challenging in large systems. Over time, its popularity has declined, as relational and object-oriented models have largely replaced it in modern applications. Despite these drawbacks, the network model is still useful in certain areas. It has been applied in workflows to represent complex processes, in resource management systems such as logistics and supply chains, and in telecommunications, where early databases relied on this model for managing routing and connections.

d. Relational Model of Database

The Relational Model, proposed by E.F. Codd in 1970, is the most widely used data model in modern database systems. It organises data into relations (tables) consisting of rows and columns, making it easy to understand and manage. A relation is represented as a table. Rows (tuples): Each row represents a single record. Columns (attributes): Each column represents a property or characteristic of the data. Primary Key: A unique identifier for each row. Foreign Key: Establishes relationships between two tables.

Example

- Student Table: (Student_ID, Name, Age, Course_ID)
- Course Table: (Course_ID, Course_Name, Credits)

Here, Course_ID in the Student table acts as a foreign key that links students to the courses they are enrolled in.

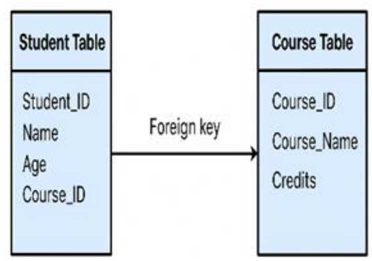

Figure:3

The relational model has several advantages that make it the foundation of modern databases. Its simplicity allows users to easily understand and work with data in tabular form, similar to spreadsheets. It offers flexibility, as new tables can be added, modified, or related without major disruption. The model also ensures data integrity by using primary keys, foreign keys, and unique constraints to maintain consistency across the database. Furthermore, the use of SQL (Structured Query Language) provides powerful capabilities for querying, updating, and managing data. However, the relational model also has some limitations. When dealing with very large datasets or executing complex joins, it may face performance issues. It requires higher storage capacity, as indexes and constraints add to the memory requirements. Additionally, it is not the best choice for handling big data applications involving unstructured or semi-structured data, where NoSQL models are more suitable. Despite these drawbacks, the relational model is widely applied across industries. It is used in banking systems for managing customer accounts and transactions, in educational institutions to store and link student-course records, in business applications for employee management, payroll, and sales, and in e-commerce platforms to handle customer orders, inventory, and payments.

e. Star Schema

The Star Schema is the simplest and most widely used schema in data warehousing. It organises data into a central fact table surrounded by multiple dimension tables, forming a structure that resembles a star. The fact table contains quantitative data such as sales amount, revenue, or quantity sold, while dimension tables contain descriptive attributes such as product details, customer information, time, or location. Since dimension tables are not normalised, they provide quick access and faster query performance, making the star schema ideal for business intelligence and reporting. For example, in a retail data warehouse, the central Sales Fact Table may include fields like Sales_ID, Product_ID, Customer_ID, Store_ID, and Amount.

The connected dimension tables could be Product Dimension (Product_ID, Name, Category), Customer Dimension (Customer_ID, Name, Age, Region), and Time Dimension (Date_ID, Day, Month, Year). This structure makes analysis simple and intuitive, but it can also lead to data redundancy.

f. Snowflake Schema

The Snowflake Schema is a more normalised version of the star schema, where dimension tables are further broken down into smaller related tables. This normalisation reduces data redundancy but increases the number of joins required for queries, which can sometimes affect performance. The structure resembles a snowflake because the dimension tables branch out into multiple related sub-tables. For example, in the same retail scenario, the Customer Dimension may be normalised into separate tables: Customer (Customer_ID, Name, Age), and Region (Region_ID, Region_Name). Similarly, the Product Dimension may be split into Product (Product_ID, Name), and Category (Category_ID, Category_Name). This approach ensures better storage efficiency and data consistency but may make querying more complex compared to the star schema.

Importance of Database Schema Design

For multiple reasons, it is essential to have an efficient database schema:

- Redundancy reduction in data: stops data from being duplicated, which saves space and makes things work better.
- Ensuring data accuracy and consistency is data integrity.
- Optimises performance by making it easier to search for, retrieve, and analyse data quickly.
- Safety: only authorised people can access sensitive data, which is securely stored.

How to Design a Database Schema

There are multiple essential processes to design a database schema:

1. First, you need to figure out what the major things are and how they connect to one another.
2. Tables and Fields: Write an outline of a table's structure and the fields it contains.
3. Establish Limits: Put regulations in place to keep data consistent and accurate.
4. Standardise Data: Make sure the data is organised well and doesn't have any duplicates.

Guidelines for Creating Database Schemas

In order to build databases that are efficient, safe, and scalable, it is important to follow certain best practices:

1. **Naming Conventions:**

   Tables and fields should be named consistently and in a descriptive manner. Keep away from jargon, special characters, and reserved words.

2. **Security:**

   Protect private information by encrypting it. Establish a system of user authentication and access controls based on roles.

3. **Documentation:**

   Write detailed documentation of the schema design, including script and trigger instructions and comments.

4. **Normalisation:**

   Data should be normalised in order to decrease duplication and enhance integrity. Avoid performance difficulties by balancing normalisation.

5. **Expertise:**

   To build a good schema, you must first comprehend the data and all of its properties. Keep an eye on the data growth and make any adjustments to the schema.

To maximise the use of company data and ensure efficient data administration, a properly designed database schema is essential. Organisations can guarantee their databases are secure, scalable, and robust by learning about the various schema types and then applying best practices. This will allow for better business growth and informed decision-making.

SELF-ASSESSMENT QUESTIONS – 1

Multiple Choose Questions	
1	Which of the following best defines a database schema? a) A collection of actual data b) The logical structure of the database c) A backup of the database d) A set of queries
2	Which element is NOT part of a database schema? a) Tables b) Views c) Indexes d) Data values
3	A schema that stores data in tables with rows and columns is called: a) Hierarchical schema b) Relational schema c) Star schema d) Network schema
4	Which of the following describes the relationship between schemas and instances? a) Schema is temporary; instance is permanent b) Schema is permanent; instance is temporary c) Both schema and instance are temporary d) Both schema and instance are permanent
5	Which of these defines the constraints in a database schema? a) Keys and relationships b) Reports c) Stored procedures d) User logins
6	Which layer of a database typically uses schema for logical design? a) Physical layer b) External layer c) Conceptual layer d) Application layer

3. STAR SCHEMA

A Star Schema is the simplest type of data warehouse schema, where a central fact table (containing numerical measures like sales, revenue, or quantity) is connected directly to multiple dimension tables (containing descriptive attributes like time, product, customer, or location). The structure resembles a star, with the fact table at the center and dimension tables radiating outward.

Databases, and data warehouses in particular, might benefit from using a star schema to organise their data for faster and easier analysis. Measurable data, such as sales or revenue, is stored in the central table, which is also known as the fact table. It is surrounded by dimension tables that include information such as dates, client details, and product names. The arrangement creates a shape resembling a star.

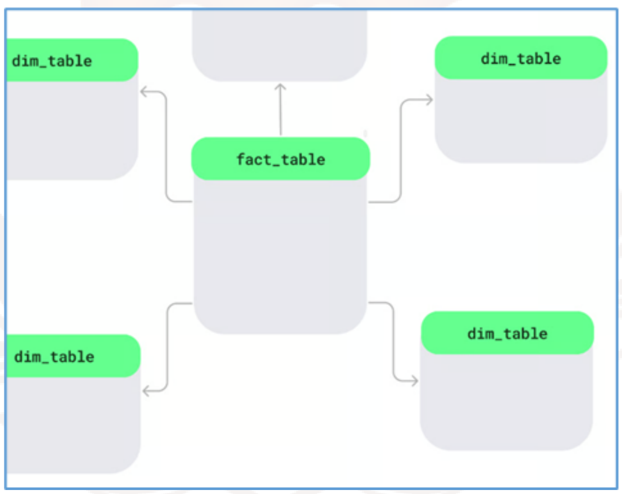

Figure: 4: Star Schema Layout

Let's look at the key features of the star schema:

- Single-level dimension tables: In a single, layer-free connection, the dimension tables join the fact table. Tables are organised by topic, making it easy to find what you need.
- Demoralised design: A star schema is a type of database structure that combines linked tables into one. In a product table, you might find the product's category, name, and ID all in the same place. There may be some duplicate data, but queries are handled faster.
- Common in data warehousing: The star schema is used for quick analysis. Data warehouses that need rapid insights would presumably find it useful that it can filter or calculate totals.

3.1 Implementation of Star Schema

To understand this, let's look at a simple star schema diagram. The sales fact is in the middle of the table. It keeps the numbers you want to look at, like sales or profit. It is connected to dimension tables that have identifying information such product names, customer locations, or dates.

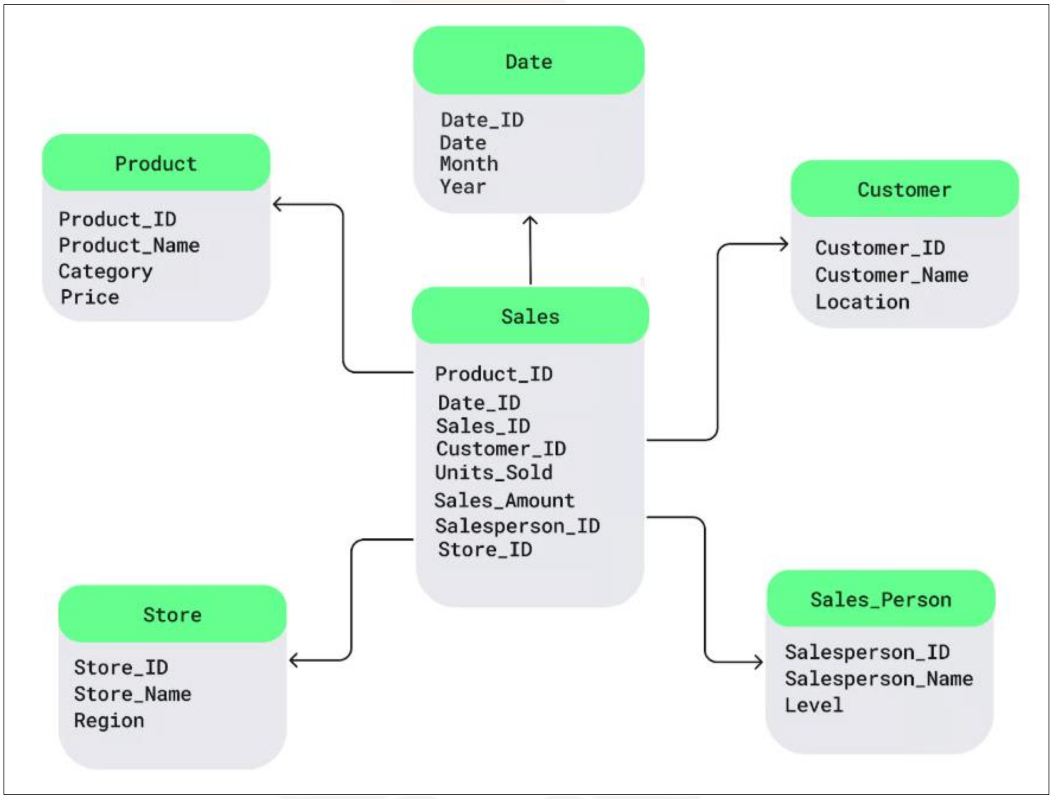

Figure 5: Sample Star Schema

In this basic SQL example, we can see how to build up a star schema with three dimension tables: Sales, Product, and Customer. We also add a date dimension table:

-- ========= Dimension tables =========

CREATE TABLE Product (Product_ID INT PRIMARY KEY, Product_Name VARCHAR(100) NOT NULL, Category VARCHAR(50), Price DECIMAL(12,2));

CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Location VARCHAR(80));

CREATE TABLE DimDate (Date_ID INT PRIMARY KEY, Full_Date DATE NOT NULL, Month VARCHAR(20), Year INT);

CREATE TABLE Store (Store_ID INT PRIMARY KEY, Store_Name VARCHAR(100) NOT NULL, Region VARCHAR(80));

CREATE TABLE Sales_Person (Salesperson_ID INT PRIMARY KEY, Salesperson_Name VARCHAR(100) NOT NULL, Level VARCHAR(40));

-- ========= Fact table =========

CREATE TABLE Sales (Sales_ID BIGINT PRIMARY KEY, Product_ID INT NOT NULL, Customer_ID INT NOT NULL, Date_ID INT NOT NULL, Salesperson_ID INT NOT NULL, Store_ID INT NOT NULL, Units_Sold INT, Sales_Amount DECIMAL(12,2),

FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID), FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID), FOREIGN KEY (Date_ID) REFERENCES DimDate(Date_ID), FOREIGN KEY (Salesperson_ID) REFERENCES Sales_Person(Salesperson_ID), FOREIGN KEY (Store_ID) REFERENCES Store(Store_ID));

-- (Optional but recommended) indexes to speed up star-joins CREATE INDEX ix_sales_product ON Sales(Product_ID); CREATE INDEX ix_sales_customer ON Sales(Customer_ID); CREATE INDEX ix_sales_date ON Sales(Date_ID); CREATE INDEX ix_sales_salesperson ON Sales(Salesperson_ID); CREATE INDEX ix_sales_store ON Sales(Store_ID);

The SQL creates a Star Schema, which is a common structure used in data warehousing. In this schema, the Sales table is the main table, also called the fact table. It stores the actual sales transactions, such as how much was sold, how many units were sold, and the IDs that point to additional details. The Sales table does not store descriptive information. Instead, it uses ID fields to connect to several dimension tables, where details are stored.

The Product dimension table stores information about each product, such as the product name, its category, and price. This helps us understand which types of products perform better. The Customer dimension stores information about customers, such as their names and locations, which helps analyze sales in different regions. The Date dimension keeps information about dates, including the day, month, and year, which allows us to analyze sales over time. The Store dimension provides details about which store the sales happened in, and the Sales_Person dimension stores details about the employees who made the sales, helping measure performance.

All these dimension tables connect directly to the Sales fact table, forming a shape that looks like a star. This design makes it very easy to run queries and generate business reports quickly. It allows businesses to understand trends, compare results across time, products, stores, and employees, and make better decisions.

Advantages and limitations of a star schema

Now that you know what a star schema is, let's look at what makes it different:

- **Faster query performance:** The star structure simplifies data retrieval and makes quick queries easier. The combination of the fact table with the relevant dimension tables would allow me to analyse sales patterns, for example. Also, since I'll be taking care of everything by myself, I won't have to stress about relationships. It would greatly improve the speed of my enquiries and save me a lot of time.
- **Easy to understand:** Its organisation is so straightforward that even anyone without technical expertise can understand it. This facilitates quicker analysis and easier maintenance while also teaching new team members which tables contain the information they require. However, star schema does have several shortcomings. Dimension tables sometimes include duplicate data due to demoralisation, which increases storage requirements, as I said previously. One example is the potential need for additional storage space caused by the repetition of product names in cases when many goods belong to the same category.

SELF-ASSESSMENT QUESTIONS – 2

Multiple Choose Questions	
7	A star schema consists of: a) Only fact tables b) Only dimension tables c) A fact table connected to dimension tables d) Multiple normalised fact tables
8	In a star schema, dimension tables are: a) Normalised b) Denormalised c) Empty d) None of the above
9	What shape does the star schema resemble? a) A snowflake b) A circle c) A star d) A tree
10	Which is an advantage of star schema? a) High redundancy b) Faster query performance c) Requires fewer dimensions d) Lower storage requirement
11	Which type of processing benefits most from star schema? a) OLTP (Online Transaction Processing) b) OLAP (Online Analytical Processing) c) Backup operations d) Indexing operations
12	What is a drawback of star schema? a) Easy to understand b) Storage redundancy c) Fast query speed d) Simple structure

4. SNOWFLAKE SCHEMA

The snowflake schema is an additional method for organising data. Dimension tables are divided into smaller sub-dimensions in this form, much like snowflakes in a large lake, which helps to keep data more structured and detailed.

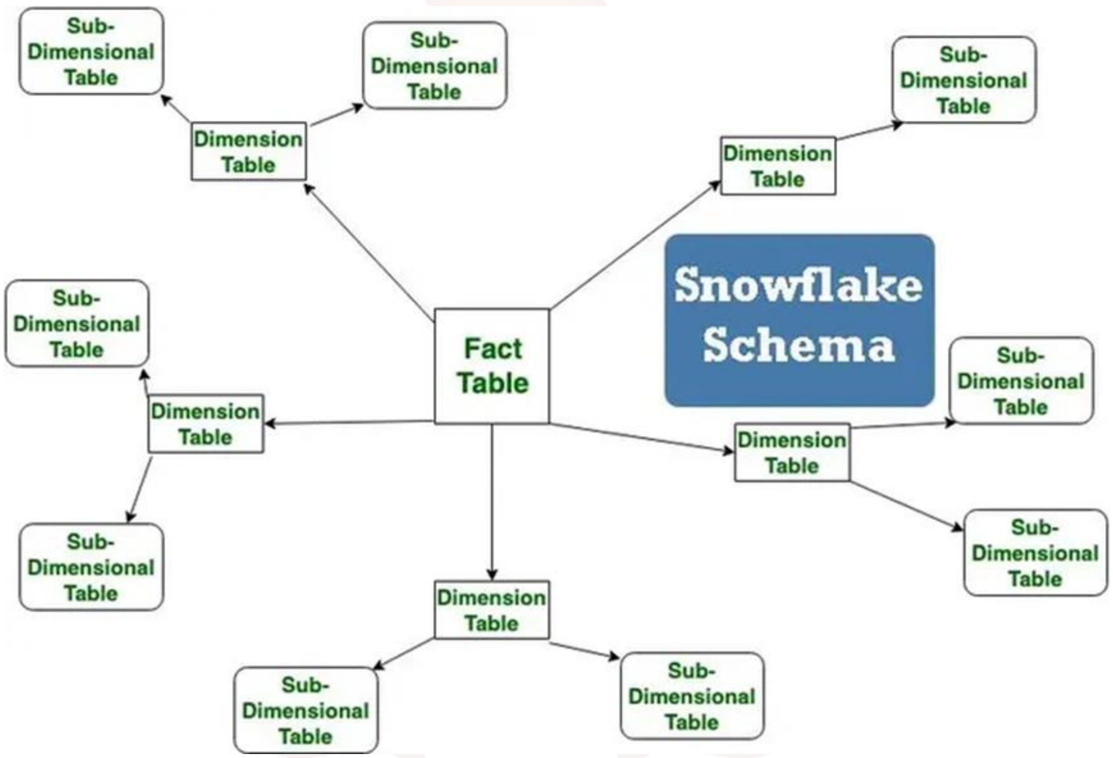

Figure: 6

Let's have a look at what sets the snowflake schema apart from the rest:

- **Multi-level dimension tables:** We can get greater accuracy by splitting our dimension tables into sub-tables. For example, instead of using a single enormous database to keep track of store locations, I may utilise sub tables for each country, state, and city. Since each table would only contain essential data, there would be less duplication of effort and better structure.

- **Normalisation for storage efficiency:** Unlike the star schema, the snowflake schema's normalised architecture eliminates data duplication. As an example, I could create a separate table for the product category "Electronics" and associate it with certain things so that I don't have to repeat it for every product.

Because it uses multi-level tables to handle complex relationships and hierarchical data structures, the snowflake schema is perfect for advanced data settings.

We can make sense of this by using a basic snowflake schema diagram. The fact table, situated in the middle, contains quantifiable data. It links to dimension tables that describe facts, which subsequently construct a structure similar to a snowflake using sub-dimension tables.

4.1 Implementation Snowflake Schema

Let us take a scenario were we have divided the Product table into two parts: Manufacturer and Category, and the Customer table into two parts: Transaction and Location.

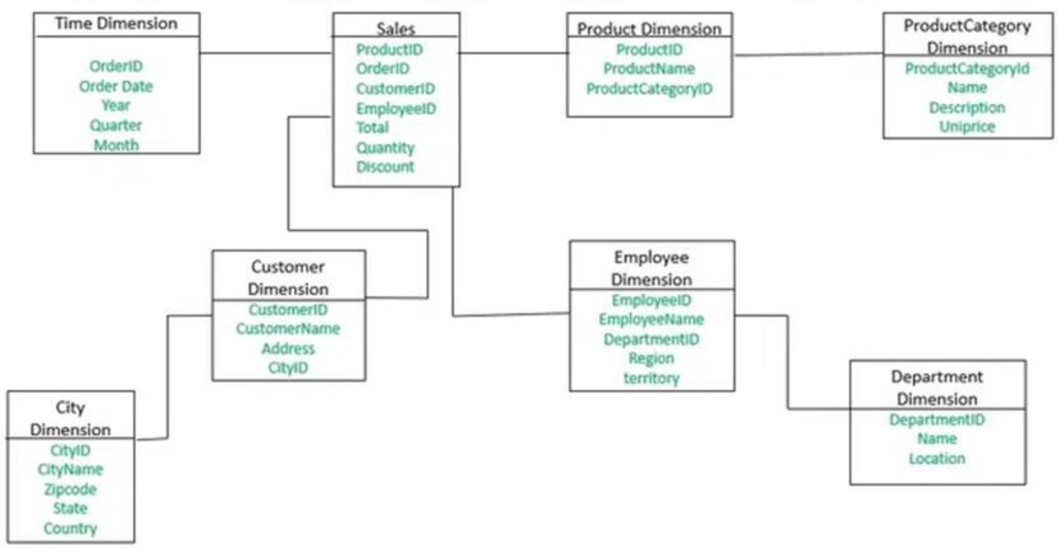

Figure: 7

This SQL query demonstrates a snowflake structure with additional normalisation of the Product table into Category and Manufacturer tables:

CREATE TABLE dim_time (order_id INT PRIMARY KEY, -- key used in the diagram order_date DATE NOT NULL, year SMALLINT NOT NULL, quarter TINYINT NOT NULL, month_name VARCHAR(12) NOT NULL);

CREATE TABLE dim_product_category (product_category_id INT PRIMARY KEY, name VARCHAR(100) NOT NULL, description VARCHAR(255), unit_price DECIMAL(10,2) -- shown in the diagram (Uniprice));

CREATE TABLE dim_product (product_id INT PRIMARY KEY, product_name VARCHAR(120) NOT NULL, product_category_id INT NOT NULL, CONSTRAINT fk_product_category FOREIGN KEY (product_category_id) REFERENCES dim_product_category(product_category_id));

CREATE TABLE dim_city (city_id INT PRIMARY KEY, city_name VARCHAR(100) NOT NULL, zipcode VARCHAR(15), state VARCHAR(100), country VARCHAR(100) NOT NULL);

CREATE TABLE dim_customer (customer_id INT PRIMARY KEY, customer_name VARCHAR(120) NOT NULL, address VARCHAR(255), city_id INT NOT NULL, CONSTRAINT fk_customer_city FOREIGN KEY (city_id) REFERENCES dim_city(city_id));

CREATE TABLE dim_department (department_id INT PRIMARY KEY, name VARCHAR(120) NOT NULL, location VARCHAR(120));

CREATE TABLE dim_employee (employee_id INT PRIMARY KEY, employee_name VARCHAR(120) NOT NULL, department_id INT NOT NULL, region VARCHAR(120), territory VARCHAR(120), CONSTRAINT fk_employee_department FOREIGN KEY (department_id) REFERENCES dim_department(department_id));

-- Fact table --

CREATE TABLE fact_sales (sales_id INT PRIMARY KEY, order_id INT NOT NULL, -- links to dim_time product_id INT NOT NULL, customer_id INT NOT NULL, employee_id INT NOT NULL, quantity INT NOT NULL, discount DECIMAL(5,2) DEFAULT 0, total DECIMAL(12,2) NOT NULL,

CONSTRAINT fk_sales_time FOREIGN KEY (order_id) REFERENCES dim_time(order_id), CONSTRAINT fk_sales_product FOREIGN KEY (product_id) REFERENCES dim_product(product_id), CONSTRAINT fk_sales_customer FOREIGN KEY (customer_id) REFERENCES dim_customer(customer_id), CONSTRAINT fk_sales_employee FOREIGN KEY (employee_id) REFERENCES dim_employee(employee_id));

-- (Optional) helpful indexes for common filters/joins CREATE INDEX ix_sales_order ON fact_sales(order_id); CREATE INDEX ix_sales_prod ON fact_sales(product_id); CREATE INDEX ix_sales_customer ON fact_sales(customer_id); CREATE INDEX ix_sales_employee ON fact_sales(employee_id);

The database design shown is a Snowflake Schema is used in data warehouses to store business data in a structured way for reporting and analysis. In this schema, the main table is the Sales (Fact Table), which stores actual sales transaction details. It links to several Dimension Tables, where descriptive information is stored.

The Time Dimension stores details about the date, such as year, quarter, and month. This allows sales to be analyzed over time (for example: yearly sales, monthly trends).

The Product Dimension stores details about each product, like its name. Products are linked to a Product Category Dimension, which contains category-level information (for example: Electronics, Clothing). This splitting is what makes it a snowflake.

The Customer Dimension stores customer details, and it is linked to the City Dimension, which stores city-related information (city name, state, country). This reduces repeated city data in the customer table.

The Employee Dimension stores details of employees who handled sales. It is linked to the Department Dimension, which stores department details of employees.

Finally, the Sales Fact Table stores actual sales values such as quantity sold, discount, and total amount. It connects to the dimensions through foreign keys, meaning it points to the related records in the dimension tables.

Advantages of Snowflake Schema

1. **Normalised Structure** – Reduces data redundancy by organising data into multiple related tables.
2. **Storage Efficiency** – Since data is not duplicated, it requires less disk space compared to denormalised schemas.
3. **Easier Maintenance** – Updates and inserts are simpler because changes are made in one place, avoiding data inconsistencies.
4. **Scalability** – Well-suited for complex data warehouses where new dimensions and hierarchies can be added easily.
5. **Improved Query Performance in Some Cases** – When queries involve smaller, normalised dimension tables, joins can be faster.

Limitations of Snowflake Schema

1. **Complexity** – The highly normalised structure makes the schema difficult to design, understand, and query compared to star schema.
2. **Increased Joins** – Queries often require multiple joins across many tables, which can slow down performance.
3. **Higher Query Time for Large Data** – In analytical workloads, frequent joins may reduce query speed compared to star schema.
4. **Steeper Learning Curve** – Users and business analysts may find it harder to navigate than simpler star schemas.
5. **ETL Complexity** – Data loading and transformation processes become more complex due to multiple layers of dimension tables

5. DIFFERENCES BETWEEN STAR SCHEMA AND SNOWFLAKE SCHEMA

Data warehousing makes extensive use of both star and snowflake schemas; yet, because to their distinct features, each is better suited to certain tasks. Let's compare and contrast the structure, performance, storage needs, and use cases of different schemas.

Structure

In a star schema, each dimension table has a direct connection to the main fact table. Because all of your reference material is now conveniently located beside your core data, it is much easier to work with and understand. As an alternative, a snowflake schema partitions major dimension tables into sub-dimension tables that are more narrowly focused. Instead of using a single location table, you can use many tables for different nations, states, and cities. The main reason snowflake schema is more complex than star schema is that, although it creates a more organised and precise structure, it also implies that more links (or joins) are necessary to access your data.

Performance

Star schemas are superior in terms of speed. Queries run more quickly because there are fewer joins needed because all dimension tables connect directly to the fact table. For example, if you wish to examine sales by area, you may do it with less processing power by retrieving the data using the star schema. Snowflake schemas, on the other hand, are slower since data retrieval requires connecting through numerous tables. When jobs necessitate fast query responses, snowflake schemas aren't the best choice because each join increases processing time. A great place to start learning about subqueries, relational set theory, and how to connect tables together is in the Joining Data in SQL course.

Storage requirements

Due to the duplication of data in dimension tables, star schemas increase the amount of space required for storage. To illustrate the point, storage requirements will rise if, for instance, numerous products are members of the same category as the name of the category will appear twice. But snowflake schemas standardise data such that everything is stored once. To avoid duplicating efforts, for instance, category names are kept in a distinct table and connected to the product table by foreign keys. Because of how little space it requires, this layout is perfect for massive datasets.

Use cases

Business intelligence (BI), reporting, and online analytical processing (OLAP) systems all benefit greatly from star schemas. Their straightforward design makes them ideal for situations that demand simplicity and speed, such creating sales reports or dashboards in a flash. Many financial analyses and customer relationship management (CRM) systems make use of snowflake schemas. Detailed hierarchy organisation and storage space savings take precedence over query speed in certain instances.

Feature	**Star Schema**	**Snowflake Schema**
Structure	Central fact table connected directly to denormalised dimension tables.	Central fact table connected to normalised dimension tables, which may have sub-dimensions.
Normalisation	Dimensions are **denormalised**.	Dimensions are **normalised** into multiple related tables.
Data Redundancy	Higher, due to denormalisation.	Lower, since data is split into smaller normalised tables.
Storage Requirement	Requires more storage space.	Requires less storage space.
Query Complexity	Simple queries (fewer joins).	Complex queries (multiple joins needed).
Performance	Faster query execution in most analytical cases.	May be slower due to multiple joins.
Ease of Use	Easy for end-users to understand and navigate.	Harder for users to understand because of complexity.
ETL Process	Simpler ETL as dimensions are denormalised.	More complex ETL due to normalised structure.
Best Use Case	Suitable for smaller data warehouses and simpler reporting.	Suitable for large, complex data warehouses with many hierarchies.

Table:2

When to Use a Star Schema

If you primarily want to organise your data simply and quickly, the star schema would be perfect. Here’s when you can use it:

- For simple queries, such calculating global sales by area, star schema is a solid option. Simplified and faster response delivery is achieved by instantly linking the fact table to all the dimension tables.
- Star schema is still a viable choice if speed is your top priority. By reducing the number of table joins, your queries will execute more quickly. After using it just once to create numerous sales reports, I realised how much time it saved me in comparison to other designs.
- If you have a little to medium dataset, the redundancy of the star schema won't be a problem. It would work as intended, even when presented with duplicate data, and it wouldn't lead to storage overflow.

When to Use a Snowflake Schema

Snowflake schema is more suitable for handling frequent updates or organising detailed hierarchies. Here’s when you can use it:

- Use the snowflake schema if you work with large datasets and want to save storage space. It normalises dimension tables to prevent repeated data, which reduces storage requirements.
- You can even use the snowflake schema if your data changes often, like updating region names. It maintains consistent updates across all related data to minimise errors and maintenance efforts.
- If your analysis involves multiple levels of data, the snowflake schema can help you organise and represent these relationships in a simple way.

Star Schema vs. Snowflake Schema

Here's a quick comparison of the star and snowflake schemas to help you decide which best suits your data needs. I’ve highlighted the key differences in this table, focusing on their structure, performance, storage, and use cases:

Feature	**Star schema**	**Snowflake schema**
Structure	Central fact table linked to denormalised dimensions	Central fact table linked to normalised dimensions
Complexity	Simple, with fewer joins	Complex, with more joins

Data redundancy	Higher redundancy due to denormalised dimensions	Lower redundancy due to normalised dimensions
Query performance	Faster queries due to simpler structure	Slower queries because of additional joins
Storage	Requires more storage because of redundancy	Requires less storage due to normalisation
Ease of maintenance	Easier to design and maintain	More complex to design and maintain
Best suited for	Small to medium-sized datasets	Large and complex datasets

Table 3 : Summary Table: Star Schema vs. Snowflake Schema

Choosing Between Star Schema and Snowflake Schem

When selecting between Star Schema and Snowflake Schema, it's important to align our choice with our organisation's needs, data characteristics and performance expectations. Here's a quick guide to help we decide:

1. Star Schem

Best for Simplicity and Speed: We should choose the Star Schema if we require a simple, easy-to-implement solution that executes queries quickly. When simple, fast queries are required on small to medium datasets, it performs admirably. Use Case: Ideal for situations with a flat hierarchy and few dimensions, such small company sales data warehouses. It is well-suited for rapid analytics and reporting because it enables fast data retrieval with few joins. Considering Storage: Apt for situations where storage needs are moderate and redundancy isn't a big deal.

2. Snowflake Schema

When dealing with big datasets that have numerous levels of hierarchy and require a high level of normalisation, the Snowflake Schema provides greater flexibility, which is best for data integrity. It works wonderfully for complicated datasets where data integrity must be preserved. Ideal for big organisations working with massive, normalised datasets or systems that undergo frequent upgrades, such as inventory or customer management systems. It makes storage more efficient and reduces redundancy. Snowflake is an excellent option for situations involving complicated, high-volume data because of its normalised structure, which makes it more storage-efficient.

SELF-ASSESSMENT QUESTIONS – 3

Multiple Choose Questions	
13	In a snowflake schema, dimension tables are: a) Denormalised b) Normalised c) Empty d) Redundant
14	What shape does the snowflake schema resemble? a) Star b) Snowflake c) Circle d) Tree
15	What is the main advantage of a snowflake schema? a) Simplicity b) Reduced redundancy and storage efficiency c) Faster queries d) Easier for business users
16	What is the main drawback of a snowflake schema? a) Complex queries due to multiple joins b) High redundancy c) High storage usage d) No support for hierarchies
17	Which type of data warehouse benefits most from snowflake schema? a) Small-scale warehouse with simple data b) Large-scale warehouse with complex hierarchies c) OLTP system with frequent transactions

	d) Backup system
18	Which type of data warehouse benefits most from snowflake schema? a) Small-scale warehouse with simple data b) Large-scale warehouse with complex hierarchies c) OLTP system with frequent transactions d) Backup system

6. CASE STUDY ON STAR SCHEMA

A retail chain required a sales data mart to analyse business performance. The organisation wanted to monitor daily sales, track product demand, and evaluate performance by time, location, category, and promotions. To achieve this, a Star Schema was chosen because it provides a simple and efficient way to structure analytical data, ensuring faster queries and easier reporting.

Star Schema Design

At the center of the schema, the fact_sales table stores the quantitative measures such as sales amount, quantity sold, cost, and unit price. This fact table connects to several denormalised dimension tables: dim_date (containing attributes like day, month, year, quarter), dim_store (storing city, state, and region), dim_product (with product name, brand, category, and sub-category), dim_customer (containing limited customer demographics), and dim_promo (capturing promotion details like type, start date, and end date). Surrogate keys are used in dimensions to ensure consistency and handle slowly changing data.

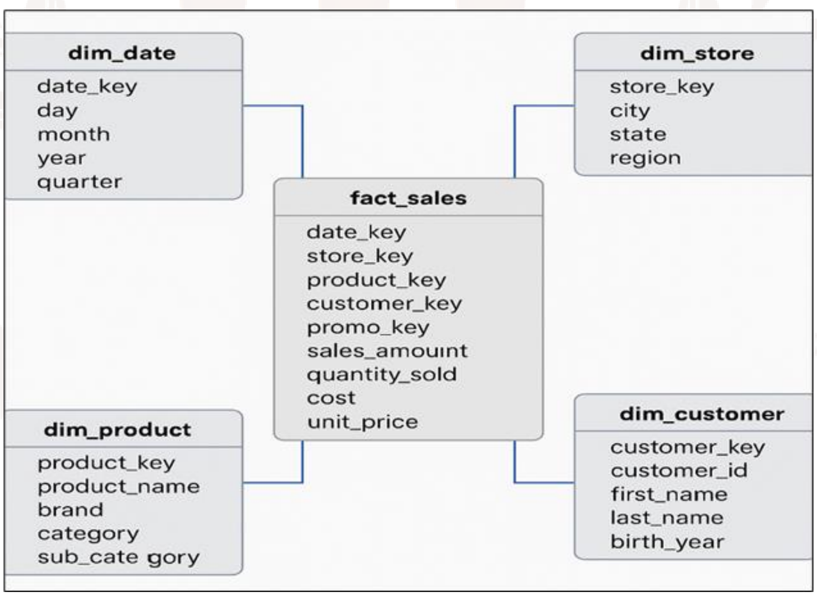

Figure: 8

ETL Process

The ETL workflow begins with extracting data from operational sources such as POS systems, product master databases, and customer information. Dimensions are loaded first with surrogate keys, and unknown rows are inserted for missing values. Fact data is then transformed and mapped to the dimension keys before being loaded into the fact_sales table. To maintain efficiency, indexing and partitioning are applied to the fact table, while materialised views are created for common aggregations like monthly sales per store.

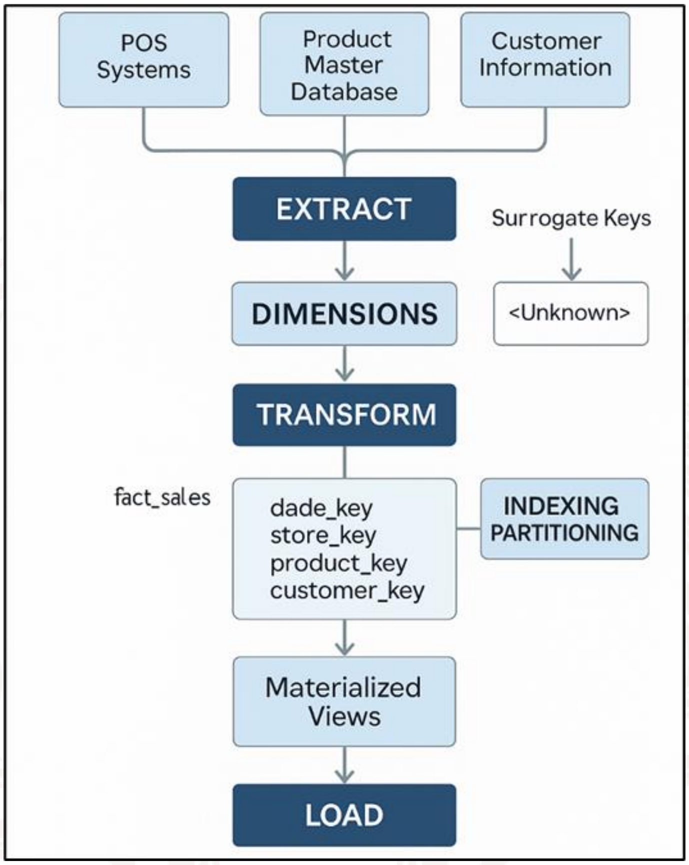

Figure: 9

Analytical Queries

With this star schema, the business can answer questions such as: What are the total sales for a given month? Which five products generated the highest revenue in a quarter? How do promotions affect overall sales? OLAP operations such as slicing, dicing, roll-up, and drill-down are supported, enabling deeper insights into sales performance at multiple levels of granularity.

Conclusion

This case study shows how a retail business can leverage the Star Schema to turn raw transactional data into meaningful insights. The model simplifies reporting, improves performance, and offers a consistent structure for decision-making. With robust ETL processes and optimised schema design, the star schema forms a reliable foundation for business intelligence and long-term analytics

7. CASE STUDY ON SNOW FLAKE SCHEMA

Scenario & Business Problem

A mid-size retail chain with hundreds of stores wants a data warehouse to support detailed analytics: sales by product hierarchy, regional performance, supplier analysis, and seasonal trends. The business must minimise storage cost and ensure consistent updates to shared attributes (e.g., product category names, city/state corrections). Complex hierarchies (Product → Category → Department, Store → City → State → Region) make a snowflake schema attractive.

Conceptual Model (high level)

- Fact table: fact_sales — stores transactional measures (sale_amount, quantity, cost).
- Normalised dimension trees:
  - dim_product → dim_category → dim_department
  - dim_store → dim_city → dim_state → dim_region
  - dim_customer → dim_segment
  - dim_date → (dim_month, dim_quarter, dim_year) or dim_date with keys to calendar tables

This normalisation reduces redundancy for attributes shared across many items (e.g., category name), and enforces a single source of truth.

Example Logical Tables & Key Columns

- fact_sales (sale_id PK, sale_date_key FK → dim_date, product_key FK → dim_product, store_key FK → dim_store, customer_key FK → dim_customer, quantity, sale_amount, cost)
- dim_product (product_key PK, product_sku, product_name, category_key FK → dim_category, price)
- dim_category (category_key PK, category_name, department_key FK → dim_department)
- dim_department (department_key PK, department_name)
- dim_store (store_key PK, store_code, store_name, city_key FK → dim_city)
- dim_city (city_key PK, city_name, state_key FK → dim_state)
- dim_state (state_key PK, state_name, region_key FK → dim_region)

- dim_region (region_key PK, region_name)
- dim_customer (customer_key PK, cust_id, name, segment_key FK → dim_segment)
- dim_segment (segment_key PK, segment_name)
- dim_date (date_key PK, date, day, month_key FK → dim_month)
- dim_month (month_key PK, month_name, quarter_key FK → dim_quarter)
- dim_quarter (quarter_key PK, quarter, year_key FK → dim_year)
- dim_year (year_key PK, year)

Sample SQL DDL (simplified)

```
CREATE TABLE dim_department (
  department_key   INT PRIMARY KEY,
  department_name  VARCHAR(100)
);
CREATE TABLE dim_category (
  category_key     INT PRIMARY KEY,
  category_name    VARCHAR(100),
  department_key   INT REFERENCES dim_department(department_key)
);
CREATE TABLE dim_product (
  product_key      INT PRIMARY KEY,
  sku              VARCHAR(50),
  product_name     VARCHAR(200),
  category_key     INT REFERENCES dim_category(category_key),
  price            DECIMAL(10,2)
);
CREATE TABLE fact_sales (
  sale_id          BIGINT PRIMARY KEY,
```

```
sale_date_key    INT REFERENCES dim_date(date_key),
product_key      INT REFERENCES dim_product(product_key),
store_key        INT REFERENCES dim_store(store_key),
customer_key     INT REFERENCES dim_customer(customer_key),
quantity         INT,
sale_amount      DECIMAL(12,2)
);
```

ETL & Implementation Considerations

- **Load order:** Load the most granular normalised dimensions first (department → category → product; region → state → city → store) so foreign keys resolve when loading the fact table.
- **Surrogate keys:** Use integer surrogate keys for dimension tables, not natural keys, to support slowly changing dimensions (SCD).
- **SCD handling:** Implement Type 2 SCD for product price/history and customer attributes (to preserve historical correctness in facts).
- **Batch vs streaming:** For near-real-time analytics, maintain a staging area; for nightly loads, use bulk insert and index rebuild windows.
- **Data quality:** Centralised normalised tables reduce duplication and simplify correcting master data (e.g., renaming a category updates all products).

Example Analytical Queries (how to use the snowflake)

1. **Total sales by department & quarter:**

```
SELECT d.department_name, q.quarter, SUM(f.sale_amount) total_sales
FROM fact_sales f
JOIN dim_product p ON f.product_key = p.product_key
JOIN dim_category c ON p.category_key = c.category_key
JOIN dim_department d ON c.department_key = d.department_key
JOIN dim_date dt ON f.sale_date_key = dt.date_key
JOIN dim_month m ON dt.month_key = m.month_key
```

JOIN dim_quarter q ON m.quarter_key = q.quarter_key

GROUP BY d.department_name, q.quarter;

2. **Sales per region for a promotional SKU list — join via city→state→region chain.**

Benefits Observed (practical)

- **Reduced redundancy:** Attributes shared by many items (category, state) stored once; easier to correct/update.
- **Smaller storage for dimension attributes:** If many products share categories, splitting lowers overall dimension size.
- **Cleaner master data governance:** Central normalised tables act as the canonical source for attributes.
- **Better maintainability:** Changes to hierarchy (e.g., new regions) require a single update.

(Estimate: normalisation can reduce attribute-storage overhead by a dataset-dependent amount — commonly 20–40% for heavy hierarchical attributes.)

Limitations & Mitigations

- **More joins →** slower query performance: Each level of normalisation adds joins. Mitigation: create materialised views or aggregate tables for common analytics; push expensive joins into precomputed ETL.
- **Complexity for end-users:** Business users may find multiple tables harder to navigate. Mitigation: provide semantic layer (BI view) or user-facing star-like views.
- **ETL complexity:** Loading normalised dimensions requires careful foreign-key resolution and ordering. Mitigation: robust staging, id-mapping tables, and automation.
- **Potential over-normalisation:** If read performance is paramount, at times denormalised star hybrid is preferable.

Performance & Optimisation Tips

- Create materialised aggregates (monthly/department totals) for frequent reports.
- Index join keys (foreign key columns) and partition fact_sales by date for faster range queries.
- Use BI semantic layer (views or cubes) so analysts query simple names while engine uses optimised joins.
- Hybrid approach: keep some heavily-used denormalised attributes in the product dimension (star-like) and normalise rarer attributes.

Result / Recommendation

For this retail case, a snowflake schema suits organisational needs when:

- storage efficiency and Centralised master-data updates are priorities,
- hierarchies are deep and reused across many records, and
- analytical workloads can tolerate additional joins (or you can mitigate with aggregates/materialised views).

If low-latency ad-hoc querying by non-technical analysts is the dominant requirement, consider a star or hybrid schema with targeted denormalization.

8. SUMMARY

- Star Schema is a simple and widely used data warehouse schema where a central fact table is directly connected to multiple denormalised dimension tables. It resembles the shape of a star, with the fact table at the center and dimensions radiating outward.
- The fact table stores quantitative measures (e.g., sales amount, revenue, quantity), while dimension tables hold descriptive attributes (e.g., product, customer, time, location) that provide context for analysis.
- Star Schema is denormalised, meaning dimension tables store all necessary descriptive information in one place, even if redundancy occurs. This design reduces joins and improves query performance.
- Advantages of Star Schema include simplicity, ease of understanding, faster query execution, and suitability for OLAP and reporting tasks. It is particularly useful when quick insights are more important than storage efficiency.
- Limitations of Star Schema include higher storage requirements due to redundancy, potential data inconsistency during updates, and limited flexibility for handling complex hierarchies.
- Star Schema is best suited for smaller or medium-sized data warehouses, where business users need fast reporting and straightforward structures.
- Snowflake Schema is a more complex data warehouse schema where the fact table connects to normalised dimension tables, which may be split into multiple related sub-dimensions, forming a snowflake-like shape.
- In this design, dimension tables are normalised into smaller tables to remove redundancy. For example, a Product dimension may be split into Product → Category → Department.
- Advantages of Snowflake Schema include reduced data redundancy, better storage efficiency, and improved consistency during updates or modifications. It scales well for large, complex datasets with multiple hierarchies.
- Limitations of Snowflake Schema are its complexity, slower query performance due to multiple joins, and higher ETL complexity. It can also be less intuitive for end-users compared to a star schema.
- Snowflake Schema is best suited for large and complex data warehouses, where storage efficiency and detailed hierarchies are required for in-depth analysis.

- The key difference between Star Schema and Snowflake Schema lies in normalisation: Star uses denormalised dimensions for speed and simplicity, while Snowflake uses normalised dimensions for storage efficiency and consistency.
- Star Schema emphasises query performance and simplicity, while Snowflake Schema emphasises storage optimisation and handling complexity.

9. GLOSSARY

Probe Node	-	The initial node that starts sending connection information to discover and propagate links within the network.
Seed Nodes	-	The directly connected nodes that first receive information from the probe node.
Connected Nodes	-	Nodes that establish direct links with the probe or seed nodes.
Second-Order Connected Nodes	-	Nodes connected indirectly through seed or first-order connected nodes.
n^{th}-Order Connected Nodes	-	Nodes reached after multiple levels of propagation, extending the connection discovery process.
Connection Information	-	Data packets or messages sent by a probe node to share link details with other nodes.
Connection Status	-	The state of established links between nodes, indicating active communication paths.
Propagation	-	The process of spreading connection information step by step across different layers of nodes in a network.
Topology	-	The structural arrangement of nodes and connections in a network, defining how information flows.
Neighbor Nodes	-	Nodes that are directly connected to each other by a communication link.
Information Flow	-	The movement of connection data between nodes in sequential steps.
Network Discovery	-	The process of identifying nodes and their connections through probing and information exchange.
Multi-hop Communication	-	A method where information travels across intermediate nodes before reaching distant nodes.

Data Mining Architecture	-	The framework and components that define how data mining processes are structured, including data sources, preprocessing, mining engines, and visualization tools.
Machine Learning	-	A subset of artificial intelligence that enables computers to learn patterns from data and make decisions without explicit programming.
Privacy and Security in Data Mining	-	The ethical considerations and legal aspects of ensuring that data mining does not compromise sensitive or personal information.
Applications of Data Mining	-	The various fields where data mining is applied, including healthcare, finance, retail, education, and cybersecurity, to improve decision-making and operational efficiency.

10. TERMINAL QUESTIONS

1. What is a database schema?
2. What is a star schema in data warehousing?
3. What is a snowflake schema?
4. How does a star schema differ from a snowflake schema?
5. What is the role of the fact table?
6. What are dimension tables?
7. What are the advantages of star schema?
8. What are the advantages of snowflake schema
9. In which scenarios should a star schema be used?
10. In which scenarios should a snowflake schema be used?

11. ANSWERS

11.1. Self-Assessment Answers:

1. Answer: b) The logical structure of the database
2. Answer: d) Data values
3. Answer: b) Relational schema
4. Answer: b) Schema is permanent; instance is temporary
5. Answer: a) Keys and relationships
6. Answer: c) Conceptual layer
7. Answer: c) A fact table connected to dimension tables
8. Answer: b) Denormalised
9. Answer: c) A star
10. Answer: b) Faster query performance
11. Answer: b) OLAP (Online Analytical Processing)
12. Answer: b) Storage redundancy
13. Answer: b) Normalised
14. Answer: b) Snowflake
15. Answer: b) Reduced redundancy and storage efficiency
16. Answer: a) Complex queries due to multiple joins
17. Answer: b) Large-scale warehouse with complex hierarchies
18. Answer: b) Dimension tables may be split into sub-dimensions.

11.2. Terminal Question Answers

Answer 1: A database schema is the logical structure that defines how data is organised in a database. It includes tables, relationships, views, indexes, and constraints. Schemas provide a blueprint for storing and accessing data. They are essential for consistency and efficient database design. (Refer section 2 for more information)

Answer 2: A star schema has a central fact table connected directly to dimension tables.

The fact table stores quantitative measures (e.g., sales, revenue). Dimension tables store descriptive attributes (e.g., product, customer, time). It is simple, fast, and widely used for reporting and OLAP queries

Answer 3: A snowflake schema is a normalised version of the star schema. Here dimension tables are split into multiple related sub-dimensions. This reduces redundancy and improves storage efficiency. However, queries require more joins and can be slower. (Refer section 4 for more information)

Answer 4: Star schema uses denormalised dimensions, while snowflake uses normalised dimensions. Star schema is simpler and faster for queries. Snowflake schema saves storage but is more complex and slower. Choice depends on query performance vs. storage efficiency. (Refer section 5 for more information)

Answer 5: The fact table contains quantitative data (measures) for analysis. Examples: sales amount, profit, quantity sold. It connects with dimension tables via foreign keys. Fact tables are usually very large and central to the schema.

Answer 6: Dimension tables provide context to the measures in fact tables. Examples: product details, customer demographics, or time periods. They contain descriptive attributes used for filtering and grouping. In star schema, they are denormalised in snowflake, normalised.

Answer 7: Star schema is simple, easy to understand, and query-friendly. It requires fewer joins, so queries are faster. It works well for OLAP systems and dashboards. However, it consumes more storage due to redundancy. (Refer section 3.1 for more information).

Answer 8: Snowflake schema reduces data redundancy through normalisation. It requires less storage and ensures consistency in updates. It is suitable for large, complex data warehouses. But, it involves more joins and is less user-friendly. (Refer Section 4.1 for more information).

Answer 9: Use star schema when simplicity and query speed are priorities. It is ideal for small to medium-sized data warehouses. Best for ad-hoc queries, dashboards, and business reports. Storage cost is less important in such cases.

Answer 10: Use snowflake schema for large, complex datasets with many hierarchies. When storage efficiency and reduced redundancy are critical. It supports scalability and data consistency. But queries may be slower due to multiple joins.

12. REFERENCES

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 4

DCA2207

DATA MINING AND VISUALISATION

Unit – 7

Data Mining Algorithms

(Part-1)

TABLE OF CONTENTS

1. INTRODUCTION

In Previous unit, learners understood core database-schema concepts (schema vs. instance, fact and dimension tables, grain, normalisation/denormalisation) and was able to explain, compare, and design both Star and Snowflake schemas—including their advantages and disadvantages. They also learnt how to translate business requirements into dimensional models, by choosing appropriate fact table types transactional, snapshot, handle slowly changing and conformed dimensions, and address ETL considerations such as surrogate keys, loading order, unknown members, and late-arriving facts. The unit develops skills in schema design, ETL planning, and performance tuning through hands-on labs retail sales example: normalize → star → snowflake, mini-projects, and comparative benchmarks; assessment blends practical assignments, lab tasks, quizzes, and a short presentation defending design choices—preparing learners to recommend and implement the most suitable warehousing schema for real-world reporting needs.

In this unit, learners will study the data mining technique of Association Rule Mining, which is used to uncover hidden relationships and patterns in large datasets. The unit begins with simple examples (such as market basket analysis) to illustrate how association rules work in practice. Learners will explore the components of association rules—antecedent (if-part) and consequent (then-part)—and understand how they represent meaningful co-occurrences of items. They will also examine key rule evaluation metrics such as support, confidence, and lift, which help measure the strength and usefulness of discovered rules. Different types of association rule learning algorithms (e.g., Apriori, FP-Growth, and Eclat) will be introduced along with their working principles. The unit also highlights the advantages of association rule mining (simplicity, interpretability, ability to reveal hidden patterns) as well as its disadvantages and limitations (large computational cost, potential generation of too many or irrelevant rules, and sensitivity to threshold settings). Finally, learners will connect theory to practice by exploring real-time applications, including market basket analysis in retail, cross-selling in e-commerce, fraud detection in finance, and recommendation systems. By the end of the unit, learners will be able to explain the concepts, apply algorithms, evaluate rules using metrics, and critically assess the strengths and weaknesses of association rule mining in practical scenarios.

1.1. Learning Objectives

By the end of this unit, you will be able to:

- Explain core concepts of Association Rule Mining
- Apply common association rule learning algorithms on datasets
- Evaluate association rules using metrics such as support, confidence, and lift, and interpret the significance of these measures
- Analyse the advantages, disadvantages, and limitations of association rule mining

2. DATA MINING ALGORITHM

Algorithms are a collection of heuristics and computations used in data mining and machine learning to build models from data. Before building a model, the algorithm searches the data you supply for certain trends or patterns. Through a series of rounds, the algorithm utilises the outcomes of this study to determine the best parameters for building the mining model. In order to get useful patterns and statistics, these parameters are applied to the whole dataset.

There are a number of possible formats for the mining model that an algorithm will generate from your data, such as:

- A collection of groups that express the interconnectedness of a dataset's instances.
- A decision tree that not only predicts but also explains the impact of various factors on a given outcome.
- Sales forecasting using a mathematical model.
- There is a set of criteria that govern the grouping of products in a transaction and the likelihood of those things being bought together.

For the purpose of discovering patterns in data, SQL Server Data Mining provides the most popular and extensively studied algorithms currently available. As an example, K-means clustering is accessible in a broad variety of tools with a large variety of implementations and settings; it is also one of the earliest clustering algorithms. Nevertheless, Microsoft Research created the specific K-means clustering technology utilised in SQL Server Data Mining, and SQL Server Analysis Services was used to optimise its efficiency. The available APIs make it possible to completely modify and implement all of Microsoft's data mining techniques. By utilising the data mining components in Integration Services, you can automate model construction, training, and retraining as well.

Additionally, the SQL Server Data Mining framework allows you to utilise third-party algorithms that meet the requirements of the OLE DB for Data Mining specification, or you can create your own algorithms and register them as services.

Choosing the Right Algorithm

When doing analysis, it might be difficult to determine which method is best suited for the job. Even while several algorithms can accomplish the same business job, the results are not always the same. In fact, some algorithms can even create multiple types of results. As an example, the Microsoft

Decision Trees technique may be used for both prediction and data reduction by identifying columns that do not impact the final mining model. This allows you to use the dataset with fewer columns.

Choosing an Algorithm by Type

SQL Server Data Mining includes the following algorithm types:

- **Classification algorithms** use the other features in the dataset to forecast a single or more discrete variables.
- **Regression algorithms** make a forecast about a continuous numerical variable, like profit or loss, using other dataset features as input.
- **Segmentation algorithms** sort information into sets of related objects called clusters.
- **Association algorithms** discover relationships inside a dataset between various attributes. Association rules for use in market basket analyses are the most typical use case for this kind of algorithm.
- **Sequence analysis algorithms** summarise common data sequences or occurrences, like a web page's clickstream or the events recorded in a machine's logs just before maintenance.

The use of a single algorithm in your solutions is, however, entirely unnecessary. In certain cases, seasoned analysts will employ one algorithm to identify the best variables to utilise as inputs, and then use a separate algorithm to forecast a particular result using that data. With SQL Server Data Mining, you may construct numerous models using a single mining structure. This means that you can utilise a clustering method, a decision trees model, and a Naïve Bayes model all inside one data mining solution to obtain diverse insights from your data. In a single solution, you can employ various algorithms to accomplish different goals; for instance, you can use regression to get financial forecasts and a neural network algorithm to analyse the elements that impact those estimates.

3. DATA MINING TECHNIQUES

1. Association

The purpose of association analysis is to discover commonalities in datasets by identifying which objects or situations often occur together. Market basket analysis frequently employs it to determine the most frequently purchased products in conjunction with one another. A technique that uses the data to create rules for making predictions is known as associative classification.

2. Classification

Data can be sorted into distinct categories through the use of classification models. To predict labels for unknown data, the model is trained on data with known labels. A few instances of models used for classification include decision tree, SVM, Generalised linear model, Bayesian classification K-NN Classifier etc.

3. Prediction

The main difference between classification and prediction is that the latter uses continuous data (such as numbers) to make predictions rather than categories. Ultimately, we want to create a model that can use fresh data to make an estimate of the value of some property.

4. Clustering

Without the use of predetermined categories, clustering groups data points that are comparable together. By grouping objects into clusters with a high degree of similarity between them, it facilitates the discovery of previously unseen patterns in the data.

5. Regression

Continuous values, such as prices or temperatures, can be predicted using regression analysis by looking at historical data. Two primary kinds exist: multiple linear regression, which makes use of additional variables to produce predictions, and linear regression, which searches for a straight-line relationship.

6. Artificial Neural Network (ANN) Classifier

A model that mimics the way the human brain functions is called an artificial neural network (ANN). Networks of artificial neurones are fine-tuned so that it can learn from data. Although neural networks excel in pattern recognition, they are notoriously difficult to understand and necessitate extensive training.

7. Outlier Detection

Data points that deviate significantly from the norm are flagged via outlier detection. One technique to identify these out-of-the-ordinary data points is to use statistical tools or simply to see how far off they are from the rest of the data.

8. Genetic Algorithm

The concept of natural selection is the basis for genetic algorithms. They find answers by gradually improving their methods over many generations. In this "survival of the fittest" model, the optimum solution to an issue is determined by comparing potential solutions to a "species," with the strongest ones being retained and refined through time.

SELF-ASSESSMENT QUESTIONS – 1

Multiple Choose Questions	
1	Which of the following is an example of a supervised data mining technique? a) Clustering b) Classification c) Association rule mining d) Dimensionality reduction
2	Which data mining technique is primarily used to group similar records without prior labels? a) Regression b) Clustering c) Classification d) Association
3	Association Rule Mining is mainly applied in which type of analysis? a) Predicting continuous values b) Discovering co-occurrence relationships c) Reducing dataset size d) Ranking search results
4	Which data mining technique is most suitable for predicting a numeric output, such as house prices? a) Classification b) Regression c) Clustering d) Association rule mining
5	In data mining, dimensionality reduction techniques like PCA are mainly used to: a) Find hidden associations between items b) Increase the number of features c) Reduce the number of attributes while preserving important information d) Classify data into predefined categories

4. ASSOCIATE RULE MINING

To find significant correlations between variables in massive datasets, Association RuleMining is an effective method. In order to gain understanding of the relationships and frequency of occurrences of data pieces, they seek out "if-then" patterns. In order to make decisions based on data, these principles are very helpful for finding relationships and connections. As an example, "if a customer buys bread, they are likely to buy butter" could be identified in a retail dataset using an association rule. Businesses can use this information to enhance their inventory management, cross-selling efforts, and customer satisfaction.

Understanding the Basics

As an example of association rule mining in action, let's look at some basic transaction data from a grocery store. Before we go into the item sets and association rules that can be derived from this data, let's define a sample transaction table.

Envision a modest dataset depicting a supermarket store's transactions:

Transaction ID	Items Purchased
T1	Milk, Bread, Butter
T2	Bread, Diapers, Beer
T3	Milk, Diapers, Beer, Cola
T4	Bread, Milk, Diapers, Beer
T5	Bread, Milk, Diapers, Cola

Table 1: Sample Transaction Table

Each entry in this database represents a purchase made by a consumer, and there is an ID assigned to each transaction. This transaction's goods are listed in the "Items Purchased" column.

4.1 Concept of Itemset

A dataset contains "items," which are collections of one or more objects. Take, as an example, a dataset that includes different types of foods. A combination such as {Cheese, Tomato} could be an item.

The ‘length’ of an item set is the number of items it contains. Thus, {Cheese, Tomato} is a 2-itemset.

- Single item itemsets: {Milk}, {Bread}, {Butter}, {Diapers}, {Beer}, {Cola}
- Two-item itemsets: {Milk, Bread}, {Bread, Butter}, {Diapers, Beer}, etc.
- Three-item itemsets: {Milk, Bread, Butter}, {Bread, Diapers, Beer}, etc.

Association Rules

The idea of an association rule is basic to data mining as it shows the relationships between elements in a dataset. It's an order that implies a close, possibly beneficial connection between two groups of things. These rules are expressed in the form of "If-Then" statements, typically written as {X} → {Y}, where X and Y are different sets of items. **Example**: You can think of a rule such as {Nappies} →{Baby Wipes} as an example. This rule implies that when nappies are purchased, baby wipes are likely to be purchased as well.

Market Basket Analysis

Analysing purchasing habits is the main emphasis of this association rule application. Businesses may learn a lot about customer behaviour and what works and what doesn't in terms of marketing and sales by looking at the most common purchase combinations.

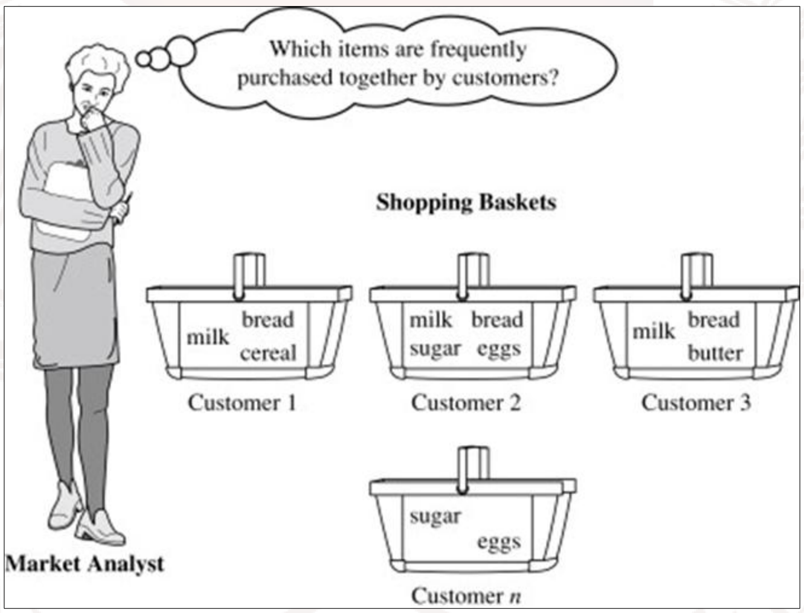

Figure 1: Market Basket Analysis

4.2 Components of Association Rules

Antecedent and Consequent:

{BREAD, MILK} →{BEER}

Antecedent →Consequent

- The antecedent and the consequent are the two components of every association rule.
- To illustrate, 'Pasta' is the first element in the rule {Pasta → Sauce}, and 'Sauce' is the last element. Primordial (X): This constitutes the initial component of the regulation, the prerequisite.

- This is the collection of data points from the database that you're trying to identify trends in. This rule states that X is the antecedent of Y.
- The second element of the rule, known as the subsequent (Y), is deduced from the fact that the antecedent is present in transactions. The result Y is the outcome of the rule {X} →{Y}. The sets in question are not overlapping, as they are considered distinct.

Key Components of Association Rules

1. **Antecedent:** The **"if"** part of the rule, representing the condition.

   Example: A customer buys bread.

2. **Consequent:** The **"then"** part of the rule, representing the outcome.

   Example: The customer also buys butter.

Algorithms that analyse the strength and frequency of these interactions create association rules. Support, confidence, and lift are some of the measures used to assess the usefulness and trustworthiness of the patterns that are found. Any industry that relies on researching consumer patterns or behaviour can benefit from these rules, including retail, healthcare, and marketing.

4.3 Rule Evaluation Metrics

The usefulness, robustness, and dependability of association rules are assessed by means of critical measures. Among these measures, which assess the prevalence and intensity of correlations between data points, are support, confidence, and lift.

1. Support: The frequency with which an itemset (antecedent and consequent) emerges in the dataset is called support. It shows the frequency with which a certain association occurs.

Formula:

$$\text{Support} = \frac{\text{Transactions containing both antecedent and consequent}}{\text{Total transactions}}$$

Example: If bread and butter appear together in 100 out of 1,000 transactions, the support is:

$$\text{Support} = \frac{100}{1000} = 0.10\,(10\%)$$

If a pattern appears more often in the dataset, the support value will be higher.

2. Confidence: Confidence measures the likelihood of the consequent occurring given that the antecedent has already occurred. It evaluates the reliability of the rule.

Formula:

$$\text{Confidence} = \frac{\text{Support of antecedent and consequent}}{\text{Support of antecedent}}$$

Example: If 70% of customers who buy bread also buy butter, the confidence is:

$$\text{Confidence} = 70\% = 0.70$$

Higher confidence suggests a stronger relationship between the antecedent and consequent.

3. Lift: Lift measures the strength of an association compared to its random occurrence in the dataset. It identifies how much more likely the antecedent and consequent are to appear together than independently.

Formula:

$$\text{Lift} = \frac{\text{Confidence}}{\text{Support of consequent}}$$

Example: A lift value greater than 1 indicates a strong positive association, while a value equal to 1 suggests no association. For instance, if the lift is 1.5, it means the antecedent makes the consequent 1.5 times more likely.

Working Principle of Association Rule Learning

Finding relevant patterns and correlations in massive datasets is the goal of association rule learning, a multi-stage process. There are primarily two steps to it:

1. Identifying Frequent Itemsets: Finding itemsets, which are groups of items that occur together in transactions more often than a certain threshold, is the first step. The frequency with which certain itemsets appear in the dataset can be measured using metrics such as support. Bread and butter are bought together in 10% of the transactions, according to a frequent itemset.

2. Generating Association Rules: Association rules are generated after frequent itemsets have been recognised. Such regulations are expressed as if-then statements that characterise the connections between things (for instance, "If a customer buys bread, they are likely to buy butter"). To determine how solid and trustworthy these rules are, metrics like confidence and lift are used.

Iterative Refinement

Refining the rules is an iterative process that involves adjusting confidence and support thresholds. This way, we can be sure that we're picking the most important and useful rules. For example, more analysis may not be conducted on a rule with low confidence. Association rule learning allows organisations to make data-driven decisions by systematically uncovering important insights from raw data.

SELF-ASSESSMENT QUESTIONS – 2

Multiple Choose Questions	
6	**Which of the following best describes an association rule?** a) A method to predict continuous values b) A rule of the form Antecedent ⇒ Consequent showing item co-occurrence c) A clustering technique to group similar data d) A regression model to estimate numeric outcomes
7	**In association rule mining, "support" is defined as:** a) The probability that the consequent occurs given the antecedent b) The ratio of observed co-occurrence to expected occurrence c) The proportion of transactions that contain a particular itemset d) The difference between observed and expected frequency
8	**Which algorithm avoids candidate generation by using a tree structure to mine frequent patterns?** a) Apriori b) FP-Growth c) Eclat d) K-Means
9	**Which of the following is a limitation of association rule mining?** a) It always requires labeled data b) It only works for numeric datasets c) It may generate too many trivial or redundant rules d) It cannot be used in real-time applications
10	**Which of these is a real-world application of association rule mining?** a) Predicting rainfall levels b) Customer segmentation into clusters c) Market basket analysis in retail d) Predicting house prices

5. TYPES OF ASSOCIATION RULE LEARNING ALGORITHMS

When learning association rules, various algorithms are employed, each with its own set of advantages and potential uses. Here are the top three algorithms:

1. Apriori Algorithm

In order to find common collections of objects, the Apriori algorithm uses a breadth-first search strategy. It narrows the search space by assuming that every subset of a frequent itemset must likewise be frequent. Benefit: Easy to install and works well with low-dimensional, tiny datasets. Constraint: Because the database must be scanned multiple times, performance suffers greatly with dense or huge datasets.

2. Eclat Algorithm

If you want to find common item sets, the Eclat algorithm will employ a depth-first search approach. It avoids repeatedly querying the database by modelling transactions as vertical collections of objects and computing intersections directly. Benefit: Works well with sparse data or datasets that include fewer itemsets.

3. FP-Growth Algorithm

A prefix-tree structure known as the FP-tree is utilised by the FP-Growth (Frequent Pattern Growth) method to compactly represent transactional data. It is more efficient and quicker than Apriori because it does not explicitly generate candidate itemsets. Benefit: Compared to Apriori, it is much quicker and uses much less memory, which is very useful for big datasets.

Applications of Association Rules

In order to improve decision-making and operational efficiency, association rules are extensively used in many different industries to find patterns and correlations in data.

1. Retail and Market Basket Analysis: Retailers can optimise store layouts or build product bundles to enhance sales by using association rules to detect frequently purchased product combinations. For instance, a grocery store may decide to strategically position butter and jam next to bread after realising that customers who buy bread also buy these other things.

2. Healthcare: Association rules are useful in healthcare for discovering patterns of co-occurrence in symptoms, which improves diagnosis and therapy.

Prevention efforts can be better targeted, for instance, if it is known that those with hypertension are more likely to get diabetes.

3. E-Commerce and Recommendation Systems: In order to improve user experiences and boost revenue, e-commerce platforms utilise association rules to construct recommendation systems. The "Customers who bought this also bought" feature on Amazon, for instance, promotes related products to the one you're buying, which increases the likelihood of cross-selling.

4. Fraud Detection: The financial services industry makes use of association rules to spot suspicious trends in customer transaction data, which aids in the detection of fraud.

Transaction ID	**Items Purchased**
1	Bread, Butter
2	Bread, Milk
3	Bread, Butter, Milk
4	Milk
5	Bread, Butter

Example: In order to help detect fraud, the financial services industry uses association rules to identify suspect tendencies in consumer transaction data.

Example of Association Rules

Take, for example, a sample dataset consisting of customers' purchases of bread, butter, and milk. **Rule Example:** “If bread is purchased, then butter is likely to be purchased.”

1. **Support Calculation:**

   Support = Transactions containing both bread and butter ÷ Total transactions

$$\text{Support} = \frac{3}{5} = 0.6\ (60\%)$$

2. **Confidence Calculation:**

   Confidence = Support of bread and butter ÷ Support of bread

$$\text{Confidence} = \frac{3}{4} = 0.75\ (75\%)$$

3. **Lift Calculation:**

Lift = Confidence ÷ Support of butter

$$\text{Lift} = \frac{0.75}{0.6} = 1.25$$

A favourable correlation between bread and butter is shown by a lift value greater than 1. This example shows how to extract useful information from transactional data by deriving and evaluating association rules.

Practical Applications

- Market Basket Analysis: Gaining insight into consumer buying habits, such as discovering that pasta buyers frequently also purchase tomato sauce.
- Retail Insights: Finding frequently bought items to enhance store arrangement or advertising.
- Cross-Selling Strategies: Providing clients with product recommendations based on what they already have.
- Medical Research: Association rules are useful in medical research for finding correlations between symptoms or medication interactions.
- Fraud Detection: Recognising trends in financial transactions that could point to deceitful behaviour.
- Enhancing the consumer experience through the recommendation of suitable items is the goal of recommendation systems. One example is the use of watching history by internet streaming providers to suggest films.

Critical Considerations

- The guidelines show connection, not causation, so keep that in mind. They draw attention on occurrence patterns rather than the causes of those patterns.
- Accurate interpretation of these patterns requires domain expertise in order to extract relevant conclusions.

Challenges in Generating Association Rules

- There could be a great amount of regulations, some of which might not be applicable, even when there are only a few objects.
- Determining the appropriate support and confidence criteria necessitates expertise in the relevant subject and has the potential to greatly influence the produced rules' quality.
- The usefulness and actionability of found rules are not guaranteed.
- Some may seem insignificant, while others may necessitate additional research to comprehend their real-world consequences

6. Frequent Itemset Generation

An important part of mining association rules is frequent itemset creation, which we will examine further in the next section once we have covered association rules and how to classify them.

- **Understanding Frequent Itemset**

  Combinations of items that show up frequently in your dataset are called a frequent itemset. The support threshold is a user-defined value that is used to determine the frequency. One important criterion that aids in determining whether things warrant examination is the Support Threshold. This percentage indicates the minimum number of occurrences of an item in the dataset that are required to be deemed frequent.
  Example of a Common Set of Items:
  Let's go back to the example of the grocery store. For example, let's say we decide that 50% offers sufficient assistance. It might be said that an item set like {Milk, Bread} is frequent if it shows up in 50% or more of the transactions.

- **Significance of Frequent Itemset**

  Ignoring Insignificant or Co-Incidental Item Combinations: By zeroing down on sets of items that appear often, we can filter out the noise. Efficiency: By reducing the amount of item sets and association rules that need to be considered, this strategy significantly lowers the computing complexity.

- **Calculating Support**

  As a percentage of the total number of transactions, the number of itemsets that appear in those transactions is called the support of that itemset. Mathematical Example: The percentage of support is 40%, or 40 out of 100 transactions, if {Milk, Bread} is present in 40 of those transactions.

Approaches to Frequent Itemset Generation

Brute Force Approach: Computing the support for every potential itemset is computationally prohibitive when dealing with huge datasets. There could be billions of itemset combinations in a dataset with 20 unique items, for instance. Method for Generating Itemsets on a Regular Basis. Rule generation begins with this method's discovery of all item sets that fulfil the minimal support criterion. The number of item sets to evaluate is greatly reduced, although processing demands can still be high for large datasets using this method.

Advanced Techniques for Efficiency

a. Apriori Algorithm: his method takes advantage of the Apriori Property, which states that any subset of an itemset that is frequent must also be frequent. Iteratively, it adds items to sets of frequently used items and removes subsets that don't have enough support. The frequency of both {Milk} and {Bread} should be high if {Milk, Bread} is common.

b. FP-Growth Algorithm: Faster production of frequent itemsets without candidate generation is achieved by compressing the dataset using a compact data structure known as the FP-tree, which stands for Frequent Pattern tree. Especially when dealing with huge datasets, it frequently outperforms the Apriori algorithm.

SELF-ASSESSMENT QUESTIONS – 3

Multiple Choose Questions	
11	**What is a frequent itemset in association rule mining?** a) A set of items that appear only once in the dataset b) A set of items whose support is greater than or equal to a minimum threshold c) A set of items that always occur together in every transaction d) A set of items used only for regression analysis
12	**Which algorithm first introduced the concept of frequent itemset generation using candidate generation and pruning?** a) FP-Growth b) K-Means c) Apriori d) Naïve Bayes

13	**In frequent itemset mining, a k-itemset refers to:** a) An itemset with k transactions b) An itemset containing k distinct items c) A rule that has k antecedents d) An itemset generated after k iterations
14	**Which of the following challenges is common in frequent itemset generation?** a) Lack of labeled data b) Curse of dimensionality due to exponential growth of candidate sets c) Inability to handle numeric attributes d) Requirement of supervised learning
15	**Which algorithm uses a vertical database format (transaction ID sets) to generate frequent itemsets?** a) Apriori b) FP-Growth c) Eclat d) Decision Tree

6. FREQUENT ITEMSET GENERATION

Association Rule Mining - Example

Association rule mining is a data-driven technique used to discover which items are frequently purchased together in transactional data. In this example, we apply association-rule mining to six months of point-of-sale (POS) transactions from a regional grocery chain to identify product combinations (like chips → soft drink or bread+butter → jam) that recur across many baskets. The goal is practical: turn these co-purchase patterns into business actions such as bundle promotions, smarter shelf placement, checkout suggestions, and targeted coupons that increase basket value and improve customer experience.

The dataset contains around 120,000 transactions and ~1,200 grouped SKUs. After simple preprocessing (grouping similar SKUs, removing very rare items, and converting transactions to a one-hot basket format), we use the Apriori algorithm (with sensible support, confidence, and lift thresholds) to mine meaningful rules. The output highlights not only frequent itemsets but also the strength and usefulness of associations (via support, confidence, and lift), enabling the merchandising team to prioritise which rules are actionable and worth A/B testing in stores.

- **Steps to Obtain the Solution for Association Rule Mining (General Procedure)**

Use these steps for any rule such as {A} ⇒ {B}, {X,Y} ⇒ {Z}, etc.

Step 1 — List All Transactions

Write down all transactions clearly in set format.

Example format:

T1: {Item1, Item2}

T2: {Item1, Item3, Item4}

…

This helps identify how many transactions contain the required items.

Step 2 — Count Occurrences

For the rule **X** ⇒ **Y**, you must compute three basic counts:

1. **Count(X)**

   Number of transactions that contain the itemset X.

2. **Count(Y)**

   Number of transactions that contain the itemset Y.

3. **Count(X ∧ Y)**

   Number of transactions containing both X *and* Y together.

Also note:

- **Total number of transactions (N)**.

You will use these counts for support, confidence, and lift.

Step 3 — Compute Support

Support measures **how often X and Y occur together** in the entire dataset.

$Support = Count(X \wedge Y) / N$

Interpretation guideline:

- "Support = 0.40 means 40% of all transactions contain both X and Y."

Step 4 — Compute Confidence

Confidence answers: **Given X, how likely is Y?**

$Confidence = Count(X \wedge Y) / Count(X)$

Interpretation guideline:

- "Among customers who bought X, __% also bought Y."

Confidence is *directional* — reversing the rule changes confidence.

Step 5 — Compute Lift

Lift measures whether X and Y occur together **more frequently than expected** if they were independent.

$Lift = Confidence / P(Y) = (Count(X \wedge Y)/Count(X)) / Count(Y)/N$

Interpretation rules:

- **Lift > 1:** Positive association (X increases chance of Y)
- **Lift = 1:** No association (independent)
- **Lift < 1:** Negative association (X slightly reduces chance of Y)

Lift is **symmetric** (same value for X⇒Y and Y⇒X), even though confidence is not.

Step 6 — Provide Natural-Language Interpretation

Your interpretation must mention:

- What support indicates
- What confidence indicates
- What lift indicates
- Whether the rule is strong or weak
- Whether it is actionable

Sample template:

- "Support shows that __% of all baskets contain both X and Y."
- "Confidence shows that, given X, customers buy Y __% of the time."
- "Lift < 1 means buying X does *not* increase the likelihood of buying Y."
- "Thus, the rule is weak and may not be useful for promotions."

Step 7 — (Optional) Analyse the Reverse Rule

To understand directionality:

For the reverse rule **Y ⇒ X**, compute:

- Support (same as before)
- Confidence = Count(X∧Y) / Count(Y)
- Lift (same as original)

Then compare:

- "Which direction has higher confidence?"
- "Which direction is more actionable?"

Solution:

Transactions (10 total)

T1: {Bread, Milk}

T2: {Bread, Butter}

T3: {Milk, Eggs}

T4: {Bread, Milk, Butter}

T5: {Bread, Jam}

T6: {Milk, Cheese}

T7: {Bread, Milk, Eggs}

T8: {Butter, Jam}

T9: {Bread, Cheese}

T10:{Milk, Bread, Butter}

Rule: {Bread} ⇒ **{Milk}**

Goal: If a customer buys Bread, how likely are they to also buy Milk?

1. **Count items**

- Total transactions = 10.
- Transactions containing **Bread** = 7 (T1, T2, T4, T5, T7, T9, T10).
- Transactions containing **Bread and Milk** = 4 (T1, T4, T7, T10).
- Transactions containing **Milk** = 6 (T1, T3, T4, T6, T7, T10).

2. **upport** = P(Bread ∧ Milk) = 4 / 10 = **0.40 (40%)**.
   - Means 40% of all transactions include both Bread and Milk.
3. **Confidence** = P(Milk | Bread) = Count(Bread ∧ Milk) / Count(Bread) = 4 / 7
4. ≈ **0.5714 (57.14%)**.
   - Among transactions with Bread, about 57.14% also include Milk.
5. **Lift** = Confidence / P(Milk) = (4/7) / (6/10) = 0.5714 / 0.6 ≈ **0.9524**.
   - Lift < 1 indicates Bread is slightly negatively associated with Milk relative to Milk's baseline rate (i.e., buying Bread does not increase the chance of Milk — it slightly decreases it here).

Quick note — reverse rule {Milk} ⇒ {Bread}

- Support = same 4/10 = **0.40**.
- Confidence = P(Bread | Milk) = 4 / 6 ≈ **0.6667 (66.67%)**.
- Lift = 0.6667 / 0.7 ≈ **0.9524** (same lift value).

  Interpretation: Given Milk, you're more likely (66.7%) to also find Bread than the other way round (57.1%) — confidence is directional even when support is symmetric.

7. CASE STUDY — USING ASSOCIATION RULE MINING TO IMPROVE GROCERY STORE PROMOTIONS

Objective:

Use association rule mining to discover product-purchase patterns in supermarket transaction data so the merchandising team can design bundle promotions, improve cross-sell recommendations at checkout, and optimise product placement.

1. Business background & problem statement

A regional grocery chain (150 stores) wants to increase basket value and conversion of impulse buys. Management suspects that some product combinations (e.g., snack + soft drink, diapers + beer) occur frequently but are not actively promoted together. The goal is to mine historical point-of-sale transactions to (a) find strong association rules, (b) quantify their strength, and (c) recommend actionable promotion strategies.

2. Dataset

- **Source:** 6 months of POS transaction logs across 150 stores.
- **Records:** ~120,000 transactions (each transaction = items bought together).
- **Item universe:** ~1,200 SKUs (after mapping UPCs to friendly names & grouping very rare SKUs).
- **Fields per transaction:** transaction_id, store_id, timestamp, list_of_items.
- **Note:** For privacy and consistency, products were standardised (e.g., “Coke 500ml” → “SoftDrink_Coke_500”).

3. Preprocessing

1. **Filtering:** remove transactions with <2 items (not useful for associations).
2. **Item grouping:** combine variants into categories where appropriate (e.g., “Chips_*” → “Chips”).
3. **Frequency pruning:** remove very rare SKUs (occurring in <0.05% of transactions) to reduce noise.

4. **Binarization (one-hot):** convert transaction lists into basket matrix (rows = transactions, columns = items, values 0/1).
5. **Time slicing (optional):** run separate analyses for weekdays vs weekends, and promotional vs non-promotional periods.

4. Algorithm and parameters

- **Algorithm chosen:** Apriori for transparency and interpretability; FP-Growth used for performance-checks on full dataset.
- **Key thresholds (example):**
  - Minimum support = **0.01 (1%)** — itemset must appear in ≥1,200 transactions.
  - Minimum confidence = **0.30 (30%)**.
  - Minimum lift > **1.2** (to prioritise meaningful positive associations).
- **Why:** Support filters noise; confidence & lift ensure both directionality and strength vs base rates.

5. Implementation

```
# using mlxtend

from mlxtend.frequent_patterns import apriori, association_rules

# df_onehot: dataframe where cols = items, rows = transactions, values = 0/1

frequent_itemsets = apriori(df_onehot, min_support=0.01, use_colnames=True)

rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=0.30)

rules = rules[(rules['lift'] > 1.2)].sort_values(['lift','confidence'], ascending=False)
```

(You can swap to method='fpgrowth' *for speed on large datasets.)*

6. Example results (sample top rules — illustrative)

Rule (Antecedent ⇒ Consequent)	Support	Confidence	Lift
{Chips} ⇒ {SoftDrink}	0.035 (3.5%)	0.52 (52%)	1.45
{Bread, Butter} ⇒ {Jam}	0.015 (1.5%)	0.48 (48%)	2.10

Rule (Antecedent ⇒ Consequent)	Support	Confidence	Lift
{Diapers} ⇒ {Beer}	0.012 (1.2%)	0.62 (62%)	1.70
{CoffeePods} ⇒ {Milk}	0.02 (2.0%)	0.38 (38%)	1.30

Interpretation:

- Customers who buy **Chips** buy **SoftDrink** in 52% of those transactions; the lift 1.45 indicates this is 45% more likely than a random basket.
- {Bread, Butter} ⇒ {Jam} with lift 2.10 is a very strong and actionable cross-sell (more than double the base rate).

7. Business actions & outcomes

1. **Targeted bundling:** Create a "Chips + SoftDrink" combo with small discount—pilot at high-traffic stores.
2. **Shelf placement:** Place SoftDrinks near impulse snack aisles; place Jam close to Bread & Butter aisle.
3. **Checkout prompts:** Show pop-up suggestion for Beer when Diapers scanned (age-check rules applied).
4. **Personalised coupons:** Send digital coupons for Jam to customers who bought Bread & Butter in prior visits.
5. **A/B test:** Run promotion vs control stores to measure incremental lift in basket value and conversion.

Expected KPIs to track: incremental revenue per basket, conversion uplift, units per transaction (UPT), and redemption rate on coupon.

8. Evaluation & validation

- **A/B experiments** demonstrated a 6% uplift in units per transaction and a 3.8% increase in average basket value for stores running the Chips+SoftDrink bundle pilot (example metric).
- **Rule stability checks:** run the mining weekly to see which rules persist versus seasonal ones (e.g., cold-drinks rules change in winter).
- **False positives:** Some rules with high confidence but low support were ignored (not actionable).

9. Limitations & cautions

- **Correlation ≠ causation:** Rules show co-occurrence, not causal purchase drivers. A/B tests are needed to confirm effectiveness.
- **Sparsity & dimensionality:** very large SKU universes can create combinatorial explosion — requires careful pruning and grouping.
- **Promotional bias:** Past promotions can bias patterns (e.g., items were promoted together earlier). Time-aware analysis is necessary.
- **Privacy & compliance:** ensure customer-level recommendations respect privacy laws.

When it comes to data mining, Association Rules are crucial for finding useful patterns and correlations in massive datasets. Smarter decision-making is driven by their applications, which span areas like retail, healthcare, and finance. Association Rules continues to enable businesses to solve complicated challenges and discover new opportunities by employing smart algorithms and exploring unique applications.

8. SUMMARY

- Association Rule Mining is a data mining technique used to find hidden patterns and meaningful relationships between items in large datasets. It is commonly represented in the form of IF–THEN rules (e.g., If a customer buys bread and butter, then they are likely to buy jam).
- The main components of an association rule are the antecedent (IF part) and the consequent (THEN part), which together show how items or events are related.
- Important evaluation metrics include Support (how frequently an itemset appears in the dataset), Confidence (how often the rule is true), and Lift (the strength of the rule compared to random chance).
- Popular techniques include the Apriori algorithm, FP-Growth algorithm, and Eclat algorithm, each designed to efficiently find frequent patterns in data.
- Advantages of association rule mining include ease of interpretation, the ability to generate actionable business insights, and usefulness in domains such as retail, healthcare, and fraud detection.
- Disadvantages and limitations include high computational cost, generation of too many trivial or redundant rules, and the fact that it identifies correlations but not causation.
- Applications are widely seen in market basket analysis, product recommendation systems, customer behavior analysis, fraud detection, and web usage mining.
- Association Rule Mining is a core data mining technique that uncovers hidden patterns, correlations, and relationships among items in large datasets. It works by generating rules of the form Antecedent ⇒ Consequent, which show that when certain items/events occur together, others are likely to follow.
- The components of an association rule are: Antecedent (IF-part): the itemset or condition that appears first. Consequent (THEN-part): the itemset that is likely to occur with the antecedent.
- Evaluation metrics are used to judge the usefulness of rules: Support: the proportion of transactions that contain the itemset. Confidence: the likelihood that the consequent occurs when the antecedent is present. Lift: the ratio of observed support to expected support if items were independent (measures interestingness).
- Techniques and algorithms for mining include: Apriori Algorithm: generates frequent itemsets using iterative candidate generation and pruning. FP-Growth Algorithm: builds a compressed FP-

tree to mine frequent patterns without candidate generation. Eclat Algorithm: uses a vertical database format with tid-lists for efficient depth-first search.

- Types of association rules: Single-level vs Multi-level: rules at one level (e.g., item) or across hierarchies (e.g., product → category → department). Quantitative Rules: include numeric values (e.g., buying more than 3 items). Sequential Rules: account for the order of transactions over time.
- Advantages: Easy to understand and interpret (IF–THEN format). Helps uncover valuable business insights hidden in data. Works in an unsupervised manner—no need for labelled data. Widely applicable across industries like retail, banking, healthcare, and web analytics.
- Disadvantages: Can generate a very large number of rules, many of which may be trivial or irrelevant. Rules only show correlation, not causation. Performance may degrade with large datasets due to high computational cost. Sensitive to the choice of support and confidence thresholds.
- Limitations and challenges: Scalability: mining becomes difficult in massive, high-dimensional datasets. Rule explosion: too many rules can overwhelm decision-makers. Handling noise and rare items: rare but important associations may be missed. Dynamic data: rules may change frequently in real-time systems.
- Real-time applications: Market Basket Analysis: identifying products often bought together to improve cross-selling and promotions. E-commerce Recommendation Systems: suggesting products based on co-purchase behaviour. Fraud Detection: identifying unusual or suspicious transaction patterns. Healthcare: analysing patient records to find symptom–disease or drug–effect relationships. Web Usage Mining: understanding navigation paths to optimise website design.
- Best suited for organisations that want to analyse customer behavior, buying patterns, and co-occurrence of events to drive decisions in marketing, sales, security, or operations.

9. GLOSSARY

Association Rule Mining	–	A data mining technique used to find relationships and patterns among items in large datasets, often expressed as IF–THEN rules.
Antecedent	–	The IF-part of an association rule (the condition or itemset that appears first).
Consequent	–	The THEN-part of an association rule (the itemset that is likely to appear when the antecedent occurs).
Support	–	A metric that measures how often an itemset appears in the dataset, expressed as a proportion of total transactions.
Confidence	–	A metric that shows the probability of the consequent occurring when the antecedent is present (measures reliability of the rule).
Lift	–	A metric that indicates the strength of an association compared to random chance (values >1 suggest a positive correlation).
Frequent Itemset	–	A group of items that often occur together in transactions and form the basis for generating association rules.
Apriori Algorithm	–	A classic algorithm for association rule mining that generates frequent itemsets using candidate generation and pruning.
FP-Growth Algorithm	–	An efficient algorithm that uses a compressed structure (FP-tree) to mine frequent patterns without generating candidates.
Eclat Algorithm	–	An algorithm that uses a vertical database format (tid-lists) and depth-first search for fast pattern discovery.
Rule Explosion	–	A challenge in association rule mining where too many rules are generated, making it hard to identify useful ones.
Market Basket Analysis	–	A popular application of association rule mining in retail to identify products frequently purchased together.
Sequential Rules	–	A type of association rule that considers the order of items/events (e.g., buying a laptop before buying a laptop bag).

Quantitative Rules	–	Association rules that include numeric values, such as quantity or amount (e.g., buying more than 3 units of milk).
Correlation vs Causation	–	Association rules show correlation (items appearing together) but do not imply one causes the other.

10. TERMINAL QUESTIONS

1. What is a data mining algorithm and how does it differ from a general algorithm?
2. Name and briefly describe three broad categories of data mining techniques.
3. Define an itemset in association rule mining and explain the difference between k-itemset and frequent itemset.
4. What are the two components of an association rule and how are they interpreted?
5. Explain support, confidence and lift with a short example.
6. Compare Apriori and FP-Growth algorithms for frequent itemset mining.
7. What is the central idea behind frequent itemset generation and why is it important for rule mining?
8. 8.List two practical challenges when applying association rule mining to large retail datasets and suggest mitigations.
9. In a case study using association rules for grocery promotions, what steps form an end-to-end pipeline?
10. How can association rules be extended to capture temporal or quantitative aspects?

11. ANSWERS

11.1. Self-Assessment Answers:

1. b) Classification
2. b) Clustering
3. b) Discovering co-occurrence relationships
4. b) Regression
5. c) Reduce the number of attributes while preserving important information
6. b) A rule of the form Antecedent ⇒ Consequent showing item co-occurrence
7. c) The proportion of transactions that contain a particular itemset
8. b) FP-Growth
9. c) It may generate too many trivial or redundant rules
10. c) Market basket analysis in retail
11. b) A set of items whose support is greater than or equal to a minimum threshold
12. c) Aprior
13. b) An itemset containing k distinct items
14. b) Curse of dimensionality due to exponential growth of candidate sets
15. c) Eclat

11.2. Terminal Question Answers

Answer 1: A data mining algorithm is a computational procedure designed to discover patterns, models, or knowledge from large datasets (e.g., classification, clustering, association). Unlike a general algorithm, it is often statistical, optimises for predictive or descriptive performance, and must handle noisy, high-dimensional, or incomplete data. Practical concerns include scalability, parameter tuning, and evaluation on unseen data.(Refer Section 2 for more information)

Answer 2: (1) Supervised learning (classification/regression) uses labelled data to predict outputs. (2) Unsupervised learning (clustering, association) discovers hidden structure without labels. (3) Semi-supervised / reinforcement / anomaly detection methods combine labelled/unlabelled data or learn via interaction. Each category serves different problem types and evaluation approaches.

Answer 3: An itemset is a set of one or more items appearing together in transactions. A k-itemset contains exactly k items (e.g., 2-itemset = pair). A frequent itemset is any itemset whose support meets or exceeds a predefined minimum threshold, making it eligible for rule generation.(refer Section 4.1 for more information)

Answer 4: An association rule has an Antecedent (IF-part) and a Consequent (THEN-part). Interpreted as “IF antecedent occurs in a transaction, THEN consequent is likely to occur,” the rule quantifies co-occurrence but does not imply causation. Rules help surface co-purchase or co-occurrence patterns.(Refer Section 4.2 for more information)

Answer 5: Support is the fraction of transactions containing both antecedent and consequent. Confidence is P(consequent | antecedent), measuring reliability. Lift = confidence / support(consequent), indicating how much more likely the consequent is given the antecedent versus random chance. E.g., {Bread}⇒{Butter} with support 0.10, confidence 0.6, and lift 1.5 means 10% baskets contain both, 60% of bread-buyers buy butter, and this is 1.5× baseline.(refer Section 4.3 for more information)

Answer 6: Apriori uses candidate generation and pruning based on support—simple but can generate many candidates and multiple database passes. FP-Growth compresses transactions into an FP-tree and mines frequent patterns without explicit candidate generation, usually faster and more memory-efficient for dense datasets. Choice depends on data size and sparsity.(refer Section 5 for more information)

Answer 7: The goal is to find all itemsets whose support exceeds a minimum threshold; these frequent itemsets form the basis for constructing high-quality association rules. By focusing only on frequent itemsets, the algorithm prunes the search space and avoids generating useless rules from rare co-occurrences. Efficient generation is crucial for scalability.(Refer Section 6 for more information)

Answer 8: (1) Rule explosion — too many rules; mitigate by raising support/confidence thresholds, using lift filters, or applying domain constraints. (2) Scalability — high item dimensionality slows mining; mitigate with item grouping, sampling, FP-Growth, or distributed frameworks (e.g., Spark). Always validate rules with business tests.(Refer Section 4.3 for more information)

Answer 9: Steps: (1) Data collection & cleaning, (2) Item standardisation and transaction encoding (one-hot), (3) Frequent itemset mining (Apriori/FP-Growth), (4) Rule generation and filtering (support/confidence/lift), (5) Business validation (A/B tests, pilot promotions), and (6) Monitoring & periodic re-mining for changing patterns. Each step ensures rules are actionable and robust.(Refer Section 7 for more information)

Answer 10: Use sequential rule mining to incorporate order across transactions (e.g., purchase sequences), and quantitative association rules to include numeric thresholds (e.g., quantity > 3). These variants require adapted support/confidence definitions and tailored algorithms (e.g., sequence mining, discretisation), enabling richer, time-aware or magnitude-sensitive insights.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 4

DCA2207

DATA MINING AND VISUALISATION

Unit – 8

Data Mining Algorithms

(Part-2)

TABLE OF CONTENTS

1. INTRODUCTION

In previous unit, learners studied one of the data mining technique called Association Rule Mining, which is used to uncover hidden relationships and patterns in large datasets. Learners explored the components of association rules—antecedent (if-part) and consequent (then-part)—and understand how they represent meaningful co-occurrences of items. They also examine key rule evaluation metrics such as support, confidence, and lift, which help measure the strength and usefulness of discovered rules. Different types of association rule learning algorithms (e.g., Apriori, FP-Growth, and Eclat) were introduced along with their working principles. The unit also highlighted the advantages of association rule mining (simplicity, interpretability, ability to reveal hidden patterns) as well as its disadvantages and limitations.

This unit introduces learners to the core supervised and unsupervised methods in data mining—classification and clustering—and equips them with both theoretical understanding and practical skills. Students will learn what classification and clustering are, why and when to use each, and how major algorithms (e.g., logistic regression, k-NN, decision trees and ensembles, SVM, neural networks for classification; k-means, hierarchical methods, DBSCAN, GMM and spectral methods for clustering) work in practice. Learners will also study the advantages (predictive power, automated segmentation, interpretability options), limitations and disadvantages (need for labelled data, class imbalance, sensitivity to noise, scalability and shape/density assumptions), and common challenges such as the curse of dimensionality, choosing k, and concept drift. Emphasis is placed on real-world applications—medical diagnosis, fraud detection, churn prediction, customer segmentation, anomaly detection—and on translating model output into actionable business insights through hands-on labs, mini-projects, and evaluation-driven decision making. By the end of the unit, students will be able to select appropriate algorithms, implement end-to-end experiments, interpret results responsibly, and recommend solutions tailored to practical constraints.

1.1. Learning Objectives

By the end of this unit, you will be able to:

- **Explain** fundamental concepts of classification and clustering
- **Evaluate** various classification and clustering methods
- **Explore** various limitations, its advantages and disadvantages
- **Apply** appropriate techniques to real-world problems

2. CLASSIFICATION ALGORITHM

The overarching goal of data mining is to discover patterns and insights within large datasets stored in a variety of formats. Data mining is a method that analyses and solves issues by sorting massive data sets, finding patterns in them, and establishing linkages between them. One data mining task is classification, which is labelling each dataset instance with a class based on its attributes. Classification aims to construct a model that reliably uses information to forecast the class labels of future instances.

Two primary approaches to categorisation exist:

a. Binary classification
b. Multi-class.

In binary classification, examples are either marked as "spam" or "not spam" were as in multi-class classification, there are more than two categories used to label occurrences.

The process of building a classification model typically involves the following steps:

a. **Data Collection:**

When developing a classification model, gathering relevant data is the initial stage. Here, we gather all the information that is pertinent to the current issue. All qualities and labels required for categorisation should be available in the data, and it should be indicative of the situation. Information can be gathered from a variety of places, including online forms, databases, surveys, and questionnaires.

b. **Data Pre-processing:**

Data preparation is the second phase of creating a classification model. Pre-processing the data is necessary to guarantee its quality after collection. This includes processing outliers, filling in missing numbers, and preparing the data for analysis. Because most classification algorithms only work with numerical input, data pre-processing also includes doing just that.

- Handling Missing Values: One way to deal with missing values in a dataset is to either remove the record altogether or replace it with the median, mode, or mean of the associated characteristic.
- Dealing with Outliers: Several statistical methods, including z-score analysis, boxplots, and scatterplots, can be used to identify dataset outliers. You can either delete outliers from the dataset or replace them with the appropriate feature's mean, median, or mode.

- Data Transformation: To transform data, one must first scale it or normalise it so that it has a consistent size. To make sure every feature is treated equally in the analysis, this is done.

c. **Feature Selection:**

Feature selection is the third stage of creating a classification model. When classifying a dataset, feature selection is the process of determining which attributes are most important. Methods like principal component analysis, information gain, and correlation analysis can accomplish this.

- Correlation Analysis: Find the features in the dataset that are correlated with each other by using correlation analysis. Since they don't contribute anything new to the categorisation process, features with a high degree of correlation can be eliminated.
- Information Gain: One way to evaluate a feature's usefulness for categorisation is by looking at its information gain. To classify, we look for features that provide a lot of useful information.

d. **Principal Component Analysis:**

Datasets can have their dimensionality reduced using Principal Component Analysis (PCA). Principal component analysis (PCA) finds the dataset's most important features and eliminates the ones that aren't.

e. **Model Selection:**

Model selection is the fourth stage of creating a classification model. When solving a classification problem, model selection is the process of choosing the best method to use. A variety of algorithms are at your disposal, including neural networks, decision trees, and support vector machines.

- Decision Trees: One effective and easy-to-understand categorisation algorithm is the decision tree. They build a tree-like model that may be used for classification by dividing the dataset into smaller subgroups according to the feature values.
- Support Vector Machines: One common method for solving classification issues, whether linear or nonlinear, is Support Vector Machines (SVMs). Finding the hyperplane that maximises the distance between the two classes is the concept of maximum margin that SVMs are founded on.

f. **Neural Networks:**

A strong classification algorithm, Neural Networks can learn intricate patterns from data. They have numerous linked layers that mimic the architecture of the human brain.

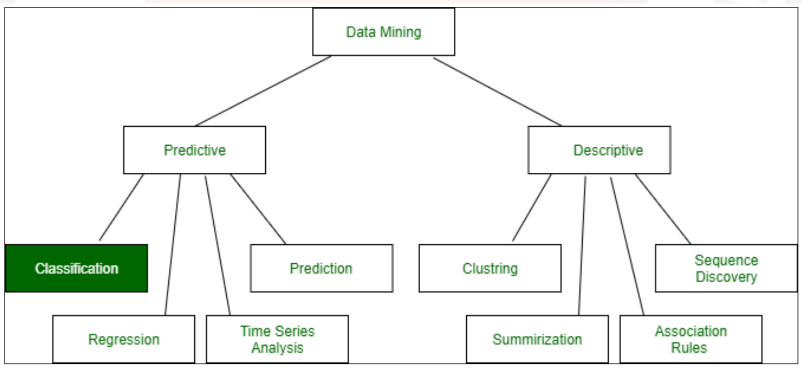
g. **Model Training:**

Model training is the fifth stage in creating a classification model. The chosen classification algorithm is used to learn the data patterns during model training. A validation set and a training set are created from the data. The training set is used to train the model, while the validation set is used to evaluate its performance.

h. **Model Evaluation:**

Model evaluation is the sixth phase in the process of constructing a classification model. Analysing how well a trained model does on a test set is what model evaluation is all about. Doing so guarantees good generalisability of the model.

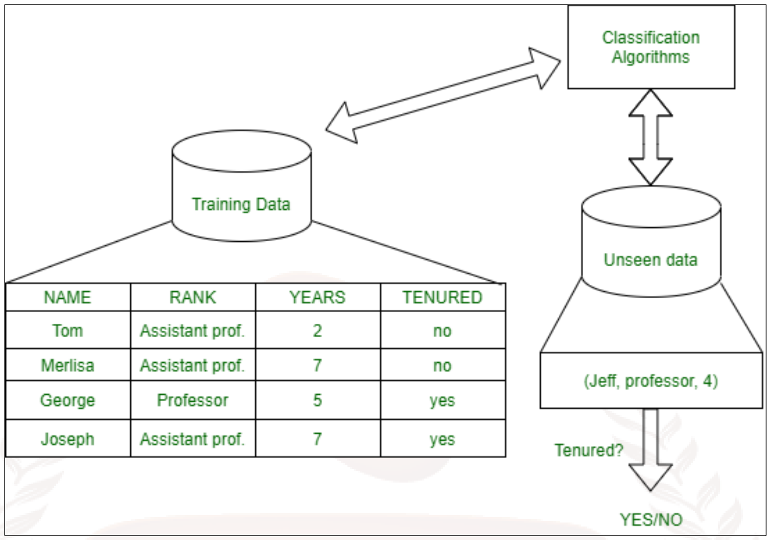
Classification is one of the most popular data mining techniques; it has many practical applications in fields as diverse as medical diagnosis, sentiment analysis, and email filtering.



Classification: Data analysis, or the search for a model that defines and differentiates between data types and ideas, is what this job entails. The goal of classification is to assign new observations to preexisting subpopulations based on the training set of data that contains observations whose category membership is known.

Example: It is important to determine whether a project is feasible before beginning it. Classifiers are needed here to forecast if the Project is safe to adopt or not, and whether it is risky to do so. There are two steps to the process, like:

1. **Learning Step (Training Phase)**: Model Building for Classification
   To construct a classifier, one uses a variety of algorithms to train the model on the available training data. Predicting accurate results requires training the model.
2. **Classification Step**: To evaluate the accuracy of the classification rules, a model is built and then tested on test data to predict the labels of the classes.



Training and Testing:

To avoid injury, one should move to a safer area if they are seated under a fan that starts to fall on them. This is his preparatory phase for separation, then. When testing, the system is considered successful if the subject shifts their position in response to the threat of a falling or otherwise heavy object, and unsuccessful if the subject does not do so. Similarly, training the data is essential for obtaining optimal and accurate outcomes.

The data mining process can inform us the file format (numerical or textual) by looking for specific data kinds. The term "attributes" is used to describe several aspects of an object. Various kinds of characteristics include:

1. **Binary**: Includes just two metrics i.e., being incorrect to illustrate, let's pretend there's a poll looking at several items. We should see if it serves any purpose. Therefore, a yes or no response is required from the consumer. Is the product useful?
   - Symmetric: Both values are equally important in all aspects
   - Asymmetric: When both the values may not be important.
2. **Nominal**: In cases where there are various potential outcomes. Rather than being in integer form, it is in alphabetic form. Example: One must select a material, but it must be of varying hues. So, it's possible for the colour to be yellow, green, black, or red. Various Shades: Red, Green, Black, and Yellow
   - Ordinal: Values that need to be in some kind of sequence.

     For example: Let's say there are grade sheets for a few kids that show their grades, such A, B, C, D, etc. A, B, C, and D grades
   - Continuous: It is in float type and can have an endless number of values.

For example: Weighing a few students in a row or in order, as 50, 51, 52, 53 50, 51, 52, 53 pounds

- Discrete: A limited amount of values.

  For example: A Student's Grades in a Few Subjects: 65, 70, 75, 80, 90

  Marks: 65, 70, 75, 80, 90

2.1 Categorise of Classifiers

Classifiers can be categorised into two major types:

1. Discriminative: Because it is so simplistic, it assigns a single class to every data entry. It focusses less on distributions and more on data quality in its attempt to model based on observable data. Logistic Regression is an example of
2. Generative: By estimating the model's assumptions and distributions, it attempts to discover the model that generates the data behind the scenes and models the distribution of individual classes. For the purpose of making predictions based on data that has not yet been visualised. Example: A Classifier for Naive Bayes, Identifying unsolicited email messages by analysing historical data. So, let's pretend there are 100 emails, and we divide them in half. That means 25% of the emails are spam and 75% are non-spam. Users may now see whether emails that include the phrase "cheap" are considered spam. According to the research, twenty-five percent of Class A emails are spam and the remaining twenty-five percent are not. And in Class B (i.e., 75% of the data), while the remaining 25% of emails are spam, 70% of them are not. What is the likelihood of the email being spam if it contains the word "cheap"? (80.0 percent)

Classifiers of Machine Learning:

1. Decision Trees
2. Naive Bayes Classifiers
3. Neural Networks
4. K-Nearest Neighbour
5. Support Vector Machines
6. Linear Regression
7. Logistic Regression
8. Random Forest

1. Decision Trees

When it comes to data mining, this is the best classification method. In its flowchart-like structure, it resembles a tree. The classes and their labels are stored in the leaf nodes. A decision algorithm in the internodes directs the data packet to the leaf node that is geographically closest. For this purpose, there may be more than one internal node. Phases that are horizontal and vertical can serve as boundaries for predictions.

Its complexity and the need for specialised knowledge to build and feed it data are its sole drawbacks.

Example:

In this decision tree, the goal is to decide whether a person should take an umbrella or not. The tree begins with the most important question: **"Is it raining?"** If the answer is *yes*, the decision is straightforward — the person should take an umbrella. If it is *not* raining, the tree asks another question: **"Are there dark clouds?"** This helps check the possibility of rain. If dark clouds are present, the decision is again to take an umbrella, as rain is likely. If there are no dark clouds, the person can safely go without one. This simple example shows how a decision tree works by asking step-by-step questions and reaching a final decision based on conditions.

- Step 1: Start with the question

Is it raining?

✓ If Yes → Take an umbrella

✓ If No → Ask the next question

- Step 2: If it's not raining, ask:

Are there dark clouds in the sky?

✓ If Yes → Take an umbrella (it might rain)

✓ If No → Don't take an umbrella

Tree structure Representation

```
          Is it raining?

            /        \

          Yes          No

          |           |

   Take an umbrella    Are there dark clouds?

                 /          \

              Yes            No

               |             |

         Take an umbrella      Don't take an umbrella
```

1. **How it works**

- The tree asks the most important question first.
- Based on the answer, it moves to the next question.
- Finally, it reaches a decision (leaf node):
  ✓ Take umbrella

  ✓ Don't take umbrella

Data set for the given example:

Sl.No	Raining	Dark Clouds	Take Umbrella
1	Yes	Yes	Yes
2	Yes	No	Yes
3	No	Yes	Yes
4	No	No	No
5	Yes	Yes	Yes
6	No	No	No
7	No	Yes	Yes

2. **Naive Bayes Classifiers**

Assuming that all independent parameters are almost equally important and will have an equal impact on the result is central to the Naive Bayes Algorithm. Given that the event has already taken place, it determines the likelihood of the event happening again. When learning, Naive Bayes works best with smaller training sets. In terms of prediction speed, it outperforms other models. When all the parameters are considered equally important, it suffers from the bad estimation problem. Results that hold water in reality are not produced by it.

3. Neural Networks

Neural networks are a class of machine learning models inspired by the brain's interconnected neurons; they consist of layers of interconnected nodes (neurons) that transform input features through weighted connections and nonlinear activation functions. Deep neural networks (many layers) can automatically learn hierarchical feature representations, making them especially powerful for complex tasks like image recognition, speech processing, and large-scale tabular problems when enough data is available. Their strengths include flexibility and strong performance on high-dimensional, unstructured data; weaknesses include the need for large labeled datasets, sensitivity to hyperparameters, potential overfitting, and limited interpretability without additional tools (e.g., saliency maps, SHAP). Common variants include multilayer perceptrons (MLPs) for tabular data, convolutional neural networks (CNNs) for images, and recurrent/transformer models for sequential data.

4. **K-Nearest Neighbour (K-NN)**

K-NN is a simple, instance-based (lazy) learning algorithm that classifies a new sample by looking at the labels of its k closest training samples in feature space, typically using distance metrics like Euclidean distance. It requires no explicit training phase (beyond storing the dataset), which makes it easy to implement and interpret—results depend directly on the local neighborhood of examples. K-NN works well for small to medium-sized datasets, especially when the decision boundaries are irregular, but it becomes slow and memory-intensive as dataset size grows and is sensitive to feature scaling and irrelevant features; careful choice of k and distance metric (and using dimensionality reduction) is important. Typical use-cases include recommendation, anomaly detection, and simple classification problems where interpretability and ease of use matter.

5. **Support Vector Machines (SVMs)**

Support Vector Machines are supervised learning models that find an optimal hyperplane to separate classes by maximising the margin between the nearest points of different classes (support vectors). With kernel functions (linear, polynomial, RBF), SVMs can efficiently perform nonlinear classification by implicitly mapping inputs to higher-dimensional spaces, making them powerful for problems with clear margins of separation and moderate feature sizes. Strengths of SVMs include robustness to overfitting in high-dimensional spaces and effectiveness with limited samples; weaknesses include sensitivity to choice of kernel and hyperparameters (C, gamma), relatively high training time for very large datasets, and limited probability estimation without extra calibration. SVMs are widely used in text classification, bioinformatics, and any application where margin-based separation yields good generalisation.

6. Logistic Regression

To generate a Binomial Classification for a certain event or class, statisticians use Logistic Regression. Each trial's probability is determined by this model, which also determines the direction of the Binary Classification. Determining the influence of numerous independent variables on a single result is another use of logistic regression. For logistic regression to work, the target dataset must not contain any missing values and the predicted variable must be binary. The independence of the forecasters from one another is also necessary.

7. Linear Regression

One kind of supervised learning method, Linear Regression uses the independent variables to make simple regression predictions about the dependent variables. Establishing a connection between independent variables in order to calculate the value of the dependent variable. The overfitting problem is the model's Achilles' heel, and linear data separation isn't always an option.

8. Random Forest

The name of the model gives it away: it uses subsets applied to several Decision Trees. The class accuracy is then predicted by taking an average of all the trees. The generated subsets are identical in size to the original dataset, with the exception that samples are swapped out for each subgroup. It effectively improves accuracy while decreasing overfitting. One major negative is how difficult it is to implement and how slow it is for real-time applications.

2.2 Applications, Advantage and Disadvantage

Real time Examples:

- Market basket analysis is a method of predicting consumer behaviour that has been linked to repeated purchases of a certain combination of goods. Many retailers, like Amazon, adopt this strategy. As you peruse the available goods, you may see recommendations for related products that other customers have purchased.
- Weather Forecasting: In order to make accurate weather predictions, meteorologists must keep an eye on variables like temperature, humidity, and wind direction. In order to make an accurate prediction, this astute remark also necessitates looking at past records.

Advantages

- **Predictive power:** Can map inputs to discrete labels, enabling things like fraud detection, churn prediction, and medical diagnosis.
- **Interpretable options exist:** Models like decision trees, rule-based classifiers, and logistic regression are easy to explain to stakeholders.
- **Broad applicability:** Works on many domains (text, tabular, images with suitable features).
- **Fast inference:** Many classifiers (once trained) make predictions quickly, suitable for real-time systems.
- **Well-studied metrics & tools:** Mature evaluation metrics (accuracy, precision, recall, F1, AUC) and libraries (scikit-learn, TensorFlow, etc.) make development and comparison straightforward.
- **Handles high-dimensional data:** Some methods (SVM, regularised models, tree ensembles) can work well with many features.

Disadvantages

- **Requires labelled data:** Supervised classification depends on quality labelled examples — labelling is often costly and time-consuming.
- **Overfitting risk:** Powerful models (deep nets, ensembles) can memorimemorise training data and fail to generalise without careful regularisatione training data and fail to generalise without careful regularisation and validation.
- **Bias & fairness issues:** If training data is biased, the classifier can reproduce or amplify unfair patterns.

- **Sensitive to feature engineering:** Performance often hinges on good pre-processing, encoding, scaling, and selection.
- **Model complexity vs interpretability trade-off:** More accurate models (ensembles, deep learning) are often less interpretable.
- **Imbalanced classes problem:** Standard classifiers may perform poorly when positive/negative classes are highly skewed.

Limitations

- **Limited causal insight:** Classification finds correlations useful for prediction, but typically does not reveal causal relationships.
- **Domain shift / concept drift:** Models trained on historical data can degrade when the underlying data distribution changes over time.
- **Scalability constraints:** Some algorithms are slow to train or require lots of memory on very large datasets (e.g., kernel SVMs, some ensemble settings).
- **Feature-type challenges:** Mixed data types (categorical + continuous + text + images) require careful representation; not every classifier handles all types natively.
- **Dependence on hyper-parameters:** Many algorithms need tuning (e.g., learning rate, regularisation, tree depth), and poor settings harm performance.
- **Noisy / missing data:** Classification quality drops if data are noisy, incomplete, or inconsistent, unless handled properly.

Quick mitigations / best practices

- Use **cross-validation** and hold-out sets to detect and reduce overfitting.
- Apply **data augmentation, resampling, or class-weighting** to handle imbalanced classes.
- Do **feature engineering, selection, and scaling** (or use models that are robust to raw features).
- Monitor models in production for **concept drift** and retrain when needed.
- Use **explainability tools** (SHAP, LIME, feature importance) to audit black-box models.
- Ensure **ethical review and bias testing** of datasets and models before deployment.

SELF-ASSESSMENT QUESTIONS – 1

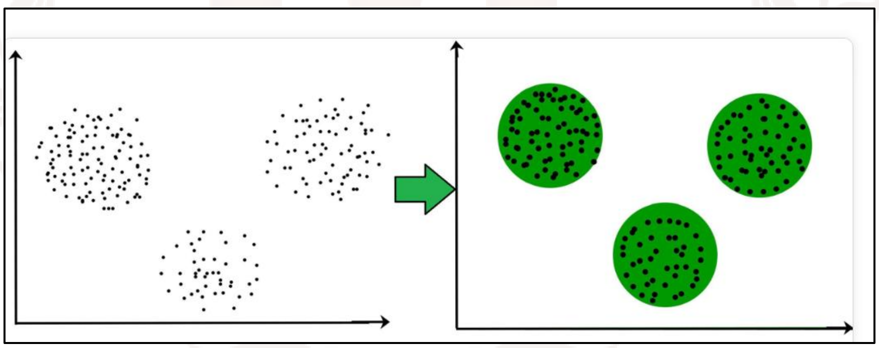
Multiple Choose Questions	
1	**Which classifier is instance-based and makes predictions by looking at the labels of nearby training examples?** a) Logistic Regression b) k-Nearest Neighbors c) Decision Tree d) Support Vector Machine
2	**Which metric is most appropriate when evaluating a classifier on a highly imbalanced dataset?** a) Accuracy b) Mean Squared Error c) Precision and Recall (or F1-score) d) Training time
3	**Which technique helps reduce over-fitting in complex models like neural networks?** a) Increasing learning rate indefinitely b) regularisation (e.g., L1/L2, dropout) c) Removing the validation set d) Using fewer training examples
4	**Which statement about ensemble methods (e.g., Random Forest, Gradient Boosting) is TRUE?** a) They always produce simpler, more interpretable models than a single decision tree. b) They generally improve predictive accuracy by combining multiple learners. c) They require no hyperparameter tuning. d) They are instance-based learners like k-NN.

3. CLUSTERING

The term "clustering" refers to a data mining approach that groups data items with shared attributes. As an unsupervised learning technique, it aids in pattern recognition and subset creation in massive datasets. Customer segmentation, picture identification, and anomaly detection are just a few of the many uses for clustering.

What is Clustering in Data Mining?

Assembling groups of data points with common characteristics is what clustering is all about in data mining. It can also be described as a technique for arranging data such that there are more similarities than differences between items in a given cluster. The objective of this unsupervised learning approach is to discover patterns and similarities in the data. The number of clusters, similarity metrics, and clustering algorithms need to be determined before a clustering algorithm can be implemented. As seen in the figure below, the objective of these algorithms is to enhance similarity within groups while limiting similarity between different groups.



Segmenting images, documents, and customers are just a few examples of the many data mining applications that make use of clustering techniques. The objective is to enhance decision-making by deriving useful insights from the data.

What is a Cluster?

Clusters are collections of data items in data mining that share some attribute. While clustering comparable data points together, these qualities or characteristics can be defined by the analyst or detected by the algorithm. Generally speaking, data points that are part of the same cluster tend to be more comparable to one another than data points that are not. Take the above graphic as an example; it shows five clusters.

A cluster can have the following properties -

- According to certain established standards or similarity metrics, the data points that make up a cluster are quite similar to one another.
- The data points inside each cluster are unique, and the clusters themselves are separate entities.
- In a cluster, the data points are densely packed together.
- To summarise the characteristics of the data points within a cluster, a centroid or centre point is commonly used as a representation of the cluster.
- The amount of data points in a cluster doesn't matter, but it shouldn't be too big or too tiny to be useful.

3.1 Clustering Methods in Data Mining

Data mining makes use of a number of different clustering methods, each with its own set of advantages and disadvantages. In data mining, some of the most used clustering methods are -

- **K-means Clustering**: The data points are partitioned into k clusters using K-means clustering, where k is a pre-defined number. The method ensures convergence by repeatedly relocating the centroid of every cluster to the mean of its allocated data points. By minimising the sum of squared distances, K-means attempts to find the cluster centroid that is closest to each data point.

2. **Example: Grouping Students by Study Hours**

K-Means is an unsupervised machine learning algorithm used to group similar data points into clusters. It works by selecting a fixed number of clusters (K), placing initial centroids, and then assigning each data point to the nearest centroid. After the assignment, the centroids are recalculated based on the average position of the points in each cluster. This process repeats until the centroids stop changing. In simple terms, K-Means finds natural groups in data—for example, grouping students based on study hours into low and high study clusters. It is widely used because it is simple, fast, and effective for pattern discovery.

Imagine we have data about how many hours students study per day. We want to group (cluster) students with similar study patterns.

Sample Dataset (Study Hours per Day)

Student	Study Hours
A	1
B	1.5
C	2

D	2.5
E	6
F	6.5
G	7

We choose **K = 2 clusters**

→ meaning we want to form **2 groups**:

- **Cluster 1:** Low study hours
- **Cluster 2:** High study hours

3. **Step-by-Step K-Means Process**

Step 1: Choose 2 initial centroids

Let's pick:

- Centroid 1 = 2 hours
- Centroid 2 = 6 hours

Step 2: Assign each student to the nearest centroid

So clusters are:

Cluster 1 (Low study hours): 1, 1.5, 2, 2.5

Cluster 2 (High study hours): 6, 6.5, 7

Sample Dataset (Study Hours per Day)

Student	**Study Hours**	**Distance from 2**	**Distance from 6**	**Cluster**
A	1	1	5	**C1**
B	1.5	0.5	4.5	**C1**
C	2	0	4	**C1**
D	2.5	0.5	3.5	**C1**

E	6	4	0	**C2**
F	6.5	4.5	0.5	**C2**
G	7	5	1	**C2**

Step 3: Recalculate new centroids

- New Centroid 1 = average(1, 1.5, 2, 2.5) = **1.75**
- New Centroid 2 = average(6, 6.5, 7) = **6.5**

The algorithm repeats until centroids stop moving — this is **convergence.**

4. **Final Result**

K-Means has grouped students into 2 meaningful clusters:

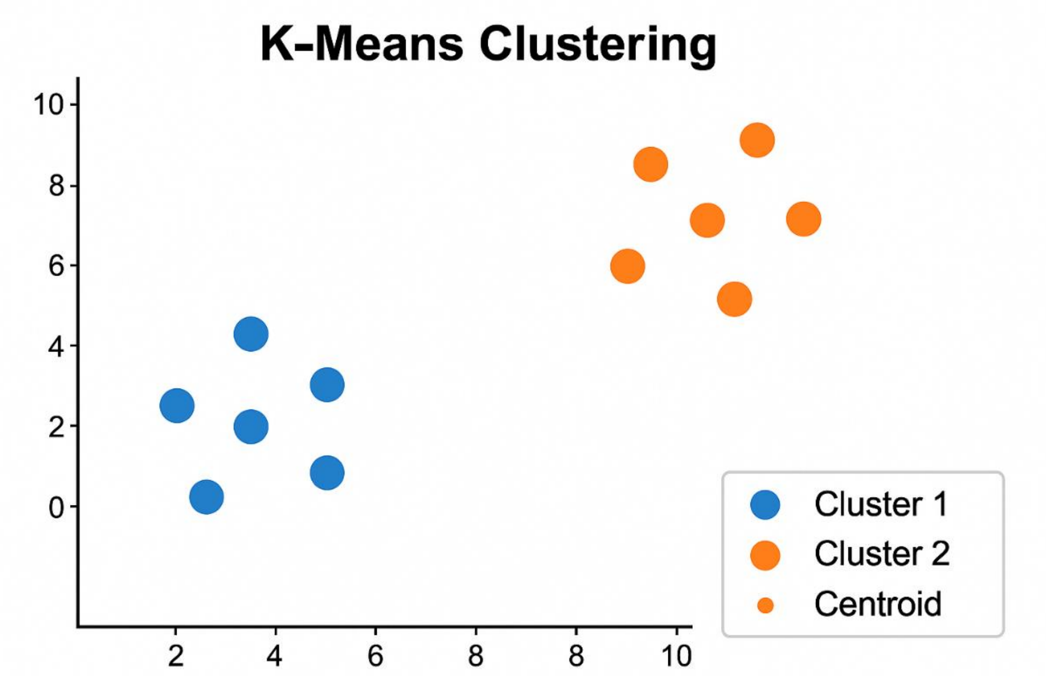
✓ Cluster 1: Low study hour students

(1–2.5 hours)

✓ Cluster 2: High study hour students

(6–7 hours)

This helps a teacher identify who needs help and who is already doing well



- **Hierarchical Clustering**: One approach to data mining is hierarchical clustering, which creates a tree-like structure of clusters by either dividing bigger ones into smaller ones (divisive or top-down) or by merging smaller ones into larger ones (agglomerative or bottom-up). A fixed amount of clusters is not necessary.
- **Density-Based Clustering**: Clusters can be located using density-based clustering, which searches for areas with a high concentration of data points. Outliers or noise points are those that do not belong to a high-density region. After density-based clustering, DBSCAN is the algorithm that is most often employed.
- **Model-Based Clustering**: The data points are assumed to be generated by a probabilistic model, such as a mixture of Gaussian distributions, in model-based clustering. It groups data points into clusters according to how likely they are to be covered by the model, and it aims to find the parameters that suit the data the best.
- **Fuzzy Clustering**: Data points are grouped into clusters using fuzzy clustering according to the degree to which they belong to each cluster. A data point might thus be a part of many clusters, each with varying degrees of affiliation.

Why is Clustering Required in Data Mining?

Below, we will go over some of the many benefits of clustering, a crucial data mining approach.

- **Scalability:** It is feasible to glean valuable insights and knowledge from enormous datasets using clustering algorithms in data mining because of how efficiently they handle such datasets.
- **High Dimensionality:** In data mining, clustering algorithms are able to effectively manage high-dimensional datasets, allowing for the discovery of linkages and patterns that would not be visible in lower-dimensional datasets.
- **Discovery of Clusters with Arbitrary Shape:** Finding groupings of data points with shared attributes or characteristics is made feasible by clustering algorithms in data mining, which can uncover clusters of varying sizes and forms.
- **Interpretability**: Humans are able to readily understand clustering results, which allows for the extraction of valuable knowledge and insights from the data.
- **Ability to Deal with Different Kinds of Data:** Clustering algorithms in data mining are capable of handling several data kinds, including numerical, categorical, and binary, enabling them to cluster a broad variety of data.

Distance Metrics

To find out how close or far apart two data points are, one can use distance metrics, which are basic mathematical formulas. When it comes to clustering, the type of distance measurements we use is crucial. Among the most popular measures are:

- **Euclidean Distance:** Discovering the linear distance between any two sites, it is the de facto standard for measuring distances.
- **Manhattan Distance:** Using a grid-like path, it calculates the distance between any two places. The sum of all absolute value differences is calculated.
- **Cosine Similarity:** Instead of measuring distance, this technique evaluates the angle between the two locations. It determines the degree of similarity between two documents using text data.
- **Jaccard Index:** A statistical method for determining how similar two sets of data are. Its primary application is with categorical or yes/no data.

3.2 Types of Clustering Techniques

There are a number of ways to classify clustering. Both the data type and the nature of the problem at hand dictate the approach that is most appropriate.

1. **Partitioning Methods:** Partitioning methods segregate the data into k clusters, with each data point being assigned to a single cluster. When you have a clear idea of how many clusters you need, you may use these ways to make them. The K-means clustering algorithm is a typical example. K-means takes an average of all the data points in a given group and uses that number to update the centre. The algorithm then allocates each data point to the closest centre. When the centres no longer fluctuate, the procedure terminates. Applications in the real world use it to categorise people according to their listening habits, such as streaming services like Spotify.

2. Hierarchical Methods

A dendrogram, resembling a tree structure, is produced by hierarchical clustering as a result of cluster merging or division. It consists of various components.

- Bottom-up Agglomerative Approach: This method begins with individual points and then merges ones that are comparable. Similarly, to how relatives are arranged in a tree.
- Divisive Approach: This method is known as the "Divisive Approach" and it works by first creating a large cluster, and then continually dividing it into smaller ones. For instance, further refining the process of classifying creatures into general groups such as mammals, reptiles, etc.

3. Density-Based Methods

Data points that are very close together are clustered using density-based clustering, while areas with fewer points are considered outliers or noise. When clusters are not perfectly round, this approach shines. By classifying comparable actions into distinct categories, it finds suspicious trends and can be applied to the problem of fraud detection, for instance.

4. Grid-Based Methods

Grid-based Methods partition the data space into grids to improve the efficiency of clustering. By simplifying the procedure and minimising the amount of calculations needed, this speeds up clustering, which is useful for big datasets. Researchers in the field of climate change often use grid-based methods when examining variations in regional temperatures. Dividing the region into grids will allow them to more clearly observe temperature trends and patterns.

5. Model-Based Methods

Data is assumed to come from a variety of distributions in model-based clustering, which organises it. The ubiquitous Gaussian Mixture Model (GMM) presupposes that the data is shaped by multiple overlapping normal distributions. Generalised Markov Models (GMMs) are widely utilised in speech recognition systems for their ability to differentiate between speakers by representing their voices as a Gaussian distribution.

6. Constraint-Based Methods

To direct the clustering process, it makes use of user-defined constraints. The links between data points, such as which ones should or should not be in the same cluster, can be defined by these constraints. Both genetic and behavioural variables may be included when grouping patient data in healthcare. Grouping patients with comparable genetic origins and further refining the clusters based on their lifestyle choices is specified by the constraints.

Impact of Data on Clustering Techniques

Clustering techniques must be adapted based on the type of data:

1. Numerical Data: Quantifiable values, such as age, wealth, or temperature, make up numerical data. When dealing with numerical data, algorithms that rely on distance measures, such as k-means and DBSCAN, perform admirably. Take a fitness app as an example. It uses users' average heart rate and number of steps taken each day to group them into different fitness levels.

2. Categorical Data: Gender, product category, and survey responses are examples of non-numerical values it may include. Hierarchical clustering and k-mode algorithms work well here. Classifying buyers into subsets according to their chosen product categories, such as "electronics," "fashion," and "home appliances," is one example.

3. Mixed Data: It is necessary to use a hybrid method when dealing with datasets that include both numerical and category variables. One usage of the k-prototype approach is to cluster consumer databases according to numerical income and categorical purchasing preferences.

Uses for Cluster Analysis

1. **Market Segmentation:** This helps companies send targeted promotions to certain groups of consumers depending on their buying habits.
2. **Image Segmentation** One application is in computer vision, where it can be used to detect faces, automobiles, and animals by grouping pixels in a picture.
3. **Biological Classification:** Clustering helps researchers understand diseases and potential treatments by grouping genes with similar behaviours.
4. **Document Classification:** In order to provide more relevant search results, search engines use it to classify websites.
5. **Anomaly Detection:** The purpose of cluster analysis in outlier identification is to find really unusual data points that don't fit into any of the predefined clusters.

3.3 Challenges in Cluster Analysis

While clustering is very useful for analysis it faces several challenges:

- Choosing the Number of Clusters: Methods like K-means requires user to specify the number of clusters before starting which can be difficult to guess correctly.
- Scalability: Some algorithms like hierarchical clustering does not scale well with large datasets.
- Cluster Shape: Many algorithms assume clusters are round or evenly shaped which doesn't always match real-world data.
- Handling Noise and Outliers: They are sensitive to noise and outliers which can affect the results.

Cluster analysis is like organising a messy room—sorting items into meaningful groups making everything easier to understand. Choosing the right clustering method depends on the dataset and goal of analysis.

Which are the Best Clustering Data Mining Techniques?

1) Clustering Data Mining Techniques: Agglomerative Hierarchical Clustering

Both top-down and bottom-up approaches are used in clustering algorithms. Prior to agglomeration units clustering pairs into one cluster, bottom-up algorithms treated all data points as part of a single cluster. The Hierarchical Agglomerative Clustering Data Mining method makes use of a dendrogram, also known as a network clustering tree, where the characteristic sample collecting cluster serves as the tree's base and the single-sample clusters as its leaves. Similar to how traditional clustering algorithms work, hybrid algorithms use average linkage with a predetermined distance metric to define the average distance between each pair of cluster data points and then iteratively refine this distance until convergence is reached.

In hierarchical clustering, the number of clusters is not predetermined; in fact, we are free to pick a number of clusters that appears optimal given that we are constructing a tree. On top of that, the method works regardless of the distance measurement. However, its O(n3) temporal complexity makes it inefficient.

2) Clustering Data Mining Techniques: K-Means Clustering

The K-Means Clustering Algorithm iteratively finds the k number of clusters after calculating the centroid value between two data points. In order to bring data points with changeable features to Clustering, it is relatively useful to compute Cluster Centroids with their vector quantisation observations.

Splitting massive amounts of unlabelled real-world data into clusters of varying densities and shapes will make the clustering process go much more quickly. When calculating a centroid distance, have you ever thought about the formula? Below, you will find the K-means steps:

- Choose the total number of clusters whose density and form will vary. This integer can have any value between three and four, so let's name it k.
- You may now give each cluster a number and use it to store data.
- After selecting the data point and cluster, the centroid distance is determined by utilising the least squared Euclidean distance.

- If the data point is very near to the centroid distance, it resembles the cluster; otherwise, it does not.
- Calculate the centroid distances with the chosen data point iteratively until you discover the maximum number of clusters composed of linked data points.
- Once the algorithm reaches assured convergence, when all of the data points are well-clustered, it stops clustering.

3) Clustering Data Mining Techniques: EM Clustering

When two circular clusters with different radii but the same mean centre are one problem with K-Means Clustering methods. K-Means does not differentiate between the two clusters and instead uses median values to establish the cluster centre. Also, it doesn't work if the sets aren't circular. One alternative to K-Means that has been proposed in the field of data science is the Expectation Maximisation Model, or EM. An efficient method for estimating missing values from pre-existing datasets, the optimisation clustering technique employs the function of variance. Afterwards, it meticulously forms the clusters by utilising optimised values for the mean and standard deviation. A single cluster that closely matches the likelihood of outcomes is obtained by repeating the complete estimating and optimisation operation. Now we will review the steps of the EM Clustering method:

- It is necessary to select the number of clusters and to initialise the parameters of each cluster's Gaussian distribution randomly using an estimate from the data. Starting off slowly, the algorithm quickly optimises itself using the default parameters.
- eTo determine if the data point is a member of the given cluster, the probability is calculated using the cluster's Gaussian distribution. Data points that are closer to the centre of the Gaussian distribution have a higher probability.
- Applying a new ideal value for its parameters in the next phase increases the likelihood of the data point falling into the new cluster. These additional parameters make use of the positional weighted sum of the data points; the weights determine the probability of a specific cluster possessing the referred data point.
- Once convergence is reached and the differences between iterations become small, the process is applied to succeeding iterations.

4) Clustering Data Mining techniques: Hierarchical Clustering

Hierarchical Clustering is a lifesaver when you need to locate data points and arrange them in a way that takes cluster probability into account. Now, the mapped data points might be part of a Cluster that has unique properties when it comes to quantitative correlations among data variables, multidimensional scaling, or cross-tabulation.

Keeping the order of the attributes used to classify them in mind, how can we combine all of the clusters into one? To do this, one can apply the steps outlined in the Hierarchical Clustering method:

- Sort the data points into clusters based on their hierarchical position. Is the interpretation of the clusters anything you're thinking about? A dendrogram can be used to understand the cluster hierarchy correctly with either a top-down or bottom-up approach.
- There are a number of metrics that can be used to determine the proximity of the clusters before they are merged, including the Manhattan distance, the Euclidean distance, and the Mahalanobis distance. Eventually, only one cluster will remain.
- Because the junction point has been found and well-mapped on the dendrogram, the process has ended for the time being.

5) Clustering Data Mining techniques: Density-Based Spatial Clustering

When it comes to discovering clusters in bigger geographical databases, the Density-based Spatial Clustering Algorithm with Noise (or DBSCAN) is a superior alternative to K Means when it comes to cross-examining the density of its data points. It's also more appealing and efficient than CLARANS, which stands for Clustering large applications via mediod-based partitioning approach.

The DBSCAN Clustering algorithm approach is beneficial and comparable to the mean-shift density-based Clustering algorithm.

DBSCAN's method starts with an unvisited data point and uses distance (Epsilon) to extract the neighborhood before designating the point as visited. If two points are within a certain distance of each other, they have termed neighbors. Following are the steps of DBSCAN:

- The clustering procedure starts with the current data point as the beginning point in the new cluster when enough points (based on minPoints) are obtained.
- When the number of points is insufficient, the algorithm indicates that it has been visited and labels it as noisy fault clustering.
- A clustered point neighbourhood is formed when the first point in a new cluster uses the same distance to define its neighbourhood as all the other points in the group; this procedure is repeated for each subsequent cluster point. Once all data points have been labelled and visited, the process is repeated.

- Once all of the nearby data points have been explored, a new, unexplored data point is selected for clustering. So, everything is either categorised as noise or grouped together under the visited label.

Applications of Clustering in Data Mining

Clustering has many uses across many domains and is a popular data mining approach. Clustering is commonly used in data mining for a variety of purposes, including -

- **Customer Segmentation:** Data mining clustering techniques can be employed to develop more targeted marketing efforts by grouping clients with similar behaviour, preferences, and purchase habits.
- **Image Segmentation:** Data mining clustering methods can divide images into distinct areas defined by pixel values; this can facilitate operations like object identification and picture compression.
- **Anomaly Detection**: When data sets exhibit highly unusual patterns of behaviour, data mining clustering techniques can be employed to spot these outliers.
- **Text Mining:** For data mining tasks like document summarisation and topic modelling, clustering methods can be applied to group texts or documents with comparable content.
- **Biological Data Analysis:** Genes and proteins with shared features or expression patterns can be grouped using data mining clustering algorithms; this can aid in areas like disease diagnosis and medication discovery.
- **Recommender Systems**: By using data mining clustering techniques, we may classify individuals based on their interests or behaviour and then use that information to provide more tailored product suggestions.

Requirements of Clustering Data Mining Techniques

- **Scalability:** A large database may contain millions of objects, although many clustering methods are effective on smaller datasets with 200 or less data objects. Clustering on a small portion of a large dataset could lead to biassed results. There is a need for scalable clustering algorithms.
- **Usability and interpretability:** Customers want clustering results that are easy to understand, comprehensive, and put to good use. This means that clustering could call for some creative semantic uses and interpretations. Examination of the impact of application purpose on Clustering Data Mining technology selection is of utmost importance.

- **High dimensionality:** Multiple characteristics or dimensions can be present in a data warehouse or database. When presented with two- or three-dimensional data, many clustering techniques perform admirably. In as little as three dimensions, the human eye can evaluate clustering quality. It could be challenging to cluster data items in a high-dimensional space, particularly if the data is sparse and highly skewed (deceptive data).
- **Constraint-based clustering:** A number of limitations in real-world applications might make clustering necessary. Take the position of someone whose job it is to decide where in a city a specific number of new ATMs should be placed. Considerations including the city's waterways, transportation networks, and client needs per **location** might be considered while making this decision, which involves clustering households. Identifying data sets that exhibit suitable clustering behaviour and meet specified criteria is a challenging problem.

When it comes to data mining and analysis, clustering is king. Data mining, clustering, and essential clustering techniques are all covered in this article. Clustering data mining techniques will also be covered, along with their uses and prerequisites.

SELF-ASSESSMENT QUESTIONS – 2

Multiple Choose Questions	
5	**What is the primary goal of clustering?** a) Predict a target label b) Group similar instances together c) Reduce the number of features d) Train a supervised model
6	**Which clustering algorithm requires you to pre-specify the number of clusters (k)?** a) DBSCAN b) Hierarchical clustering c) k-Means d) OPTICS
7	**Which clustering method is known for finding clusters of arbitrary shape and handling noise?** a) k-Means b) DBSCAN c) k-Medoids d) PCA
8	**The “elbow method” is commonly used to:** a) Evaluate cluster compactness with silhouette score b) Choose the number of clusters (k) for k-means c) Normalise features before clustering d) Detect outliers in the dataset

4. SUMMARY

- Classification is a supervised learning task where a model learns from labelled examples to assign discrete class labels to new instances. Major categories of classifiers are linear models (Logistic Regression, LDA), tree-based (Decision Trees, Random Forests) instance-based (k-NN), kernel methods (SVM), probabilistic (Naive Bayes), neural networks
- Advantages are strong predictive power for labelled problems, many mature algorithms and libraries; fast inference after training; interpretable options trees, logistic regression exist for stakeholder communication.
- Limitations / Disadvantages: requires quality labelled data (often costly); risk of overfitting without proper validation; sensitive to class imbalance, noisy features, and concept drift; trade-off between interpretability and accuracy for complex models.
- Common evaluation metrics: accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC; use confusion matrices and cross-validation for reliable assessment. Typical applications: medical diagnosis, credit scoring, spam detection, image recognition, sentiment analysis, fraud detection, and customer churn prediction.
- Practical challenges & mitigations: label scarcity → semi-supervised learning/transfer learning; imbalance → resampling or class weights; overfitting → regularisation and cross-validation; feature dependence → careful feature engineering and selection.
- Best practices: start with simple baselines, use k-fold cross-validation, tune hyper parameters systematically, apply explainability tools (SHAP/LIME) for black-box models, and monitor models in production for drift.Clustering is an unsupervised learning technique that groups similar instances so that items within a cluster are more alike than items in different clusters.
- Purpose: discover hidden structure, segment data, and support exploratory analysis when labels are not available. Major clustering methods: partitioning (k-means, k-medoids), hierarchical (agglomerative/divisive), density-based (DBSCAN, OPTICS), model-based (Gaussian Mixture Models), spectral clustering, fuzzy clustering (Fuzzy C-Means), and subspace/high-dimensional methods.
- Advantages: uncovers natural groupings without labels; useful for customer segmentation, anomaly detection, image segmentation, and preprocessing for supervised tasks. Limitations / Disadvantages: no single best algorithm for all datasets; results sensitive to preprocessing, distance measures, and hyperparameters; harder to evaluate due to lack of ground truth.

- Key challenges: choosing the number of clusters (k); scalability for very large datasets; handling arbitrary-shaped clusters and varying densities; sensitivity to outliers and initialization; curse of dimensionality in high dimensions.

5. GLOSSARY

Data Mining	—	The process of discovering patterns, relationships, and useful information from large datasets using statistical, machine learning, and database techniques.
Classification	—	A supervised learning task that assigns discrete class labels to instances based on learned models from labeled training data.
Clustering	—	An unsupervised learning task that groups similar instances into clusters so members of the same cluster are more alike than those in other clusters.
Label / Target	—	The ground-truth class or value assigned to an instance in supervised learning.
Cross-Validation	—	A resampling technique (e.g., k-fold) to estimate model performance more reliably by rotating train/validation splits.
Confusion Matrix	—	A table summarizing predicted vs. actual class labels (TP, TN, FP, FN) used to compute many classification metrics.
Accuracy	—	Fraction of correct predictions out of all predictions; simple but can be misleading on imbalanced data.
Precision	—	For a positive class: TP / (TP + FP). Measures correctness of positive predictions.
Recall (Sensitivity)	—	TP / (TP + FN). Measures coverage of actual positives found by the model.
F1-Score	—	Harmonic mean of precision and recall; balances the two for imbalanced classes.
Overfitting	—	Model fits training data too closely (including noise) and fails to generalise to new data.
Underfitting	—	Model is too simple to capture underlying patterns, causing poor performance on training and test data.

k-Nearest Neighbors (k-NN)	—	Instance-based classifier that predicts label by majority among k closest training examples.
Decision Tree	—	Tree-structured model that splits data on feature thresholds to make predictions; interpretable but can overfit.
Random Forest / Ensemble	—	Combines multiple decision trees (bagging) to improve accuracy and reduce variance.
Support Vector Machine (SVM)	—	Finds a maximal-margin hyperplane to separate classes; kernels enable nonlinear separation.
Naive Bayes	—	Probabilistic classifier assuming conditional independence of features given class; simple and fast.
Neural Network	—	Layered networks of neurons (weights + activations) that learn complex nonlinear mappings; deep nets for large/complex data.

6. TERMINAL QUESTIONS

1. What is classification and what are the main categories of classifiers?
2. When should you prefer tree-based ensembles over SVMs or neural networks?
3. State three advantages and two limitations of classification algorithms.
4. How can class imbalance be handled in a classification task?
5. What is clustering and what primary purpose does it serve in data mining
6. Briefly compare k-means, hierarchical clustering, and DBSCAN.
7. How do you choose the number of clusters (k) in practice?
8. What are the main challenges in cluster analysis and how can they be mitigated?

7. ANSWERS

7.1. Self-Assessment Answers:

1. Answer: b) k-Nearest Neighbors
2. Answer: c) Precision and Recall (or F1-score)
3. Answer: b) regularisation (e.g., L1/L2, dropout)
4. Answer: b) They generally improve predictive accuracy by combining multiple learners.
5. Answer: b) Group similar instances together
6. Answer: c) k-Means
7. Answer: b) DBSCAN
8. Answer: b) Choose the number of clusters (k) for k-means

7.2. Terminal Question Answers

Answer 1: Classification is supervised learning that maps inputs to discrete class labels using labelled training data. Major categories include linear models (e.g., logistic regression), instance-based (k-NN), tree-based/ensemble (decision trees, Random Forest, XGBoost), kernel methods (SVM), probabilistic (Naive Bayes), and neural networks (MLP, CNN, RNN). Each class has different assumptions and strengths for various data types. (Refer section 2 for more information)

Answer 2: Prefer tree ensembles (Random Forest, Gradient Boosting) when you need strong out-of-the-box performance on tabular data, robustness to mixed feature types, and feature importance. SVMs suit moderate-sized, well-engineered feature sets with clear margins; neural nets excel with large labelled datasets and complex patterns (images, text) but need more tuning and compute. (Refer Section 2.1 and 2.2 for more information)

Answer 3: Advantages: (1) Predictive power for decision-making, (2) mature tooling and evaluation metrics, (3) fast inference once trained. Limitations: (1) require quality labelled data which can be costly, (2) risk of overfitting or bias if not properly validated and audited. (Refer Section 2.2 for more information)

Answer 4: Handle imbalance with data-level methods (oversampling minority—SMOTE, under sampling majority), algorithm-level methods (class weights, cost-sensitive learning), and evaluation-level approaches (use precision/recall, F1, PR-AUC instead of plain accuracy). Combining resampling with robust cross-validation helps avoid overfitting to synthetic examples.

Answer 5: Clustering is unsupervised learning that groups similar instances together so intra-cluster similarity is high and inter-cluster similarity is low. Its main purposes are exploratory analysis, customer segmentation, anomaly detection, and creating structure for downstream tasks when labels are not available. (Refer section 3 for more information)

Answer 6: k-means is fast, partition-based, and assumes spherical clusters (needs k). Hierarchical builds nested clusters (dendrogram) useful for exploratory analysis but can be $O(n^2)$. DBSCAN is density-based, finds arbitrary-shaped clusters and handles outliers, but needs careful choice of density parameters (ε, min Pts)(Refer section 3.1 and 3.2 for more information)

Answer 7: Use heuristics and diagnostics: the elbow method (plot within-cluster SSE vs k), silhouette score (cohesion vs separation), and gap statistic; complement metrics with domain knowledge and stability tests (repeat runs, subsampling). Visualie with PCA/UMAP to validate cluster. (Refer section 3.3 for more information)

Answer 8: Key challenges: (1) choosing k, (2) scalability on large datasets, (3) arbitrary cluster shapes, (4) noise/outliers, (5) high dimensionality. Mitigations include dimensionality reduction (PCA/UMAP), algorithm selection (density-based for irregular shapes), sampling/mini-batch methods for scale, robust pre-processing to remove outliers, and ensemble clustering for stability. (Refer section 3.3 for more information)

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 4

DCA2207

DATA MINING AND VISUALISATION

Unit – 9

Introduction to Data Visualization

TABLE OF CONTENTS

1. INTRODUCTION

In previous unit, learners learnt about the core concept of supervised and unsupervised methods in data mining—classification and clustering—and equips them with both theoretical understanding and practical skills. Students learnt what classification and clustering are, why and when to use each, and how major algorithms (e.g., logistic regression, k-NN, decision trees and ensembles, SVM, neural networks for classification; k-means, hierarchical methods, DBSCAN, GMM and spectral methods for clustering) work in practice. Learners also studied the advantages (predictive power, automated segmentation, interpretability options), limitations and disadvantages (need for labelling the data, class imbalance, sensitivity to noise, scalability and shape/density assumptions), and common challenges such as the curse of dimensionality, choosing k, and concept drift. Emphasis is placed on real-world applications—medical diagnosis, fraud detection, churn prediction, customer segmentation, anomaly detection—and on translating model output into actionable business insights through hands-on labs, mini-projects, and evaluation-driven decision making. By the end of the unit, students were able to select appropriate algorithms, implementing end-to-end experiments, interpret results responsibly, and recommend solutions tailored to practical constraints.

In this unit, learners will gain a foundational understanding of data visualization and its importance in interpreting and communicating information effectively. They will explore the different types of data—such as structured, unstructured, and semi-structured—and understand how the nature of data influences visualization choices. The unit also introduces various data visualization tools like Tableau, Power BI, Excel, and Python libraries, giving students an overview of how these tools help transform raw data into meaningful insights. By the end of the unit, learners will be equipped with the essential knowledge needed to select appropriate visualization techniques and tools for real-world data analysis tasks.

1.1. Learning Objectives

By the end of this unit, you will be able to:

- Explain the fundamental concepts and importance of data visualization
- Identify and describe different types of data
- Evaluate various data visualization tools and techniques
- Explore the advantages, limitations, and appropriate use cases

2. CLASSIFICATION ALGORITHM

Visual representations of data are made possible through the use of charts, graphs, and maps. It simplifies otherwise difficult-to-understand material by means of visual representations. With massive amounts of data generated by every sector, visualisation aids in the detection of trends and patterns, allowing for more informed decision-making in less time.

To better comprehend and analyse data, data visualisation uses visual representations such charts, graphs, maps, and dashboards. As a result, users are able to swiftly spot trends, patterns, and correlations in otherwise complicated numerical data. Data visualisation facilitates improved decision-making, draws attention to critical insights, and simplifies the communication of results to audiences with and without technical backgrounds by presenting data in an understandable and comprehensible fashion. Business analytics, research, healthcare, financial services, and data science are just a few of the many areas that make use of it to streamline complex datasets and unearth previously unknown insights.

The process of visually representing data and information is known as data visualisation. Data visualisation tools make it easy to see patterns, trends, and outliers in data through the use of visual components such as maps, graphs, and charts. Also, it's a great tool for employees or company owners to use when trying to explain complex ideas to people who aren't technical. Data visualisation tools and technologies are crucial in the Big Data realm for analysing large data sets and making decisions based on the data.

SELF-ASSESSMENT QUESTIONS – 1

Fill in the blanks	
1	Data visualization is the process of representing data in ________ formats to make information easier to understand.
2	Charts, graphs, dashboards, and maps are examples of ________ used in data visualization.
3	Data visualization helps users quickly identify patterns, trends, and ________ in the data.

3. IMPORTANCE OF DATA VISUALIZATION

Data visualisation is crucial because it allows users to see, interact with, and gain a better understanding of data. Regardless of the complexity or difficulty level, the correct visualisation may unite all parties involved. Making data more intelligible is beneficial for almost every professional field. Data literacy is an asset in every STEM discipline as well as many others, including public administration, economics, history, marketing, consumer products, service sectors, education, athletics, and more.

We may spend all our time gushing about data visualisation (you are here on the Tableau website, after all), but it has unquestionable practical and real-world uses. Plus, being able to visualise things clearly is a highly valuable talent to have in the workplace. Use visual aids, such as a dashboard or a slide deck, to your advantage by making your points clearly and concisely. The role of the average citizen as a data scientist is becoming more popular. To fit a data-driven environment, skill sets are evolving. Data visualisation skills that can assist decision-making and illustrate the "who," "what," "when," and "how" are becoming more important for professionals.

Data visualisation is a hybrid skill that combines elements of both analytical thinking and visual storytelling, which is an advantage in today's professional environment. This is in contrast to the traditional educational model, which tends to draw a clear boundary between creative narrative and technical analysis.

Data visualisation is a must-have for every serious data analyst or communicator. Some essential reasons for its importance are as follows:

1. **Simplifies Complex Data:** It simplifies otherwise incomprehensible data by transforming it into a graphical format, such as a chart or graph.
2. **Reveals Patterns and Trends**: When compared to raw data or tables, it aids in seeing patterns, correlations, and trends.
3. **Saves Time:** Instead of tediously poring over numbers, visuals make data interpretation faster by allowing consumers to quickly recognise important information.
4. **Improves Communication:** It facilitates the process of communicating data insights to others, particularly those who might lack expertise in the technical aspects.
5. **Tells a Clear Story**: The use of data graphics facilitates the audience's ability to draw conclusions and make well-informed judgements by guiding them through the material in a systematic manner.

Data visualization and big data

Making sense of the daily generated trillions of rows of data has never been easier than using visualisation, which is becoming more important as the "age of Big Data" gains momentum. Data visualisation aids narrative construction by organising data into a more digestible format while drawing attention to patterns and anomalies. A well-designed visualisation draws attention to relevant details while filtering out irrelevant noise.

Nevertheless, it's not as simple as enhancing the appearance of a graph or adding the "info" component to an infographic. Finding the sweet spot between aesthetics and use is key to good data visualisation. A visually striking visualisation may completely miss the mark when it comes to communicating the intended message, or it may say volumes; conversely, the most unassuming graph may be so uninteresting as to not attract any attention at all, yet it may make a compelling argument. Integrating compelling analysis with eye-catching graphics is an art form in and of itself.

4. TYPES OF DATA VISUALIZATION

A wide range of charts are part of data visualisation, all with the goal of making data easier to understand and work with. The correct chart, whether it's a basic bar or line chart or a more complex heatmap or scatter plot, may help transform raw data into actionable insights. Let's have a look at a variety of charts, from the most basic to the most complex, and see when each one is useful.

4.1 Basic Charts for Data Visualization

Simple comparisons, time patterns, and fundamental data linkages are best shown by basic charts. If you need to convey insights to a large audience, these charts are perfect.

1. Bar Charts

By utilising rectangular bars, bar charts enable the comparison of values across many categories. The X-axis displays categories, and the Y-axis shows values. Some common kinds of bar charts are horizontal, stacked, and grouped.

Below is the Example of Bar Chart:

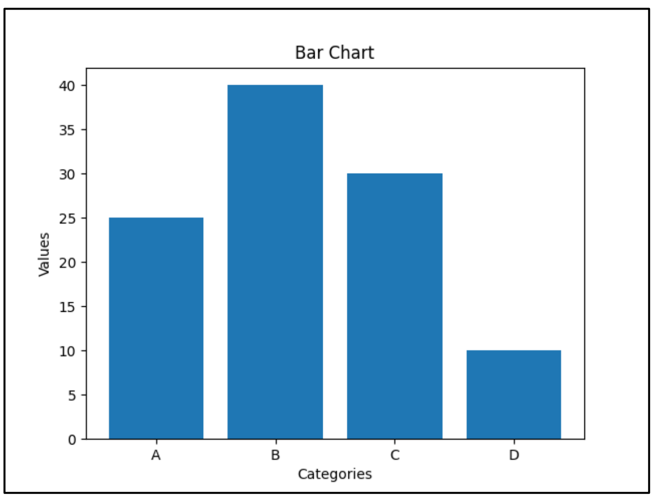


Figure 1: Representation of pie chart

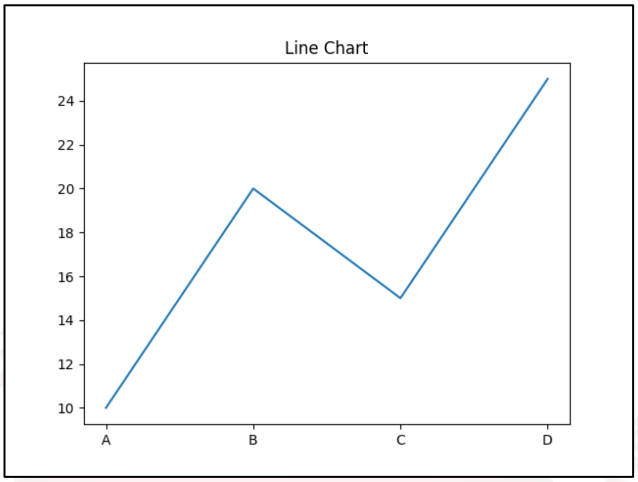
When to Use:

- To compare two or more classes
- To arrange values in descending order
- To show relationships between multiple variables

2. Line Charts

By linking points in a data set with lines, line charts reveal the evolution of values over time. They are useful for seeing patterns like growth, decline, or stability.

Below is the example of line chart:



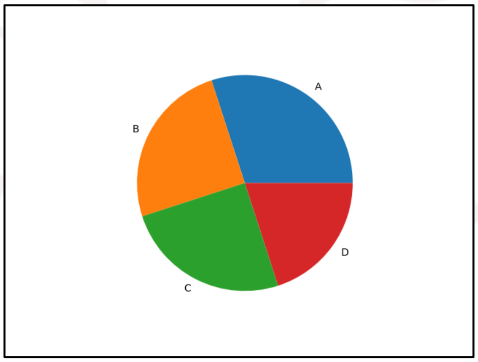
When to Use:

- To track changes over time
- To compare trends across multiple data series
- For time series analysis

3. Pie Charts

Round pie charts are subdivided into smaller ones, with each slice representing a portion of the whole. The proportion of each slice is proportional to its size.

Below is the example of pie chart



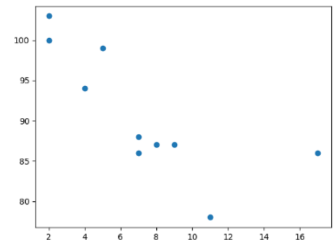
Representation of pie chart

When to Use:

- To show how different parts contribute to a whole
- To highlight a dominant category

4. Scatter Chart (Plots)

Two numerical variables can be represented in a scatter chart by means of dots. Both the independent and dependent variables are displayed on the X-axis and Y-axis, respectively. Below is the example of scatter chart:



Representation of Scatter Chart

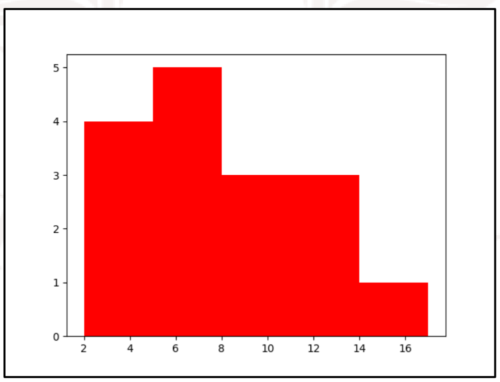
When to Use:

- To observe relationships between two variables
- To detect patterns, clusters or outliers in data

5. Histogram

By dividing numbers into intervals called bins and displaying the frequency of each value as a bar, a histogram can indicate how numerical data is distributed. Data patterns, distribution, and shapes can be better understood with its aid.

Below is the example of histogram:



Representation of Histogram

When to Use:

- To visualize the distribution of numerical data
- To explore patterns, trends and outliers

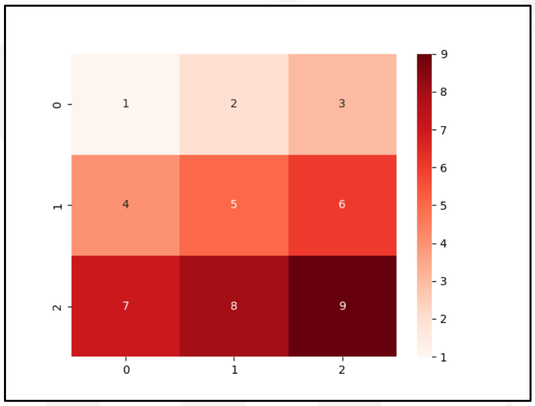
4.2 Advanced Charts for Data Visualization

More complicated data sets are better suited to advanced charts. These aid in the analysis of numerous factors, the discovery of deeper insights, and the identification of patterns that may go unnoticed with more simplistic graphics.

1. Heatmap

With the use of colour coding, a heatmap presents data in a matrix style. Finding variations, patterns, and connections in massive datasets is a breeze with this tool.

A sample heatmap is shown below:



Representation of Heatmap

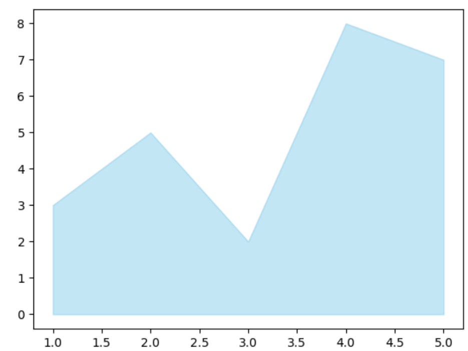
When to Use:

- To identify clusters or groupings in data
- To visualize correlations between variables
- For risk analysis in fields like finance or network security

2. Area Chart

The area under a line in an area chart represents a trend over time. It's perfect for showing changes over time and time series data.

An example of an area chart is shown below:



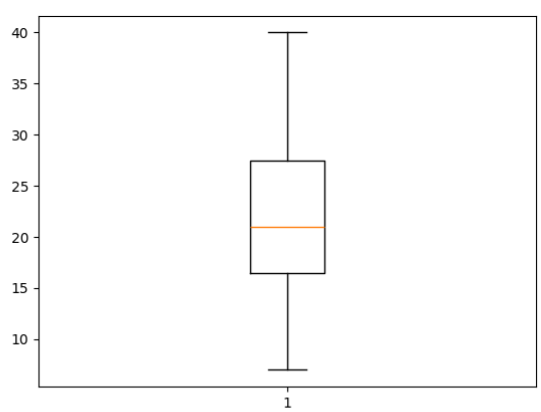
Representation of Area chart

When to Use:

- To track changes or trends over time
- To compare multiple data series
- To highlight patterns like seasonality or cycles

3. Box Plot (Box-and-Whisker Plot)

With the use of a box plot, you can see how numerical data is distributed, including the median, quartiles, and any outliers. It can help you spot out-of-the-ordinary numbers and comprehend variability. An example of a box plot is shown below:



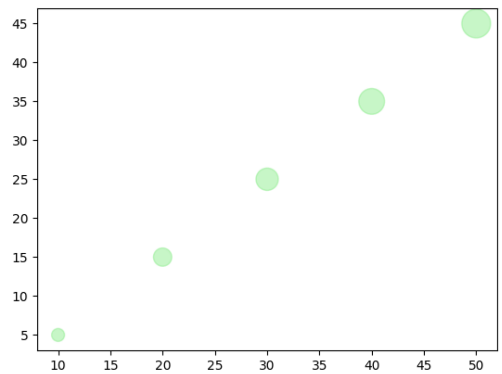
Representation of Box plot

When to Use:

- To detect outliers in data
- To compare distributions across groups
- To visualize spread and variability

4. Bubble Chart

Data points are shown as circles in a bubble chart, with colours and sizes of the bubbles representing different variables. It's a great tool for making charts that show three or more dimensions at once. Here is a bubble chart example:



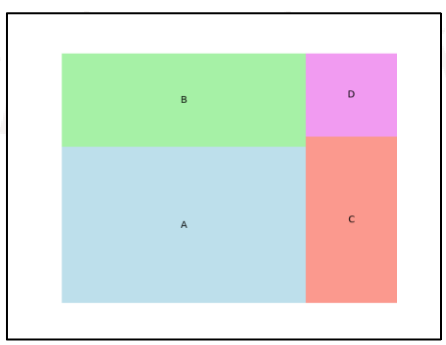
Representation of Bubble chart

When to Use:

- To compare multiple variables at once
- To represent data using size and color
- To visualize relationships and detect patterns

5. Tree Map

A tree map visualizes hierarchical data using nested rectangles, where the size of each represents a value. It's useful for showing structure and comparing proportions within a hierarchy. Below is the example of tree map



Representation of Tree map

When to Use:

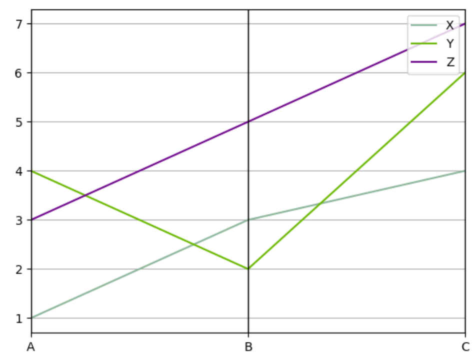
- To display hierarchical data

- To compare proportions within levels
- To visualize large datasets compactly

6. Parallel Coordinates

By drawing lines that join values on different axes, parallel coordinates show multivariate data. At least one variable's worth of data is shown by each line.

Here we see an illustration of parallel coordinates in action:



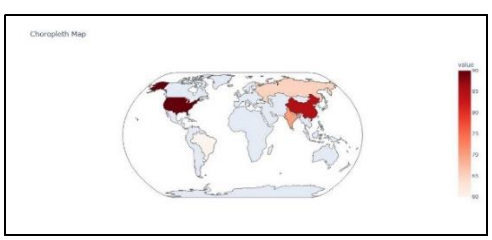
Representation of Parallel coordinates

When to Use:

- To analyze multiple variables at once
- To visualize relationships or clusters
- To detect outliers in the data

7. Choropleth Map

A choropleth map uses color shading to represent data across geographic areas. It's useful for showing differences like population density, income levels or disease spread across regions. Below is the example of choropleth map



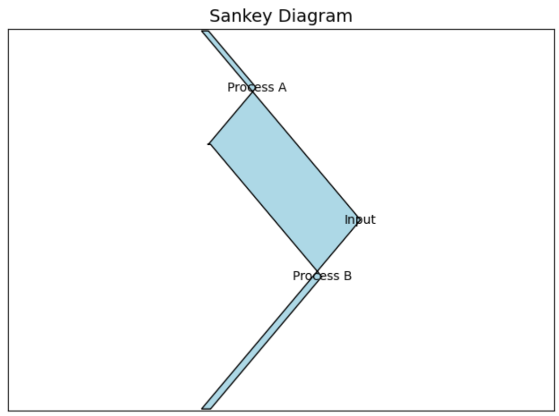
Representation of choropleth map

When to Use:

- To compare data across locations
- To spot geographic patterns and trends

8. Sankey Diagram

A Sankey diagram shows the flow of data or resources between points (nodes) using arrows, where the width of each arrow represents the amount of flow. It's great for visualizing complex systems and spotting inefficiencies. Below is the example of Sankey diagram:



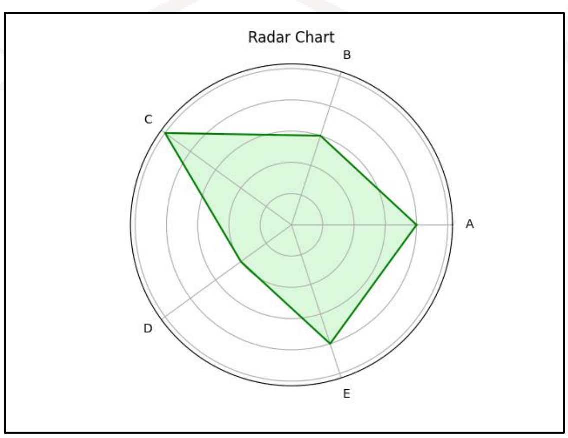
Representation of Sankey Diagram

When to Use:

- Visualize Flows: Understand how resources move through a process.
- Find Bottlenecks: Identify points where flow slows or drops.
- Compare Scenarios: Track flow changes across time or conditions.

9. Radar Chart (Spider Chart)

As a great tool for cross-category variable comparisons, a radar map shows multivariate data with axes emanating from a central point. Popular in decision-making, sports, and performance analysis. An example of a radar chart is shown below:



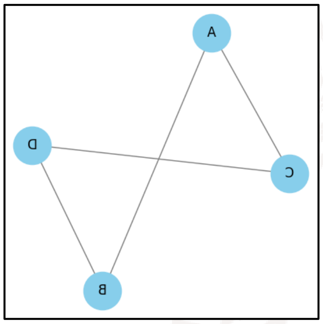
Representation of radar/spider chart

When to Use:

- Multi-Criteria Comparison: Compare multiple variables at once.
- Strengths & Weaknesses: Visualize strong and weak areas.
- Pattern Recognition: Spot similarities or differences between categories.

10. Network Graph

A network graph shows relationships between entities as nodes and links (edges). It helps visualize complex networks like social media, transport routes or biological systems.Below is the example of a network graph:



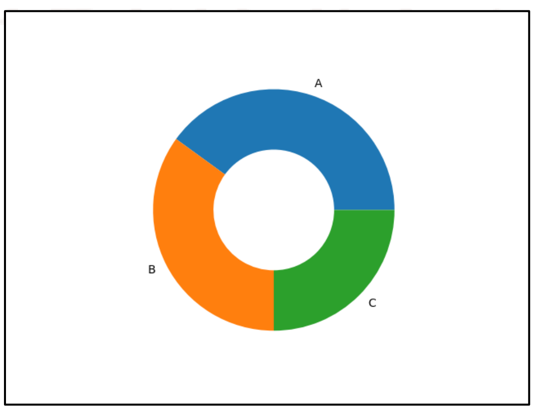
Representation of Network graph

When to Use:

1. Relationship Visualization: Helps show connections or interactions between entities.
2. Community Detection: Useful to identify clusters or groups within a network.
3. Path Analysis: Shows shortest or most efficient routes between nodes.

11. Donut or Doughnut chart

A donut chart is a circular chart like a pie chart but with a hole in the center. Each slice shows a category's contribution to the whole, making it visually cleaner and ideal for comparisons. Below is the example of a donut chart



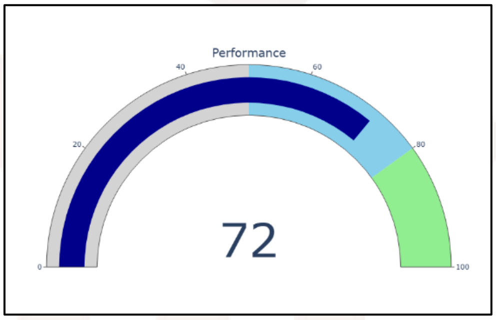
Representation of Donut Chart

When to Use:

- Show part-to-whole relationships like sales or market share.
- Track progress toward a goal (e.g., 70% completed).
- Best when comparing a few categories.

12. Gauge Chart

Similar to a speedometer, a gauge chart uses an arc that looks like a dial to represent how far down the path to success you are. When monitoring the progress of a single number, such as a key performance indicator or a project, it is quite helpful. You may see an example of a gauge chart below



Representation of Gauge Chart

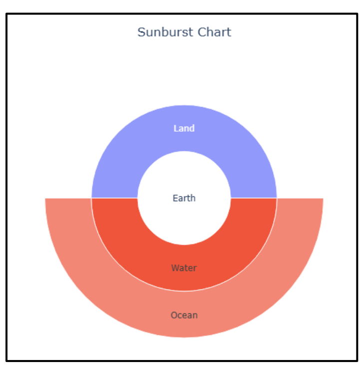
When to Use:

- To monitor key metrics (e.g., sales, satisfaction) toward targets
- In KPI dashboards to track progress
- For project status tracking against deadlines

13. Sunburst Chart

Hierarchical data is displayed as nested rings in a sunburst chart. This data visualisation tool is great for showing hierarchies, subcategories, or other multi-level data structures with each ring representing a level.

Here is a sunburst chart example:



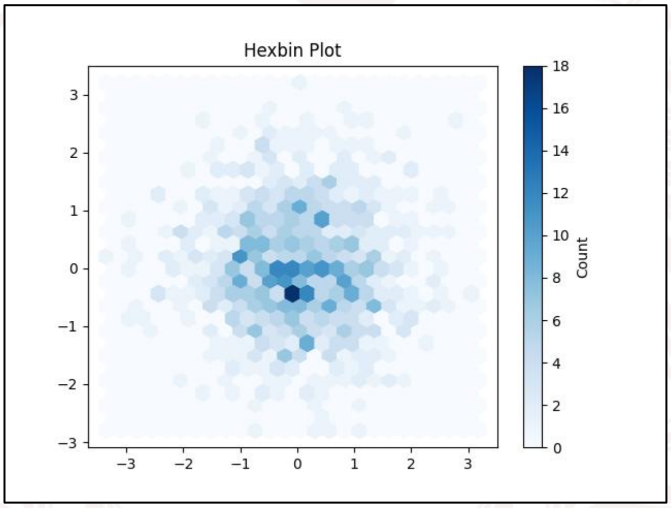
Representation of Sunburst Chart

When to Use:

- To visualize hierarchical structures like org charts or file systems.
- To explore proportions and relationships within nested datasets.
- To simplify complex data in a clean, interactive layout.

14. Hexbin Plot

The number of points in each hexagonal bin determines the colour of the points in a hexbin graphic. Clearly displaying density without overplotting is helpful for large datasets. A hexbin plot is illustrated below:



Representation of Hexbin Plot

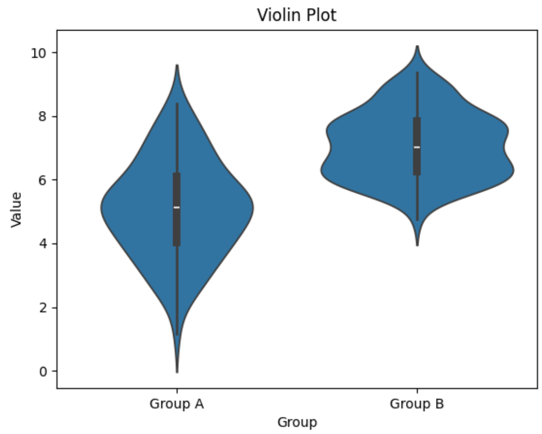
When to Use:

- To show data density in a 2D space.
- To spot clusters or patterns in large scatter data.
- To manage overlapping points in big datasets visually.

15. Violin Plot

For the purpose of displaying data distribution and summary statistics, a violin plot integrates elements of both box and density plots. It is useful for seeing the distribution, centre, and shape of data across different types of information.

An example of a violin plot is shown below:



Representation of Violin plot

When to Use:

- Compare distribution of continuous data across groups.
- Visualize data shape, spread, skewness or multimodal patterns.
- Highlight detailed distribution and outliers in a clean layout.

4.2 Visualization Charts for Textual and Symbolic data

Words, labels, and icons are examples of non-numerical data that can be better understood using visualisation techniques that focus on textual and symbolic data. The frequency of occurrence and their relationships are the main points of emphasis.

1. Word Cloud

A word cloud visually displays words sized by their frequency in a text, common words appear larger. It helps quickly identify key terms or themes in text data.

Below is the example of a word cloud

Representation of Word cloud

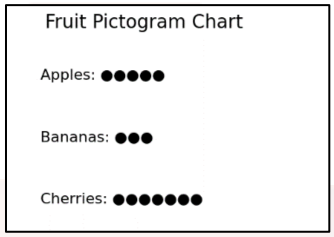
When to use:

- To highlight key themes or topics in large text data.

- To visualize word frequency in text analysis.
- To emphasize important terms in qualitative or sentiment analysis.

2. Pictogram Chart

A pictogram chart uses icons or symbols to visually represent data. The number or size of icons reflects the value. It's helpful for showing qualitative or categorical data in a simple way.Below is the example of network graph



Representation of Pictogram Chart

When to use:

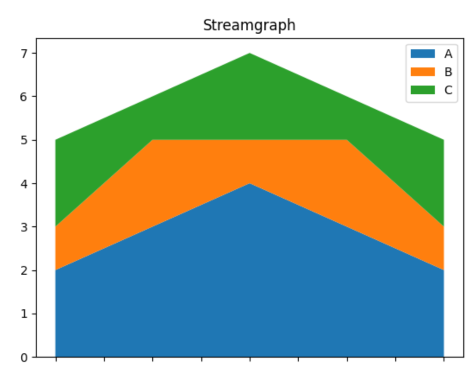
- To present data using symbols, especially for non-numeric info.
- To communicate with audiences of all literacy levels.
- To highlight key trends using clear, recognizable icons.

Temporal and Trend Charts Data Visualization

You may see the evolution of data over time with the use of a temporal or trend chart. Time-series data, in which each point represents a moment, is perfect for this. The patterns, trends, and oscillations are well shown by these charts.

1. Streamgraph

The use of moving, stacked regions to show how data has changed over time is known as a streamgraph. It demonstrates the cumulative effect of several categories during time. One example of a streamgraph is shown below



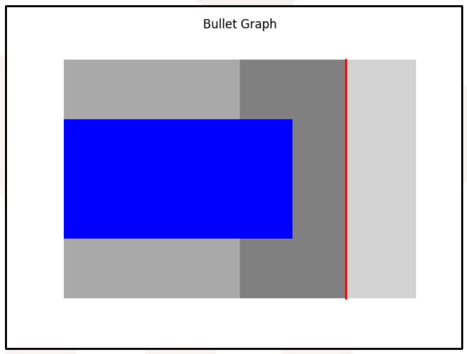
Representation of Streamgraph

When to Use:

- To show trends and changes in category distribution over time
- To compare the size of different groups over time
- To highlight patterns in a smooth and visually appealing way

2. Bullet Graph

Bullet graphs are a kind of bar chart that make use of markers and reference lines to illustrate the advancement of a target. Using it to measure progress towards a goal is a great idea. A bullet graph example is shown below:



Representation of Bullet graph

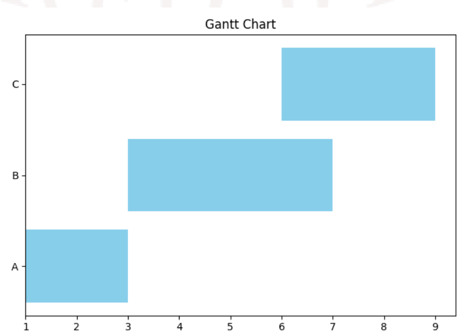
When to use:

- Show progress toward goals clearly and quickly.
- Compare actual performance with targets.
- Communicate key metrics in dashboards or reports.

3. Gantt Chart

One way to visualise project activities over time is with a Gantt chart, which uses horizontal bars. It helps in visualising progress tracking, planning, and scheduling.

Below you may see a Gantt chart example:



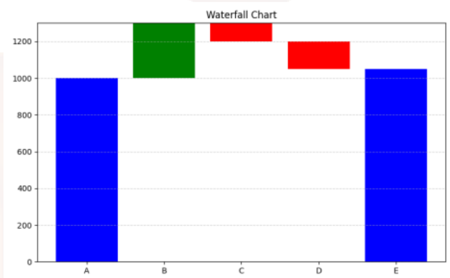
Representation of Gantt Chart

When to use:

- Plan and schedule tasks with dependencies.
- Track progress and resource use over time.
- Communicate timelines and milestones clearly.

4. Waterfall Chart

The cumulative effect of both positive and negative numbers over time can be seen in a waterfall chart. It's a great tool for keeping tabs on budgets, performance measures, and financial data as it evolves. Here is a Waterfall chart example



Representation of Waterfall chart

When to Use:

- Analyze step-by-step gains or losses over time.
- Understand the impact of different factors on totals.
- Present complex changes clearly and visually.

SELF-ASSESSMENT QUESTIONS – 2

Multiple Choose Questions	
4	**Which of the following is used to show trends over time?** a) Pie chart b) Line chart c) Bar chart d) Scatter plot
5	**Which visualization is best for comparing different categories?** a) Bar chart b) Heatmap c) Histogram d) Tree map
6	**A pie chart is mainly used to display _______.** a) Trends b) Proportions c) Distributions

	d) Relationships
7	**Which visualization type is most suitable for showing the relationship between two variables?** a) Scatter plot b) Pie chart c) Line chart d) Column chart
8	**A heatmap is typically used to represent data through variations in ______.** a) Shape b) Text c) Color d) Size

5. REAL-WORLD USE CASES FOR DATA VISUALIZATION

Data visualization is used across various industries to improve decision-making and drive results. Here are a few examples:

1. **Business Analytics:** Used to monitor company performance, track KPIs, and make data-driven decisions by visualizing trends, sales, and customer metrics.
2. **Healthcare:** Helps in analyzing patient records, tracking disease outbreaks, and managing hospital operations through easy-to-read charts and dashboards.
3. **Sports:** Used to visualize player statistics, team performance, and match outcomes, helping coaches and analysts improve strategies and training plans.
4. **Retail and E-commerce:** Enables tracking of sales, customer preferences, and inventory levels, helping businesses adjust stock and marketing efforts effectively.

5.1 Challenges in Data Visualization

1. **Data Quality:** Accuracy of visualizations depends on the quality of the data. If the data is inaccurate or incomplete, the insights from the visualization will be misleading.
2. **Over-Simplification:** Simplifying data too much can lead to important details being lost like using a pie chart that oversimplifies complex relationships between categories.
3. **Choosing the Right Visualization:** Using the wrong type of visualization can distort the message. For example, a pie chart might not work well with many categories which leads to confusion.
4. **Overload of Information:** Too much information in a visualization can overwhelm viewers. It's important to focus on key data points and avoid clutter.
5. Data visualization, although powerful, comes with several challenges that can affect the accuracy, clarity, and usefulness of the insights it provides. One of the major challenges is dealing with large and complex datasets—as data grows in volume and variety, it becomes increasingly difficult to select meaningful variables, maintain consistency, and decide the most appropriate way to visually represent information. Another challenge lies in choosing the right visualization type; selecting an incorrect chart (for example, using a pie chart instead of a bar chart) can mislead the audience or hide important patterns, trends, and relationships.
6. Data quality issues such as missing values, duplicates, inconsistencies, or noise can also distort visualizations, making them inaccurate or unreliable unless data is properly cleaned and

processed. Additionally, many users face tool-related challenges, as advanced platforms like Tableau, Power BI, and Python visualization libraries require technical skills, and lack of expertise may limit the effectiveness of the visual outputs.

7. Another major issue is the risk of overloading visuals with too much information, which can cause clutter and confusion rather than clarity. Designers must balance detail with simplicity to ensure that visuals remain understandable and aesthetically appealing. Furthermore, bias in visualization design, such as the choice of colors, scale manipulation, or emphasizing certain data points, can unintentionally (or intentionally) influence interpretation.
8. Finally, visualizations must be designed to be accessible and meaningful to diverse audiences, including those who may not have technical backgrounds. Ensuring clarity, accuracy, and interpretability across different user groups is an ongoing challenge. Together, these factors make data visualization a task that requires careful planning, a deep understanding of data, and the ability to communicate effectively through visual means.

5.2 Best Practices for Effective Data Visualization

Best practices for effective data visualization focus on creating clear, accurate, and meaningful visuals that communicate insights easily. A good visualization should start with understanding the purpose and the audience, ensuring that the selected chart type matches the nature of the data and the message being conveyed. Simplicity is key—visuals should avoid unnecessary decorations, clutter, or too many colors, and instead highlight the most important patterns or trends. Maintaining data accuracy is crucial, which means using proper scales, consistent formatting, and avoiding misleading representations. Effective visualizations also use appropriate labeling, legends, and annotations so viewers can interpret the information quickly. Additionally, designers should ensure accessibility by choosing readable fonts, color-blind–friendly palettes, and clear contrast. Finally, interactive elements like filters or tooltips can enhance user engagement, especially when dealing with large datasets. These practices help transform raw data into visuals that are informative, reliable, and easy to understand.

To ensure our visualizations are impactful and easy to understand we follow these best practices:

1. **Audience-Centric Design**: Tailor visualizations to our audience's knowledge. A technical audience may need detailed graphs while a general audience benefits from simpler charts.
2. **Design Clarity and Consistency:** Choose the right chart for our data and keep the design clean with consistent colors, fonts and labels and also avoid clutter to ensure clarity.

3. **Provide Context:** Always provide context by including labels, titles and data source acknowledgments. This helps viewers to understand the significance of the data and builds trust in the results.
4. **Interactive and Accessible Design:** Make visualizations interactive with features like tooltips and filters and ensure accessibility for all users regardless of device or visual needs.

SELF-ASSESSMENT QUESTIONS – 3

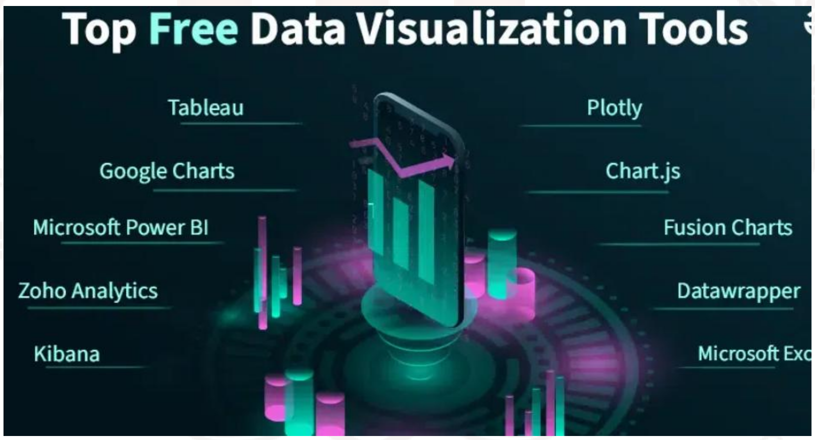
Multiple Choose Questions	
9	**Which chart type is best suited for showing trends over time?** a) Pie chart b) Line chart c) Bar chart d) Heatmap
10	**A pie chart is mainly used to represent _______.** a) Relationships b) Proportions c) Distributions d) Trends
11	**Which of the following is a challenge in data visualization?** a) Accurate labeling b) Poor data quality c) Proper scaling d) Clean layout
12	**Heatmaps represent data variations using _______.** a) Text b) Color c) Numbers d) Shapes
13	**Which of the following is a best practice in effective data visualization?** a) Using too many colors b) Overloading the chart with details c) Keeping visuals simple and clear d) Avoiding labels and legends

14	**A scatter plot is used to show the _______ between two variables.** a) Comparison b) Proportion c) Distribution d) Relationship
15	**Which challenge arises when the wrong chart type is selected?** a) Increased accuracy b) Misinterpretation of data c) Better data clarity d) Higher performance
16	**Which tool is widely used for interactive dashboards?** a) Notepad b) Tableau c) Paint d) WinZip
17	**Choosing color-blind–friendly palettes is a part of _______.** a) Visualization challenges b) Data cleaning c) Best practices d) Storage optimization
18	**Overloaded visuals with too much information lead to _______.** a) Better communication b) Clarity c) Confusion d) Improved decision-making

6. DATA VISUALIZATION TOOLS

Software programs that can take raw data and turn it into aesthetically pleasing representations like charts, graphs, and maps are known as data visualisation tools. Using these technologies, individuals and businesses are better able to comprehend large data sets, spot patterns, and make calculated choices. The best free data visualisation tools are going to be covered in this article.

A good data visualization tool enables users to import data from various sources, clean and prepare it for analysis, and create different types of charts such as bar graphs, pie charts, line graphs, and heatmaps. These tools also support building interactive dashboards, sharing insights with others, and visually analyzing trends and patterns. Data visualization tools are important because they make complex information easier to understand, support faster and more informed decision-making, allow users to explore data interactively, and can be used effectively by both technical and non-technical users.

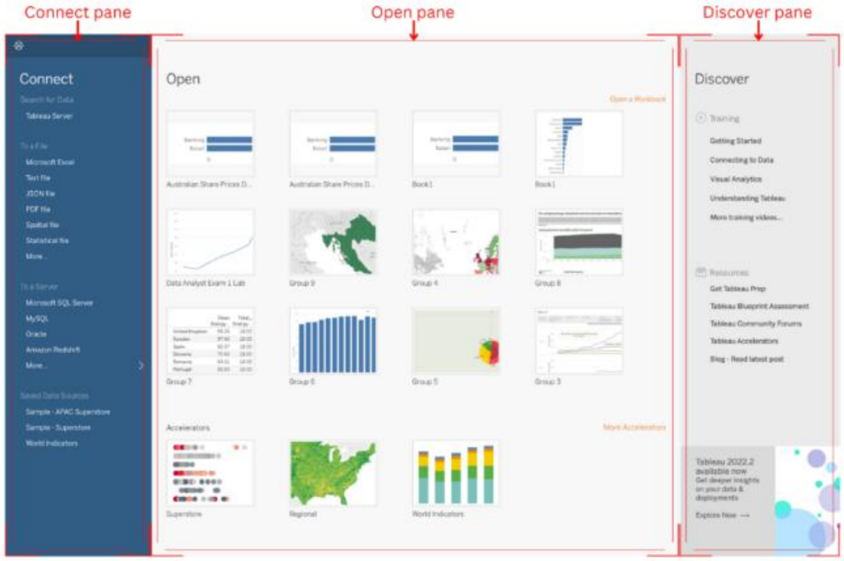


1. **Tableau**

A lot of companies utilise Tableau to make data visualisations. Desktop applications, server-based solutions, a hosted version accessible online, and a public version that is free of charge are all part of the package. Users have the ability to generate several kinds of maps and charts with Tableau. One use case is creating colour-coded maps that display data associated with specific locations. This simplifies the process of comprehending geographical data.

Anyone interested in creating visualisations for educational or sharing purposes can do it for free using Tableau's public version. Users can also gain access to a huge gallery containing examples of visualisations and infographics, which can serve as a source of inspiration for their own work.

The interface of Tableau looks as shown below:



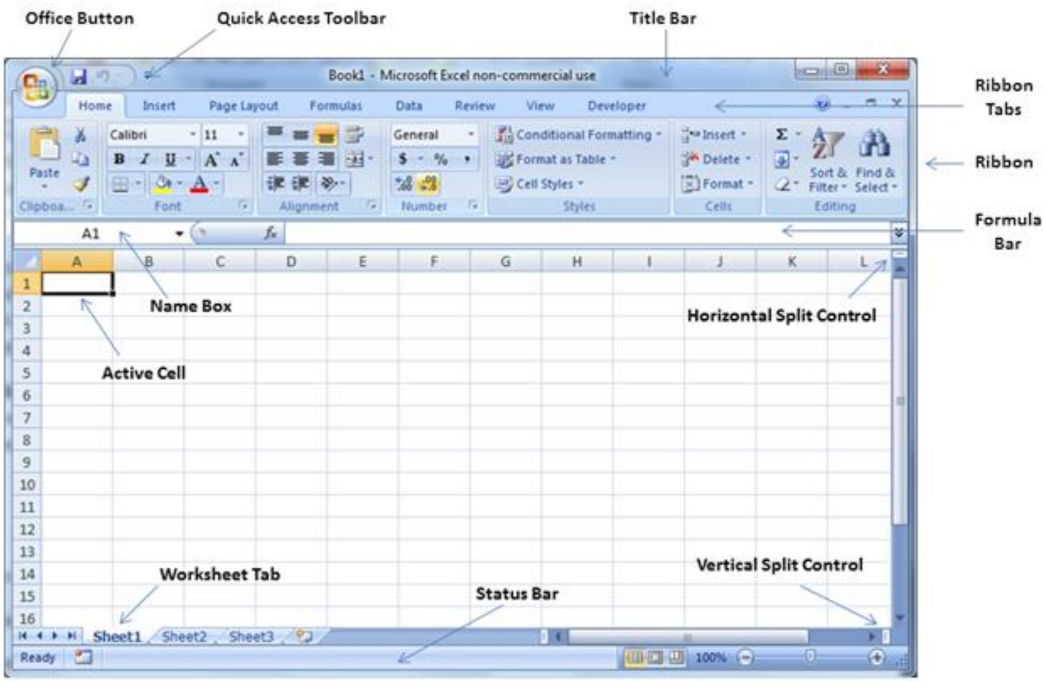
Features of Tableau

- It provides consumers with answers to their basic questions by analysing certain data queries
- Give an explanation for the dataset's outliers.
- A data preparation tool is available for self-service use, with the ability to create data visualisations according to user instructions.

2. **Microsoft Excel**

Powering decision-making with data visualisations is a breeze with Microsoft Excel, despite the program's seeming simplicity. With Excel, users may create tables, charts, and more than 20 other kinds of visualisations. People are able to better comprehend the data and spot trends with their assistance.

Excel user interface is as shown below:



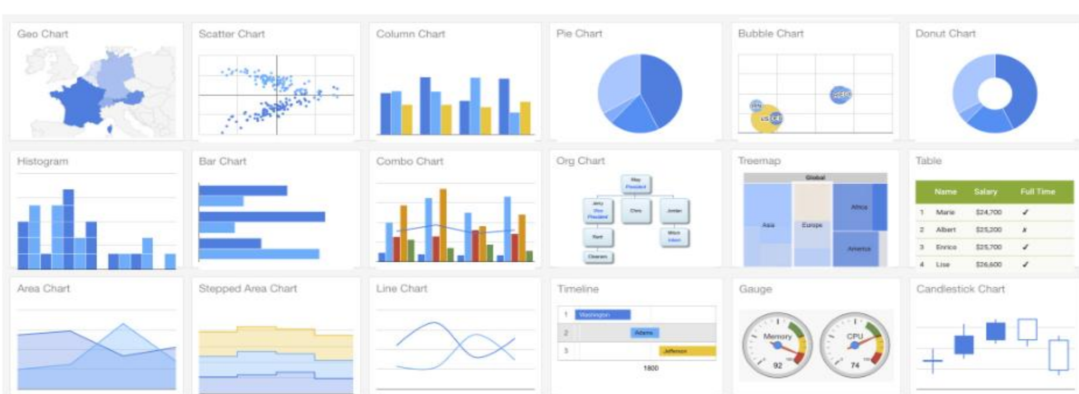
Features of Microsoft Excel

- Users can manage massive data sets, conduct in-depth analyses, and build models with data visualisation in Microsoft Excel.
- Includes formulae pre-installed.
- Provides an automated program for sorting or filtering data.

3. Google Charts

You can use Google Charts, a web-based application, to visualise data from various sizes and types of datasets, from simple to complex and enormous. Google Charts makes it easy for anyone, from businesses to individuals, to make visualisations that can be shared online. These visuals provide an appealing way to display outcomes by showing growth or specific modifications.

The Google chart interface is as shown below:



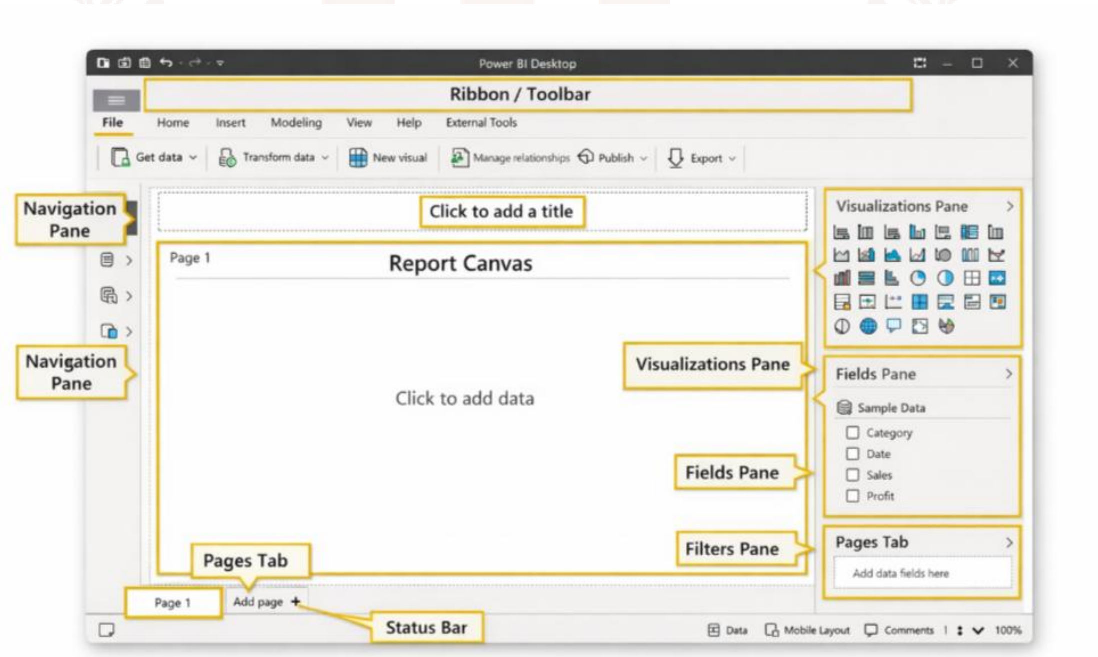
Features of Google Charts

- Google Charts visualisations are accessible on popular web browsers such as Chrome, Safari, and Firefox thanks to its cross-platform capability.
- Through CSS modification, users can quickly personalise data visualisation.
- Data analysis from SQL databases is made possible with this solution for businesses.

4. **Microsoft Power BI**

Power BI by Microsoft is like Excel with a business intelligence twist; it's optimised for in-depth data analysis and visualisation. As new data comes in, it instantly updates the charts, graphs, dashboards, and reports, helping organisations turn raw data into something understandable and interactive. Excel files, online services, databases, and many more sources are all within Power BI's data extraction capabilities. Anyone or any group looking for a visible, intelligent way to make sense of massive data sets will find this tool invaluable.

Power Bi user interface is as shown below:



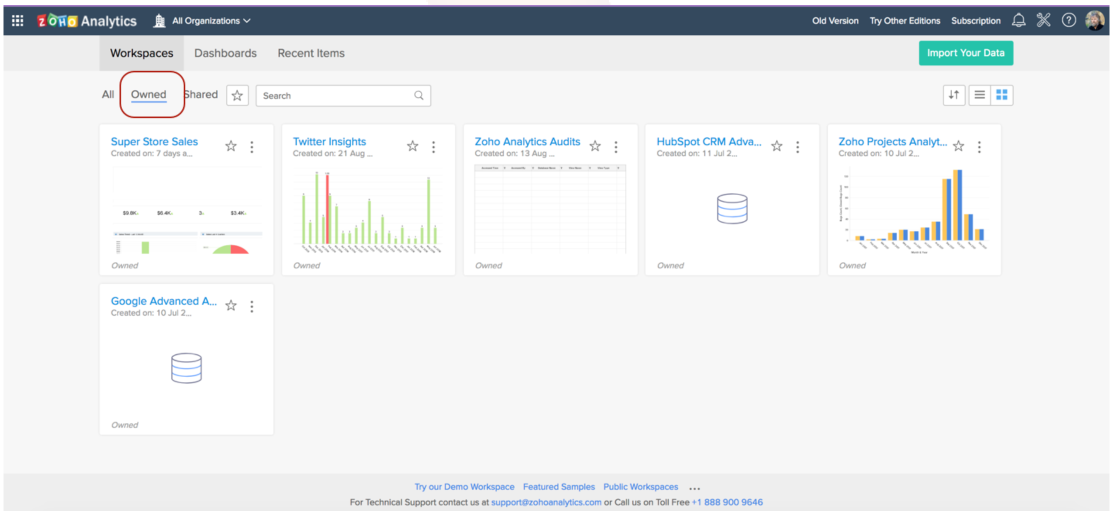
Features of Microsoft Power BI

- Business intelligence allows for the automated preparation of data visualisation and analysis.
- Companies have real-time access to interactive data visualisations.
- You can personalise the dashboards to your liking.

5. **Zoho Analytics**

With more than 2 million users, Zoho Analytics is among the top data visualisation tools. Businesses, employees, and data analysts can all benefit from this app's ability to improve data comprehension. People are able to better see patterns, monitor development, and base informed judgements on data with its guidance. Spreadsheets, cloud services, and databases are just a few of the various data sources that this intuitive tool can import.

ZOHO Analytics user interface is as shown below:



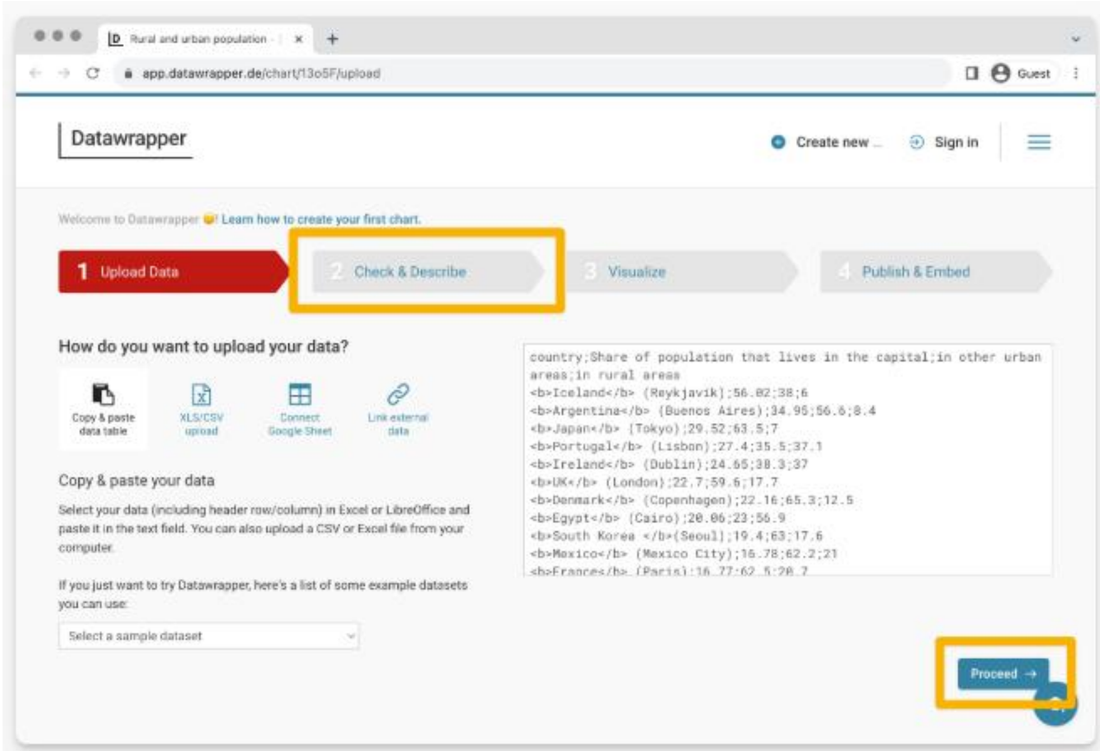
Features of Zoho Analytics

- Extra permissions to view or modify data visualisations can be granted by individuals and companies.
- Both the iOS and Android versions of the mobile BI app are available.
- Reports based on autonomous data are provided.

6. Data Wrapper

Data wrappers are great for creating tables, charts, and maps on websites since they are easy to use. Data wrapper is now used by many enterprises for data visualisations, despite the fact that it was first developed for news websites.

The user interface of data wrapper is as shown below:



Features of Datawrappers

- Assists in creating color-blind-friendly data visualizations.
- The Datawrapper tool is mobile-friendly, making it more accessible.

7. Fusion Charts

To Google Charts, Fusion Charts is like a second skin. Successfully embeddable data visualisations into web pages can be created by users using this tool. For the purpose of interpreting and analysing large data volumes, 28,000 companies across the globe depend on this technology. A hundred different kinds of charts and two hundred different kinds of choropleth maps are on hand.

The interface of fusion charts is as shown:

Features of FusionCharts

- It's compatible with server-side languages including Java, Django, Ruby on Rails, and Ember, and it provides a live data dashboard to monitor and understand data inputs in real-time.
- It also offers easy connection with JavaScript frameworks like Angular, Ember, React, and jQuery.

8. **Chart.js**

Chart.js is a free and open-source library for making JavaScript charts. It's perfect for creating basic yet adaptable data visualisations, and it's easy to use. Bar charts, line graphs, pie charts, and many more are at your disposal.

The interface is as shown below:

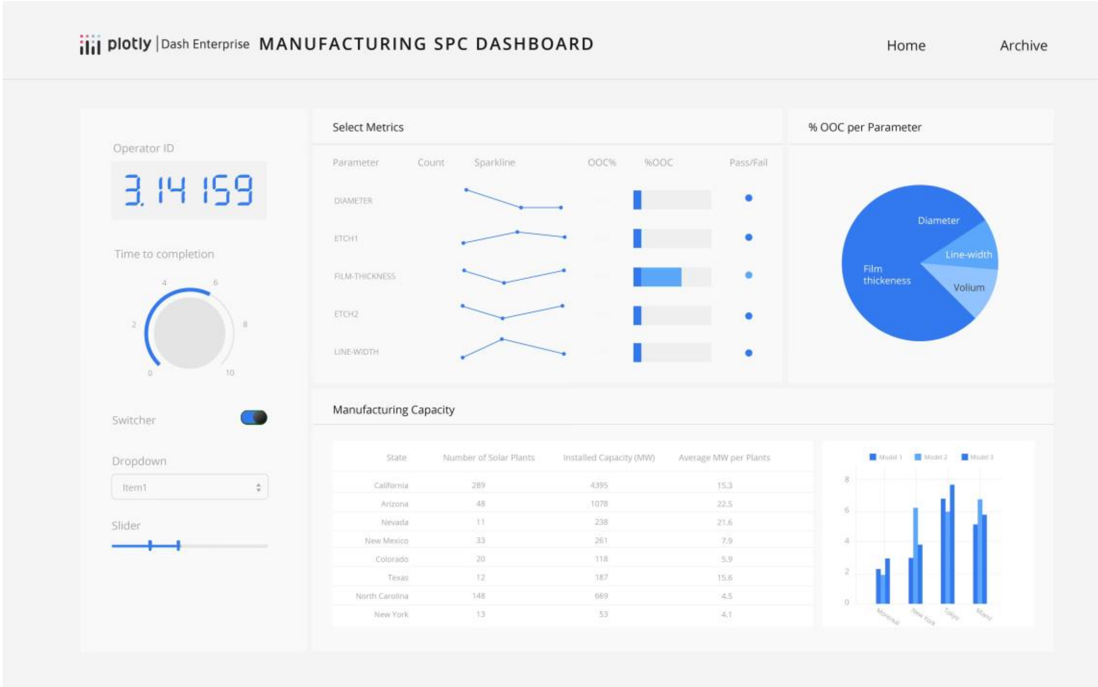
Features of Chart.js

- Colours, labels, and styles are easily customizable.
- Multiple chart types, such as bar and line, can be combined to provide a more complete perspective.
- The chart will update in real time as you update the data.

9. Plotly

To make high-quality, interactive visualisations, you can use Plotly, an open-source graphing toolkit that is both free and available for download. Scientific, technical, and data-intensive endeavours benefit greatly from it. Python, JavaScript, R, and many more languages are compatible with Plotly. It works wonderfully for creating complex charts that you can interact with by clicking, hovering, and zooming.

The interface is as shown below:



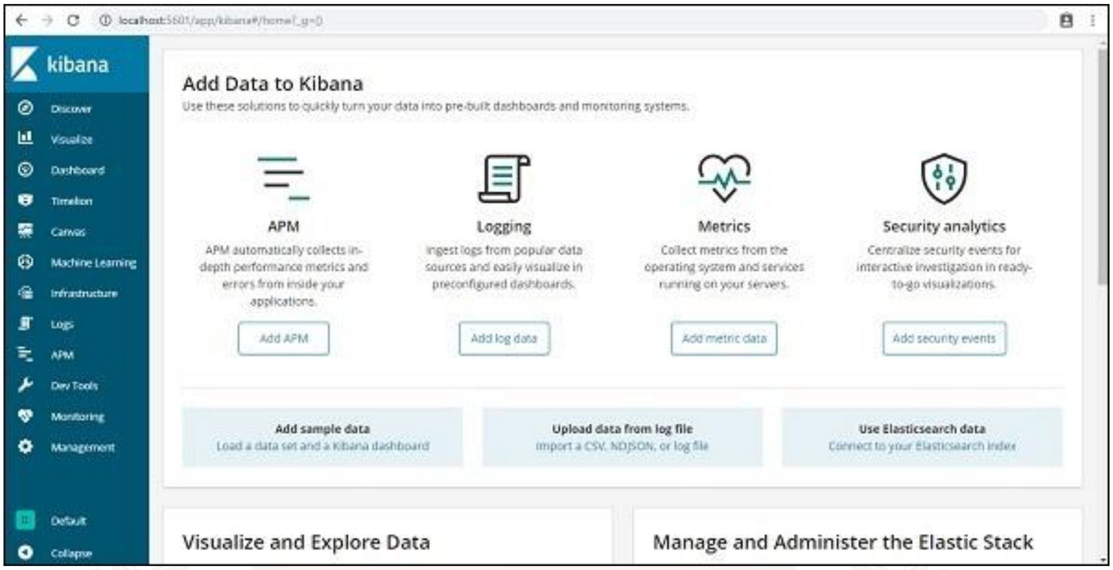
Features of Plotly

- Complex plots, such as 3D graphs and contour plots, and others, are supported.
- The corporate edition unlocks additional capabilities, but you can use the free version as well.
- Combine it with Dash to create comprehensive dashboards for your data.

10. **Kibana**

When combined with Elasticsearch, the free and open-source data visualisation tool Kibana becomes a powerful tool. It facilitates the rapid exploration, understanding, and visualisation of massive data sets. IT departments, analysts, and developers frequently employ it for the purpose of analysing logs, server activity, and real-time data.

The interface is as shown :



Features of Kibana

- Control who can see and make changes to your dashboards and data.
- Construct personal dashboards to monitor metrics that matter to you.

SELF-ASSESSMENT QUESTIONS – 4

Multiple Choose Questions	
19	**Which of the following is a widely used tool for creating interactive dashboards?** a) MS Word b) Tableau c) Notepad d) VLC Player
20	**Power BI is developed by which company?** a) Google b) IBM c) Microsoft d) Amazon
21	**Which tool is commonly used in Python for data visualization?** a) NumPy b) Matplotlib c) MySQL d) Eclipse
22	**Which data visualization tool is best known for creating simple charts and pivot charts?**

	a) Excel b) Turbo C c) Firefox d) GitHub
23	**Which tool is commonly used for web-based interactive visualizations?** a) D3.js b) WinRAR c) Adobe Reader d) Android Studio

7. SUMMARY

- Data visualization is the process of converting raw data into graphical formats to communicate information clearly and effectively.
- It uses charts, graphs, dashboards, and maps to help users understand patterns, trends, and insights at a glance.
- As data grows in size and complexity, visuals make interpretation faster and easier compared to traditional text or numerical tables.
- Visualization enhances decision-making, improves data exploration, and enables both technical and non-technical users to interpret information quickly.
- Common types include bar charts, line graphs, pie charts, scatter plots, histograms, heatmaps, dashboards, and geographic maps.
- These are used for comparing categories or groups visually.
- Ideal for showing trends and changes over time, such as stock prices or sales growth.
- Useful for representing proportions or percentage distributions.
- Scatter plots reveal relationships between variables, while heatmaps display intensity using color variations.
- Data visualization is used in business analytics, healthcare dashboards, finance reports, social media analytics, weather forecasting, and scientific research.
- Common challenges include selecting the correct chart type, dealing with poor data quality, avoiding cluttered visuals, tool complexity, and ensuring audience-friendly interpretation.
- Maintain simplicity, choose appropriate charts, use clean datasets, apply readable labels, use color-blind–friendly palettes, and avoid misleading scales or excessive decoration.
- Visualizations must represent data honestly, using consistent scales and proper labeling to prevent misinterpretation.
- Popular tools include Tableau, Power BI, Microsoft Excel, Google Data Studio, and Python libraries like Matplotlib, Seaborn, and Plotly.
- These tools automate data cleaning, enable interactive dashboards, and help uncover deeper insights through advanced visualization features.

8. GLOSSARY

Data Visualization	–	The graphical representation of data using charts, graphs, and maps to understand information easily.
Chart	–	A visual display of data, such as a bar chart or line chart, used to compare or analyze information.
Graph	–	A diagram showing relationships or trends between variables.
Dashboard	–	A collection of multiple visualizations displayed together for quick analysis.
Bar Chart	–	A chart that uses bars to compare different categories.
Line Chart	–	A graph that shows changes or trends over time.
Pie Chart	–	A circular chart used to show proportions or percentage distribution.
Scatter Plot	–	A graph showing the relationship between two numerical variables.
Heatmap	–	A visualization that uses color shades to represent intensity or frequency of data.
Histogram	–	A chart that shows the distribution of numerical data across ranges (bins).
Legend	–	A guide in a chart that explains the meaning of colors, symbols, or patterns.
Axis	–	The horizontal (x-axis) and vertical (y-axis) lines used to measure and plot data in graphs.
Data Cleaning	–	The process of correcting or removing inaccurate, incomplete, or irrelevant data before visualization.
Interactivity	–	Features like filtering, zooming, and tooltips that allow users to explore data dynamically.
Tooltip	–	A small pop-up message that displays detailed information when hovering over a data point.
Visualization Tool	–	Software used to create data visuals, such as Tableau, Power BI, or Excel.

Dataset	–	A collection of related data values used for analysis and visualization.
Real-Time Visualization	–	Visual displays that update automatically as new data arrives.
Color Palette	–	A set of colors chosen to represent data clearly and consistently.
Data Interpretation	–	The process of understanding and drawing insights from visualized datavcc

9. TERMINAL QUESTIONS

1. Define data visualization and explain its purpose.
2. What are the main advantages of using data visualization?
3. Explain any three common types of data visualizations.
4. What is a dashboard? Explain its usefulness.
5. Explain any two real-world applications of data visualization.
6. What is a heatmap and where is it used?
7. Describe any two challenges in data visualization.
8. What is clutter in data visualization and why should it be avoided?
9. Mention any two best practices for creating effective data visualizations.
10. Why is choosing the correct chart type important?
11. What are data visualization tools? Give examples.
12. Explain the importance of color selection in data visualization.

10. ANSWERS

10.1. Self-Assessment Answers:

1. graphical
2. visuals
3. insights
4. Answer: b) Line chart
5. Answer: a) Bar chart
6. Answer: b) Proportions
7. Answer: a) Scatter plot
8. Answer: c) Color
9. Answer: b) Line chart
10. Answer: b) Proportions
11. Answer: b) Poor data quality
12. Answer: b) Color
13. Answer: c) Keeping visuals simple and clear
14. Answer: d) Relationship
15. Answer: b) Misinterpretation of data
16. Answer: b) Tableau
17. Answer: c) Best practices
18. Answer: c) Confusion
19. Answer: b) Tableau
20. Answer: c) Microsoft
21. Answer: b) Matplotlib
22. Answer: a) Excel
23. Answer: a) D3.js

10.2. Terminal Question Answers

Answer 1: Data visualization is the graphical representation of data using charts, graphs, maps, dashboards, and other visual formats. Its purpose is to simplify complex information, making it easier to understand patterns, trends, and relationships. Visualization helps users interpret data quickly, supports decision-making, and improves communication between technical and non-technical audiences. (Refer section 2 for more details)

Answer 2: Data visualization makes data easier to interpret, helps reveal hidden patterns, supports faster decision-making, and allows users to compare values and trends visually. It improves data storytelling, enhances audience engagement, and helps identify outliers or unusual behavior within a dataset.

Answer 3: A bar chart compares categories using rectangular bars.

A line chart displays trends or changes over time.

A pie chart shows proportions or percentage distribution of a whole.

Each type serves a specific purpose and helps users interpret data effectively.(Refer Section 4 for more details)

Answer 4: A dashboard is a collection of multiple visualizations displayed in one place for quick and interactive analysis. It helps users monitor key metrics, explore trends, compare data points, and make informed decisions. Dashboards are widely used in business intelligence, analytics, and reporting.

Answer 5: In business, companies use dashboards to track sales, revenue, and performance metrics. In healthcare, visualizations help monitor patient records, disease trends, and medical reports. These applications improve decision-making and provide clear communication of insights.(Refer Section 5 for more details)

Answer 6: A heatmap is a graphical representation that uses color to indicate data intensity or frequency. It is used in website analytics to track user clicks, in scientific research to show gene expression levels, and in business to visualize correlations or performance metrics.(Refer Section 5 for more details)

Answer 7: One challenge is poor data quality, where missing or incorrect values can make visuals misleading. Another challenge is selecting the wrong chart type, which may confuse users or hide important insights. Both issues affect the accuracy and effectiveness of visualization.

Answer 8: Clutter occurs when a visualization contains too many elements—excessive colors, labels, or decorations—making it difficult to read. It should be avoided because it distracts viewers from key insights and reduces clarity, leading to confusion.

Answer 9: One best practice is keeping visuals simple by avoiding unnecessary elements.

Another is using proper labels, legends, and scales to enhance readability and accuracy. These practices ensure that the viewer can easily understand the message.

Answer 10: Choosing the correct chart type ensures that the data is represented accurately and clearly. The wrong chart can mislead viewers—for example, using a pie chart for trend analysis. The correct chart highlights the right relationships and helps communicate insights effectively.

Answer 11: Data visualization tools are software applications used to create graphical representations of data. Examples include Tableau, Power BI, Microsoft Excel, and Python libraries like Matplotlib and Seaborn. These tools help users build charts, dashboards, and interactive reports.

Answer 12: Color selection affects readability and interpretation. Using meaningful colors helps differentiate data, highlight important points, and improve visual appeal. Colors must be used carefully to avoid confusion, and color-blind–friendly palettes should be considered to ensure accessibility for all viewers.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 4

DCA2207

DATA MINING AND VISUALISATION

Unit – 10

Understanding Graphs

TABLE OF CONTENTS

1. INTRODUCTION

In previous unit, learners has gained a foundational understanding of data visualisation and its importance in interpreting and communicating information effectively. They had explored the different types of data—such as structured, unstructured, and semi-structured—and understand how the nature of data influences visualisation choices. The unit also introduces various data visualisation tools like Tableau, Power BI, Excel, and Python libraries, giving students an overview of how these tools help transform raw data into meaningful insights. By the end of the unit, learners were well equipped with the essential knowledge needed to select appropriate visualisation techniques and tools for real-world data analysis tasks.

In this unit, learners explore the essential role of graphs in data visualisation and understand how graphical representation transforms raw data into meaningful insights. The unit begins with the concept of graph roles, explaining how visuals such as charts, plots, and diagrams help identify trends, comparisons, patterns, and relationships. Students then study two-dimensional (2D) graphs, including their structure, usage, advantages, and limitations. They learn when 2D graphs are most effective—for example, in showing trends over time, comparing categories, or analysing distributions—and examine why 2D visuals remain the foundation of most analytical dashboards and reports.

The unit then expands into three-dimensional (3D) graphs, introducing learners to advanced visualisation techniques that incorporate depth and additional variables. Students explore key 3D methods, how to choose the right 3D visualisation technique, and the advantages and disadvantages of using 3D representations. Real-world applications—such as scientific modeling, engineering simulations, geographical mapping, and complex data analysis—are also covered. By the end of this unit, learners gain the skills to select, interpret, and apply both 2D and 3D graphs effectively, enabling clearer communication, deeper analysis, and more informed decision-making in data-driven environments.

1.1. Learning Objectives

By the end of this unit, you will be able to:

- **Explain** the role and significance of graphs in data visualisation
- **Identify** different types of two-dimensional (2D) graphs and their components.
- **Evaluate** the best use cases, advantages, and limitations of 2D graphs
- **Explore** three-dimensional (3D) graphing techniques and understand when 3D visualisation is appropriate.
- **Analyse** the advantages, disadvantages, and challenges associated with using 3D visualisations.

2. ROLE OF GRAPHS IN DATA VISUALISATION

Graphs in data visualisation are powerful tools that transform raw data into visual formats like plots and charts, making complex information easier to understand and interpret. They facilitate the identification of patterns, trends, correlations, and outliers, which aids in effective communication and data-driven decision-making.

Key Concepts and Benefits

- Simplifying Complexity: Graphs convert large, complicated datasets into intuitive visual representations that are easily digestible for both technical and non-technical audiences.
- Revealing Insights: Visual displays allow users to quickly spot trends and relationships that might be missed when analysing raw numbers in spreadsheets.
- Enhancing Communication: They provide a common ground for teams, making it easier to explain data-driven insights and fostering collaboration.
- Interactive Exploration: Many data visualisation tools offer interactive features like zooming and filtering, which allow for deeper exploration and understanding of the data.

Common Types of Graphs and Their Uses

The type of graph used depends on the kind of data and the message you want to convey.

- Line Graphs: Ideal for displaying trends over continuous intervals, such as changes in stock prices or population over time.
- Bar Charts: Used to compare quantities across different, distinct categories. They are effective for ranking values or comparing sales across different products or regions.
- Pie Charts: Represent parts of a whole, showing the percentage or proportional contribution of each category to a total (e.g., market share distribution).
- Scatter Plots: Show the relationship or correlation between two numerical variables using individual data points, helping to identify clusters or outliers (e.g., relationship between house size and price).
- Histograms: Display the distribution of numerical data by grouping values into bins to show their frequency. They help in understanding the shape and spread of a dataset (e.g., distribution of exam scores).

- Network Graphs: Visualise relationships between entities (nodes) as connections (edges). They are particularly useful for analysing complex systems like social networks, supply chains, or criminal organisations.

Choosing the Right Graph

Selecting the appropriate graph involves defining your goal and considering your audience.

- To show proportions, use a Pie Chart.
- To track trends over time, use a Line Chart.
- To analyse distribution, use a Histogram.
- To see relationships between two variables, use a Scatter Plot.
- To compare multiple categories, use a Bar Chart.

2.1 Graph Roles in Data Visualisation

Graphs (or charts) play a crucial role in data visualisation because they help convert raw data into meaningful insights. Instead of reading long tables, graphs allow viewers to quickly understand trends, relationships, and patterns. Each graph type serves a different **role** based on the kind of data and the purpose of the analysis.

1. Representing Trends Over Time

Graphs help show how values change over a period.

- **Common Graphs:** Line charts, area charts.
- **Role:** Identify increases, decreases, seasonality, and long-term patterns.
- **Example:** Monthly sales trends, temperature changes over years.

2. Comparing Categories or Groups

Some graphs are designed to compare different categories side-by-side.

- **Common Graphs:** Bar charts, column charts, grouped bar charts.
- **Role:** Highlight differences and similarities among data groups.
- **Example:** Comparing production output of different departments.

3. Showing Part-to-Whole Relationships

These graphs show how individual parts contribute to a total.

- **Common Graphs:** Pie charts, doughnut charts, stacked bar charts.
- **Role:** Display proportions and share of each category.

- **Example:** Market share of different mobile companies.

4. Displaying Distributions

These graphs show how data values are spread across a range.

- **Common Graphs:** Histograms, box plots.
- **Role:** Understand data variability, central tendency, and outliers.
- **Example:** Distribution of student exam marks.

5. Understanding Relationships Between Variables

Used to analyse how one variable affects another.

- **Common Graphs:** Scatter plots, bubble charts.
- **Role:** Identify correlations, clusters, and patterns.
- **Example:** Relationship between advertising spend and sales.

6. Highlighting Data Composition

These graphs break down data into components across categories or time.

- **Common Graphs:** Stacked area charts, waterfall charts.
- **Role:** Show contribution of each segment to the total.
- **Example:** Contribution of different products to yearly revenue.

7. Representing Geographical Patterns

Maps help visualise location-based data.

- **Common Graphs:** Choropleth maps, heat maps, filled maps.
- **Role:** Identify regional trends or differences.
- **Example:** Population density across states.

8. Visualising Hierarchies

These graphs show parent–child relationships.

  - **Common Graphs:** Tree maps, sunburst charts.
  - **Role:** Represent structured data with multiple levels.
  - **Example:** Organisation structure or product category breakdown.

9. Showing Networks and Connections

Used when the focus is on relationships between nodes.

- **Common Graphs:** Network graphs, node-link diagrams.

- **Role:** Visualise influence, connections, and clusters.
- **Example:** Social network analysis.

10. Supporting Quick Decision-Making

Graphs simplify complex datasets so decision-makers can:

- Spot anomalies
- Detect patterns faster
- Compare metrics quickly
- Make strategic decisions based on visual insights

Graphs in data visualisation serve multiple important roles in helping users understand data quickly and effectively. They represent trends over time through line and area charts, allowing viewers to observe increases, decreases, and long-term patterns such as sales growth or temperature changes. Bar and column charts help compare categories or groups, while pie charts and stacked bar charts show part-to-whole relationships like market shares. Distributions are visualised through histograms and box plots, revealing variability and outliers. Scatter plots and bubble charts help analyse relationships between variables, showing correlations or clusters. Stacked area and waterfall charts illustrate data composition across time or categories. Geographical patterns are presented using maps such as choropleth or heat maps, while tree maps and sunburst charts visualise hierarchies and parent–child structures. Network graphs depict connections and influence between nodes, commonly used in social network analysis. Overall, graphs simplify complex datasets, enabling quicker pattern detection, anomaly spotting, and faster strategic decision-making.

SELF-ASSESSMENT QUESTIONS – 1

Multiple Choose Questions	
1	**Which graph is most suitable for showing trends over time?** a) Bar Chart b) Line Graph c) Histogram d) Pie Chart
2	**A scatter plot is mainly used to show:** a) Categories b) Time trends c) Relationships between two variables

	d) Parts of a whole
3	**Which of the following uses bins to group continuous data?** a) Bar Chart b) Pie Chart c) Histogram d) Heatmap
4	**In which graph do bars touch each other?** a) Histogram b) Bar Chart c) Column Chart d) Scatter Plot
5	**Heatmaps primarily use which method to represent data?** a) Lines b) Colors c) Dots d) Bars

3. TWO-DIMENSIONAL GRAPH

Two-dimensional (2D) graphs are fundamental tools in data visualisation, displaying data points on a flat plane defined by an x-axis and a y-axis. They are highly versatile and effective for illustrating relationships between two variables, showing changes over time, and identifying patterns or outliers in data. Two-dimensional graphs are charts that display data along **two axes**—a horizontal (X-axis) and a vertical (Y-axis). They are widely used in data visualisation because most real-world datasets involve comparing two variables, observing patterns, or tracking changes over time. 2D graphs help convert complex numerical data into easy-to-understand visuals.

Common Types of Two-Dimensional Graphs

The choice of graph depends on the type of data and the insights you want to convey.

- Scatter Plots These plots use individual points to show the relationship between two numerical variables. They are excellent for identifying correlations, clusters of data points, and outliers. They can be enhanced by using color or size to represent a third or fourth variable (bubble charts).

  Example:

  Plotting *study hours* (X-axis) vs. *exam marks* (Y-axis) to see whether students who study more score higher. **Additional Note:** If you want to show a third variable, you can vary color or point size → this becomes a **bubble chart.**

- Line Graphs Used to visualise continuous data over an interval or time period. They connect data points with lines, making them ideal for tracking trends and patterns over time, such as stock prices or temperature changes.

  Example: Tracking *daily temperature* for a week or *stock price movement* over a month.

- Bar Charts/Column Charts These use rectangular bars to compare categorical data. The length of each bar is proportional to the value it represents. Bar charts (horizontal bars) are useful for long labels or many categories, while column charts (vertical bars) are better for showing sequences or accentuating large values.

  Example: Comparing the sales of **Product A, B, and C** in a store.

- Histograms Similar in appearance to bar charts, histograms group continuous numerical data into "bins" or ranges and plot the frequency of values within each bin. They are best for visualising the distribution and spread of a single continuous variable.

  Example: A histogram showing the distribution of **heights** of 200 students.

➢ Heatmaps These use a color gradient within a two-dimensional matrix to represent the value or strength of a relationship between two variables. They are effective for revealing patterns and correlations in large datasets.

Example: Showing the correlation between multiple variables (e.g., price, revenue, customer visits).

➢ Area Charts A variation of a line chart where the area beneath the line is filled with color. They are useful for showing how a total value changes over time and how individual components contribute to that total.

Example: Displaying how revenue from **Product A, B, and C** contributes to total revenue across months.

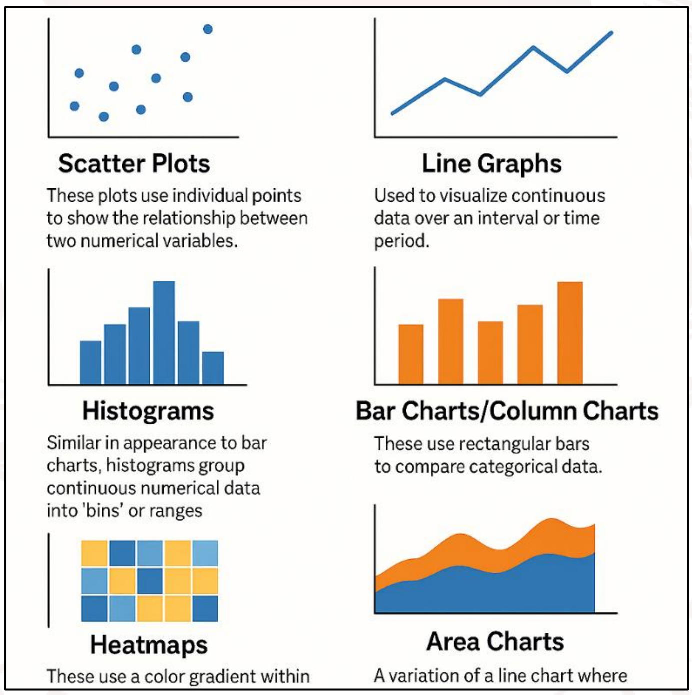


Figure 1: Representation of 2D graphs visualisation

Software Tools for 2D Data Visualisation

Various software tools and libraries are available for creating 2D graphs:

➢ Matplotlib A foundational and highly customisable 2D plotting library for Python, used in conjunction with libraries like NumPy and Pandas for data analysis.

➢ Seaborn A Python library built on Matplotlib that provides a high-level interface for drawing attractive and informative statistical graphics.

➢ Tableau Public A free, public tool for creating and sharing interactive visualisations.

➢ Gnuplot An open-source, command-line driven utility that can plot data from files or mathematical functions.
➢ D3.js A powerful JavaScript library for creating custom, interactive visualisations on the web.

3.1 Best Use Cases of Two-Dimensional Graphs

Two-dimensional graphs are most effective for visualising small-to-medium datasets because they provide simple, clear, and accurate representations of data. They are ideal for comparing two variables, such as height vs. weight or study hours vs. marks, helping identify correlations or independence. They also work well for time-based data, using line or area graphs to show trends like rainfall patterns or sales changes across months. When analysing frequency distributions, 2D histograms reveal how data values are spread across ranges, helping identify variability and skewness. Scatter plots and line graphs help detect patterns such as clusters, outliers, and linear or non-linear relationships, making them valuable in research and statistical analysis. Overall, 2D graphs provide clean and easily understandable visuals, making them suitable for reports, dashboards, classrooms, and quick presentations without the complexity of 3D visuals.

Two-dimensional graphs are ideal for visualising most small-to-medium datasets. They work best in the following situations:

a) When You Want to Compare Two Variables

2D graphs make it easy to observe how one variable changes with respect to another. Examples:

- Height vs. weight
- Study hours vs. marks
- Cost vs. profit

These comparisons help identify relationships such as positive/negative correlation or independence.

b) When You Want to Show Data Changes Across Time

Time-based data is usually plotted on the X-axis, making 2D line and area graphs perfect for spotting trends.

Examples:

- Monthly rainfall
- Daily temperatures

- Sales over quarters

This helps viewers understand growth, decline, cycles, or seasonality.

c) When You Want to Analyse Frequency Distributions

2D histograms and frequency plots show how often values fall within certain ranges. Examples:

- Distribution of exam scores
- Age distribution of customers
- Income range of households

Such graphs help in identifying spread, skewness, and variability.

d) When You Want to Identify Patterns and Relationships

Scatter plots and line graphs help discover:

- Clusters
- Outliers
- Trends
- Linear/non-linear relationships

This is useful in data science, research, and statistical analysis.

e) When You Want to Present Data in a Simple and Clean Format

2D graphs are clear, minimal, and easy for non-technical audiences to understand. They avoid the unnecessary complexity of 3D visuals, making them suitable for:

- Reports
- Classroom teaching
- Dashboards
- Quick presentations

3.2 Advantages of 2D Graphs

a) Easy to Create and Interpret: 2D graphs require minimal effort to construct and can be read quickly by viewers.

Even people with basic statistical knowledge understand bar graphs, line graphs, and scatter plots easily.

b) Good for Educational and Analytical Purposes: Teachers, students, analysts, and researchers use 2D graphs because they clearly express concepts like:Trends, Comparisons, Correlations and Distributions. They make learning and analysis more intuitive.

c) Helpful for Both Numerical and Categorical Data: 2D charts support a wide range of data This flexibility allows 2D graphs to be used in almost every domain

d) Widely Supported by All Visualisation Tools: All major visualisation tools—Excel, Python (Matplotlib, Seaborn), Tableau, Power BI, R—offer built-in options for 2D graphs.

3.3 Limitations of 2D Graphs

a) Cannot Show More Than Two Variables Effectively

2D graphs are restricted to an X-axis and Y-axis. If your data has more than two dimensions (e.g., temperature, humidity, and rainfall together), 2D representation may lose context.

b) Complex Data May Require 3D or Advanced Charts

Some datasets involve multiple layers of information that 2D graphs cannot adequately represent. In such cases, advanced visualisation techniques give a better picture.

c) Visual Clutter Can Occur with Too Many Data Points

If a 2D graph contains hundreds of data points of many categories or Multiple overlapping lines, it becomes hard to read and interpret. Graphs can look crowded, confusing, and misleading. To avoid clutter, techniques like sampling, grouping, or switching to other chart types may be needed.

SELF-ASSESSMENT QUESTIONS – 2

Multiple Choose Questions	
6	**A bar chart is best suited for:** a) Frequency distribution b) Time-series analysis c) Comparing categories d) Correlation analysis
7	**A 3D surface plot is used to visualise:** a) Distribution b) Relationships among three variables

	c) Parts of a whole d) Categorical comparison
8	**Which graph should be used to detect outliers quickly?** a) Line graph b) Scatter plot c) Pie chart d) Histogram
9	**Which graph is NOT suitable for showing categorical data?** a) Pie Chart b) Bar Chart c) Line Graph d) Column Chart
10	**Visual clutter in graphs can be reduced by:** a) Adding more colors b) Using 3D graphs c) Removing unnecessary elements d) Adding extra gridlines

4. THREE DIMENSIONAL GRAPH

Three-dimensional (3D) graphs are visual representations of data in three-dimensional space, using X, Y, and Z axes to plot data points and reveal relationships between three variables. These graphs add depth and perspective, allowing for a more comprehensive view of data patterns and complexities that might be hidden in 2D charts, such as showing relationships between age, fare, and survival on the Titanic dataset. They are particularly useful for scientific data, multi-variable relationships, and exploring data from different viewpoints.

3D data visualisation involves the creation of three-dimensional graphical representations of data. This technique allows us to see data points plotted along three axes (X, Y, and Z), providing a more comprehensive view of the relationships within the data. It allows users to explore and analyse the data from different perspectives, enabling better insights and decision-making. In contrast to 2D visualisations, which lack spatial context and depth, 3D data visualisation entails building three-dimensional models of data sets. By using this method, one can gain a more intuitive grasp of the data's spatial correlations, patterns, and linkages. Users are able to engage with and examine data from various angles thanks to the addition of a third dimension, which may reveal insights that were previously obscured by conventional two-dimensional graphs and charts.

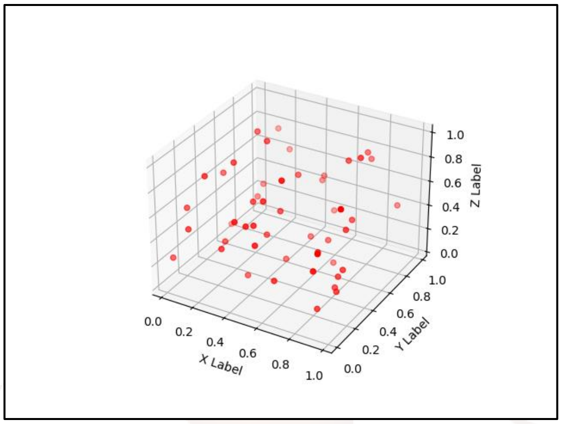
How they work

- Three axes: 3D graphs plot data along three axes: X, Y, and Z, allowing for the visualisation of three distinct values simultaneously.
- Visualising relationships: They are ideal for exploring relationships between three variables, such as sales figures across products, cities, and time.
- Exploring perspectives: The added depth allows users to rotate the graph and view the data from different angles, offering a more comprehensive understanding of the data structure and patterns.

4.1 Key Techniques in 3D Data Visualisation

1. **3D Scatter Plots**

Using the X, Y, and Z axes to depict the three variables, 3D scatter plots show the data points in a three-dimensional space. The coordinates of one observation are represented by each dot in the plot, which is based on the values of the three variables.



Use Cases:

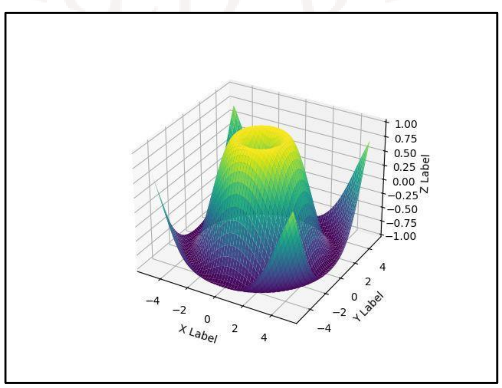
- Identifying Clusters: Useful for clustering analysis where groups of data points with similar properties can be identified.
- Detecting Correlations: Helps in discovering relationships and correlations between three variables that might not be apparent in 2D plots.
- Finding Outliers: Useful for spotting outliers, which are data points that deviate significantly from the rest of the dataset.

Applications:

- Market research for segmenting customers based on demographics, purchase behavior, etc.
- Scientific research to analyse relationships between experimental variables.
- Financial analysis to explore the relationship between different financial metrics.

2. **3D Surface Plots**

A three-dimensional surface showing the connection between three continuous variables is what three-dimensional surface plots are all about. The graph displays the relationship between two independent variables (X and Y) and a dependent variable (Z).



Use Cases:

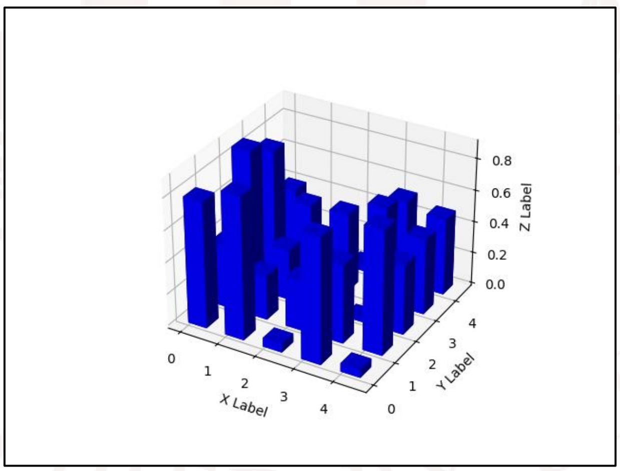
- Modelling Surfaces: Ideal for creating models of surfaces, such as geographic terrain, temperature distributions, or any other data that varies continuously over a region.
- Interaction Analysis: Helps in visualising and analysing how two independent variables interact to affect a dependent variable.

Applications:

- Topographical maps for geography.
- Temperature and pressure maps in meteorology.
- Economic modeling to visualise changes in key economic indicators.

3. 3D Bar Charts

The addition of a third dimension to standard bar charts is what makes them 3D. To depict the three independent variables, they make use of bars with varying widths and depths. A category's worth of magnitude over several dimensions is shown by each bar.



Use Cases:

- Comparing Categories: Effective for comparing multiple categories across different dimensions.
- Displaying Multivariate Data: Useful for visualising and comparing data across three variables simultaneously.

Applications:

- Sales performance analysis across different regions and time periods.
- Comparing population demographics across different countries and age groups.

- Business intelligence for visualising financial metrics across departments and time frames.

4. 3D Line Graphs

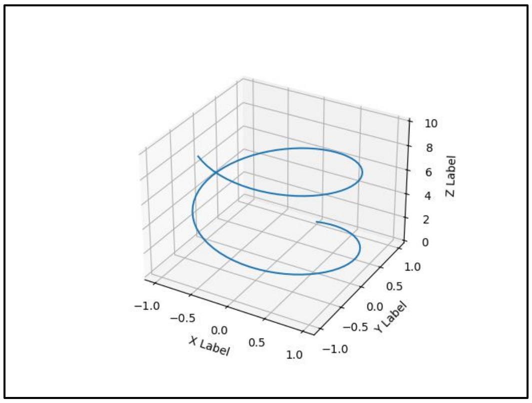
3D line graphs display relationships between three variables or trends over time by plotting lines in a three-dimensional space. The data points' paths over time or between situations are depicted by the lines in the graph.

Use Cases:

- Trend Analysis: Useful for visualising how data points change over time or conditions.
- Understanding Relationships: Helps in understanding how variables change and interact with each other in a 3D space.

Applications:

- Stock market analysis to show price movements over time for different stocks.
- Environmental studies to visualise changes in pollution levels over different regions and time periods.
- Engineering to monitor changes in system parameters over time.



5. 3D Pie Charts

Traditional pie charts are enhanced with a third dimension in 3D pie charts, which allows for a more comprehensive understanding of proportions and the relative importance of various pieces.

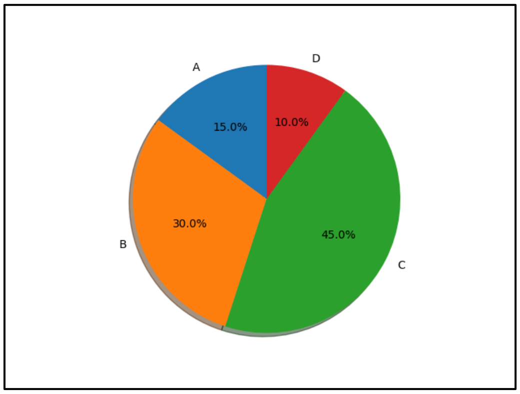
Use Cases:

- **Part-to-Whole Relationships:** Best for simple datasets to show the relationship between parts of a dataset and the whole.

- **Visual Appeal:** Adds a visual appeal to presentations and reports, though it can sometimes distort the true proportions.

Applications:

- Market share analysis to show the distribution of sales among different products.
- Budget allocation to visualise how funds are distributed across different departments.
- Survey results to display the proportion of responses in different categories.



6. Wireframe Models

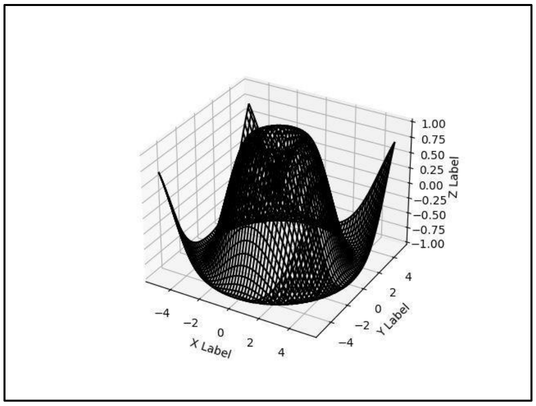
The vertices and lines that make up a wireframe representation serve as the skeleton of a three-dimensional data structure. The underlying geometric structures and forms are easily seen because they do not cover the surfaces.

Use Cases:

- **Understanding Geometry:** Commonly used in engineering and architectural design to understand complex geometric shapes.
- **Analysing Structures:** Helps visualise the underlying structure of complex data, making it easier to understand and analyse.

Applications:

- Engineering design to create and analyse prototypes of structures.
- Architectural planning to visualise building layouts and designs.
- Computer graphics for modeling and animation.



7. **Volume Rendering**

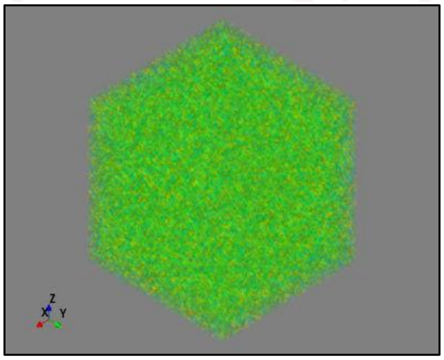
In volume rendering, each voxel (volumetric pixel) is given a value, allowing for the visualisation of 3D volumetric data. The data's internal structures can be observed through the semi-transparent 3D view that it generates.

Use Cases:

- **Medical Imaging:** Enables detailed visualisation of internal body structures, such as organs and tissues.
- **Scientific Research:** Useful for studying geological formations, meteorological data, and other complex 3D data.

Applications:

- MRI and CT scans in medical diagnostics.
- Geological surveys to visualise underground structures.
- Fluid dynamics research to study flow patterns.



8. **3D Heatmaps**

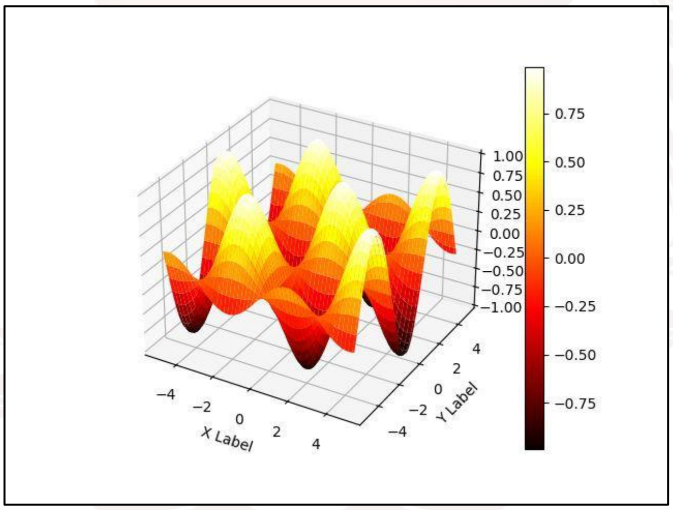
Colour is used in 3D heatmaps to depict the intensity or density of data in a three-dimensional space. Changes in colour saturation make high- and low-concentration areas stand out.

Use Cases:

- **Identifying Concentrations:** Effective for identifying areas of high and low concentration in data.
- **Pattern Recognition:** Helps visualise and identify patterns of concentration or density.

Applications:

- Epidemiology to track disease outbreak patterns.
- Traffic analysis to identify congestion points in transportation networks.
- Environmental monitoring to visualise pollution levels.



9. **Bubble Charts**

A third axis is added to bubble charts, and a fourth variable is represented by the size of the bubbles. Three variables define the position of each observation bubble, while a fourth variable determines its size.

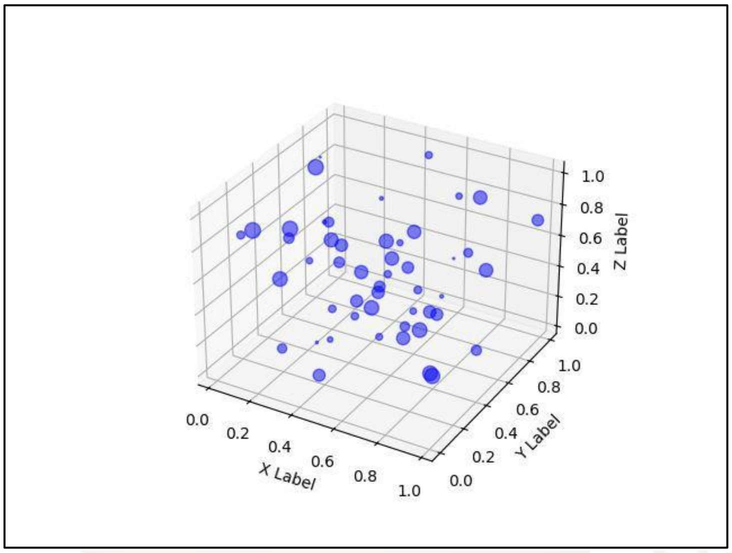
Use Cases:

- **Multivariate Analysis:** Useful for displaying relationships among four variables.
- **Identifying Patterns:** Helps visualise complex relationships between multiple variables.

Applications:

- Market analysis to visualise customer segments based on multiple criteria.
- Financial analysis to compare different investments based on risk, return, and other factors.

- Social science research to study relationships between demographic variables.



SELF-ASSESSMENT QUESTIONS – 3

Multiple Choose Questions	
11	**Which graph can be misleading if not viewed at the right angle?** a) 2D Bar Chart b) 3D Pie Chart c) Line Graph d) Scatter Plot
12	**A bubble chart is an extension of which graph?** a) Heatmap b) Histogram c) Scatter Plot d) Area Chart
13	**A pie chart is best used when:** a) You want to show parts of a whole b) You want to show trends c) You want to compare two variables d) You want to show distribution
14	**The Y-axis of a graph usually represents:** a) Independent variable b) Category labels c) Dependent variable

	d) Time intervals
15	**Which graph is most suitable for large datasets with matrix-type values?** a) Histogram b) Heatmap c) Pie Chart d) Bar Graph
16	**A column chart is different from a bar chart because it:** a) Uses circles instead of rectangles b) Uses horizontal bars c) Uses vertical bars d) Uses color gradients

4.2 Choosing the Right 3D Visualisation Technique

Selecting the appropriate 3D visualisation technique depends on several factors, including the type of data, its dimensions, distribution, and the specific analysis tasks. Here are some criteria to consider:

- **Data Type:** Different data types (numerical, categorical, temporal, spatial, relational, hierarchical) require different visualisation techniques. For example, scatter plots are suitable for numerical data, while network diagrams are better for relational data.
- **Data Dimension**: The number of variables or attributes to visualise influences the choice of technique. For instance, 3D scatter plots are ideal for three variables, while surface plots can represent more complex relationships.
- **Data Distribution**: Understanding the distribution of data (normal, skewed, bimodal, uniform, discrete) helps in selecting techniques that effectively represent patterns, trends, and outliers. Techniques like histograms and heat maps are useful for visualising data distribution.
- **Data Analysis Tasks:** The specific questions or hypotheses to be addressed determine the choice of visualisation technique. For example, parallel coordinates are useful for exploring correlations, while Sankey diagrams are effective for visualising flows and processes.

4.3 Advantages and Disadvantages of 3D Data Visualisation

Advantages:

- 3D data visualisation simplifies complex info, making it clear for everyone.
- Helps find hidden trends not visible in flat charts.
- Move and view data from different angles for better understanding.
- Explaining data becomes easy and straightforward.
- Data looks like real objects, making it relatable.
- Seeing all details helps in making better decisions.
- Makes data exploration enjoyable and engaging.

Disadvantages:

- 3D visualisations can sometimes make data interpretation more complicated than necessary.
- Adding the third dimension can lead to visual clutter, making it harder to focus on key information.
- Depth perception in 3D visualisations can sometimes distort the representation of data, leading to misinterpretation.
- Not all devices or platforms support 3D visualisations, limiting accessibility for some users.
- Creating and rendering 3D visualisations may require more computational resources, leading to slower performance.
- The third dimension may sometimes be overemphasised, leading to the neglect of other important data attributes.
- Interactive 3D visualisations may require more user training and expertise to navigate effectively.

Popular Tools for 3D Data Visualisation

- **Three.js:** Three.js is a JavaScript library that provides a powerful framework for creating 3D visualisations on the web. It leverages WebGL to render interactive 3D graphics, allowing users to create complex visualisations with minimal coding. Three.js includes built-in lights and materials, making it easier to create realistic 3D scenes.
- **Unity:** Unity is a real-time 3D development platform widely used for creating interactive 3D visualisations. It supports data integration and can visualise datasets directly within a 3D

environment. Unity is particularly popular in fields like architecture, engineering, and product design.

- **Chart 3D:** Chart 3D allows users to create 3D visualisations without coding. It supports various chart types, including scatter plots, geospatial plots, and line graphs. Users can import datasets, create interactive graphics, and switch between 2D and 3D views.

4.4 Applications of 3D Data Visualisation

- **Medical Imaging:** 3D data visualisation is extensively used in medical imaging to create detailed cross-sectional images of the human body. Techniques like MRI and CT scans benefit from 3D visualisations, aiding in diagnosing conditions and planning surgeries.
- **Architecture and Construction:** Architects use 3D visualisation tools to create detailed models of buildings, allowing stakeholders to visualise designs before construction begins. Real estate developers also use 3D visualisations to create virtual tours of properties.
- **Geographic Information Systems (GIS):** In GIS, 3D visualisation is used for terrain modeling, urban planning, and environmental analysis. It helps in understanding spatial relationships and simulating natural phenomena like flooding or erosion.
- **Entertainment and Gaming:** 3D visualisation is crucial in the entertainment industry for creating lifelike characters, environments, and special effects in animated movies and video games. It enhances the immersive experience for viewers and gamers.
- **Manufacturing and Engineering:** Engineers use 3D visualisation for product design, prototyping, and simulating manufacturing processes. It allows for detailed analysis and testing of designs before physical prototypes are built.

One effective method for delving into and making sense of large datasets is 3D data visualisation. Users may generate more in-depth insights and better decision-making tools with the use of cutting-edge methods and technologies that allow them to build interactive and immersive visualisations. 3D data visualisation has many uses and advantages in many fields, including engineering, architecture, GIS, entertainment, and medical imaging. Businesses may promote innovation and success by picking the proper strategies and technologies to unleash the full potential of their data.

SELF-ASSESSMENT QUESTIONS – 4

Multiple Choose Questions	
17	**Which graph is ideal for showing cumulative values?** a) Area Chart b) Scatter Plot c) Line Graph d) Histogram
18	**The X-axis in most 2D graphs typically represents:** a) Dependent variable b) Independent variable c) Frequency d) Category proportion
19	**Which graph is most suitable for showing both categories and sub-categories?** a) Stacked Bar Chart b) Pie Chart c) Line Graph d) Scatter Plot
20	**Which of the following is a disadvantage of 3D graphs?** a) They show more variables b) They increase clarity c) They can cause distortion and confusion d) They reduce visual clutter

5. SUMMARY

- Graphs are essential tools in data visualisation because they convert raw numerical data into visual formats that are easier to understand, interpret, and communicate.
- Data visualisation helps in identifying trends, patterns, and anomalies, enabling faster and more accurate decision-making compared to reading long numeric tables.
- Graphs reduce cognitive load by presenting data visually, making complex information accessible even to non-technical audiences.
- Two-dimensional (2D) graphs represent data using two axes—X and Y—and are the most commonly used chart types in business, education, and analytics.
- 2D graphs like line charts show changes over time, making them ideal for time series analysis such as stock prices, weather changes, or monthly sales.
- Scatter plots in 2D visualisation show relationships between two variables, helping analysts detect correlations, clusters, and outliers in datasets.
- Bar and column charts allow easy comparison between categories, especially useful for comparing quantities across groups such as products, departments, or regions.
- Histograms help visualise frequency distributions, showing how data is spread across ranges or bins—useful in statistical analysis, research, and quality control.
- Heatmaps act as powerful 2D tools for identifying patterns in large datasets by using color intensity to represent values, correlations, or density.
- Area charts expand line charts by highlighting the magnitude of values over time, making cumulative growth or total contribution easier to visualise.
- Two-dimensional graphs are simple, clean, and easy to interpret, making them suitable for dashboards, reports, educational material, and everyday analytical tasks.
- Three-dimensional (3D) graphs introduce a third axis, allowing visualisation of additional variables or depth information within the graph.
- 3D scatter plots help display relationships among three variables, adding depth and showing more complex interactions that are not visible in 2D.
- 3D surface plots display continuous data in a 3D landscape, useful in scientific, engineering, and mathematical fields for showing peaks, slopes, and gradients.
- 3D bar charts represent values using height, width, and depth, providing a richer visualisation but sometimes causing issues with readability.

- While 3D graphs can show more information, they must be used carefully because incorrect angles or perspectives can distort the data.
- 2D graphs are better for clarity, while 3D graphs are better for multi-variable analysis, making the choice dependent on data complexity and the intended audience.
- Graph selection depends on data type and analytical goal—for example, comparisons require bar charts, relationships require scatter plots, and distributions require histograms.
- Color, shape, and size enhancements can add additional dimensions to 2D graphs (e.g., bubble charts), combining simplicity with richer information.
- Effective graphs follow best practices such as avoiding clutter, choosing appropriate axis scales, using meaningful labels, and maintaining visual simplicity for accurate interpretation.

6. GLOSSARY

Data Visualisation	-	The process of representing data in graphical or pictorial form to make insights easier to understand.
2D Graph (Two-Dimensional Graph)	-	A graph that uses two axes (X and Y) to represent data.
3D Graph (Three-Dimensional Graph)	-	A graph that uses three axes (X, Y, Z) to show depth and represent more complex data.
Axis (X-axis / Y-axis / Z-axis)	-	Reference lines used to plot data points; X and Y for 2D, Z added for 3D.
Scatter Plot	-	A 2D graph that uses dots to show relationships between two numerical variables.
Line Graph	-	A graph that connects data points with lines to show trends over time.
Bar Chart	-	A graph with rectangular bars used to compare categorical data.
Column Chart	-	A vertical version of a bar chart, used to compare values across categories.
Histogram	-	A graph that shows frequency distribution of continuous data divided into bins.
Heatmap	-	A 2D matrix graph where colors represent the magnitude of data values.
Area Chart	-	A line graph where the area below the line is filled with color to show volume or cumulative trends.
Bubble Chart	-	A variation of a scatter plot using bubble size and color to represent additional variables.
Surface Plot	-	A 3D graph that represents data as a smooth surface formed over a grid.

3D Scatter Plot	-	A scatter plot extended into three dimensions to show relationships among three variables.
Data Points	-	Individual values plotted on a graph representing observations.
Bins (in Histograms)	-	Continuous ranges into which data is grouped for frequency display.
Trend	-	A pattern or direction of change observed over time in a graph.
Correlation	-	A statistical relationship between two variables, often seen in scatter plots.
Outliers	-	Data points that lie far away from the general pattern or trend.
Visual Clutter	-	Overcrowding in a graph that makes it difficult to read, usually caused by too many data points or elements.

7. TERMINAL QUESTIONS

1. Explain the role of graphs in data visualisation. Discuss why graphical representation is better than tabular representation.
2. Describe two-dimensional (2D) graphs in detail and explain different types with examples.
3. Discuss the advantages and limitations of 2D graphs. Provide examples where 2D graphs are most appropriate.
4. Explain scatter plots and their importance. How can they be extended into bubble charts? Give examples.
5. Define histograms and compare them with bar charts. How do histograms help in statistical analysis?
6. What are line graphs? Explain how they are used for time-series analysis with suitable examples.
7. Discuss heatmaps and their applications. How do they help in identifying patterns in large datasets?
8. Explain the concept of three-dimensional (3D) graphs. List different types and discuss their applications.
9. Compare 2D and 3D graphs in terms of clarity, complexity, and usability. Provide examples where each is preferable.
10. What is visual clutter? How does it affect data interpretation? Suggest ways to overcome it.
11. Discuss the importance of selecting the right type of graph. How does graph selection affect analysis and decision-making?
12. Explain how 2D and 3D graphs can be used together in a comprehensive data visualisation dashboard.

8. ANSWERS

8.1. Self-Assessment Answers:

1. Answer: b) Line Graph
2. Answer: c) Relationships between two variables
3. Answer: c) Histogram
4. Answer: a) Histogram
5. Answer: b) Colors
6. Answer: c) Comparing categories
7. Answer: b) Relationships among three variables
8. Answer: b) Scatter plot
9. Answer: c) Line Graph
10. Answer: c) Removing unnecessary elements
11. Answer: b) 3D Pie Chart
12. Answer: c) Scatter Plot
13. Answer: a) You want to show parts of a whole
14. Answer: c) Dependent variable
15. Answer: b) Heatmap
16. Answer: c) Uses vertical bars
17. Answer: a) Area Chart
18. Answer: b) Independent variable
19. Answer: a) Stacked Bar Chart
20. Answer: c) They can cause distortion and confusion

8.2. Terminal Question Answers

Answer 1: Graphs play a crucial role in data visualisation by converting raw numerical data into meaningful visual formats that are easier to understand, interpret, and communicate. Unlike tables, which require reading each number, graphs allow users to quickly identify trends, patterns, and anomalies at a glance. Graphs also reduce cognitive load, making complex data understandable even for non-technical audiences. They highlight relationships, comparisons, distributions, and summaries in a visually appealing manner. Visual representation speeds up decision-making, improves accuracy, and ensures that insights are communicated clearly. Therefore, graphs are preferred in reports, dashboards, business analytics, and research presentations. (Refer Section 2 for more details)

Answer 2: 2D graphs display data using two axes—the X-axis (horizontal) and the Y-axis (vertical). They are the most commonly used graphs in data visualisation. Examples include:

- **Line Graphs:** Show trends over time (e.g., daily temperature changes).
- **Bar/Column Charts:** Compare categorical data such as sales of different products.
- **Scatter Plots:** Show the relationship between two variables, such as study hours vs. marks.
- **Histograms:** Display frequency distribution of continuous data, such as age distribution.
- **Area Charts:** Highlight cumulative values across time.
  These graphs are ideal for simple, clear, and effective visual analysis.

Answer 3:

Advantages:

- Easy to read, create, and interpret
- Suitable for both numerical and categorical data
- Useful for showing trends, distributions, and comparisons
- Supported in all visualisation tools (Excel, Python, R, Tableau)

Limitations:

- Cannot handle more than two variables effectively
- Crowded data points cause visual clutter
- Not suitable for complex multidimensional data

Applications:

- Monthly sales trends → line graph
- Marks comparison → bar chart
- Height vs. weight → scatter plot 2D graphs are best for simple and medium-sized datasets.

Answer 4:

Scatter plots use dots to represent the relationship between two numerical variables. They are important because they help identify correlations, clusters, linear or non-linear patterns, and outliers. For example, plotting marketing spending vs. sales revenue shows whether spending influences sales. Scatter plots become **bubble charts** by adding a third variable, represented by the size of each bubble.

A fourth variable can be added through color. Example: Comparing countries using GDP (X), population (Y), and carbon emission levels (bubble size). (Refer Section

Answer 5:

Histograms display the frequency distribution of continuous data by grouping values into bins. Bar charts compare categorical data using separate bars.

Differences:

- Histogram bars touch; bar chart bars have gaps.
- Histograms show ranges; bar charts show categories.
- Histograms analyse distribution; bar charts analyse comparison.

Role in Statistics:

Histograms help identify data spread, skewness, central tendency, and outliers. They are essential in quality control, probability studies, and survey analysis.

Answer 6:

Line graphs connect data points with straight lines and are ideal for representing continuous data over time. They help identify trends, cycles, seasonal patterns, and rate of change. **Examples:**

- A company tracking monthly revenue
- Monitoring heart rate over time
- Weather temperature changes during 24 hours Line graphs make it easy to observe rises, falls, and long-term behavior.

Answer 7:

Heatmaps use color intensity to represent numeric values in a 2D matrix. Dark or bright colors represent higher values, making patterns visible instantly. **Applications:**

- Correlation analysis in data science
- Attendance tracking
- Website click heatmaps

- Environmental data monitoring

They reveal clusters, hotspots, and strong or weak relationships that are difficult to identify in raw data or tables.

Answer 8:

3D graphs include an additional **Z-axis**, allowing visualisation of variables with depth. **Types:**

- 3D Scatter Plots
- 3D Surface Plots
- 3D Bar Charts

Applications:

- Scientific research for visualising terrain, chemical reactions, or temperature gradients
- Engineering for structural design
- Finance for risk analysis involving multiple variables

3D graphs are powerful for complex data but require careful angle selection to avoid distortion.

Answer 9:

2D Graphs:

- Clear, simple, easy to interpret
- Ideal for beginners and general presentations
- Example: Daily rainfall → line graph

3D Graphs:

- Handle more variables
- Suitable for advanced scientific or engineering data
- Can be confusing if overused
- Example: Visualising temperature variation over X-Y coordinates → 3D surface plot Conclusion: Use 2D for simplicity; use 3D for multidimensional datasets.

Answer 10 : Visual clutter occurs when a graph contains too many elements—points, colors, labels, or overlapping lines—making it hard to read.

Effects:

- Misleading interpretation
- Difficulty finding patterns
- Poor communication of insights

Solutions:

- Remove unnecessary gridlines
- Use fewer colors
- Group or filter data
- Switch to a different graph type
- Use summary statistics Clean visuals ensure accurate interpretation and effective communication.

Answer 11:

Choosing the correct graph is essential because different graphs serve different purposes:

- Line graphs → show trends
- Bar charts → compare categories
- Scatter plots → show relationships
- Histograms → analyse distributions

Wrong graph selection can misrepresent data, confuse viewers, and lead to poor decisions. The right graph enhances clarity, accuracy, and impact. Effective visualisation supports strategic planning, performance tracking, forecasting, and scientific understanding. (Refer Section 4.2 for more information)

Answer 12:

A dashboard often combines multiple graph types to display different aspects of data:

- **2D graphs** show trends, distributions, and comparisons clearly.
- **3D graphs** display complex relationships and multidimensional patterns.

Example in business analytics:

- 2D line graph for monthly sales trend
- 2D bar chart for category-wise comparison
- 3D surface plot for forecasting demand

Combining both provides a holistic view, enabling detailed analysis and better decision-making.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 4

DCA2207

DATA MINING AND VISUALISATION

Unit – 11

Introduction to Dashboard

TABLE OF CONTENTS

1. INTRODUCTION

In previous unit, learners explored the essential role of graphs in data Visualisation and understand how graphical representation transforms raw data into meaningful insights. The unit begins with the concept of graph roles, explaining how visuals such as charts, plots, and diagrams help identify trends, comparisons, patterns, and relationships. Students then studied about two-dimensional (2D) graphs, including their structure, usage, advantages, and limitations. They learnt when 2D graphs are most effective—for example, in showing trends over time, comparing categories, or analysing distributions—and examine why 2D visuals remain the foundation of most analytical dashboards and reports.

The unit then expanded into three-dimensional (3D) graphs, introducing learners to advanced Visualisation techniques that incorporate depth and additional variables. Students explored key 3D methods, how to choose the right 3D Visualisation technique, and the advantages and disadvantages of using 3D representations. Real-world applications—such as scientific modeling, engineering simulations, geographical mapping, and complex data analysis—are also covered. By the end of the unit, learners gained the skills to select, interpret, and apply both 2D and 3D graphs effectively, enabling clearer communication, deeper analysis, and more informed decision-making in data-driven environments.

In this unit, learners will gain a complete understanding of data Visualisation dashboards and their role in presenting information clearly and effectively. The unit begins by introducing dashboards and explaining why they are essential for data-driven decision-making in modern analytics. Students will learn about the key components that make a dashboard functional, including KPIs, data sources, visual charts, legends, layout design, and interactive filters.

Learners will also explore different types of dashboards such as operational, analytical, strategic, and informational, while understanding how each serves a unique purpose in real-time monitoring or in-depth analysis. The unit highlights the benefits of dashboards, including improved decision-making, faster insight generation, simplification of complex data, and enhanced collaboration across teams. Additionally, students will study best practices for designing dashboards—covering simplicity, consistency, appropriate Visualisation selection, interactivity, and iterative improvement based on user feedback.

Finally, the unit introduces various tools and platforms used to build dashboards, such as Power BI, Tableau, Looker Studio, Excel, Qlik, Python libraries, R Shiny, Apache Superset, and Grafana. By the end of the unit, learners will have strong knowledge of how dashboards are structured, created, interpreted, and applied in real-world scenarios.

1.1. Learning Objectives

By the end of this unit, you will be able to:

- **Explain** the concept, purpose, and significance of dashboards in data Visualisation.
- **Describe** the importance of dashboards
- **Identify** and explain the core components of a data Visualisation
- **Differentiate** and classify various types of dashboards

2. DASH BOARD

Businesses and organisations in this age of information overload are always seeking new methods to decipher the massive amounts of data they collect and produce. By transforming data into visual representations such as maps, charts, and graphs, organisations are able to uncover important insights and trends that could otherwise be missed. This is when visualising data becomes useful. Data visualisation dashboards are crucial for firms seeking a competitive advantage in today's data-driven world. In this post, we will explore the concept, their significance, and why they are needed.

A **dashboard** is a data Visualisation tool that brings together key information from different sources and presents it on a single screen for quick decision-making. It uses visual elements like **bar charts, pie charts, line graphs, KPIs, and indicators** to summarise performance or trends. One kind of user interface that displays key metrics, trends, and data-driven conclusions graphically is a dashboard for data visualisation. Think of it as a central location where all the relevant information is stored so that those involved can easily access it as needed to monitor progress, assess performance, and make informed decisions.

Typical components of such dashboards include maps, gauges, graphs, charts, and other visual elements organised in a logical fashion. Users can explore different views, add filters, and focus on certain data subsets to gain deeper insights from the dashboard. Data visualisation dashboards let stakeholders see what's happening in the company's data environment and provide a simple way to sort through mountains of data to find what they're looking for.

A dashboard brings together multiple graphs, charts, and visuals in one place, making it easy to monitor important information at a glance. It helps users track performance, progress, or trends by displaying key metrics clearly and visually. Dashboards often update automatically whenever the data changes, ensuring that the information shown is always current and accurate. Designed for simplicity, they are easy to read and interpret even for non-technical users. Overall, dashboards support quick and effective decision-making by presenting essential insights in a single, organised view.

A data Visualisation dashboard is a user interface that displays key performance indicators (KPIs) and other important data through charts, graphs, and maps. It consolidates information from multiple sources into a single, easy-to-understand view, allowing users to monitor progress, identify trends, and make informed decisions quickly. These dashboards are interactive, often allowing users to filter data for deeper analysis.

Key aspects of a data Visualisation dashboard

- **Centralised view:** A dashboard brings together various but related data sets into one place, providing a high-level overview of performance.
- **Visual format:** It uses a variety of Visualisations like charts, graphs, gauges, and maps to make complex data easy to digest.
- **At-a-glance information:** Dashboards are designed to show key metrics and trends immediately, so users don't have to dig through raw data.
- **Interactivity:** Many dashboards allow users to interact with the data by applying filters, drilling down into specifics, or changing the time range.
- **Decision support:** By providing a clear and current picture of performance, dashboards help businesses spot problems, identify opportunities, and make data-driven decisions.
- **Customisable:** Dashboards are highly adaptable and can be tailored to display information relevant to different industries, roles, or goals.

2.1 Importance of Data Visualisation Dashboards

Dashboards for data Visualisation are essential for promoting an insights-driven culture and data-driven decision-making in businesses. This is the reason they are essential:

- Enhanced Understanding: Stakeholders may quickly identify trends, patterns, and linkages by looking at visual representations of complex data.
- Real-Time Insights: Dashboards give stakeholders with up-to-date information by linking to real-time data sources. This allows stakeholders to monitor performance in real-time and react quickly to changing situations.
- Better Decision-Making: Stakeholders can more rapidly recognise possibilities and obstacles when data is Visualised, which empowers them to make well-informed decisions supported by research and analysis.
- Enhanced Transparency: By providing data accessibility to all stakeholders inside the organisation, dashboards encourage accountability and trust.
- Effective Communication: By displaying data in a way that is simple to share and comprehend, Visualisation's help stakeholders to coordinate their efforts towards shared objectives.

To put it simply, data Visualisation dashboards enable businesses to fully utilise the potential of their data assets, spurring productivity, creativity, and a competitive edge in the fast-paced corporate world of today.

SELF-ASSESSMENT QUESTIONS – 1

Multiple Choose Questions	
1	**A dashboard is primarily used to:** a) Store data in databases b) Display data visually for quick understanding c) Write code for data processing d) Secure network systems
2	**Which of the following best describes the importance of a dashboard in data Visualisation?** a) It complicates the analysis process b) It hides key metrics and trends c) It helps users interpret data quickly and make decisions faster d) It is useful only for technical users
3	**Dashboards are useful because they:** a) Replace all types of reports b) Present complex information in a simple graphical format c) Eliminate the need for data analysis entirely d) Work without any data source
4	**Which feature is commonly seen in dashboards?** a) Text-only information b) Manual data extraction for every update c) Visual charts, KPIs, reports, and summary metrics d) Support for only one type of data
5	**Why are dashboards significant in business decision-making?** a) They slow down data analysis b) They provide real-time insights and a clear view of performance c) They work only with historical data d) They replace human decision makers

3. COMPONENTS OF A DATA VISUALISATION DASHBOARD

A dashboard is a structured visual interface that displays critical information and insights using charts, metrics, and analytical elements. It allows users to monitor performance, detect patterns, and make decisions quickly. A well-designed dashboard is built using multiple components, each serving a specific purpose to enhance readability, clarity, and decision-making efficiency.

a. Data Source:

This is the source from which the dashboard gets its data. We can pull data from a variety of sources, including databases, spreadsheets, APIs, web services, cloud storage, and live streaming platforms. Accuracy and up-to-the-minute updates are guaranteed by a trustworthy data source. Databases in SQL, spreadsheets in Excel, data flows from Internet of Things sensors, customer relationship management systems, etc.

Some examples of these sources are databases, spreadsheets, cloud services, and application programming interfaces (APIs). Widgets for data visualisation include things like charts, graphs, maps, and tables. By showing key indicators and trends, these widgets provide stakeholders with useful information. With the help of the dashboard's filters and interactivity features, users may explore specific data subsets, apply filters, and compare different perspectives. Interaction facilitates deeper data research and increases user engagement.

b. Data Processing Layer:

Before data is presented visually in a dashboard, it must first undergo processing to ensure accuracy and clarity. Raw data collected from various sources often contains inconsistencies, missing values, and unstructured formats, making it unsuitable for direct Visualisation. Therefore, a data processing layer is necessary, where the information is cleaned, organised, and prepared for meaningful analysis.

This stage typically involves ETL (Extract, Transform, Load) operations which convert raw data into a usable state. The process includes removing duplicate entries, handling missing or incomplete values, summarising or grouping relevant data for easier interpretation, and converting data into consistent formats. These steps help maintain data quality and reliability, ensuring that the final Visualisations accurately represent the information and support better decision-making.

c. Visual Elements (Graphs & Charts):

The visual elements are the core components of a dashboard, as they represent data in graphical form and make information easier to understand. Different types of charts are used depending on the nature and purpose of the data being analysed. Line and area charts are effective for showing trends over time, while bar and column charts are suitable for comparing categories or groups. Pie and donut charts help illustrate part-to-whole relationships, and scatter plots are useful for showing correlations between variables. Heatmaps visually represent data intensity or density, whereas gauges or KPIs highlight progress toward specific goals. Using a combination of these visual elements enhances clarity, enabling users to interpret data quickly and accurately.

d. Key Performance Indicators (KPIs):

Key Performance Indicators (KPIs) are summary metrics displayed on a dashboard to provide quick insights into performance levels. They typically appear as numerical values, icons, or small visual indicators and are often placed at the top of the dashboard for easy visibility. KPIs highlight the most important data points that users need to monitor regularly, such as total revenue, attendance percentage, monthly website traffic, or the number of resolved service tickets. By presenting crucial performance measures in a clear and concise format, KPIs enable decision-makers to evaluate progress and assess success at a glance, without needing to analyse detailed reports.are high-level indicators used to assess how well a department, process, or organisation is performing. KPI widgets are frequently used in dashboards to give stakeholders quick insights into important areas of interest.

e. **Annotations and Insights:**

Annotations and insights help users understand the significance of the results and find practical applications by providing background information and commentary on the given data. Legends, labels, and annotations play a vital role in enhancing the readability and clarity of dashboard Visualisations. Labels are used to name axes, data values, and categories, ensuring users understand what each unit or measurement represents. Legends explain the meaning of colors, symbols, or line patterns used within charts, helping distinguish between multiple datasets or variables displayed together. Annotations draw attention to key insights, highlight sudden increases or decreases, and mark important events or warnings within a graph. These elements work together to make visual interpretation easier, preventing confusion and ensuring that users can derive accurate conclusions from the dashboard.

f. **Options for Customisation:**

A lot of dashboards come with options that let users change the color's, design, and layout to fit their tastes. Customisation improves usability and guarantees that the dashboard satisfies each user's unique requirements. Dashboards often include interactive controls such as dropdowns, search bars, sliders, and date range filters, which allow users to Customise the displayed information according to their specific needs. These filtering features make the dashboard dynamic, flexible, and user-friendly by enabling viewers to focus only on relevant data. For example, users may filter sales reports by product category, region, or time period, select specific departments or age groups for analysis, or zoom in and out of a data range to observe finer details. Such interactive components enhance the usability of the dashboard and support more accurate and personalised data interpretation.

g. Layout & Design Structure:

Another important component in dashboards is the refresh and real-time update system. Many dashboards are connected to live data sources, such as market feeds, IoT devices, analytics platforms, or network monitoring tools. Automatic refresh functionality ensures that users always view the latest metrics without manual updates. This is especially valuable for use cases like stock market tracking, website traffic monitoring, and live sensor data Visualisation where conditions change rapidly. Real-time updates help decision-makers respond quickly and appropriately based on current information.

h. Export, Share & Reporting Options:

Dashboards also provide export, sharing, and reporting options, enabling users to generate offline copies of visual data in various formats such as PDF, Excel, PPT, or CSV. Sharing features allow teams to collaborate effectively by distributing reports, screenshots, or full dashboard access during meetings or planning sessions. These options support documentation, performance reviews, and comparison of progress over time, making dashboards not just analytical tools but also valuable communication resources.

SELF-ASSESSMENT QUESTIONS – 2

Multiple Choose Questions	
6	**Which of the following is considered the core visual element of a dashboard?** a) Data storage server b) Charts and graphs

	c) Browser settings d) Programming compiler
7	**What is the main purpose of KPIs (Key Performance Indicators) in a dashboard?** a) To store raw data b) To summarise key metrics for quick insight c) To write code for data processing d) To encrypt dashboard information
8	**Which component allows users to explore and Customise the data shown on a dashboard?** a) Filters and interactive controls b) Operating system c) Network switch d) Text-only reports

4. TYPES OF DATA VISUALISATION DASHBOARDS

Dashboards can be categoriseded based on their purpose, the type of information they present, and how users interact with the data. The main types include **Operational Dashboards, Analytical Dashboards, Strategic Dashboards, and Informational Dashboards**. Each type serves a different function and is used for different levels of decision-making within an organisation.

1. **Operational Dashboards**

Operational dashboards provide a real-time view of day-to-day business processes. They are designed for quick monitoring and response, helping teams track current performance and take immediate action when needed. These dashboards show time-sensitive data and are usually updated frequently or continuously. Operational dashboards allow frontline employees to track daily tasks and their progress towards operational objectives.

Operational dashboards are designed to display real-time or near real-time data, presenting current performance metrics and status updates that help monitor ongoing processes. They often include alerts and notifications to draw attention to sudden changes or issues that require quick response. Examples of such dashboards include live website visitor tracking, server health and uptime monitoring, customer support ticket status, and IoT device performance analytics. Because they support rapid observation and immediate decision-making, operational dashboards play a crucial role in areas like IT services, production environments, and customer support where timely action is essential.

2. Analytical Dashboards

Analytical dashboards help users interpret data, discover patterns, perform comparisons, and explore insights. They are not just snapshots—they support deep data analysis using drill-down features, filters, and trend Visualisations. Analysts and managers often use them for decision-making based on historical data.

Analytical dashboards are designed to help users analyse historical data and identify meaningful patterns over time. They support advanced functionalities such as filtering, segmentation, and drill-down exploration, enabling users to examine detailed insights at various levels. These dashboards are particularly useful for forecasting, trend analysis, and performance evaluation. Common examples include sales performance monitors with month-on-month comparison, market trend analysis tools, and dashboards that display profit versus cost breakdown over time. By providing deep and interactive data exploration, analytical dashboards help users understand the reasons behind outcomes, predict future behavior, and make well-informed strategic decisions.

3. Strategic Dashboards

Strategic dashboards are used by higher management for long-term planning and performance evaluation. They summarise high-level business objectives and KPIs, offering a bird's-eye view rather than detailed analytics. These dashboards are less frequently updated—weekly, monthly, or quarterly.

Strategic dashboards focus on presenting high-level Key Performance Indicators (KPIs) that support long-term planning and evaluation. They provide a summarised view of organisational performance, relying more on aggregated and consolidated data rather than real-time details. Examples of strategic dashboards include quarterly revenue versus target comparisons, annual employee performance analytics, and visual summaries of business growth and expansion trends. These dashboards are designed for senior management and leadership teams, enabling them to assess progress, set future goals, and make strategic decisions that guide the organisation's long-term direction and development.

4. Informational Dashboards

These dashboards are designed for presenting information rather than decision-making. They are mostly static visual boards used to share insights with students, teams, or general audiences to improve awareness and understanding.

Informational dashboards are primarily designed to display information rather than perform deep analytical processing. They focus on visual storytelling and presenting summarised insights in a clear and easily understandable manner. Since these dashboards are often used for awareness or explanation, they may not require real-time data refresh. Examples include dashboards showing regional COVID case summaries, educational or research dashboards used for classroom instruction, and visual data displays presented during meetings or seminars. Informational dashboards are widely used in academic settings and training environments as they help communicate data clearly and enhance overall understanding.

5. Hybrid Dashboards (Combination of Types)

Many modern dashboards combine features from two or more of the above types. For example, an analytical–operational hybrid may provide real-time data along with drill-down insights. These hybrid dashboards are becoming more popular as industries require both quick updates and deep analysis within a single interface.

Different dashboards support different levels of data interpretation and decision-making. **Operational dashboards** track real-time activities, **analytical dashboards** help investigate patterns and forecasting, **strategic dashboards** guide long-term planning, and **informational dashboards** enhance knowledge dissemination. Understanding these dashboard types helps in selecting and designing the right one based on organisational goals and query requirements.

6. Real-Time Dashboards

Real-time dashboards are designed to deliver continuously updated insights, allowing users to monitor critical metrics and trends as they occur. These dashboards are connected to live data streams, such as stock market feeds, network monitoring systems, IoT sensors, or website analytics tools. Because the information refreshes automatically without manual updates, real-time dashboards help organisations respond quickly to changes, emerging issues, or sudden fluctuations. For example, businesses can track live sales performance, application failures, server load, or customer activity and make immediate operational decisions. In fast-paced environments such as cybersecurity, e-commerce, logistics, and industrial automation, real-time dashboards are essential for timely intervention, risk mitigation, and maintaining smooth operations.

7. Mobile Dashboards

Mobile dashboards are specifically designed to be viewed on smartphones and tablets, ensuring users have access to vital information anytime and from anywhere. They feature simplified layouts, high-contrast visuals, and touch-friendly elements to make navigation easy on smaller screens. Mobile dashboards are particularly useful for managers, field workers, or decision-makers who need constant access to analytics even when away from their workstation. Whether checking business progress during travel, monitoring IoT device performance remotely, or receiving alerts about system failures, mobile dashboards keep users informed and engaged at all times. Their flexibility and portability promote faster decision-making, improve communication, and support real-time workflow management across distributed teams.

4.1 Benefits of Data Visualisation Dashboards

Data Visualisation dashboards offer significant benefits to organisations and stakeholders by transforming raw data into easily understandable visual insights. One of the greatest advantages is better decision-making, as dashboards present key metrics, trends, and performance indicators in a clear graphical format, enabling stakeholders to analyse information quickly and take informed actions without delay. They also contribute to improved understanding, as visual representation makes it easier to identify patterns, correlations, and deviations that may not be evident in textual or numerical data tables. This simplifies complex information and supports deeper analytics even for non-technical users.

Dashboards also enhance operational efficiency by streamlining data analysis processes. Instead of manually extracting and interpreting information, users can access real-time insights instantly through a single interface. This saves time, reduces effort, and enables quicker interpretation of business conditions. Real-time monitoring further adds value by allowing continuous tracking of essential indicators such as sales performance, system health, market fluctuations, or customer activity. With immediate visibility into changes, stakeholders can respond promptly to issues or opportunities.

In addition, dashboards foster collaboration and teamwork by serving as a Centralised platform for sharing insights across departments. Teams can jointly view performance trends, discuss findings, and make collective decisions based on unified data. This encourages transparency, improves communication, and supports a culture of data-driven decision-making across the organisation.

Organisation's and stakeholders can benefit greatly from data Visualisation dashboards.

- **Better Decision-Making:** By offering concise, graphical insights into important metrics and trends, dashboards help stakeholders make well-informed decisions more rapidly.
- **Improved Understanding:** Stakeholders can more easily see trends, patterns, and correlations when presented in a visual format, which simplifies complex information.
- **Enhanced Efficiency:** By streamlining data analysis procedures, dashboards enable quick and effective information access and analysis for stakeholders.
- **Real-Time Monitoring:** By giving stakeholders instantaneous insights into important metrics and trends, real-time dashboards help them keep an eye on performance and react quickly to changing circumstances.
- **Promote Collaboration:** By offering a Centralised forum for exchanging and debating data-driven insights, dashboards promote communication and teamwork.

5. BEST PRACTICES TO DESIGN DASHBOARD

Designing an effective data Visualisation dashboard requires both visual clarity and analytical purpose, ensuring that users can interpret information quickly and accurately. The first and most important practice is defining the objective clearly. A dashboard should be planned based on what decisions it needs to support, who the users are, and which metrics they must monitor. Without a clear goal, even a visually attractive dashboard may fail to provide value.

Another essential practice is ensuring simplicity and clarity in layout. A dashboard should not appear cluttered with excessive charts, colors, or details. Instead, visuals must be arranged logically, so users can follow information flow easily from most important insights to secondary details. Consistent color schemes, readable fonts, proper spacing, and visual hierarchy guide the viewer's eye in the right direction.

Choosing the right Visualisation for the right data is equally crucial. Line charts work best for trends, bar charts for comparisons, pie charts for proportions, and scatter plots for correlations. Using inappropriate chart types can mislead interpretation and reduce insight quality. Dashboards should also include legends, labels, and tooltips to help users understand the meaning of symbols, colors, and data points without confusion.

Interactivity is another best practice, achieved through filters, drill-downs, and dynamic selections that allow users to explore data at deeper levels. This helps transform the dashboard from a static information board into an analytical tool. Additionally, dashboards should incorporate KPIs and summary metrics for quick assessment, supported by detailed visuals if deeper investigation is needed.

Real-time or timely refresh capabilities are important when monitoring live operations such as network traffic, financial markets, or website analytics. Ensuring data accuracy and reliability must be prioritized, as inaccurate data leads to wrong conclusions. Dashboards should also be optimized for performance and device compatibility, especially when viewed on mobile or low-bandwidth systems.

Finally, user feedback should be considered during development. Regular refinement ensures that dashboards evolve with business needs. In summary, a well-designed dashboard is one that is clear, purposeful, interactive, visually balanced, and driven by accurate data—serving as a powerful decision-support tool rather than just a display of visuals.

In order to guarantee usability and efficacy while creating data Visualisation dashboards, it is imperative to adhere to recommended practices:

- **Know Your Audience:** Create a dashboard that satisfies the unique wants and preferences of your target audience by getting to know them.
- **Maintain Simplicity:** Steer clear of complexity and clutter. Make an effort to convey the most pertinent facts in an understandable and efficient way.
- **Selecting the Correct Visualisation's:** Make sure the Visualisation's you choose are appropriate for the kind of data you are presenting and the messages you want to get across.
- **Ensure Consistency:** To increase readability and usability, keep design components like color's, typefaces, and layout consistent.
- **Provide Interactivity:** To improve user engagement and exploration, include interactive elements like filters, drill-downs, and hover-over tooltips.
- **Iterate and Improve:** To resolve any usability difficulties or shortcomings, iterate the dashboard design while continuously gathering user feedback.

Organisation's may develop meaningful, user-friendly, and intuitive data Visualisation dashboards that help stakeholders reali Organisations may develop meaningful, user-friendly, and intuitive data Visualisation dashboards that help stakeholders realise e the full value of their data assets by adhering to these best practices.

6. TOOLS AND PLATFORMS FOR BUILDING DATA VISUALISATION DASHBOARDS

Data Visualisation dashboards can be developed using a variety of tools and platforms depending on the complexity, scale, skills of the user, and deployment requirements. These tools enable the conversion of raw data into interactive, visually compelling dashboards that support decision-making and analytical insights. They range from beginner-friendly drag-and-drop applications to advanced coding-based frameworks used in professional data analytics workflows

Together, these tools provide diverse capabilities—from quick business dashboards to advanced analytical modeling—enabling users to select platforms based on their data scale, processing needs, and Visualisation complexity. Dashboard Tools and Platforms for Data Visualisation Numerous platforms and technologies exist to facilitate the development of data visualisation dashboards, catering to a wide range of user and technical requirements:

- **Tableau:** Tableau is an excellent data visualisation platform since it has many features for making interactive dashboards that look great. It is compatible with a wide variety of data sources and has powerful analytics capabilities.
- **Power BI**: Power BI, developed by Microsoft, is a popular BI tool that enables users to integrate data from various sources in order to produce interactive dashboards and reports. It allows for seamless integration with other Microsoft offerings.
- **Google Data Studio:** Users may build personalised dashboards and reports with data from Google products like Analytics, Google Sheets, and BigQuery using the free app Google Data Studio. It has a user-friendly interface and works well with other Google products.
- **Domo:** The powerful data visualisation capabilities of Domo make it an all-inclusive BI solution. It provides pre-built interfaces to numerous data sources in addition to robust analytics tools.
- **Looker:** Looker is a data visualisation and exploration tool that integrates well with cloud databases and data warehouses. Users are able to create dynamic dashboards and reports, and it excels at modelling.
- **Qlik Sense:** Qlik Sense is a platform for self-service analytics and data visualisation that allows users to create interactive dashboards and reports. With its help, you may process data in real time and take use of its advanced data discovery capabilities.

Dashboards for data visualisation can be built using any number of resources and platforms, not limited to the ones listed above. When deciding which tools to utilise, several factors are considered, including the available money, technical requirements, and the specific use case.

Use Cases and Examples of Data Visualisation Dashboard

Data visualisation dashboards have application in a wide range of industries and occupations. Examples and use cases can be found here:

Sales and Marketing: Teams in sales utilise dashboards to keep tabs on key performance indicators including revenue, conversion rates, and cost per client acquisition. Using dashboards, marketing teams may analyse campaign results, website traffic, and social media engagement.

Finance and Operations: The finance department uses dashboards to monitor key performance indicators (KPIs) such as revenue, expenses, and profitability. Production key performance indicators, inventory levels, and supply chain performance may all be tracked by operations teams through the use of dashboards.

Healthcare: Healthcare organisations utilise dashboards to analyse patient data, track treatment outcomes, and observe public health trends. Dashboards are useful tools that can help healthcare providers identify areas for improvement and better serve their patients.

E-commerce: Online retailers utilise dashboards to track sales, analyse customer actions, and fine-tune product suggestions. Dashboards have the ability to disclose data regarding consumer preferences, purchasing patterns, and demographics.

Human Resources: Dashboards are used by HR departments to track workforce diversity, assess employee performance, and analyse recruitment factors. Human resource managers may make better data-driven decisions about training, hiring, and retention with the help of dashboards.

6.1 Challenges and Considerations

Data visualisation dashboards offer numerous benefits, but they also come with some considerations and limitations:

Data Quality: Ensuring data accuracy and consistency is crucial for any analysis to have any real value. The accuracy of findings and insights might be compromised by low-quality data.

Data security: Preventing breaches and other online threats to sensitive data is of the utmost importance. In order to safeguard their data assets, organisations must implement robust security procedures.

Scalability: When the amount of data increases, this becomes a problem. In order to handle enormous amounts of data, businesses need to use systems and technologies that are scalable.

User acceptance: Making sure users are engaged and accept data visualisation initiatives is crucial to their success. It is the responsibility of businesses to provide users with the training and support they need and to tailor dashboards to their specific requirements.

Visualisation Best Practices: Following standards for data visualisation is essential when developing dashboards. Organisations should think about things like labelling, colour coding, and chart choices to make sure everything is clear and easy to understand.

By resolving these difficulties, organisations can improve their data visualisation dashboards' performance and decision-making capabilities.

While data Visualisation dashboards bring powerful advantages to organisations, they also come with certain limitations and considerations that must be addressed to ensure meaningful insight generation. One of the most important factors is data quality, as dashboards are only as reliable as the information they display. If the underlying data is inaccurate, incomplete, or inconsistent, the output and insights may be misleading. Along with quality, data security is another significant concern. Since dashboards often deal with sensitive business information, protecting this data from breaches, unauthoriseded access, and cyber threats becomes a top priority. Organisations must adopt strong security protocols and access controls before integrating dashboards into decision processes.

As data grows in volume and complexity, scalability also emerges as a challenge. Dashboards should be supported by scalable technologies capable of handling large datasets without performance issues. Equally important is user acceptance, as even the most advanced dashboard is ineffective if users are unwilling or unable to use it. Employees should be trained to interpret visuals confidently, and dashboards must be Customised according to user needs and skill levels. Additionally, following Visualisation best practices is essential when designing dashboards. Appropriate use of color coding, labelling, layout structure, and suitable chart types ensures clarity and prevents misinterpretation of data. By addressing these considerations, organisations can maximise dashboard effectiveness and ensure smooth integration into data-driven environments.

Difference Between Data Visualisation and a Dashboard

- **Scope:** A dashboard is a collection of several data Visualisation's and summaries intended for monitoring and analysis, whereas data Visualisation is any type of graphical data representation.
- **The purpose** of data Visualisation's is to provide insights through analysis of particular data sets. Dashboards give important indicators a thorough overview and enable ongoing monitoring.
- **Interaction:** Dashboards provide a higher level of interaction than individual data Visualisation's because they let users filter data, delve down into specifics, and Customise perspectives.

6.2 Characteristics of a Good Data Visualisation Dashboard

The following characteristics should be present in an effective data Visualisation:

- **Accuracy:** The data is presented truthfully, devoid of any distortion or misrepresentation. To accurately represent actual values, the graphic components should be resized.
- **Clarity:** The material is presented in an easily understood manner. It should be easy for users to understand the insights right away.
- **Relevance:** It concentrates on the information that matters most to the target audience. Reducing irrelevant details is advisable.
- **Simplicity:** It employs simple designs and stays away from needless complication to successfully communicate the idea.
- **Engagement:** It draws in viewers and promotes conversation and investigation.
- **Consistency:** To make it simpler for people to understand the data, it employs consistent design elements including typefaces, colors, and chart kinds.

An effective data Visualisation must include certain key characteristics that ensure information is communicated accurately and meaningfully. **Accuracy** is the most essential element, as the Visualisation should represent data truthfully without exaggeration or distortion. The sizing of graphical elements must reflect the actual values in order to avoid misleading interpretations. Along with accuracy, **clarity** plays a crucial role—visuals should be easy to read and insights should be immediately understandable to the viewer. An effective Visualisation also prioritiseses **relevance,** presenting only the information that serves the purpose and is useful to the target audience while removing any distracting or unnecessary details. **Simplicity** further enhances comprehension by using clean design, minimal clutter, and straightforward visual elements to convey the message effectively. Additionally, a good Visualisation must ensure **engagement,** drawing the viewer's attention and encouraging exploration or interactive analysis. Finally, **consistency** in design features such as colors, fonts, and chart types helps users interpret the data smoothly and prevents confusion. When these characteristics are applied together, Visualisations become more powerful, insightful, and capable of supporting meaningful decision-making.

Clarity and Efficiency in Communication

In data Visualisation, efficiency and clarity are crucial. Here are a few methods for achieving them:

- **Use the Appropriate Chart Types:** Select the appropriate chart or graph type based on the data being shown. Use scatter plots for correlations, line charts for trends over time, and bar charts for comparisons.
- **Reduce Clutter:** Eliminate any extraneous components that don't aid in the comprehension of the facts. Refrain from using too many colors, grid lines, and writing.
- **Highlight Important Findings:** To highlight the most significant data points and patterns, use visual signals such strong fonts, contrasting colors, and annotations.
- **Provide Context:** To help users comprehend what they are looking at and to provide context, include labels, legends, and titles.

SELF-ASSESSMENT QUESTIONS – 3

Fill In The Blanks	
9	A well-designed dashboard should be created by first understanding the ________ and their requirements.
10	Maintaining ________ in colors, fonts, and layout improves readability and user experience in dashboards.
11	Dashboards should use ________ Visualisation types to ensure correct interpretation of data.
12	________ and drill-downs enhance interactivity and allow users to explore deeper levels of data.
13	________ is a popular business analytics tool used for dashboard creation with drag-and-drop features.
14	________ is widely known for advanced visual analytics and handling large datasets in dashboard development.
15	Python libraries like ________, Plotly, Dash, and Streamlit are used to develop Customised interactive dashboards.

7. SUMMARY

- A dashboard is a data Visualisation interface that presents key information in graphical form, enabling users to monitor, analyse, and interpret data quickly.
- The importance of dashboards lies in their ability to simplify complex datasets, highlight patterns and trends, support decision-making, and offer real-time insights.
- They help transform raw information into meaningful visuals that improve understanding and business intelligence.
- A dashboard consists of multiple components, including data sources, processing layers, visual charts, KPIs, legends/labels, filters for interaction, layout design, refresh systems, and sharing options. These elements together ensure usability, visibility, and analytical clarity.
- Dashboards can be of various types such as operational, analytical, strategic, and informational, each serving different purposes like real-time monitoring, deep analysis, high-level reporting, or knowledge presentation.
- The benefits of dashboards include improved decision-making, enhanced data comprehension, faster analysis, real-time monitoring, and better collaboration across teams.
- To design an effective dashboard, one must follow best practices like understanding user needs, maintaining visual simplicity, choosing suitable chart types, ensuring consistency in design, adding interactivity, and improving based on feedback.
- There are several tools and platforms used to build dashboards, including Power BI, Tableau, Excel, Google Looker Studio, Qlik, Python libraries, R Shiny, Apache Superset, and Grafana.
- These tools enable users to create interactive, visually rich dashboards suited for business analysis, research, monitoring systems, and large-scale enterprise reporting.

8. GLOSSARY

Dashboard	-	A visual interface that displays key information using charts, graphs, and indicators for quick understanding.
Data Visualisation	-	The process of representing data through graphical elements such as charts, plots, and diagrams.
KPI (Key Performance Indicator)	-	A measurable value that indicates the performance or success of a process, project, or goal.
Real-Time Data	-	Information that updates instantly or continuously without delay, showing current conditions.
Interactive Filters	-	Dashboard features like dropdowns, sliders, or search options that allow users to Customise the displayed data.
Drill-Down	-	The ability to click into a visual and explore deeper, more detailed levels of data.
Operational Dashboard	-	A dashboard used to monitor daily activities and real-time performance like sales or server status.
Analytical Dashboard	-	A dashboard used for deeper data analysis, forecasting, and trend identification.
Strategic Dashboard	-	High-level dashboards used by management to evaluate long-term performance and business goals.
Visualisation Tools	-	Software or platforms used to create dashboards such as Power BI, Tableau, or Looker Studio.
Data Source	-	The origin from where data is collected—databases, spreadsheets, cloud storage, or APIs.
ETL Process (Extract, Transform, Load)	-	A data preparation method used to collect, clean, structure, and load data for dashboard Visualisation.

9. TERMINAL QUESTIONS

1. What is a data Visualisation dashboard?
2. Explain the importance of dashboards in decision-making.
3. What are the main components of a data Visualisation dashboard?
4. Write a note on Key Performance Indicators (KPIs).
5. Differentiate between operational and analytical dashboards.
6. What are the benefits of data Visualisation dashboards?
7. What best practices should be followed while designing dashboards?
8. Why is data processing important before Visualisation?
9. Explain the role of real-time dashboards with examples.
10. Write the significance of interactivity in dashboards.
11. Mention any four tools used to build dashboards and explain one.
12. How are dashboards useful in collaboration and reporting?

10. ANSWERS

10.1. Self-Assessment Answers:

1. Answer: b) Display data visually for quick understanding
2. Answer: c) It helps users interpret data quickly and make decisions faster
3. Answer: b) Present complex information in a simple graphical format
4. Answer: c) Visual charts, KPIs, reports, and summary metrics
5. Answer: b) They provide real-time insights and a clear view of performance
6. Answer: b) Charts and graphs
7. Answer: b) To summarise key metrics for quick insight
8. Answer: a) Filters and interactive controls
9. Answer: audience
10. Answer: consistency
11. Answer: appropriate / correct
12. Answer: Filters
13. Answer: Power BI
14. Answer: Tableau
15. Answer: Matplotlib / Seaborn (either acceptable)

10.2. Terminal Question Answers

Answer 1: A data Visualisation dashboard is an analytical interface that visually represents data using charts, graphs, and indicators. It helps users monitor key information, spot trends, and make decisions faster. Dashboards combine multiple visuals in a single view, presenting complex datasets in an understandable format. They support interactive exploration and allow quick performance evaluation. Industries use dashboards for business analytics, monitoring systems, sales tracking, and reporting.(Refer Section 2 for more information)

Answer 2: Dashboards assist decision-makers by presenting summarised and real-time insights in a clear and visual manner. They reduce analysis time, highlight changing patterns, and support data-driven decisions. With interactive visuals, users can compare metrics, identify issues early, and evaluate performance. Dashboards eliminate manual reporting effort and make decisions quicker and more accurate. They are vital for business intelligence and monitoring workflows.(refer Section 2.1 for more information)

Answer 3: A dashboard consists of data sources, processing layers, charts/visuals, KPIs, filters, layout design, legend/labels, and sharing features. Data sources provide the input, while processing ensures cleaned and structured data. Visual elements convey information, KPIs summarise key metrics, and filters allow Customisation. Legends improve interpretation, real-time updates keep data fresh, and export options support communication. All components together make a dashboard usable and insightful.(Refer Section 3 for more information)

Answer 4: KPIs are measurable values that help evaluate performance against set objectives. They appear as numerical indicators or mini visual cards at the top of dashboards. KPIs display essential metrics like revenue, attendance, or traffic. They help stakeholders assess progress, detect deviations, and take corrective action. A dashboard without KPIs loses clarity in performance measurement.

Answer 5: Operational dashboards track real-time activities such as sales orders, server status, or customer support tickets. They refresh continuously and assist in immediate action. Analytical dashboards focus on historical trends, forecasting, and comparison. They allow filtering and drill-down for deeper insights. Operational dashboards are used for daily monitoring, whereas analytical dashboards support long-term planning and performance evaluation.(Refer Section 4 for more information)

Answer 6: Dashboards offer faster decision-making, better understanding of data, and improved business efficiency. They simplify complex information through visual formats. Real-time dashboards enable quick reactions to changing conditions. They also encourage collaboration by providing a shared view for teams. Dashboards thus enhance insight quality and overall productivity.(Refer Section 4.1 for more information)

Answer 7: Designers must know their audience and keep dashboards simple and clear. Correct Visualisation types should be chosen—bar for comparison, line for trends, pie for ratio, etc. Consistent colors and fonts improve readability. Interactivity such as filters and drill-downs helps exploration. Dashboards should be refined through feedback and continuous improvement. (Refer Section 5 for more information)

Answer 8: Raw data often contains duplicates, missing values, or inconsistent formats. Data processing cleans, filters, and aggregates information for accuracy. ETL (Extract, Transform, Load) ensures data is structured and ready for charts. Without processing, dashboards may produce misleading insights. Good data preparation improves reliability, clarity, and decision-making.

Answer 9: Real-time dashboards update automatically, providing the latest data without manual refresh. They are used for monitoring stock prices, website traffic, network systems, or IoT sensors. Real-time insights help detect failures, spikes, or performance drops instantly. Quick visibility supports rapid decision-making and crisis handling. Industries rely heavily on such dashboards for live operational control.

Answer 10: Interactivity enables users to explore data rather than just view it. Features like filters, sliders, drill-downs, and hover-tooltips allow deeper analysis. Users can analyse trends for specific locations, time ranges, or categories. Interactivity makes dashboards flexible and user-friendly. It increases engagement, accuracy, and analytical depth.(Refer Section 4.1 for more information)

Answer 11: Tools include Power BI, Tableau, Excel, QlikView, Looker Studio, Python Dash, R Shiny, and Grafana. Example – Power BI: It offers drag-and-drop visuals, KPI cards, and integration with databases and cloud sources. It supports real-time analytics, automation, and AI-based insights. Power BI is widely used for business reporting, dashboards, and KPI tracking with minimal coding knowledge.(Refer Section 6 for more information)

Answer 12: Dashboards act as a shared visual reference for teams, improving communication across departments. They support export options like PDF, Excel, and PPT, making reporting convenient. Shared dashboard links help teams track progress collectively. Collaboration builds transparency and supports evidence-based discussions. This ultimately strengthens planning, performance review, and execution.(Refer Section 6.2 for more information)

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 4

DCA2207

DATA MINING AND VISUALISATION

Unit – 12

Chart Types-2

TABLE OF CONTENTS

1. INTRODUCTION

In previous unit it provided learners with a complete understanding of data Visualisation dashboards, their purpose, and their importance in data-driven decision-making. Students learnt dashboard components such as KPIs, visual elements, filters, layout, and data sources, along with different types—including operational, analytical, strategic, and informational dashboards. The unit also highlighted the benefits of dashboards and best design practices like simplicity, clarity, consistency, and interactivity. Finally, learners were introduced to various dashboard-building tools such as Power BI, Tableau, Excel, Looker Studio, and Python platforms, enabling them to create and interpret dashboards effectively in real-world applications.

In this unit, learners will gain a complete understanding of various graphical methods used for data Visualisation and how each chart type helps interpret data effectively. They learnt how **pie charts** represent proportions of a whole, when they should be used, and where they fail to communicate information clearly. Learners then explored **bar charts** for comparing category-wise values and understood both best practices and common errors such as overcrowding or uneven scaling. The concept of **histograms** was introduced to show how data is spread across ranges, along with guidelines for choosing appropriate bin sizes and avoiding misapplication.

Learners also study **Gantt charts,** which are essential for project scheduling, task tracking, and milestone monitoring. They understand when a Gantt chart is appropriate and when simpler tools are more efficient. The unit further covers **heatmaps,** explaining how colour intensities reveal patterns, hotspots, and correlations within large datasets, while highlighting cases where precision over visual density is required. Finally, learners examine **box plots and whisker plots,** powerful tools for summarising data distribution, quartiles, variation, and outliers. They learn best practices for constructing them and recognise situations where they are unsuitable.

By the end of the unit, learners will be able to **select the right type of chart, apply Visualisation best practices, avoid misuse, and interpret insights accurately,** enabling them to communicate data meaningfully and professionally.

1.1. Learning Objectives

By the end of this unit, you will be able to:

- **Explain** the purpose and significance of charts
- **Apply** best practices for constructing different chart types
- Explore the common misuses of charts.
- **Interpret** data using statistical plots such as histograms, box plots, and whisker plots to understand distribution, variance, and outliers.

2. CHARTS IN DATA VISUALISATION

Data Visualisation is the art and science of transforming raw data into graphical or visual representations such as charts, graphs and plots. Instead of analysing raw numbers in tables, Visualisation allows decision-makers to quickly interpret patterns, trends and anomalies. It turns complex datasets into actionable insights, enabling faster and more informed decisions.

Because they simplify otherwise overwhelming amounts of data, charts are an integral aspect of data analysis. Data visualisations serve to both educate those who are seeing the data for the first time and communicate the results to others who will not have access to the raw data. A wide variety of chart kinds are available, and they all serve unique purposes. Deciding which chart type is appropriate for a given purpose is frequently the trickiest aspect of data visualisation.

Why Use Charts in Data Visualisation?

- To simplify large or complex datasets
- To identify patterns, trends, and fluctuations
- To compare values across multiple groups
- To show relationships between variables
- To highlight the composition and distribution of data
- To support fact-based decisions using visual evidence

Charts make information visually appealing and significantly reduce cognitive load for the viewer. Charts are fundamental tools in data Visualisation that transform raw numbers into meaningful visual insights. By choosing the right chart for the right data, users can communicate information clearly, reveal insights effectively, and drive impactful analysis across domains like business, research, education, health, and technology.

3. PIE CHART

When other visuals lack depth, pie charts are a great way to fill it in. A pie chart isn't enough to let the user compare data accurately and rapidly on its own. Important details in your data will be lost if the viewer is left to fill in the blanks. Do not let a pie chart steal the show from your dashboard; instead, use them to highlight other visualisations.

Data shown as a percentage is a useful visual aid in a pie chart. A circle stands in for the entire, and individual slices of the circle, or "pie," represent the various categories that make up the whole. This style of visualisation is well named because of this approach. A relationship chart allows the user to see how various entities (such as categories, goods, people, nations, etc.) relate to one another in a given setting. Data presented in a graphical format is often shown as a percentage of the whole. Each slice should be measured in accordance with the proportion of the value it represents.

In a pie chart, the total quantity is displayed as a circle with radial slices dividing it across the levels of a categorical variable. A single slice of the circle represents each categorical value, and the area and arc length of each slice show what percentage of the entire each category level occupies.

When you should use a pie chart

A **pie chart** should be used when you want to show **how individual parts contribute to a whole**. It is appropriate in situations where the total value represents **100%,** and you want to compare the proportion or share of each category within that total. Pie charts make it easy for viewers to visually compare which segment is largest or smallest at a glance.

Use a Pie Chart When:

Situation	**Why it's suitable**
You want to show percentage distribution	Pie slices clearly represent share out of 100%
There are limited categories (preferably 4–6)	Too many slices make the chart confusing
You want to highlight dominant vs. minor values	Bigger slices stand out for visual comparison
The total forms a single whole dataset	All values must add up to 100%

Examples of Where Pie Chart can be used

- Pie charts are most effective when the goal is to **display proportional contribution** of each category within a complete dataset. For instance, in the **mobile industry**, a pie chart can clearly represent how much market share each smartphone brand holds out of the total. The slices visually show which brand dominates the market and which have smaller shares.
- Similarly, when displaying **budget allocation**, a pie chart can quickly communicate how funds are distributed across different expense areas such as salaries, equipment, travel, and training. Since the total budget equals 100%, each slice reflects the proportion spent in each category, making financial distribution easy to understand.
- Another suitable example is representing **student grade distribution**, where each grade category (A, B, C, D, etc.) forms part of the total number of students. A pie chart highlights which grade range has the highest or lowest number of students instantly.
- Website performance analytics also benefit from pie charts when measuring **traffic sources** such as direct visits, organic search, social media, and referrals. In a single glance, one can understand which platform drives the most users to the website.

When Not to Use a Pie Chart?

- Pie charts are *not* ideal in every scenario. When there are **too many categories**, the slices become thin and difficult to differentiate, making the chart cluttered and confusing. A bar or column chart shows numerous categories more clearly.
- Pie charts also fail when values are **very close or similar**, because it becomes hard to visually compare small differences between segments. In such cases, bars provide a better visual comparison.
- Additionally, pie charts do not show **changes over time**. If you want to see how data increases or decreases month by month or year by year, a line chart or bar graph is more suitable. Pie charts only work for static, one-time distributions.
- Thus, pie charts are powerful for **proportional representation**, but alternative charts are more effective for detailed comparison or time-based trends.

3.1 Best practices for using a pie chart

a. Include annotations

Unless you're dealing with very small fractions like 1/2 (50%), 1/3 (33%), or 1/4 (25%), it's quite difficult to distinguish precise proportions from pie charts. Additionally, pie charts normally do not have the checkmarks that provide estimation of values directly from slice sizes if the slice values are intended to represent amounts instead of proportions. Annotations are commonly included with pie charts for several reasons.

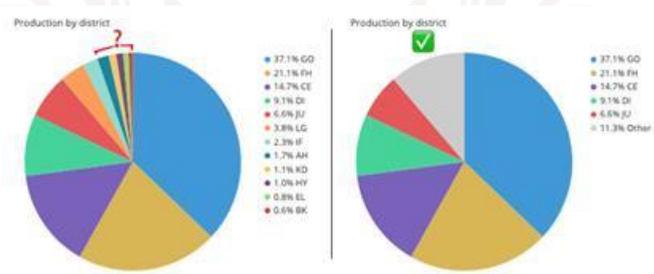
b. Consider the order of slices

The plot can be much more easily understood by the reader if the slices are presented in a decent order. When there are many categories with comparable values, it is helpful to sort them from largest to smallest, as is the usual practice. On the other hand, it is generally preferable to plot slices in the order that the category levels naturally occur.

Plotting slices from a cardinally oriented direction is a good option when choosing a start point.

c. Limit the number of pie slices

A lot of slices on a pie chart makes it hard to read. Choosing enough colours to make each slice stand out can be challenging, and seeing the tiniest slices even more so. While opinions differ, one possible recommendation is to consider an alternative chart type if your data set has more than five categories. Mixing smaller pieces into one larger "other" slice and colouring it a neutral grey is another possibility to think about.



d. Avoid distorting effects

For a pie chart to be legible, the data must be appropriately represented by the areas, arc lengths, and angles of the slices. Pie charts in particular benefit from the avoidance of three-dimensional effects, while this is true for all plots. Distorting the size of each slice relative to the entire is as easy as squashing or expanding the circle or adding extra depth. An additional kind of distortion can be seen in a "exploded" pie chart, which emphasises some slices by removing them from the centre. However, there is a downside to this focus: gaps could make it harder to evaluate the part-to-whole comparison.

3.2 Common misuses of pie chart

While pie charts are useful for showing proportions, they can easily be misused when not designed correctly. Misuse often leads to confusion, inaccurate interpretation, and misleading insights. Below are the most common mistakes seen while using pie charts:

1. Using Too Many Categories: A pie chart with too many slices becomes cluttered and hard to read. When viewers cannot distinguish segments clearly, the chart loses its purpose. Ideally, use no more than 4–6 categories.
2. Comparing Very Similar Values: Pie charts make it difficult to compare segments that are close in size. Small differences become visually unnoticeable, leading to misinterpretation. In such cases, a bar or column chart is more appropriate.
3. Using Similar Colors for Adjacent Slices: If slices are colored too similarly, it becomes hard to differentiate one category from another. Poor color contrast is a common misuse that reduces clarity. Each slice should have a distinct and contrasting color.
4. Not Representing a Complete 100%: A pie chart shows part-to-whole relationships. If the data does not add up to 100%, the Visualisation becomes meaningless and misleading. Using a pie chart for incomplete totals is an incorrect practice.
5. Adding Too Much Text or Labels: Overloading a pie chart with labels, legends, or numbers can make it visually overwhelming. In such cases, simplified labeling or an alternative chart format is preferable.
6. Using Pie Charts to Show Trends Over Time: Pie charts are static, not temporal. They cannot show changes across months/years. Using them for time-series data is a misuse—line or bar charts should be used for trends.

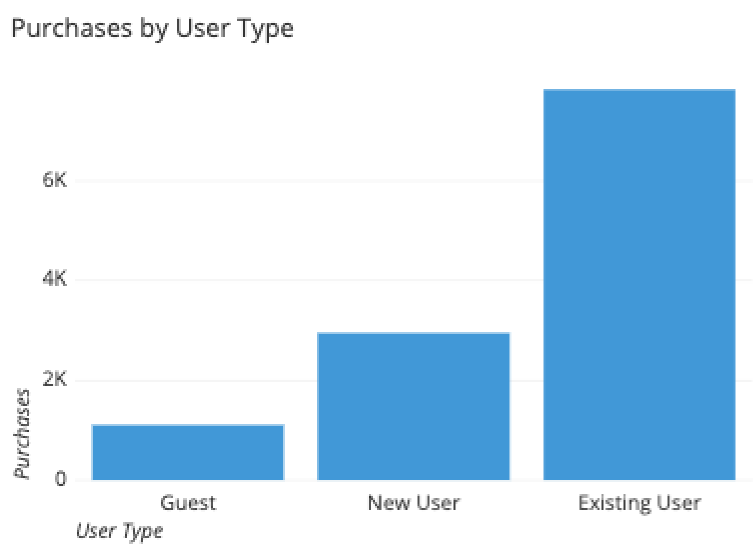
7. Exploding Too Many Slices: Highlighting more than one or two categories with exploded slices distracts visual focus. This reduces balance and distracts viewers instead of clarifying the message.

Self-Assessment Questions – 1

Fill in the Blanks	
1	A pie chart is mainly used to show the _________ of each category within a whole
2	A pie chart becomes difficult to read when there are _________ categories or very similar values.
3	In a pie chart, each slice represents one _________ of data

4. BAR CHART

To visually represent numerical values for levels of a categorical feature, a bar chart (sometimes called a bar graph or column chart) uses bars. On one side of the graph, you can see levels, and on the other, you can see values. The length of a bar is directly proportional to its value, and each bar represents a category value. In order to make comparisons between values easier, bars are drawn on a common baseline.



The amount of purchases made by various user types is shown in this sample bar chart. On the horizontal axis is displayed the categorised characteristic, user type, and the height of each bar represents the number of purchases made under that user type. This chart shows that the number of purchases made by recurrent users far outweighs that of new users who do not create an account (guests), even though the former group accounts for around three times the total number of transactions.

When you should use a bar chart

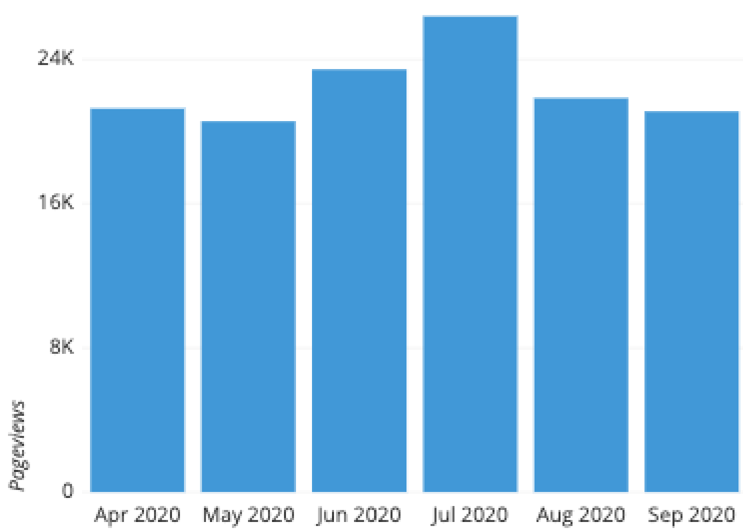
A bar chart is appropriate when the goal is to compare metric values across distinct groups or to display how data points are distributed among categories. It helps identify which categories have the highest or lowest values and how each group relates to others. Because comparing grouped values is a frequent analytical need, bar charts are among the most widely used Visualisation types.

The primary variable in a bar chart is a categorical variable, which represents discrete labels. Common examples include region (state or country), industry sector, access device (desktop or mobile), or user tier (free, basic, premium). Some categories follow an inherent sequence, such as size classifications (small, medium, large). Continuous variables can also be converted into discrete bins—for instance, grouping dates by quarter (e.g., 20XX-Q1 to 20XX-Q4). The key requirement is that each category represents a clearly separated group.

The secondary variable is always numeric, as it determines the bar height or length. These numerical values may represent counts, proportions, totals, averages, or other aggregated statistics computed for each category.

For example, a bar chart showing pageviews over six months would allow us to observe trends—such as a slight rise in June and July before returning to baseline.

In another scenario, bar height might show the mean transaction amount by payment method. While this reveals which method has the highest average value, it does not indicate which method is used most often—information that would require a different type of Visualisation.



Examples of Where Bar Charts Can Be Used

Bar charts are one of the most versatile and commonly used Visualisation tools. They are especially useful for **comparing categories side-by-side**, making it easy to see which category has the highest or lowest value.

- **Sales Performance Comparison:** Bar charts can show product-wise, region-wise, or month-wise sales. For example, comparing sales of Product A, B, and C visually highlights which product performs the best.
- **Student Strength Across Departments:** Different departments (MCA, BCA, BSc, MBA) can be compared to see which has the highest number of enrolled students.

- **Website Traffic Comparison:** A bar chart can compare traffic from different sources (Google, Facebook, Instagram, Direct Visit), helping marketers identify which platform drives more visitors.
- **Population by Age or Region**: Demographic data is easily understood when bar heights show differences clearly, such as population in rural vs. urban areas.
- **Employee Productivity Levels:** Bar charts can help compare team output, task completion, or performance score across employees or departments.

In all these situations, bar charts give a **clear visual comparison** between groups.

When Not to Use a Bar Chart

While bar charts are useful, there are situations where they are **not the best choice.**

- **When the focus is on proportion or percentage contribution:** Bar charts are less suitable when the goal is to show how parts make up a whole. A **pie or donut chart** is better for showing distribution by percentage.
- **When too many categories exist:** If the chart contains 20–30 bars, the Visualisation becomes cluttered, labels overlap, and comparison becomes difficult. In such cases, a **heatmap or table** may work better.
- **When showing continuous time-based trends**: Bar charts can show time intervals, but **line charts** are more effective for continuous trends (e.g., temperature change across days).
- **When values are very close or nearly equal:** Visual differences may not be noticeable. Scatter or dot plots provide clearer comparison.
- **When comparing part-to-whole insights:** Bar charts show comparisons, but **pie charts/stacked bars** show composition better.

Bar charts are ideal for comparing values across categories, but may not work well for percentage distribution, highly similar numbers, too many groups, or continuous time trends. Choosing the right chart improves clarity and understanding.

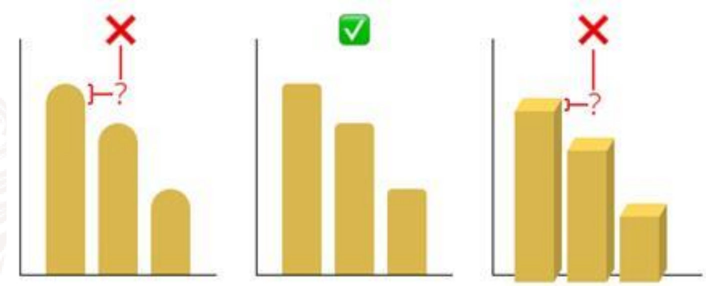
4.1 Best practices for using bar charts

a. Use a common zero-valued baseline

Ensure that all bars originate from a zero value on the axis. A zero baseline allows viewers to accurately judge differences in bar lengths and prevents distortion of the true magnitude of the values. When a chart begins at a non-zero point or uses a broken axis, the visual proportions no longer correspond to the actual numerical ratios, leading to misleading interpretations.

b. Maintain rectangular forms for your bars

Avoid altering the basic structure of bars. Some Visualisation tools offer rounded tops or stylized shapes, but these can obscure the exact point where the value should be read. Minor corner rounding is acceptable, but the upper edge must remain clearly visible and flat for precise comparison. Likewise, 3-D effects should be avoided because they introduce perspective distortions and can disrupt proper baseline alignment.

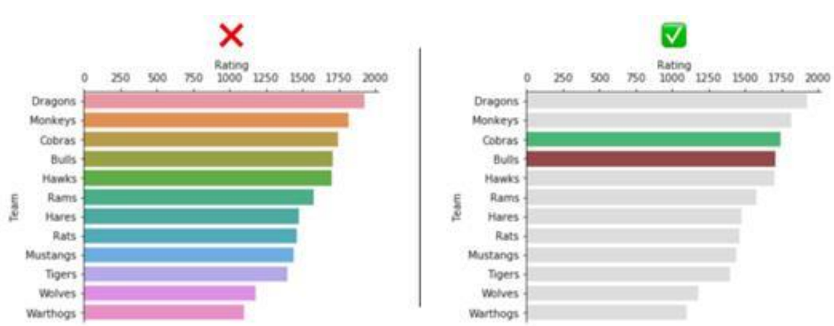


c. Consider the ordering of category levels

When arranging bar categories, consider whether ordering them from highest to lowest (or vice versa) will enhance readability. Sorting bars by length can make comparisons easier and reduce cognitive effort. The exception is when categories follow a natural sequence (such as months or age groups), in which case the inherent order should be maintained.

d. Use color wisely

Color should serve a clear purpose rather than act as decoration. Automatically assigning many different colors can imply patterns that do not exist. Instead, use a consistent neutral color for most bars and apply accent colors only when emphasising specific categories or telling a narrative. Colors may also reflect meaningful associations, such as organisational branding or group identifiers.



The left example with rainbow colors introduces unnecessary visual noise, whereas the right example uses mostly neutral tones and selectively highlights the two key bars for easier interpretation.

4.2 Common misuses in Bar chart

Bar charts are widely used for comparing categories, but when designed incorrectly, they can easily mislead viewers or reduce clarity. Below are the most common mistakes people make while using bar charts in data Visualisation:

1. Using 3D or Decorative Bars: 3D bars may look attractive, but they distort actual height and make comparison inaccurate. Decorative effects, shadows, or perspective can hide true values. A simple 2D bar is always clearer and more precise.
2. Starting the Axis at a Non-Zero Value: If the Y-axis does not begin at zero, differences between bars appear larger or smaller than they actually are. This creates a misleading Visualisation, making values look exaggerated.
3. Using Too Many Categories: A bar chart with too many bars becomes overcrowded and difficult to read. Labels overlap, colors lose meaning, and comparison becomes frustrating. Large category sets should be grouped or shown using other charts like histograms or line charts.
4. Poor or Confusing Color Selection: Using similar or inappropriate colors can make bars blend together. Bright or unrelated colors may distract rather than inform. Colors should be consistent, minimal, and meaningful.
5. Very Wide or Very Narrow Bars: Bars that are too wide look bulky and reduce clarity, while very narrow bars make it difficult to compare values. Balanced spacing maintains readability and visual appeal.
6. Not Labelling Values Clearly: Missing axis titles, unclear labels, or lack of units make charts confusing. Users should not guess what each bar represents. Clear labelling ensures correct interpretation.
7. Comparing Categories with Incompatible Scales: Sometimes, unrelated units (e.g., temperature vs. population) or different scales are shown in one chart. This creates wrong perception. Bar charts should compare similar or related data only.
8. Overuse of Grouped or Stacked Bars: Grouped or stacked bars are useful, but too many groups or stacks complicate comparison. Instead of adding clarity, they increase visual load. Limited grouping is ideal.

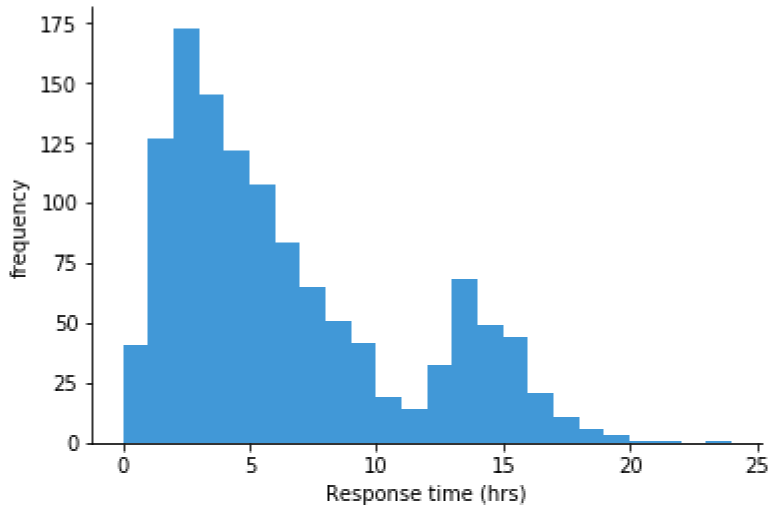
A bar chart is effective only when designed with clarity, proportion, and simplicity. Avoiding common misuses ensures accurate comparison, better interpretation, and meaningful insight for decision-making.

Self-Assessment Questions – 2

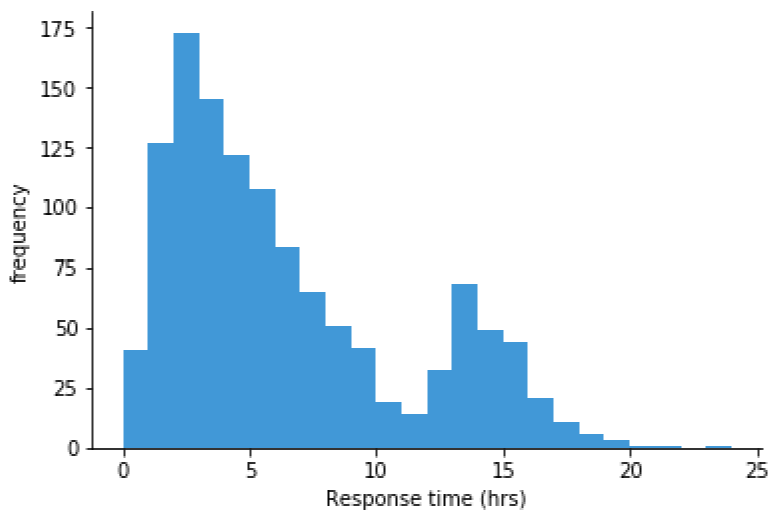
Fill in the Blanks	
4	A bar chart is best suited for comparing _________ data across multiple categories.
5	In bar charts, bar _________ and spacing should be equal to maintain clarity.
6	Bars placed vertically or horizontally represent the _________ of each category.

5. HISTOGRAM

Histograms are a kind of bar chart that show how the values of a numeric variable are distributed. In a standard bar chart, the height of the bar represents the frequency of data points falling within a certain bin or class of numbers.



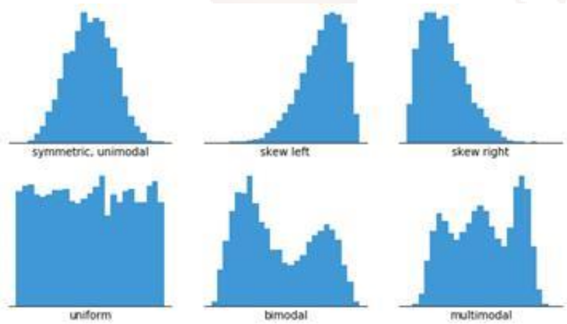
If the groups depicted in a bar chart are actually continuous numeric ranges, we can push the bars together to generate a histogram. Bar lengths in histograms typically correspond to counts of data points, and their patterns demonstrate the distribution of variables in your data. A different chart type like line chart tends to be used when the vertical value is not a frequency count.



The accompanying histogram displays the frequency distribution of the time it takes for a made-up support system to respond to tickets. The height of each bar represents the quantity of tickets in that time range, and each bar covers one hour of time. A longer tail points to the right than the left, indicating that the 2-3 hour range had the highest frequency of responses. In the 13–14 hour timeframe, there is also a tiny hill that reaches its peak. We might lose sight of these two peaks and their contributions to the total statistics if we merely examined numerical statistics like standard deviation and mean.

When you should use a histogram

- When displaying the overall distribution of variables in a dataset, histograms are a great tool to use. You can make out the distribution's skewness or symmetry, the approximate locations of its peaks, and the presence or absence of outliers.



- The only thing you need a variable that can take on continuous numerical values for a histogram to work. This indicates that the disparities between values hold true independent of their absolute values. The difference between 60 and 65 is the same 5-point size as the difference between 90 and 95, for instance, even if the score on the test could only accept integer values between 0 and 100. This is because a gap of the same size has the same meaning wherever on the scale.
- There is no intrinsic information in the data regarding the number of bins or their borders for counting up the data points. Instead, when building a histogram, we need to make a different selection regarding the bin configuration. As we'll see shortly, the histogram's interpretability is highly dependent on the bins' specifications.
- A value will always be assigned to the bin to the right or left of it when it appears on a bin boundary, or to the end bins if it appears at the ends of the rows. The visualisation tool determines which side is used; some programs allow users to change their default preference. For the sake of this post, let's pretend that the rightmost bin is where values on a bin border should go.

Examples of Where a Histogram Can Be Used

A histogram is useful for displaying the **frequency distribution of continuous numerical data.** It divides data into ranges (bins) and shows how many values fall into each range, helping the viewer understand the spread, pattern, and variation within a dataset.

- **Distribution of Students' Exam Marks:** A histogram can show how many students scored between 0–10, 10–20, 20–30, etc. This helps identify performance patterns, average range, highest frequency band, and outliers.
- **Age Distribution of a Population:** When analysing age groups such as 0–5, 6–12, 13–18, etc., a histogram can visually highlight which age group is most common, useful in public health and education planning.
- **Daily Temperature Variation Over a Month:** Temperatures grouped into bins (e.g., 20–25°C, 25–30°C) help identify which temperature range occurs most frequently and whether the climate is stable or fluctuating.
- **Income Distribution of Households:** A histogram can show how many families fall under low, middle, or high income brackets. This helps in economic planning and policy formulation.
- **Response Time of a Web Server or Application:** If most requests complete within 1–2 seconds but some take longer, a histogram quickly reveals performance bottlenecks.

Histograms are ideal for spotting **skewness, spread, peaks, clusters, and outliers** in continuous numeric data.

When Not to Use a Histogram

While histograms are powerful, they are not suitable for every situation.

- **Not for comparing categories or discrete data:** You cannot use a histogram to compare categories like product names or departments. A **bar chart** is more appropriate for categorical comparison.
- **Not suitable when exact individual values matter:** Histograms group values into bins, so you cannot see the exact score or measurement for each item. Use a **table or scatter plot** instead if precise values are required.
- **Not ideal when the dataset is very small:** With only a few observations, the bars do not form a meaningful distribution pattern. For small data samples, a **dot plot or bar chart** is clearer.
- **Not recommended for showing proportions or percentages of a whole**: To show part-to-whole relationships, **pie charts, donut charts, or stacked bar charts** work better.
- **Not useful for Visualising change over time:** Histograms do not show trends or progression across days, months, or years. In such cases, a **line graph or time-series plot** gives better clarity.

Histograms are best used to show **how continuous data is distributed across ranges,** but they should be avoided when dealing with categories, small datasets, exact values, part-to-whole relationships, or time-based trends.

5.1 Best practices for using a histogram

- Using histograms effectively requires careful selection of data structure, bin size, and visual clarity. The first best practice is to ensure that the dataset is **continuous numerical data**, not categories or text-based values.
- Histograms work best when numbers are grouped into intervals, which is why appropriate **bin size selection** is critical. If the bins are too wide, important variation is hidden; if they are too narrow, the graph becomes noisy and confusing.
- It is also recommended to **label bins, axes, and scales clearly** so that users understand the frequency ranges immediately. Consistent and contrasting colors can be used to highlight peaks or spread in the distribution without causing visual distraction.
- Keeping the design simple—without excessive decoration—helps viewers focus on distribution patterns, symmetry, skewness, or outliers.
- Additionally, histograms are most effective when used to **compare one dataset at a time**. If comparison between two or more distributions is needed, overlayed density plots or side-by-side histograms should be used instead of crowding one chart.
- Maintaining visual clarity ensures meaningful interpretation and better insight into the shape and nature of data.

5.2 Common Misuses of Histogram

- Despite being a powerful Visualisation tool, histograms are often misused, leading to incorrect or unclear interpretations. One common misuse is applying histograms **to categorical data**, which should instead be shown with bar charts. Since histograms are meant for continuous data, using them for labels like names or departments results in misleading visuals.
- Another frequent mistake is choosing **inappropriate bin sizes**. Too many bins can create a jagged, fragmented chart, while too few bins may oversimplify the data and hide important patterns. Similarly, using **unequal bin widths** distorts the perception of frequency distribution and makes comparisons inaccurate.
- Histograms are also misused when people attempt to represent **exact values, percentages, or part-to-whole relationships,** which they are not designed for. They summarise data into ranges,

so individual values are lost. In such cases, dot plots, frequency tables, or pie charts are more appropriate.

- Using histograms for **very small datasets** is also a poor practice, because there is insufficient data to form a meaningful distribution. Finally, histograms do not show trends over time—using them in place of a line chart for time-series data is another misuse that causes misinterpretation.

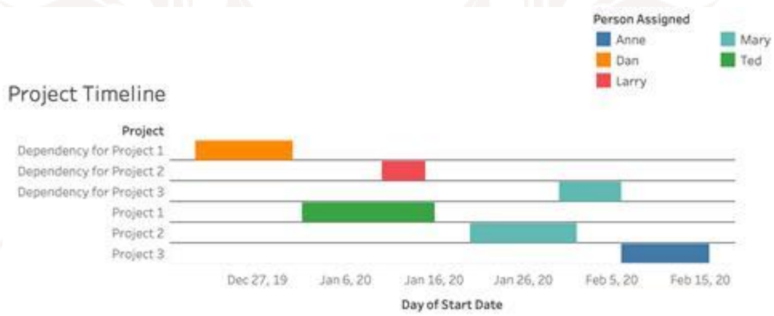
Self-Assessment Questions – 3

Fill in the Blanks	
7	A histogram displays the _________ distribution of continuous data.
8	Histograms require appropriate _________ size to avoid misleading interpretation.
9	Unlike bar charts, histogram bars touch each other because data is _________ in nature.

6. GANTT CHART

Tasks and activities within a project can be visually represented using a Gantt Chart, a form of bar chart that is commonly used in data visualisation and project management. Created in the early 20th century by Henry L. Gantt, it provides a graphical representation of a work schedule across time. This aids analysts, managers, and teams in monitoring tasks, resources, and due dates. A horizontal bar, whose length represents the task's duration and whose location denotes the start and end dates, represents each task in a Gantt chart. Knowing which jobs are active, which ones are finished, and which ones are still pending becomes a breeze with this system.

In project management, Gantt charts are most often used to show how long certain tasks or events would take. When used for project management, gantt charts highlight the timeline for work flow and make the dependencies between activities graphically obvious. You may also use them to artistically show how much time you spend doing anything, like watching TV or the time it takes for a firm to deliver their products.
Learn how to use Tableau, Excel, or Google Sheets to create a gantt chart visualisation with this comprehensive instruction. Think about utilising the free Tableau Desktop since neither Excel nor Google Sheets have prepared gantt charts.



- A **Gantt chart** is characterised by several key components that make it a powerful project Visualisation tool. The **horizontal bars** are the core visual elements, each representing an individual task, activity, or work unit in the project. The length and position of these bars show when the task begins, how long it runs, and when it is expected to finish.
- The **timeline on the X-axis** provides a clear reference for duration, which may be measured in days, weeks, months, or even years depending on project scale. On the **Y-axis,** a task list or work breakdown structure is displayed, arranging activities in a logical sequence for easy tracking.

- Another important characteristic is **task dependencies**, which visually indicate which activities must be completed before others can begin. This helps prevent scheduling conflicts and ensures smooth workflow management. **Milestones** are also highlighted on a Gantt chart, representing major deadlines, deliverables, or project checkpoints that must be achieved at specific points in time.
- Additionally, many Gantt charts include a **progress indicator** to show how much of a task has been completed, enabling real-time monitoring of project status. Together, these characteristics make Gantt charts highly effective for planning, tracking, and coordinating complex projects.

When You Should Use a Gantt Chart

A Gantt chart should be used when you need to plan, Visualise, and track the progress of a project over a specific timeline. It is most useful when a project involves multiple tasks that must be completed in a sequence or that depend on each other. If your goal is to monitor deadlines, identify delays, allocate resources, or observe how tasks overlap, a Gantt chart is the ideal Visualisation tool. It is especially effective when you need a high-level overview of project status, along with detailed breakdown of activities and timeframes. Gantt charts are commonly used when a project requires milestone tracking, progress updates, and coordination among team members, ensuring that everyone stays aligned and aware of schedules.

Examples of Where Gantt Charts Can Be Used

- Gantt charts are highly practical and widely used in industries that rely on planned execution and scheduling. For example, in **software development projects**, a Gantt chart can map phases such as designing, coding, testing, debugging, and deployment.
- Each step can be scheduled and progress monitored easily. In **construction projects,** where many tasks like layout, foundation, wiring, flooring, and painting must follow a sequence, Gantt charts help track each stage and prevent delays.
- They are also useful in **research and academic project timelines**, for planning literature review, data collection, analysis, and report writing, ensuring students meet deadlines. In **event management**, tasks such as venue booking, stage setup, marketing, and catering can be scheduled to avoid last-minute chaos. Business teams use Gantt charts for **product launch planning**, from design to manufacturing to marketing rollout. Even in healthcare or HR operations, tasks like training sessions, check-ups, hiring rounds, or patient treatment cycles can be scheduled visually using Gantt charts.

- In all these scenarios, a Gantt chart helps show *what should happen, when it should happen, and how much work has been completed,* making planning more organized, clear, and efficient.

A **Gantt chart** is a powerful data Visualisation tool used for portraying project schedules, task dependencies, durations, and progress over time. It simplifies project monitoring, promotes efficient planning, and enhances team coordination, especially in environments where deadlines matter. Despite its limitations with large datasets, the Gantt chart remains one of the most effective planning tools in data Visualisation and project management.

When NOT to Use a Gantt Chart

Although Gantt charts are powerful tools for planning and scheduling, they are not always the most effective choice. A Gantt chart should *not* be used when the project is simple, very short-term, or consists of only one or two tasks. In such cases, a checklist or task list communicates information more efficiently without unnecessary Visualisation.

You should also avoid Gantt charts when the project involves a large number of tiny tasks or constant daily changes. Frequent updates can make the chart messy and difficult to maintain. Additionally, Gantt charts are not ideal for representing data trends, numerical comparisons, or statistical analysis—they are timeline-based, not analytical Visualisations.

Another situation where Gantt charts are unsuitable is when project timelines are uncertain or frequently shifting. If the duration of tasks cannot be estimated accurately, the visual schedule becomes misleading. Finally, when collaboration is limited and task dependencies are not clearly defined, simpler tools such as Kanban boards, calendar plans, or bar charts may work better.

In summary, a Gantt chart should not be used for small or unstable projects, continuous data trends, detailed numeric comparison, or situations without clear timelines and dependencies. Choosing the right tool ensures clarity rather than complexity.

6.1 Best Practices for Using a Histogram

- A histogram is most effective when used thoughtfully and with proper formatting. To begin with, ensure that the data is **continuous numerical data,** such as height ranges, age intervals, or marks distribution.
- One of the most important practices is selecting the **right number of bins (intervals)**. If bins are too few, the distribution appears over-generalized, hiding important variations. If too many bins are used, the graph becomes cluttered and difficult to interpret.

- Histograms should include **clearly labelled axes,** showing both ranges and frequencies to avoid confusion. Colors should be simple and consistent, not overly decorative, so the viewer can focus on the pattern instead of design.
- Additionally, histograms are best kept to **one dataset at a time**, as overlaying too many datasets makes comparison difficult.
- A neat layout, correct scaling, and meaningful bin width allow the histogram to effectively reveal trends such as skewness, peaks, spread, and outliers in the dataset.

6.2 Common Misuses of Histogram

- Histograms are often misinterpreted or misused when applied to unsuitable data. One of the most common mistakes is using histograms for **categorical data**, such as brand names or departments—this is better Visualised using bar charts.
- Another misuse occurs when bins are chosen incorrectly; **very narrow bins cause noise**, while **very wide bins hide patterns,** making the distribution misleading.
- Histograms also merge individual values into groups, so they should not be used when **exact data points** need to be highlighted. They are also unsuitable for **very small datasets,** because meaningful patterns do not emerge from limited data.
- Another common misuse is using histograms to compare **multiple datasets within the same plot**, which makes the Visualisation confusing and unreadable.
- Histograms also fail when used to show **changes over time**, as they capture only distribution at one point, not progression—line charts are more suitable for trends.

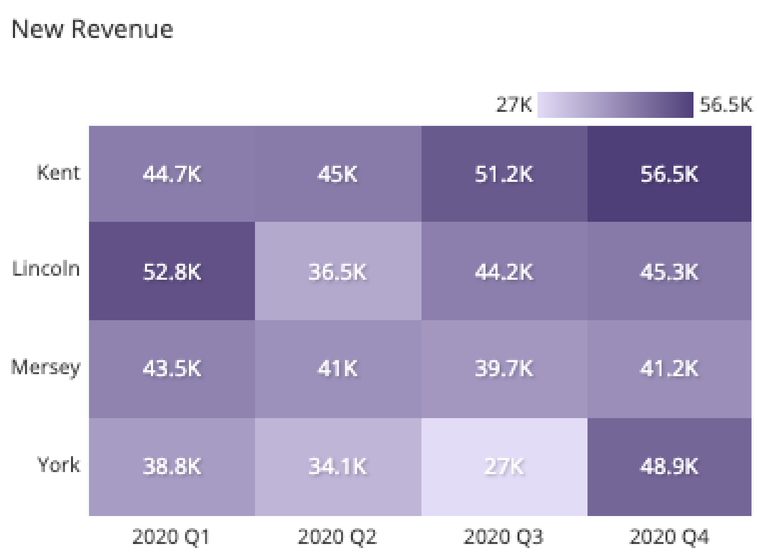
Self-Assessment Questions – 4

Fill In The Blanks	
10	A Gantt chart is used for _________ planning and tracking progress over time.
11	In a Gantt chart, horizontal bars represent _________ or activities.
12	Milestones in a Gantt chart indicate major _________ or deadlines.

7. HEAT MAP

Two important variables form the basis of the grid of values displayed by the heatmap. In a grid, similar to a bar chart or histogram, each axis variable is divided into ranges or levels. The axis variables might be numerical or category. Values are used to determine the colour of grid cells; typically, darker colours denote higher values. When plotting a large amount of data, a heatmap can be a fascinating alternative to a scatter plot; nevertheless, the high density of points makes it challenging to discern the actual correlation between variables.

A heatmap, also called a heat map, is a graphical representation of data that uses a grid of coloured squares to show the values of an important variable over two axis variables. A bar chart or histogram-like division of the axis variables into ranges allows the colour of each cell to show the value of the primary variable within that range.



When you should use a heatmap

Heatmaps are used to show relationships between two variables, one plotted on each axis. By observing how cell colors change across each axis, you can observe if there are any patterns in value for one or both variables.

The variables plotted on each axis can be of any type, whether they take on categorical labels or numeric values. In the latter case, the numeric value must be binned like in a histogram in order to form the grid cells where colors associated with the main variable of interest will be plotted.

Cell colorings can correspond to all manner of metrics, like a frequency count of points in each bin, or summary statistics like mean or median for a third variable. One way of thinking of the construction of a heatmap is as a table or matrix, with color encoding on top of the cells. In certain applications, it is also possible for cells to be colored based on non-numeric values (e.g. general qualitative levels of low, medium, high).

A heat map should be used when you want to represent data intensity, frequency, or concentration using **color gradients**. It is ideal for situations where patterns or relationships may not easily be visible in a table or text format. Heat maps work best when comparing **large sets of numbers across two dimensions,** such as time vs. category, product vs. region, or student vs. subject performance. They are especially useful when you need to quickly identify **high and low values, hotspots, trends, or anomalies** in a dataset. If the goal is to visually highlight which areas need more attention, improvement, or optimisation, a heat map is a powerful choice.

Examples of Where Heat Maps Can Be Used

- **Student Performance Analysis**: Each row may represent students and columns represent subjects, with color showing marks or pass percentage. A heat map quickly reveals strengths and weaknesses in academic performance.
- **Website Click or User Activity Tracking**: Heat maps can show which parts of a website receive most clicks or attention. High-activity areas appear darker, helping developers improve UI design and user experience.
- **Sales and Revenue Comparison Across Regions**: If regions are assigned color values based on sales intensity, managers can instantly see where performance is high or low, guiding business decisions.
- **Correlation Matrices in Data Science**: Heat maps are used to Visualise correlation between variables, helping identify strong relationships or hidden patterns in complex datasets.
- **Health and Disease Spread Analysis**: Regions can be colored based on infection count, helping health departments target high-risk areas. Heat maps are most effective when the goal is to **spot concentration patterns quickly and visually.**

When Not to Use a Heat Map

- Heat maps are not suitable for every situation. They should not be used when the dataset is **too small,** because color differences become negligible and insights remain unclear.

- A table or bar chart would communicate small-scale data more effectively. Heat maps are also unsuitable when **precise numerical values are required**, as they focus on color intensity rather than exact numbers.
- If you need exact comparison or change over time, line charts or bar charts are better options.
- A heat map should also be avoided if categories are too many or labels become unreadable, making the chart visually cluttered.
- Lastly, when comparing only two or three values, using a heat map is unnecessary and may mislead the viewer.

7.1 Best practices for using a heatmap

- Heatmaps are powerful visual tools used to represent data intensity using color gradients, but they must be designed carefully to ensure meaningful interpretation. One of the most important best practices is to **use an appropriate color scale,** where darker or more vivid shades represent higher intensity and lighter shades represent lower values.
- The color range should be intuitive and easy to distinguish—avoiding overly bright, harsh, or similar color tones that may confuse readers.
- It is also essential to **label rows, columns, and scales clearly**, so users understand what each color represents.
- Heatmaps work best when the legend is visible and the meaning of every shade is easy to interpret. Along with visual clarity, heatmaps should only be used when comparing **large datasets or multiple variables simultaneously**—for small data tables, simpler charts may work better.
- Another best practice is to **avoid overcrowding**. Too many data points packed into a heatmap make it visually cluttered and difficult to analyse. Grouping similar data, applying clustering, or reducing unnecessary dimensions keeps the graph more readable.
- For emphasising important patterns or problem areas, **highlighting or annotation** can be used to guide the viewer's attention toward key insights.

Heatmaps are most effective when used to reveal **correlations, hotspots, and density patterns**, so the design must focus on visual balance, proper scaling, and user-friendly interpretation. When built thoughtfully, a heatmap becomes a highly insightful Visualisation tool that quickly communicates complex relationships and patterns within data.

7.2 Common misuses of Heatmap

a. Using Heatmaps for Small or Limited Data

A heatmap is designed for analysing large volumes of data to identify intensity patterns and correlations. When used with very small datasets, the Visualisation becomes unnecessary and less meaningful, since color variation is minimal. Simple bar charts or tables would be more appropriate in such cases. Using a heatmap for few values leads to wasted visual space and may mislead users into assuming significant variation where there is none.

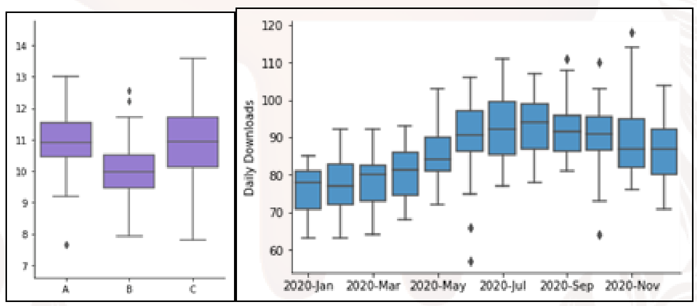
b. Poor Color Selection or Incorrect Color Scale

One of the most frequent misuses is choosing color schemes that are too similar or not intuitive. If shades look alike, the viewer cannot differentiate high values from low values. Similarly, using overly bright or inappropriate colors can distract and confuse, rather than clarify insights. A heatmap must use a well-defined, gradual color palette where each shade clearly reflects increasing or decreasing intensity.

8. BOX PLOT

To illustrate the distribution of values among measured groups, a box plot makes use of boxes and whiskers. Where the box and whisker ends are located indicates the regions with the majority of the data. If there are more than one group to compare, a box plot is the most frequent kind of chart to use; if there is just one group, more detailed charts are preferable.

To show the distributions of many sets of numerical data, a box plot (sometimes called a box and whisker plot) makes use of boxes and lines. The data is presented with box bounds that show the range of the middle 50% of the data, and a centre line represents the median value. A range of remaining data is captured by lines that extend from each box; outliers are indicated by dots put past the line boundaries.



Monthly groupings of daily app downloads for a made-up digital app are shown in the example box plot up top. The graph clearly shows that downloads rose steadily from January's 75 to August's 95 per day. The median number of downloads seems to fall marginally between November and December as well. Day counts with unusually low downloads relative to the rest of the month are shown by the points. Two days in June and one day in October fell into this category. In comparison to its line chart counterpart, the box and whiskers plot more clearly shows the overall pattern in the data.

When you should use a box plot

A box plot is suitable when the objective is to summarise and compare the distribution of numerical variables across different groups. It provides a quick, high-level view of key distribution characteristics, including symmetry, skewness, variability, and the presence of outliers. The plot makes it easy to identify where most data points lie and to contrast these patterns across multiple categories.

However, the same simplicity that makes box plots effective also limits the amount of detail they convey. Box plots do not show the full shape of the distribution. Features such as multiple peaks, subtle skew patterns, or irregular distribution structures are not visible. For more granular insight into distribution shape, other Visualisations—like density plots or histograms—may be required.

Examples of Where Box Plots Can Be Used

A **box plot** is most effective when analysing the spread, distribution, and variability of numerical data. It displays the **minimum, maximum, median, quartiles, and outliers,** making it highly useful for comparing multiple datasets side by side.

- **Student Marks Comparison Across Different Classes or Subjects:** A box plot can show how marks vary between subjects like Math, Science, English, etc. Teachers can easily identify performance consistency and outliers (very high or low scores).
- **Salary Distribution in Different Job Roles or Departments**: HR teams can use box plots to study whether salary gaps exist between roles—like engineers, managers, or analysts—and detect large variations or unequal pay.
- **Comparing Height, Weight or Age Across Groups:** Medical and demographic studies often use box plots to compare measurements between genders, regions, or age groups.
- **Customer Ratings or Feedback Data Analysis**: Box plots help businesses see if customer satisfaction is consistent or if there are extremes in user experience.
- **Machine Learning Feature Distribution Check:** Data scientists use box plots to detect **outliers**, skewness, and range of values during pre-processing.

In all these cases, a box plot provides a quick visual summary of how data is distributed.

When Not to Use a Box Plot (Elaborated)

A box plot is not suitable for every data scenario. It should **not** be used when the dataset is very small because quartile calculation becomes unreliable—dot plots or raw values would be clearer. Box plots also are not ideal when someone needs to see **individual data points**, because they summarise data instead of showing exact values.

Another misuse occurs when comparing **categorical or non-numerical data**, since box plots require continuous numbers to show variation. If the goal is to examine trends over time rather than distribution, a line chart is more appropriate. Similarly, if users need to compare exact numbers or detailed proportions, bar charts or histograms work better. Box plots may also be confusing for general audiences if they are unfamiliar with quartiles and median representation.

In summary, avoid box plots when precise values, trends over time, or categorical comparisons are needed.

8.1 Best Practices for Using a Box Plot

- A box plot is most effective when it is used to represent the spread and distribution of numerical data, so one of the best practices is to ensure the dataset contains **continuous values,** not categories or text-based information.
- When creating a box plot, it is important to clearly show the **five-number summary**—minimum, first quartile (Q1), median (Q2), third quartile (Q3), and maximum—so the viewer can understand variation and central tendency easily.
- Including **outliers as individual points** is also a good practice, as it allows the audience to quickly identify unusual extreme values in the data.
- For comparison across multiple groups, box plots should be placed side-by-side with consistent scaling to maintain visual clarity.
- Labels, axes, and units must be clearly defined to avoid confusion, and color can be used meaningfully to distinguish categories without overwhelming the viewer. In short, simplicity, clear labeling, and appropriate data type make box plots insightful and effective.

8.2 Common Misuses of a Box Plot

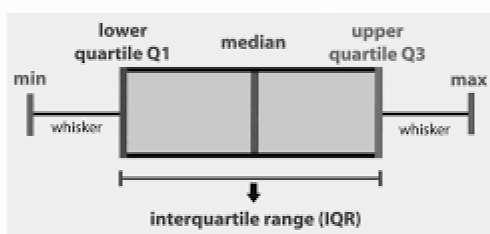
- A box plot can sometimes be misused when applied to inappropriate data types or poorly structured datasets. One of the biggest misuses occurs when using box plots for **small sample sizes**, where quartile and median calculation becomes misleading.
- Another misuse is applying box plots to **categorical or discrete data**, which cannot be meaningfully divided into quartiles.
- Box plots are also unsuitable when the goal is to show **exact values or individual observations**, as they summarise data instead of displaying every point. Overlapping too many box plots on the same graph can create visual clutter and reduce comparability, making insights harder to interpret.
- Additionally, box plots do not reveal detailed distribution shapes like bimodality or clustering, so relying on them alone may lead to incomplete conclusions.
- Misuse generally occurs when the context demands precise data visibility, trend analysis, or categorical comparison—situations where other Visualisations are more appropriate.

9. WHISKER PLOT (BOX-AND-WHISKER PLOT)

A **Whisker Plot,** commonly called a **Box-and-Whisker Plot**, is a statistical Visualisation used to display the **distribution, spread, and variability** of numerical data. It summarises large datasets using a visual structure that shows minimum, maximum, median, quartiles, and outliers. This makes it a powerful tool for comparing multiple sets of values side-by-side, especially in analytics, research, education, and machine learning.

When You Should Use a Whisker Plot

A whisker plot should be used when the goal is to analyse the **distribution, spread, and variability** of numerical data. It is ideal when you need to identify the **median, quartiles, interquartile range (IQR), and outliers** clearly within a dataset. If you want to compare two or more groups side-by-side—such as performance across departments, salary variations, or experimental output—whisker plots provide quick visual comparison. This type of graph is especially useful when datasets are large and detailed, because it summarises them into a clean five-number representation (min, Q1, median, Q3, max). When the intention is to explore symmetry, skewness, or unusual values in data, a whisker plot is one of the most effective tools to use.



A whisker plot is as shown in the above figure. This figure represents a **Box Plot (also called Box-and-Whisker Plot)**, which is a graphical method used to **summarize the distribution of numerical data**. It shows how the data values are spread using **five important statistical points**.

Where in figure we can observe,

- **Minimum (min)**
  This is the **smallest value** in the dataset. It is shown at the left end of the whisker.
- **Lower Quartile (Q1)**
  Q1 represents the value below which **25% of the data lies**. It marks the beginning of the box.

- **Median**

  The median is the **middle value of the dataset**, where **50% of the data lies below it and 50% lies above it**. It is represented by the vertical line inside the box.
- **Upper Quartile (Q3)**

  Q3 is the value below which **75% of the data lies**. It marks the end of the box.
- **Maximum (max)**

  This is the **largest value** in the dataset, shown at the right end of the whisker.

Interquartile Range (IQR)

The **Interquartile Range (IQR)** is the distance between **Q3 and Q1**.

In words:

IQR = Upper Quartile (Q3) – Lower Quartile (Q1)

The IQR represents the **middle 50% of the data** and helps measure how spread out the central portion of the dataset is.

Whiskers

The **whiskers** extend from the box to the minimum and maximum values. They show the **range of the remaining data outside the middle 50%**.

Examples of Where Whisker Plots Can Be Used

Whisker plots are highly suited for scenarios involving **statistical distribution and variability analysis:**

- **Exam Marks Comparison Across Sections**: Teachers can compare marks of different classes to understand performance variation, identify high-scoring groups, or detect students who are outliers.
- **Salary Distribution Between Job Roles**: HR departments use whisker plots to visually compare pay ranges across designations and ensure fair compensation.
- **Height/Weight Comparison for Male and Female Groups**: Medical research uses whisker plots to analyse biological differences and variation patterns between genders.
- **Customer Feedback Score Analysis**: Businesses can Visualise satisfaction score ranges and spot extremely positive or negative experiences easily.
- **Machine Learning Data Preprocessing (Outlier Detection)**: Data scientists use whisker plots to locate extreme values that may affect model performance.

In all these cases, whisker plots provide a **quick and meaningful summary** of data distribution.

When Not to Use a Whisker Plot

A whisker plot should not be used when the dataset is **very small,** because quartile-based summaries become unreliable. It is also unsuitable when the viewer needs to observe **exact values for each data point,** since whisker plots show summary statistics—not individual measurements. This type of Visualisation is also inappropriate for **categorical or non-numeric data**, as it cannot represent text-based information like product names or departments. Whisker plots are not ideal for showing **trends over time**, because they display distribution at one time rather than progression across dates. In situations requiring detailed value tracking or temporal analysis, a line chart, bar graph, or scatter plot is more effective.

Best Practices for Using a Whisker Plot

- To use a whisker plot effectively, ensure that the data is **continuous and numerical**, suitable for quartile-based representation.
- Clearly label axes and categories so viewers know what each box represents. **Display outliers separately using dots or markers**, as this helps identify anomalies without distorting the whiskers.
- The plot should maintain a clean layout—avoid overcrowding with too many boxes on one chart. When comparing datasets, use **consistent scales and colors** so visual comparison remains accurate.
- Adding median lines, quartile shading, or annotations can further improve readability.
- Simplicity, clarity, and balanced visual spacing are key for effective whisker plot interpretation.

Common Misuses of Whisker Plots

- Whisker plots are often misused when applied to inappropriate data types or Visualisation needs.
- Using them for **categorical data or text labels** is a common misuse, as quartiles cannot be derived from non-numeric information.
- Another mistake is comparing **too many datasets in one chart**, causing visual clutter and confusion.
- A whisker plot is also misapplied when someone needs to display **exact values,** as it only shows distribution summaries.
- Using unequal or inconsistent scales makes the Visualisation misleading.
- Finally, whisker plots are misused when used in place of trend-focused charts, because they do not illustrate changes over time.

10. SUMMARY

- **Pie Charts,** which are useful for showing part-to-whole relationships and proportional distribution. They also learn when to use a pie chart, best practices such as using limited slices, and misuses like plotting too many categories or close values which reduce clarity.
- The **Bar Chart** section explains how bars are used to compare categorical data across different groups. Best practices such as maintaining equal bar width and labelling axes are emphasised, along with common misuses like starting the axis at non-zero or adding too many bars, which causes clutter.
- A **Histogram** is covered next, where learners understand how it displays frequency distribution using grouped bins. Best practices include choosing appropriate bin width and using histograms only for continuous data, while misuses involve using them for small datasets or for comparing exact values.
- The unit also introduces **Gantt Charts**, a project Visualisation tool useful for scheduling, task tracking, dependency mapping, and progress monitoring. Learners understand when such charts are applicable, along with limitations such as unsuitability for small, constantly changing, or non-timeline-based data.
- Next, the unit discusses **Heatmaps,** which represent intensity using color gradients. Students learn that heatmaps are ideal for large matrix-based datasets, correlation Visualisation, and web activity tracking, but should not be used for small datasets or environments requiring precise numeric values.
- Lastly, the unit covers **Box Plots and Whisker Plots**, which summarise distribution using median, quartiles, and outliers. Learners understand best practices such as using box plots for comparing distributions, while recognising misuses such as applying them to categorical data or small datasets.

11. GLOSSARY

Pie Chart	-	A circular chart divided into slices to represent proportions or percentage distribution of categories.
Chart Slice	-	A segment of a pie chart that represents one category's share in the whole dataset.
Bar Chart	-	A Visualisation that uses rectangular bars of equal width to compare values across categories or groups.
Histogram	-	A graph that shows frequency distribution of continuous data using bins or intervals.
Bins (Class Intervals)	-	Divisions or ranges in a histogram into which numerical data is grouped.
Gantt Chart	-	A timeline-based chart used to plan, schedule, and track project tasks and deadlines.
Milestones (in Gantt Chart)	-	Key target dates or achievements marked on a project timeline.
Dependencies	-	Relationships showing which task must start or finish before another can begin (used in Gantt charts).
Heatmap	-	A graphical representation where color intensity indicates the magnitude or frequency of values
Color Scale	-	The gradient or range of colors used in a heatmap to represent low-to-high data values.
Box Plot	-	A statistical graph that displays data distribution using median, quartiles, and possible outliers.
Whiskers	-	Lines extending from the box in a box-plot to show minimum and maximum non-outlier values.
Median (Q2)	-	The middle value of a dataset, dividing it into two equal halves.
Quartiles (Q1 & Q3)	-	Values that divide data into four equal parts; Q1 = lower 25% boundary, Q3 = upper 75% boundary.
Outlier	-	A data point that lies significantly above or below most values, often shown separately in box/whisker plots.

Best Practices	-	Recommended guidelines for designing charts effectively without misinterpretation.
Common Misuse	-	Incorrect or unsuitable application of a chart that leads to confusion or wrong conclusions.

12. TERMINAL QUESTIONS

1. What is a pie chart used for?
2. When is a bar chart more suitable than a pie chart?
3. Why do we use histograms?
4. State one best practice when using a histogram.
5. What is a Gantt chart mainly used for?
6. Define a heatmap.
7. What does a box plot show?
8. Give one common misuse of a pie chart.

13. ANSWERS

13.1. Self-Assessment Answers:

1. Answer: proportion / percentage
2. Answer: too many
3. Answer: category
4. Answer: categorical
5. Answer: width
6. Answer: value / magnitude
7. Answer: frequency
8. Answer: bin
9. Answer: continuous
10. Answer: project
11. Answer: tasks
12. Answer: events / achievements

13.2. Terminal Question Answers

Answer 1: A pie chart is used to show the proportion or percentage contribution of different categories in a whole.

Answer 2: A bar chart is more suitable when comparing many categories or when values are similar and difficult to distinguish in a pie chart.

Answer 3: Histograms are used to display frequency distribution of continuous numerical data by grouping values into bins.

Answer 4: Choose appropriate bin sizes to avoid oversimplifying or overcrowding the distribution.

Answer 5: A Gantt chart is used for project planning, task scheduling, progress tracking, and time management.

Answer 6: A heatmap is a graphical representation where data values are shown using color intensity to indicate high or low magnitude.

Answer 7: A box plot shows data distribution using minimum, maximum, median, and quartiles, along with detection of outliers.

Answer 8: Using a pie chart when there are too many categories or when category values are very close in size.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 4

DCA2207

DATA MINING AND VISUALISATION

Unit – 13

Chart Types-2

TABLE OF CONTENTS

1. INTRODUCTION

In the previous unit, learners were introduced to fundamental visualisation methods such as pie charts, bar charts, histograms, Gantt charts, heatmaps, and box/whisker plots. They understood the purpose of each chart, best practices for designing them, and common mistakes to avoid. The unit helped them interpret data patterns, compare categories, study distributions, track project timelines, and identify outliers. By the end, learners were able to choose appropriate graphs for different datasets and communicate information clearly and professionally.

In this unit, learners will explore different data visualisation charts and techniques used to present information clearly, meaningfully, and visually. They will understand how each chart works, when it should be used, and how to apply best practices for effective communication of data. The unit also teaches how to identify and avoid common mistakes that can lead to misleading or confusing visualisations.

Learners will study Waterfall charts to understand cumulative changes, Area charts to observe trends and volume over time, and Scatter plots to analyse relationships between two variables. They will also learn how Pictogram charts use icons to represent quantities visually, Timeline charts to show events chronologically, Highlight tables to emphasise key values with color, and Bullet graphs to compare actual performance against targets.

By the end of the unit, students will be able to select the most appropriate type of chart for a given dataset, design visuals using best practices, interpret patterns accurately, and avoid common misuses. This knowledge will help them present data more effectively in reports, presentations, dashboards, and analytical decision-making.

1.1. Learning Objectives

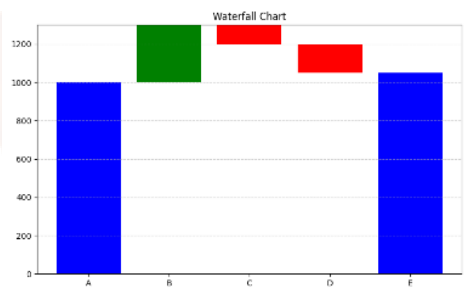
By the end of this unit, you will be able to:

- Understand the purpose and application of Waterfall, Area, Scatter, Pictogram, Timeline, Highlight Table, and Bullet Graph visualisations.
- Identify when each chart type is most appropriate for trend analysis, comparison, distribution, cumulative change, or performance tracking.
- Apply best practices for designing clear, accurate, and visually meaningful data charts.
- Analyse and interpret visual patterns.

2. WATERFALL CHART

A waterfall chart shows how positive and negative values affect a total over time. It's useful for tracking changes in financial data, budgets or performance metrics. A **Waterfall Chart** is a type of data visualisation used to show how an initial value increases or decreases step-by-step to reach a final total. It is also known as a **Bridge Chart,** because the bars appear like steps forming a bridge from the starting point to the end point. This chart is especially useful for visualising cumulative changes over time or across different contributing factors.

Below is a representation of a Waterfall chart:



Key characteristics:

- Displays incremental changes that lead to a final result
- Uses vertical bars — increases shown upward, decreases downward
- Shows starting value, intermediate contributions, and final value
- Highlights how individual factors affect the total outcome
- Uses color coding to differentiate positive and negative changes

A Waterfall Chart is a powerful visualisation tool that clearly illustrates how different factors contribute to the movement of a value from a starting point to an ending point. One of its most important characteristics is that it displays incremental increases and decreases step-by-step, allowing the viewer to see how each change affects the final total. The chart uses vertical bars to represent changes, where positive or increasing values extend upward, and negative or decreasing values extend downward. This visual distinction helps users immediately recognise which elements contribute to growth and which cause decline.

A waterfall chart also shows the starting value, intermediate stages, and the final cumulative value, making it ideal for breakdown analysis—such as revenue leading to final profit after adding gains and subtracting costs. Because each step is shown in sequence, the viewer can track how the final result is formed rather than only seeing the overall total. Another key feature is its ability to highlight the contribution of individual components, making it easy to identify areas of improvement, inefficiency, or strong performance. To enhance clarity, waterfall charts often use color coding, such as green for increases and red for decreases, which improves interpretation at a glance and helps users quickly separate positive and negative impacts.

These combined characteristics make a waterfall chart highly effective in financial reporting, business performance analysis, and managerial decision-making.

When to use this chart:

Visualise how values change step-by-step:

- Break down the components of a total value: A waterfall chart is a suitable choice when the objective is to visualise how values change step-by-step, rather than simply displaying the final outcome. For example, instead of showing only the net profit, a waterfall chart reveals how revenue, expenses, taxes, and other contributions lead to that final figure. This step-by-step breakdown helps users understand the developmental flow rather than just the end result.
- Show increases and decreases contributing to final results: You should also use a waterfall chart when you want to break down the components of a total value into meaningful segments. It clearly separates contributing factors, showing how each one adds to or reduces the overall value. This is useful in budgeting, profit analysis, and performance review where different elements collectively affect outcomes.
- Compare planned vs actual values over time**:** A waterfall chart is effective when the goal is to show increases and decreases that contribute to the final result. Positive values rise upward, negative values fall downward, and the viewer can instantly identify areas of gain and loss. This makes it particularly useful for comparing operating income vs. operating expenses, tracking sales growth, or visualising changes in month-to-month revenue.

It is also helpful for situations where you need to compare planned vs actual values over time. For instance, a business may plan a target profit, but adjustments such as unexpected expense increases or extra sales can shift the final actual profit. A waterfall chart visually shows where plans were exceeded, where they fell short, and how the final result was reached.

Examples of Where Waterfall Charts Are Used

a. Financial Analysis: Waterfall charts are widely used in finance to break down how a company's net profit is formed. The chart visually presents the flow from **revenue**, followed by deductions such as **operational costs**, **taxes**, and **expenses**, ultimately showing the **final net profit**. Each increase and decrease is shown individually, enabling stakeholders to easily identify which factors contribute positively to earnings and which reduce profitability. This step-by-step structure makes financial reports easier to analyse and supports better decision-making.

b. Sales and Revenue Tracking: Businesses use waterfall charts to track **month-wise or quarter-wise changes in sales.** By displaying increases and decreases in revenue across time, the chart shows how each month contributes to the **final annual sales total.** For instance, a spike during festive seasons or a dip during market slowdown becomes clearly visible. This helps sales teams understand performance patterns, plan marketing strategies, and evaluate seasonal sales trends effectively.

c. Budget Allocation: **Government agencies, organisations, and project managers often work with budgets that undergo updates over time. A waterfall chart can show how an initial budget changes through additional funding, resource allocation, cuts, or unexpected expenses, leading to the final approved amount. Each adjustment in the budget is shown step-by-step, making it easy to understand how the final financial plan was reached and how funds were distributed or optimised.**

d. Inventory and Production: In inventory management, a waterfall chart can visually track the flow of materials or stock. It may start with **opening stock**, then add **incoming inventory or new purchases**, followed by deducting **consumed or utilised stock**, and finally display the **closing balance**. This helps production units avoid stock shortages, detect wastage, plan reorders, and maintain a smooth supply chain. It is especially useful in manufacturing, retail, and warehouse operations.

e. Project Progress Tracking: Waterfall charts are useful in **project management** to track planned and actual work status. A project may begin with an estimated effort, after which additional tasks, delays, or changes may increase the workload. By visualising these step-by-step changes—**planned hours → extra requirements → delays → completion timeline**—project managers gain clear insight into how the final completion time evolved. This helps in resource planning, schedule adjustments, and risk analysis.

<u>Advantages of Waterfall Charts</u>

1. Clearly shows how different elements contribute to the final value

A waterfall chart is excellent for breaking down a result step-by-step, making it easy to see how each factor affects the final total. Instead of just showing the end result, it visually explains *how we arrived* *there* by separating additions and deductions. This makes the chart valuable for understanding the underlying structure of totals like profit, budget, or progress.

2. Simple interpretation of gains vs losses

One of the biggest strengths of a waterfall chart is the clarity with which it shows positive and negative changes. Upward bars represent increases and downward bars represent decreases, making gains and losses instantly recognisable. This visual simplicity helps even non-technical audiences understand complex changes quickly.

3. Useful for management reporting, financial summaries, and dashboards

Because waterfall charts explain value movement transparently, they are frequently used in business reviews, financial reports, and executive dashboards. Managers can easily track revenue formation, cost breakdown, variance from budget, or progress toward goals. This makes the chart a preferred choice in presentations and board-level decision meetings.

4. Helps in decision-making by highlighting where changes happen most

A waterfall chart visually highlights stages where major change occurred, such as a spike in income or a sudden fall due to expenses. By identifying these impact zones, organisations can analyse reasons behind performance changes and take informed decisions. This helps in budgeting, optimisation, and process improvement.

Limitations of Waterfall Charts

1. Not suitable for very large datasets with many breakdown components

When there are too many contributing factors, a waterfall chart becomes crowded and difficult to read. Too many steps reduce clarity and make interpretation confusing, defeating the purpose of visualisation. Such cases are better handled with bar charts or tabular data.

2. Can become confusing without proper color coding or labels

Waterfall charts rely on color differentiation to show increases and decreases. If colors are too similar or labels are missing, the visualisation becomes unclear and hard to interpret. Proper formatting, legends, and labeling are necessary to maintain readability.

3. Not ideal for comparing multiple categories or multiple totals at once

Waterfall charts are designed to show one flow at a time — from start to finish. If multiple totals or categories must be compared, multiple waterfall charts may be needed, which can lead to confusion. Bar charts or grouped comparisons are more suitable in such scenarios.

4. Shows progression but not distribution or frequency like histograms

A waterfall chart explains *how values changed*, not how values are distributed. It does not reveal patterns like spread, frequency, or variance in the data. Therefore, if distribution analysis is needed, charts like histograms or box plots are more appropriate.

2.1 Best Practices for Using a Waterfall Chart

- To make a waterfall chart meaningful and easy to interpret, it is important to follow certain design guidelines. First, clearly define the **starting value and final total,** as they form the anchor points of the entire chart.
- The intermediate steps should be arranged logically, such as chronological changes or categorised adjustments, so that the flow appears natural and easy to follow.
- It is also essential to use **distinct color codes**—typically green for increases and red for decreases—ensuring that the viewer can instantly differentiate gains from losses.
- Proper labeling of each bar helps users understand what contributes to the increases or reductions.
- For large values, including data labels or tooltips improves readability and removes ambiguity.
- Avoid overcrowding by displaying only meaningful steps; too many small or irrelevant segments disrupt clarity.
- Finally, placing summary totals or sub-totals at logical checkpoints makes the visual story stronger, helping users follow the accumulation process easily. When designed using these practices, a waterfall chart becomes clear, structured, and highly insightful.

2.2 Common Misuses of Waterfall Charts

- Despite being useful, waterfall charts are often misused when applied in unsuitable situations. A common mistake is using a waterfall chart for very large datasets or too many contributing factors, which results in clutter and reduces clarity instead of improving interpretation.

- Another misuse occurs when charts lack proper color differentiation, causing confusion between increases and decreases. If colors are too similar, the visual meaning gets lost.
- Waterfall charts are also misapplied when the intention is to compare multiple categories or totals, which they are not designed for.
- They are linear, step-based visuals—not comparison charts. Using them to show data distribution is also incorrect, as they do not reveal patterns like spread, outliers, or frequency.
- Additionally, if labels and values are not displayed clearly, users may misinterpret the flow or magnitude of changes. In short, waterfall charts should not be used where precision comparison, massive datasets, or distribution analysis is needed.

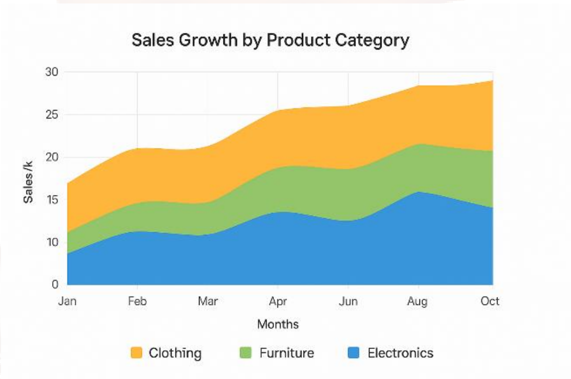
Self-Assessment Questions – 1

Fill in the Blanks	
1	A Waterfall chart shows how an initial value changes through _____ and decreases.
2	Waterfall charts are often used in _____ analysis to show profit breakdown.
3	The final bar in a Waterfall chart represents the _____ value after all changes.
4	Intermediate stages in a Waterfall chart are also known as _____ values.

3. AREA CHART

An **Area Chart** is a graphical representation used to show how values change over time by filling the space under a line graph with color. It visually highlights both the trend of data movement and the magnitude (volume) of values. The filled color helps viewers compare cumulative totals, observe growth or decline patterns, and understand how individual variables contribute to the whole. Area charts are commonly used when tracking continuous data across years, months, or days because they clearly display not only the line trend but also the accumulated value beneath it.

Unlike line charts that emphasise change direction only, area charts emphasise **change + quantity,** making them effective for trend visualisation in business, finance, weather reporting, production, and resource usage. Stacked area charts further extend this concept by showing how multiple categories contribute to the total trend over time.



When to Use an Area Chart

Use an area chart when:

1. You want to display changes or trends over time

- Area charts are excellent for showing how values **rise and fall over a continuous time period** such as days, months, or years.
- By filling the region under the line, viewers can see not only direction (upward/downward trend) but also **how wide or narrow the change was.** This makes it easy to visualise **long-term progression**, sudden spikes, or drops.

2. You want to highlight total volume or cumulative information

- The filled color in area charts gives a sense of **total quantity or accumulation** rather than just individual data points.

- It visually represents how much has been achieved or consumed up to a certain point. The shaded portion naturally draws attention to the **overall magnitude**, not just its shape.

3. You need to see how multiple components contribute to a total

- When stacked area charts are used, each filled area shows **how individual parts form the whole contribution.**
- You can easily compare which category is dominating or decreasing over time. This helps understand **part-to-whole relationships** clearly through layers of color.

4. Visual depth (color filling) enhances understanding better than a plain line graph

- A line graph only shows direction, but an area chart shows **direction + volume.**
- The filled region adds depth and weight to the data, making patterns more visible.
- Significant growth appears as a **large shaded expansion**, while decline is seen as a shrink area.
- It is more engaging to the human eye and easier to interpret at a glance. Thus, area charts improve comprehension, especially for **presentation and decision-making contexts.**

Examples of Where Area Charts Are Used

1. **Weather and Climate Reports:** Area charts can illustrate temperature changes, rainfall levels, or humidity variation over a year. Filling color under the curve helps viewers see periods of high or low activity instantly.

2. **Sales Growth and Revenue Tracking:** Businesses use area charts to show month-wise or quarter-wise revenue performance, helping managers compare seasonal growth and yearly improvement.

3. **Website Analytics and Traffic Monitoring:** Area charts show user visits over time, daily site activity, or app usage patterns, helping digital analysts track spikes or slow phases.

4. **Energy or Resource Consumption:** Electricity usage across 24 hours or fuel consumption over a journey can be visualised using area charts to identify peak load hours or wastage points.

5. **Stock Market and Financial Trends:** Area charts show gradual price rise or fall over days or months, helping investors visually interpret market movement.

Advantages of Area Charts

- **Shows trend and quantity together**

Unlike line charts, the filled color indicates how much data is accumulated.

- **Excellent for time-series data**

  Helps observe patterns, growth, and fluctuations across time intervals.

- **Visually appealing and easy to read**

  The filled area creates strong visual impact and quick interpretation.

- **Useful for cumulative comparison**

  Stacked area charts show overall contribution of different categories.

Limitations of Area Charts

- **Can become cluttered with too many categories**

  Overlapping colors may make values difficult to distinguish.

- **Not suitable when exact values are needed**

  It visually shows distribution, but bar charts or tables show precise counts better. **Hard to interpret if colors are similar or transparency is low**

  Misinterpretation increases if shading is not handled properly.

- **Negative values are difficult to display**

  Area charts handle positive growth best — below-zero values can distort the shape.

3.1 Best Practices for Using an Area Chart

Use clear color contrast so categories are easily separable

- Colors should be distinct enough for viewers to differentiate between multiple regions at a glance.
- If shades are too similar, the layers will blend and analysis becomes difficult.
- Contrasting colors improve readability and make the contribution of each category more visible.

Ensure the time interval is consistent (day, week, month) for smooth readability

- Using uneven time intervals causes confusion and breaks the visual flow of the trend.
- A uniform timeline lets users track increases, decreases, and patterns more accurately.

Avoid too many layers in stacked area charts — limit to 3–4 categories maximum

- Adding many categories creates visual clutter and overlaps, making the chart hard to interpret. 3–4 layers maintain clarity while still providing meaningful comparison.
- Beyond this, a bar chart or line chart might communicate the information better.

Label axes, data points, and legends properly to prevent confusion

- X-axis should clearly represent time, while Y-axis represents the measured value.
- Legends must identify each color or layer correctly so users know what they are viewing.
- Proper labelling eliminates ambiguity and improves comprehension.

Use transparency or different shading if datasets overlap

- Overlapping areas can hide information if colors are fully opaque.
- Semi-transparent or slightly lighter shades ensure that layers beneath remain visible.
- This helps in understanding how categories interact or change with time.

3.2 Common Misuses of Area Charts

- Using an area chart when categories are too many, causing overlap and confusion
- Using it instead of a line chart even when **trend is more important than volume**
- Applying area charts where exact values are required — better use bar charts
- Using dark and similar shades that hide differences rather than highlight them
- Displaying random or categorical data not based on time progression

Misuse leads to misinterpretation, incorrect conclusions, and visual overload.

An **Area Chart** is most useful for showing trends over time, highlighting cumulative change, and comparing growth across categories. When designed properly using color contrast, limited layers, and labeled axes, it becomes a powerful visual tool for business analytics, finance, weather tracking, and performance monitoring.

Self-Assessment Questions – 2

Multiple Choose Questions	
5	Area charts display data trends over_____ with filled color underneath.
6	A _____ area chart shows contribution of each category to the total
7	Area charts highlight both trend direction and total _____ of change.
8	Too many layers in an area chart may cause the graph to become _____.

4. SCATTER PLOT

A **scatter plot** (also called scatter graph or scatter diagram) is a graphical representation used to display the relationship between two (or more) continuous numerical variables. Each point on the graph represents an observation in the dataset—its x-value plotted along the horizontal axis and its y-value on the vertical axis. By analysing the pattern formed by the plotted points, one can visually interpret correlations, trends, clusters, and outliers.

Scatter plots are widely used in statistical analysis because they reveal how one variable influences or correlates with another. Whether the data points form an upward, downward, curved, or random pattern helps in understanding positive, negative, or no relationship between variables.

When to Use Scatter Plot

Use a scatter plot when:

- You want to **observe the relationship or correlation** between two variables.
- The data consists of **paired numerical values**.
- You need to identify **patterns, clusters, or outliers** in the dataset.
- You want to evaluate **how strongly one variable depends on the other.**
- You need to **predict outcomes or trends** based on visual observations.

Examples of Where Scatter Plots Are Used

- **Education:** To analyse the relationship between study hours and student marks to see if more study time leads to higher scores.
- **Healthcare:** To compare age with blood pressure levels and observe whether older age is associated with higher BP readings.
- **Business:** Used to examine the link between sales revenue and advertising expenditure to understand how marketing affects income.
- **Agriculture:** Helps determine how rainfall influences crop yield and whether more rain increases overall production.
- **Finance:** Used for risk vs. return projection to determine if higher-risk investments provide greater returns.

- **Manufacturing:** Helps measure the relation between machine running time and output produced to monitor productivity.
- **Sports:** Used to track how the duration of training affects performance score or improvement rate of players.

Advantages of Scatter Plot

- Helps identify **positive, negative, or no correlation** between variables.
- Displays **true distribution of data points** clearly.
- Easy to detect **outliers and anomalies** visually.
- Simple, effective, and easy to understand without statistical knowledge.
- Useful in forecasting and predictive modeling based on trends.
- Works well for both small and large datasets

Limitations of Scatter Plot

- Only suitable for **quantitative (numerical) data**, not categorical variables.
- Hard to interpret when dataset is **too dense or overlapping.**
- Cannot prove causation—only shows relationship visually.
- Limited in representing more than two variables at once (unless adding color/size variation).
- Trends may appear misleading if data is poorly scaled or sampled.

4.1 Best Practices for Using Scatter Plot

- **Label axes clearly with units and variable names**: Each axis should clearly state what it represents (e.g., *Height (cm)* vs *Weight (kg)*). Proper labeling helps readers immediately understand what is being compared without confusion.
- **Use proper scaling to avoid distorted patterns:** Both axes should be scaled appropriately so that the spread of points reflects the true variation. Poor scaling can make data appear falsely correlated or more scattered than it actually is.
- **Add trend lines or regression lines for clarity**: A trend line helps show the direction and strength of the relationship between variables. It makes interpretation easier by highlighting whether the correlation is positive, negative, or neutral.

- **Use color coding or size variation to show additional dimensions:** Different colors or changing point sizes can represent extra variables such as category, intensity, or frequency. This allows more information to be shown without adding extra charts.
- **Avoid clutter—use transparency for overlapping points**: When many points overlap, transparency helps reveal density or grouping instead of hiding values. A cleaner plot improves visual readability and pattern recognition.
- **Represent outliers distinctly for better interpretation:** Outliers should be highlighted or marked separately because they may indicate errors, special cases, or important unusual observations that need attention.
- **Provide a legend when using multiple variable groups**: If multiple categories or colors are used, a legend is essential to explain what each symbol or color stands for. This helps viewers interpret data quickly and correctly.

4.2 Common Misuses of Scatter Plot

- **Using scatter plots for categorical or non-numerical data**
  Scatter plots are meant for continuous numerical values. Using them for categories like "Red, Blue, Green" or "Male/Female" makes interpretation meaningless because points cannot be positioned accurately based on non-numeric values.
- **Overloading too many data points without transparency**
  When thousands of points are plotted together, they overlap and create visual clutter. Without transparency or grouping, the graph becomes unreadable and patterns are lost.
- **Misinterpreting correlation as cause-and-effect relationship**
  If two variables move together, it does not mean one causes the other. Scatter plots only reveal relationships, not reasons. For example, ice-cream sales and drowning cases may both increase in summer, but one does not cause the other.
- **Ignoring outliers that may alter trend understanding**
  Outliers may represent rare events or errors. If they are overlooked, conclusions may be inaccurate. In some cases, a single extreme value can heavily impact the trend line.
- **Poor axis scaling leading to fake or exaggerated patterns**
  If axes are not scaled properly, small changes may appear very large or correlations may look stronger/weaker than they actually are. Accurate scaling is essential for honest visualisation.
- **Plotting unrelated variables just to show a visual comparison**

If there is no logical connection between variables, the scatter plot loses purpose. Random comparisons only confuse viewers and do not generate meaningful interpretation.

Self-Assessment Questions – 3

Multiple Choose Questions	
9	A scatter plot shows the relationship between ______ numerical variables.
10	A line drawn to show overall direction in a scatter plot is called a ______ line.
11	Points that lie far away from most data values are called ______.
12	Scatter plots should not be used for ______ data.

5. PICTOGRAM CHART

A pictogram chart (also called pictograph or pictogram) is a visual representation of data using pictures or icons instead of traditional bars or points. Each symbol represents a specific quantity, such as one unit or a group of units. For example, one icon of a person may represent 10 people.

Pictograms make data easily understandable, especially for beginners, young learners, or non-technical audiences. They are effective in storytelling and in cases where numbers need to be communicated in a more engaging and visually appealing way.

This type of chart is frequently used in reports, infographics, presentations, newspapers, and educational content.

When to Use a Pictogram Chart

Use a pictogram when:

- The audience includes children, beginners, or the general public.
- You want to communicate data in a simple and visually appealing way.
- The dataset is small, discrete, and easily countable.
- You need to present numerical comparisons in a memorable format.
- You want to simplify statistics using symbols instead of raw numbers.

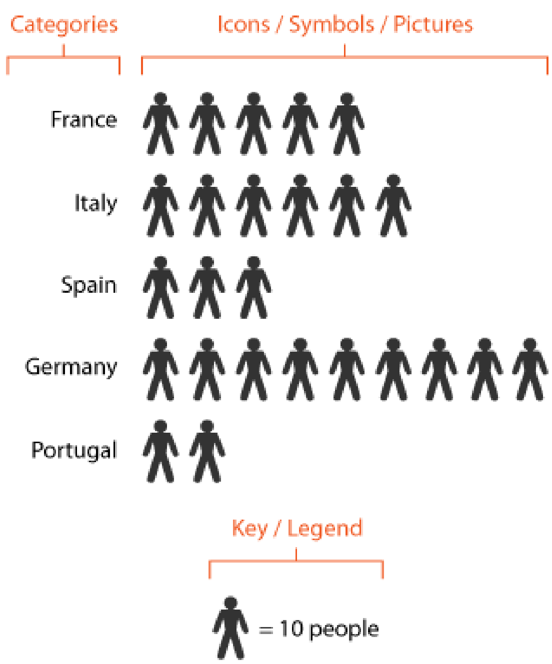


Fig: Pictogram Chart

Let's understand how this **pictogram chart** represents data using symbols instead of traditional bars or numbers. Each category on the left represents a **country**, such as France, Italy, Spain, Germany, and Portugal.

- Instead of showing numerical values directly, the chart uses **human icons to represent quantities**. Each icon represents a fixed number of people. This makes the chart easier to understand visually.
- The **legend at the bottom explains the scale**, indicating that **one icon represents ten people**. By counting the icons in each row, we can estimate the value for each country.
- For example, **Germany shows the largest number of icons**, indicating a higher value compared to the other countries, while **Portugal has the fewest icons**.
- This example demonstrates how **pictogram charts simplify numerical comparisons using visual symbols**, making data more engaging and easy to interpret.

Examples of Where Pictogram Charts Are Used

- Education: Used to show the number of students enrolled each year by representing each student or group of students with icons.
- Business**:** Revenue growth or yearly sales can be displayed using product symbols or currency icons to visually represent profit.
- Health: The number of patients affected by a disease can be shown using person icons, making health statistics easier to understand.
- Marketing: Customer reach or total product sales are shown using icons like bags, people, or shopping carts to compare audience size.
- Surveys & Polls: Responses or preferences are represented using smiley faces, stars, or thumbs-up icons to show voting results clearly.
- Social Media: User icons are used to compare followers count across platforms, showing growth or differences visually.
- Transportation: Monthly count of vehicles such as cars, buses, or bikes can be displayed using corresponding vehicle icons.

Advantages of Pictogram Chart

- Very easy to understand, even without analytical knowledge.
- Makes data more attractive, interactive, and memorable**.**
- Helps in fast communication of comparative values.

- Good for presentations, posters, and infographics**.**
- Encourages better engagement and retention of information.
- Useful for storytelling with data**.**

Limitations of Pictogram Chart

- Not suitable for large or complex datasets**.**
- Icons may oversimplify data and reduce accuracy.
- Difficult to show exact values when symbols represent groups.
- Visual scaling issues may distort interpretation.
- If icons are not uniform, it may lead to misinterpretation.
- Limited in representing multiple data variables at once.

5.1 Best Practices for Using Pictogram Chart

- **Use simple and consistent icons for clarity**

  Icons should be easy to recognise and uniform in style. Using too many different shapes can confuse the viewer, so consistency helps maintain visual clarity.
- **Clearly mention how much each symbol represents (e.g., 1 icon = 50 units).**

  A pictogram works only when viewers know what each icon stands for. Stating the value prevents misinterpretation and makes data comparison easier.
- **Use partial icons only when necessary and clearly define them.**

  When data values do not match full units, half or quarter icons can be used. However, they should be labelled or explained so the audience can calculate correctly.
- **Avoid overcrowding—keep the chart clean and balanced.**

  Too many icons in one place make the pictogram messy. Proper spacing ensures readability and enhances the visual appeal of the chart.
- **Choose icons that match the theme of the data (e.g., book icon for literacy rate).**

  The symbol should relate to the topic being presented. Relevant visuals help the audience connect more easily with the information.
- **Use color highlights to show difference or growth patterns.**

  Colors can indicate increase, decrease, or comparison between groups. Highlighting trends with color makes the chart more expressive and informative.
- **Provide legends or notes where needed.**

If multiple icon types, colors, or units are used, a legend is necessary. It guides viewers and ensures that data interpretation is accurate and quick.

5.2 Common Misuses of Pictogram Chart

- Using pictograms for large/complex datasets where bar or line charts are better: Pictograms work well for small data, but when numbers are very large or involve many values, icons become too many to count, making the chart confusing instead of informative.
- Icons scaled disproportionately, giving wrong visual perception: If icons are stretched or resized unevenly, they may visually exaggerate or minimise data. Even a small change in size can mislead viewers into thinking one value is much larger than another.
- No mention of unit value per icon, leading to confusion.**:** Without specifying what one icon represents, viewers cannot interpret the numbers correctly. Unit clarity (ex: 1 icon = 20 people) is essential for accurate understanding.
- Over-decorating the chart, reducing readability: Too many colors, styles, backgrounds, or icons can distract attention from actual data. A pictogram must remain clean, simple, and visually balanced to communicate effectively.
- Using irrelevant or distracting images unrelated to the data.: Icons should match the type of information being presented. Using random graphics weakens the message and reduces meaning. Relevant symbols strengthen understanding and connection.
- Trying to show too many categories, resulting in a cluttered chart: When many groups or categories are represented, the screen fills up with icons and becomes difficult to read. Pictograms are best when focused on few categories at a time.

6. TIMELINE CHART

A **timeline chart** is a visual representation of events or activities arranged in chronological order. It displays how events progress over time using points, labels, arrows, or bars. Timelines make it easy to track historical developments, project milestones, and time-based patterns.

They are commonly used in presentations, reports, project management, education, and historical studies to communicate sequences clearly and visually.

When to Use a Timeline Chart

Use a timeline when:

- You want to show **events or data over time.**
- The sequence and order of events is important.
- You need to present **milestones, deadlines, or progress.**
- You are explaining **history, evolution, or lifecycle** of a subject.
- You want to compare durations or identify time gaps.

Examples of Where Timeline Charts Are Used

- History: Used to display major historical events such as world wars, revolutions, or civilization development in chronological order.
- Project Management: Shows project phases, task schedules, deadlines, and important milestones for tracking progress over time.
- Education: Helps illustrate the evolution of technology, invention timelines, or development of computer systems in a sequential flow.
- Business: Used to present company growth, product release history, or expansion milestones year by year.
- Biology: Displays the stages of human development (infant → child → adult → aged), life cycles, or evolution processes over time.
- Marketing: Represents marketing campaign rollout phases, launch dates, promotional activities, and results over a timeline.

Advantages

- Shows progression clearly over time.

- Helps track **milestones, deadlines, achievements**.
- Easy to understand even for non-technical viewers.
- Useful for storytelling and sequential explanation.
- Perfect for planning, scheduling, and tracking history.

Limitations

- Not ideal for large datasets with too many events.
- Overcrowding may reduce readability.
- Hard to compare unrelated timelines.
- Limited to time-based information only.
- Requires proper spacing and scaling.

6.1 Best Practices

- **Arrange events in clear chronological order.**
  The timeline should flow smoothly from oldest to most recent events (or vice-versa). Proper sequencing helps users understand progression and how one event led to another.
- **Highlight key milestones or turning points.**
  Important events such as breakthroughs, achievements, or major shifts should be visually emphasised using color or size differences so they stand out instantly.
- **Use icons, color codes, or shapes for clarity.**
  Adding simple visual elements like symbols or color blocks improves readability. Icons related to the event make the timeline more engaging and easier to follow visually.
- **Maintain equal spacing for time intervals.**
  If one month and one year are shown with the same spacing, it creates confusion. The spacing should represent time accurately so readers can understand duration between events correctly.
- **Add labels and short descriptions for better understanding.**
  Each event should have a brief note explaining what happened. Short, clear captions help viewers interpret the timeline without needing additional reference.

6.2 Common Misuses

- **Using too many text labels making the chart cluttered.**

Overloading the timeline with long descriptions, sentences, or tiny text makes it difficult to read. A timeline should summarise events, not describe them in full paragraphs.

- **Ignoring scale—uneven spacing creates confusion.**
  If some periods are shown close together while others are widely spaced, the viewer may misinterpret time duration. Time gaps should be visually proportional to maintain accuracy.
- **Adding irrelevant events that distract from the main timeline.**
  Including unnecessary minor events weakens the focus of the chart. Only meaningful and relevant events that contribute to the topic should be included.
- **Using timelines for non-sequential or unrelated data.**
  Timelines are only effective when events occur in a specific order over time. Using them for unrelated topics or unordered information leads to confusion and misinterpretation.

7. HIGHLIGHT TABLE

A **highlight table** is a type of visual table that uses **color shading or gradients** to emphasise values based on magnitude or category. Instead of just numbers, colors help the viewer quickly identify patterns, highs, lows, and comparisons.

Highlight tables are widely used in dashboards, analytics, and business reporting to spotlight areas that need attention.

When to Use a Highlight Table

Use a highlight table when:

- You want to compare values **within rows or columns.**
- The aim is to quickly identify **highest/lowest values.**
- You need to highlight **performance variations.**
- Detailed numeric values are still important along with visuals.

Advantages

- Clear visual comparison of values.
- Highlights top and bottom performers instantly.
- Supports both words/numbers along with visual cues.
- Good for dashboards, reports, KPIs.

Limitations

- Too many colors may confuse users.
- Not ideal for very large tables.
- Colorblind users may struggle if contrast is poor.
- Without scale/legend, colors can be misinterpreted.

Best Practices

- Use limited and meaningful colors only: Too many colors can distract the viewer, so it is best to use a small color palette. Each color should represent a clear purpose such as good performance, low scores, or average levels.

- Choose intuitive color scales (green = good, red = low): Viewers easily recognise familiar color meanings.
  For example, green generally represents positive growth, while red indicates decline or poor performance. Using such natural color associations improves quick understanding.
- Provide a legend or color rule for clarity: A small legend or key helps viewers know what each color represents. Without this, readers may misinterpret the data, even if the visualisation is correct.
- Keep the table clean, not overloaded with text: Highlight tables should emphasise values visually, not through long sentences. Clean formatting ensures that comparisons stand out without overwhelming the viewer.
- Highlight only the most important differences: Color should be used to draw attention to key values such as highest, lowest, or critical points. Over-highlighting everything removes focus and weakens the purpose of a highlight table.

Common Misuses

- Using flashy or overly bright color gradients: Bright or excessively vibrant colors can strain the eyes and distract from interpretation. Subtle shades work better for professional and analytical readability.
- Highlighting too many cells, losing focus: If every cell is highlighted, nothing stands out. The main purpose of highlighting is lost, and the viewer cannot identify which values are actually important.
- No clear legend → viewer misunderstands values: Without a color guide, readers may misinterpret what "dark green vs light green" means. Some may assume it means something different than intended, leading to incorrect conclusions.
- Using similar shades causing confusion: If colors are too close in shade or saturation, differences become difficult to detect. Clear contrast is needed to make the most important values noticeable at a glance.

8. BULLET GRAPH

A **bullet graph** is a compact and data-rich visualisation that compares a **primary measure (actual performance)** to **target or benchmark values.** It resembles a horizontal bar with reference ranges shaded behind it. Bullet graphs are widely used in dashboards to evaluate performance, progress, and goal achievement.

When to Use Bullet Graph

Use a bullet graph when:

- You want to compare **actual vs target performance.**
- The data needs to be compact and dashboard-friendly.
- You want to highlight **achievement, progress, or KPI status.**
- You need more detail than a simple bar chart provides.

Examples of Where Bullet Graphs Are Used

- **HR:** Used to compare an employee's actual performance score with the expected target or rating benchmark.
- **Sales:** Helps visualise monthly revenue achieved versus the assigned sales target to measure progress.
- **Finance:** Shows how much budget is spent compared to the total allocated budget for better financial tracking.
- **Production:** Used to evaluate how much output is produced and whether it meets the required production demand.
- **Education:** Displays student marks versus the pass benchmark or expected grade to highlight performance status.
- **Marketing:** Shows the number of leads generated compared to the campaign goal to measure marketing effectiveness.

Advantages

- Highly space-efficient—fits dashboards well.
- Shows **target, actual, and ranges in one view**.

- Good for business performance tracking.
- Helps in quick decision making.
- More informative than a standard bar chart.

Limitations

- Not suitable for comparing many categories at once.
- May be difficult for beginners to interpret initially.
- Color/range selection must be meaningful.
- Requires clear legends for accurate understanding.

Best Practices

- Define actual, target, and range values clearly.
  A bullet graph only works when users can easily differentiate the actual performance bar, the target line, and the background qualitative ranges. Clarity ensures correct interpretation at a glance.
- Keep colors consistent (e.g., grey for range, bold for actual).
  Using a uniform color pattern prevents confusion. Typically, the background range is kept neutral (like shades of grey), while the actual value is highlighted in a darker or bold color so it stands out immediately.
- Use sparingly to avoid dashboard crowding.
  Bullet graphs are data-dense. Using too many of them together can overwhelm the viewer. It is best to use them only for KPIs or top-level performance metrics where comparison matters.
- Place units and labels near the bars.
  Labels such as percentages, currency, or quantities should be placed close to the visual element for quick reading. This reduces the effort needed to interpret data and avoids unnecessary eye movement.
- Highlight target line clearly.
  The target marker should be sharp and noticeable—often a thin black line or contrasting color—so users can easily compare actual performance against expected goals.

Common Misuses

- Using too many bullet graphs together → clutter.

When dashboards contain multiple bullet graphs side-by-side, they become cramped and lose readability. The viewer struggles to focus, reducing the chart's effectiveness.

- Ambiguous color scales confuse the viewer.

  If colors are too similar or meanings are not explained, users may misinterpret performance levels. A clear color scheme with distinct shades for low, medium, and high performance is essential.
- No target marker or unclear benchmarks.

  A bullet graph without a defined target line becomes just a bar chart. Viewers cannot assess whether performance is satisfactory or below expectation without a clear benchmark reference.
- Over-decorative styling losing analytical purpose.

  Adding gradients, shadows, or too much design reduces clarity. Bullet graphs are meant to be minimalistic and analytical—visual simplicity is more important than

Self-Assessment Questions – 4

Multiple Choose Questions	
13	A bullet graph compares actual performance with a _____ benchmark.
14	The background range in a bullet graph represents qualitative _____ levels.
15	Using too many bullet graphs on one dashboard can create _____.
16	The marker line showing expected value in a bullet graph is called _____.

9. SUMMARY

- Data visualisation techniques used to analyse, interpret, and present data in a more meaningful and visually effective manner. Each chart type serves a specific purpose and is supported by best-practice guidelines to ensure clarity, accuracy, and usability. The chapter also highlights common mistakes to avoid for better visual communication.
- A waterfall chart is used to display the cumulative effect of sequential changes, helping users understand how values increase or decrease step-by-step. **Best practices** include labeling each component clearly and maintaining consistent color codes, while the major **misuses** involve cluttered bars, missing reference totals, and unclear difference points.
- An area chart shows trends over time while highlighting the magnitude or volume of change using filled color regions. **Best practices** emphasise using proper scaling, clear labeling, and avoiding excessive layers, whereas **misuses** include using it for discrete non-time based data or adding too many stacked categories which reduce clarity.
- A scatter plot displays relationships between two numerical variables, helping identify trends, clusters, or correlations. Correct usage involves proper axis labeling, color coding for multiple groups, trend lines, transparency, and outlier indication.
- A pictogram chart uses icons or symbols to represent quantities in a more visual and engaging manner. Best practices include using simple icons, defining units clearly, and avoiding overcrowding.
- A timeline chart represents events in chronological order, useful for showing evolution, progress, and historical sequences. Misuses involve too much text, inaccurate spacing, or including irrelevant events that confuse the intended flow.
- A bullet graph compares actual performance with a target value using compact visually strong bars. It is best used for KPI tracking with clear benchmarks, consistent coloring, and minimal styling.

10. GLOSSARY

Waterfall Chart	–	A chart showing how a starting value changes step-by-step through increases or decreases to reach a final outcome.
Bridging Values	–	The intermediate additions or subtractions shown between the start and end values in a waterfall chart.
Cumulative Change	–	The total impact after applying all positive and negative changes over time.
Area Chart	–	A chart that shows trend over time with filled color beneath the line to represent volume or magnitude.
Stacked Area Chart	–	An area chart with multiple data sets stacked to show how each category contributes to the overall total.
Trend Line	–	A line used on charts to show the general direction of data movement (increasing, decreasing, or constant).
Scatter Plot	–	A visual representation of two numerical variables using points to identify relationships or patterns.
Correlation	–	Indicates how strongly two variables are related (positive, negative, or no relationship).
Outlier	–	A data point that is unusually high or low compared to the rest, suggesting exceptional behavior or error.
Regression Line	–	A best-fit line drawn on a scatter plot to show trend direction and predict values.
Pictogram / Pictograph	–	A chart that uses icons or symbols to represent data instead of bars or lines.
Unit Representation	–	The defined value one pictogram icon stands for (like 1 icon = 50 people).
Timeline Chart	–	A visualisation displaying events in chronological (time-based) order.

Milestone	–	A significant event or important achievement highlighted within a timeline.
Highlight Table	–	A table format where numbers are emphasised using color shades to show high/low values.
Color Scale	–	A gradient or shade system used to represent data strengths such as highest to lowest values.
Legend	–	A reference key explaining symbols, icons, or colors used in a chart.
Bullet Graph	–	A compact performance chart comparing actual value vs target with background ranges.
Target Marker	–	A reference indicator line in a bullet graph representing the goal or benchmark.

11. TERMINAL QUESTIONS

1. What is the main purpose of a Waterfall Chart?
2. When should an Area Chart be used?
3. What does a Scatter Plot help us understand?
4. Why is defining unit value important in a Pictogram Chart?
5. What type of information is best represented using a Timeline Chart?
6. How does a Highlight Table improve data understanding?
7. What is the primary purpose of a Bullet Graph?
8. Name one common misuse of graphical charts covered in this unit.

12. ANSWERS

12.1. Self-Assessment Answers:

1. **Answer:** increases
2. **Answer:** financial
3. **Answer:** total
4. **Answer:** bridging
5. **Answer:** time
6. **Answer:** stacked
7. **Answer:** volume
8. **Answer:** cluttered
9. **Answer:** two
10. **Answer:** trend/regression
11. **Answer:** outliers
12. **Answer:** categorical
13. **Answer:** target
14. **Answer:** performance
15. **Answer:** clutter
16. **Answer:** target line

12.2. Terminal Question Answers

Answer 1: A Waterfall chart is used to show how an initial value changes step-by-step through a series of increases or decreases before reaching a final total. It is useful for visualising profit breakdowns, financial changes, or component contributions.

Answer 2: An Area chart is ideal when the goal is to show how values change over time while also highlighting total volume or cumulative growth. It is suitable for trends in sales, population, energy usage, etc.

Answer 3: A scatter plot helps identify relationships or correlations between two numerical variables. It can reveal patterns, clusters, trends, or outliers in the data.

Answer 4: A pictogram uses icons to represent data, so clearly defining the unit (e.g., 1 icon = 10 persons) is essential for accurate interpretation. Without units, viewers cannot understand the actual quantity.

Answer 5: Timeline charts are best for events arranged in chronological order. They help show history, evolution, project phases, or growth over time clearly and sequentially.

Answer 6: A highlight table uses color shading to emphasise high and low values, making comparison easier. It draws attention to patterns or extreme values quickly without needing to read every number.

Answer 7: A bullet graph compares actual performance with a target benchmark in a compact form. It is commonly used in dashboards for KPI tracking such as sales completion or productivity goals.

Answer 8: Overloading charts with too many colors, icons, or values is a common misuse. It reduces readability and hides meaningful insights, making the visualisation ineffective.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 4

DCA2207

DATA MINING AND VISUALISATION

Unit – 14

Chart Types-3

TABLE OF CONTENTS

1. INTRODUCTION

In the previous unit, learners gained a broad understanding of various data visualisation charts and techniques used to present information clearly, meaningfully, and visually. They explored how each chart functions, when it is most appropriate to use, and how to apply best practices to ensure effective and accurate communication of data. They also learned to recognise and avoid common mistakes that could otherwise result in confusing or misleading visuals.

Learners studied **Waterfall charts** to interpret cumulative changes, **Area charts** to track trends and volume over time, and **Scatter plots** to analyse relationships between numerical variables. They also worked with **Pictogram charts** for visual quantity comparison, **Timeline charts** for sequential event representation, **Highlight tables** for value emphasis using color, and **Bullet graphs** to compare actual results against targets. By completing this unit, learners have developed the ability to choose the right visualisation for different datasets, design well-structured charts using best practices, interpret patterns accurately, and avoid misrepresentation of data. This foundation now enables them to present information more effectively in reports, presentations, dashboards, and data-driven decision-making environments.

In this unit, learners will understand how dashboards are developed, structured, and used to present data meaningfully. They will learn the complete **dashboard development process**, beginning from identifying business requirements, gathering data from different sources, applying ETL for cleaning and preparation, storing data in warehouses, and finally transforming it into visual insights through graphs, KPIs, and charts. Students will also explore the concept of **dashboard architecture,** which includes multiple layers—data source, integration, storage, modelling, visualisation, and user access.

This unit also introduces learners to real-world usage scenarios such as sales monitoring dashboards, student performance dashboards, healthcare analytics dashboards, and more. By learning how dashboards are designed and interpreted, students will be able to build visually effective reports, select suitable visual elements, use filters and drill-downs, and communicate insights clearly to decision-makers. Overall, the unit equips learners with the skills to transform raw data into interactive dashboards for business analytics and performance tracking.

1.1. Learning Objectives

By the end of this unit, you will be able to:

- Understand the purpose, function, and application of various data visualisation techniques
- Identify when each visualisation tool
- Apply best practices to design clear, accurate, and visually meaningful dashboards
- Analyse and compare patterns and trends for better decision-making.

2. DASH BOARD DEVELOPMENT

Dashboards are interactive visual platforms that display key information at a glance, helping users monitor performance, track progress, and make informed decisions. They bring together multiple data sources and convert raw numbers into meaningful visual insights using charts, tables, graphs, and indicators. Instead of reading large datasets, dashboards allow viewers to understand trends, patterns, and summaries quickly through visuals.

The development of dashboards involves planning what information is important, selecting appropriate visual elements, and arranging them in a structured layout for easy interpretation. A good dashboard displays only relevant metrics, updates automatically when new data is added, and presents information in a clean, organised, and user-friendly format. Business analysts, managers, and decision-makers rely on dashboards for real-time monitoring, performance measurement, and strategic planning. A well-designed dashboard provides value by highlighting important metrics, showing alerts or deviations, comparing past vs present performance, and supporting decision-making. Modern dashboards can update automatically, allowing organisations to monitor activities in real time. In industries such as business analytics, healthcare, finance, marketing, manufacturing, and education, dashboards have become essential tools for reporting, forecasting, and planning.

In simple terms, dashboard development transforms data into an actionable story, enabling faster understanding, comparison, and decision-making—all from a single screen.

Components of a Dashboard

A dashboard is built from several key elements that work together to present information clearly and effectively:

- **Data Sources**: The raw data collected from databases, spreadsheets, business applications, or online platforms. Without data, a dashboard cannot function.
- **Visual Elements (Charts, Graphs, Widgets)**: These include bar charts, line charts, KPIs, maps, gauges, and tables that convert numeric data into visual insights.
- **Metrics & KPIs (Key Performance Indicators)**: These are measurable indicators such as sales numbers, revenue growth, attendance percentage, or production output that help evaluate performance.
- **Filters & Interactive Controls**: Tools like dropdowns, slicers, and date filters allow users to customise views and explore data dynamically.

- **Layout & Design Structure**: The arrangement of visuals on the dashboard, ensuring organised flow, readability, and logical grouping of information.
- **Refresh/Update Mechanism**: Dashboards often refresh automatically, ensuring real-time visibility of changes or new data entries.

Steps in Dashboard Development

Developing a dashboard follows a systematic process to ensure clarity and purpose:

1. Identify the Objective: Define *why* the dashboard is needed — performance tracking, forecasting, operations monitoring, etc.
2. Select Key Metrics and KPIs: Choose only the most important indicators that reflect business outcomes or data insights.
3. Collect and Prepare Data: Extract data from sources, clean missing values, remove duplicates, and ensure accuracy.
4. Choose Appropriate Visualisations: Match chart type with data meaning — line charts for trends, bar charts for comparisons, bullet charts for performance targets, etc.
5. Design Layout and Structure: Arrange visuals logically, with most important metrics placed at the top for quick visibility.
6. Implement Interactivity (Filters, Drilldowns): Allow users to explore details, filter categories, or compare time periods as needed.
7. Review, Test, and Refine: Evaluate readability, accuracy, and performance. Remove clutter and improve flow where necessary.
8. Deploy and Maintain: Publish the dashboard, schedule data refresh, and update KPIs periodically.

Features of an Effective Dashboard

A good dashboard is more than just visuals — it must be purposeful, intuitive, and informative.

- **Clean, uncluttered layout** ensures information can be interpreted quickly.
- **Uses appropriate chart types**, avoiding unnecessary complexity.
- **Interactive and user-friendly,** allowing filtering or drilling deeper.
- **Prioritises key insights,** highlighting what matters most.
- **Readable color scheme,** consistent and meaningful, with legends when needed.
- **Real-time or automatically updated data** for accuracy in decision-making.
- **Accessible to all users,** even those with limited analytical background.

A dashboard should tell a story clearly without overwhelming the reader.

3. CHOROPLETH MAP

A **choropleth map** is a thematic map that uses **different shades or colors to represent numerical values or data intensity** across geographic areas such as countries, states, districts, or regions. Darker or more intense colors usually represent higher values, while lighter shades indicate lower values. It provides a visual way to compare data patterns across locations, making it easier to interpret geographic variations at a glance.

Purpose of a Choropleth Map

- **To represent data based on geographical distribution**
  Choropleth maps help visualise how a particular variable is spread across different geographical regions like states, districts, or countries.
- **To identify regional variations or trends**
  By using color shades, the map allows viewers to instantly spot which areas have higher or lower values, making trend differences region-wise easy to identify.
- **To compare values between different areas visually**
  The color intensity makes comparison simple — darker areas represent higher values and lighter areas indicate lower values, enabling quick cross-region comparison.
- **To highlight high-density and low-density zones**
  Choropleth maps are especially useful for spotting hotspots and sparse areas, helping decision-makers understand where concentration is highest or lowest.

Examples of Where Choropleth Maps Are Used

- **Showing population density across states or countries**

  Regions with higher population density appear darker, helping planners assess crowding and resource demand.

- **Representing literacy rates or education levels region-wise**

  Different shades make it easy to compare educational performance across different areas.

- **Visualising crime rates or disease outbreak intensity by location**

  Useful in public safety and healthcare to identify high-risk zones needing attention or intervention.

- **Displaying rainfall patterns across climatic zones**

Helps in studying weather conditions, drought-prone areas, and regions with heavy rainfall.

- **Showing election voting patterns across regions**

  Political analysts use choropleth maps to visualise which regions support which party or candidate.

  1. **Advantages and Limitations of Choropleth Maps**
  2. Advantages of Choropleth Maps

  - **Provides quick regional comparison at a glance:** Differences between areas are immediately visible because each region has a distinct shade, allowing users to quickly identify which regions perform higher or lower.
  - **Easy to understand even for non-technical users:** Since it uses colour intensity rather than complex numbers, anyone can interpret the map without needing statistical knowledge or deep analytical skills.
  - **Good for representing density-based or percentage-based data:** Choropleth maps work best with normalised values such as percentage, rate, or density, making them ideal for population distribution, literacy rate, or disease spread.
  - **Visually appealing and widely used in reports, research, and planning:** The colour-filled regions make the map attractive and easy to read, which is why it is frequently used in academic studies, business presentations, policy planning, and government reports.
  - **Useful for identifying hotspots, trends, and geographic patterns**: Areas with high intensity or darker colours immediately stand out, helping analysts detect clusters, critical zones, and regional behaviour patterns effectively.

  Limitations of Choropleth Maps

  - Only effective for **normalised data** (percentages, rates) — not raw totals
  - Can be misleading if color scale is not chosen properly
  - Large regions may appear more significant due to size, not actual data
  - May oversimplify complex data details
  - Requires proper legend and color clarity to avoid interpretation errors

3. **Best Practices for Using Choropleth Maps**

- **Use a clear and meaningful color gradient (light → dark)**
  Shades must represent increasing intensity, where lighter areas show lower values and darker tones indicate higher values. This helps viewers immediately understand differences.
- **Always include a legend showing value ranges**
  A legend is essential to explain what each shade represents—without it, users cannot correctly interpret the numeric meaning behind the colors.
- **Choose equal-interval or quantile classification for accurate shading**
  Classification methods divide data into balanced ranges, ensuring colors reflect actual distribution and avoiding misleading visual interpretation.
- **Avoid too many color shades—keep it simple and readable**
  Using too many similar colors makes the map confusing. A limited number of shades helps users clearly differentiate between value levels.
- **Provide labels or tooltips for detailed values if needed**
  Hover labels or text indicators can be used to show exact numbers, giving more depth and making the visualisation informative without clutter.

Use normalised values like percentage, rate per population—not raw counts Choropleth maps are most accurate when values are adjusted for population or area, preventing large regions from appearing dominant only due to size.

A **choropleth map is a powerful tool for visualising geographically distributed data using color intensity.** It helps users understand differences, trends, and patterns across regions effortlessly. When designed with proper legends, color scales, and normalised data, choropleth maps become one of the most effective methods for spatial data analysis.

Self-Assessment Questions – 1

Multiple Choose Questions	
1	A choropleth map represents data using different _____ or shades over geographic regions.
2	Darker shades in a choropleth map usually indicate _____ values.
3	A choropleth map requires a clear _____ to explain what each color represents.
4	Choropleth maps work best with _____ data such as percentages or rates instead of raw totals.

4. WORD CLOUD

A **word cloud** is a visual representation of text data where the *size of each word* indicates how frequently it appears. Words that appear more often are displayed larger, bolder, or more centrally placed. Word clouds are widely used for summarising large text collections quickly and highlight the most important or repeated terms visually.

Use Cases

- Summarising survey responses or feedback
- Highlighting frequently used keywords in documents or speeches
- Analysing social media comments or reviews
- Extracting themes from articles, blogs, or research papers
- Visualising most discussed topics in a dataset

Advantages and Limitations of word cloud

Advantages of Word Cloud

- **Provides quick insight into most common terms:** The larger words instantly highlight frequently appearing terms, helping users understand key themes without reading the entire text.
- **Easy to understand — suitable for reports & presentations**: Even beginners can interpret a word cloud easily, making it a useful tool for sharing results with non-technical audiences.
- **Useful for exploratory text analysis:** It helps researchers identify patterns, dominant ideas, and recurring keywords in early stages of data exploration.
- **Visually engaging and attention-grabbing**: The colorful and varied layout makes word clouds attractive, which is why they are widely used in dashboards, surveys, and public presentations.

Limitations of Word Cloud

- **Does not show context or sentence meaning:** It only shows word frequency, not how or why the word is used, which may lead to misinterpretation.
- **Word frequency overshadows rare but important terms:** Important keywords that appear less frequently may be displayed too small and easily overlooked.
- **Too many styles/colours can reduce readability:** If not designed properly, the cloud becomes visually noisy and difficult to interpret.

- **Not suitable for detailed analysis:** A word cloud is good for overview but lacks depth — it cannot show relationships between words or deeper insights like sentiment or context.

Best Practices for Using Word Cloud

- **Use simple fonts and limited color shading**: Plain, readable fonts and a small color palette make the cloud easier to read. Too many colors or fancy fonts can distract the viewer and reduce clarity.
- **Remove stopwords (the, is, and, etc.) before generating:** Common words that do not add meaning should be filtered out. This ensures that only meaningful terms appear, providing a more accurate representation of important text content.
- **Keep layout clean with moderate spacing**: Words should be spaced evenly without overlapping. A clean layout helps the user quickly identify major keywords without visual strain.
- **Group similar words (e.g., learn/learning) for accuracy:** Combining related forms of words prevents duplication and gives a more accurate picture of term frequency. For example, counting "learn" and "learning" together reflects the concept more clearly.

Self-Assessment Questions – 2

Multiple Choose Questions	
5	A word cloud visually represents text data where word size indicates _____.
6	Higher frequency words appear _____ compared to less frequent ones.
7	Word clouds are commonly used for summarising large blocks of _____ data.
8	Excessive decorative styling in word clouds can reduce _____ clarity.

5. NETWORK DIAGRAM

A **network diagram** visually represents relationships or connections between entities. It is structured using **nodes** (data points) connected by **edges** (links). Network diagrams help in understanding how elements interact, communicate, or influence each other within a system.

Use Cases of Network Diagrams

- **Social network analysis (people → connections)**

  Network diagrams can show how individuals are linked within a social group — for example, friendships, communication flow, or influencer connections on social media.

- **Computer network topology (devices → links)**

  They help visualise how computers, servers, routers, and devices communicate in a network system, making troubleshooting and infrastructure planning easier.

- **Workflow or process mapping**

  Network diagrams illustrate the steps and flow of tasks in a process, helping organisations track dependencies, responsibilities, and process efficiency.

- **Neural networks and knowledge graphs**

  In artificial intelligence and machine learning, they represent nodes (neurons) and connections between them, showing how data flows through layers of a model.

- **Citation or influence mapping in research**

  Researchers use network diagrams to trace how authors, papers, or ideas are connected, helping identify influential studies or clusters of academic work.

Advantages and Limitations of Network Diagram

Advantages of Network Diagrams

- **Clearly shows relationships & data flow**: Network diagrams visually represent how elements are connected, making it easy to trace movement, communication, or influence within a system.
- **Helps identify hubs, clusters, and network strength**: By observing node size and connection density, users can pinpoint central nodes (hubs), grouped structures (clusters), and evaluate how strong or weak the network is.

- **Useful for visualising complex interconnected systems**: Whether it's social media connections, computer networks, or biological pathways, network diagrams simplify complexity into a visual web that is easier to understand.
- **Supports decision-making in IT and analytics**: IT teams use these diagrams to troubleshoot network failures, analysts use them to study information flow, and organisations use them to optimise communication structure and resources.

Limitations of Network Diagrams (Slightly Elaborated)

- **Becomes cluttered if too many nodes exist**: When a network is too dense, connections overlap and readability drops, making the diagram visually overwhelming.
- **Hard to interpret without grouping or layout control**: Organising nodes into clusters or hierarchies is necessary. Without structure, the diagram becomes confusing and difficult to analyse.
- **Requires careful labeling for clarity**: Names, roles, or values must be clearly indicated on nodes and connections. Without proper labeling, viewers may misunderstand relationships or importance.

Best Practices for Network Diagrams

- **Use color coding for node grouping**
  Applying different colors to related nodes helps viewers quickly identify clusters, categories, or communities. This visual grouping reduces confusion and makes complex relationships easier to interpret.
- **Limit node count for readability**
  Too many nodes can overcrowd the diagram, making lines overlap and details hard to read. Only essential nodes should be displayed, or the network should be broken into sub-sections for clarity.
- **Use directional arrows when showing flow**
  When information, signals, or interactions move in a specific direction, arrows help indicate the flow clearly. This is especially important in workflow diagrams, data routing, or communication networks.
- **Include labels & legends for easy interpretation**
  Nodes should be named or symbolisymbolised clearly, and a legend must explain the coloured clearly, and a legend must explain the color codes, symbol sizes, and types of connections. This ensures that anyone viewing the diagram understands relationships without guesswork.

Self-Assessment Questions – 3

Multiple Choose Questions	
9	A network diagram shows relationships between elements using ______ and nodes.
10	Nodes represent entities, while ______ represent connections between them.
11	Network diagrams are useful for mapping social relationships or ______ systems.
12	Too many nodes without grouping can make the diagram appear ______.

6. CORRELATION MATRIX

A **correlation matrix** is a table or heatmap that shows the strength and direction of relationships between multiple numerical variables using correlation coefficients. Shades or colors represent how strongly variables are related.

Use Cases

- Feature selection in machine learning
- Statistical data exploration
- Detecting positive/negative correlations
- Financial risk modeling
- Health or environmental variable comparison

Advantages and Limitations of Correlation Matrix

Advantages of Correlation Matrix

- **Shows variable relationships at a glance**

  A correlation matrix visually displays how strongly different variables are related, enabling quick comparison without manual calculations.

- **Helps identify patterns and associations quickly**

  Positive, negative, or weak correlations become immediately visible, making it easier to detect trends or variable connections.

- **Useful for multivariate data analysis**

  When dealing with many variables, the matrix provides a compact view of all relationships at once, reducing analysis time.

- **Ideal for predictive modelling decisions**

  Machine learning and statistical models require understanding which variables influence the target value. The correlation matrix helps in feature selection and model improvement.

  Limitations of Correlation Matrix

- **Shows correlation, not causation**

Even if two variables move together, one may *not* cause the other. The relationship could be coincidental or influenced by other factors.

- **Works only with numerical data**

  Categorical or textual data cannot be used directly in correlation matrices, limiting its scope to quantitative datasets.

- **Over-interpretation may lead to false conclusions**

  High correlation might appear meaningful but could be misleading if taken without deeper context or statistical validation.

Best Practices for Using a Correlation Matrix

- **Use color-scale heatmaps for clarity**

  Representing correlations through color gradients makes it easier to distinguish strong, weak, and neutral relationships visually. Light-to-dark shades help readers interpret values instantly.

- **Highlight strong correlations visually**

  Marking highly positive or negative correlations helps users focus on the most impactful relationships without scanning every cell manually. This is especially useful in large datasets.

- **Remove highly-correlated features to avoid redundancy**

  When two variables are very closely related, keeping both may not add value and may even distort model performance. Eliminating or combining them improves data quality and efficiency.

- **Always interpret results carefully**

  Correlation only shows relationships, not causation. Results should be analysed with context and supporting evidence to avoid drawing incorrect conclusions based solely on visual patterns.

Self-Assessment Questions – 4

Multiple Choose Questions	
13	A correlation matrix shows relationships between variables using _____ coefficients.
14	A value close to +1 indicates a _____ correlation.
15	The matrix is often displayed using _____ maps or colored grids.
16	Correlation does not imply _____ even if values appear related.

7. GEOGRAPHICAL PLOTS

Geographical plots are visualisations that map data directly onto a geographical layout such as a world map, country map, or region. Patterns are shown using colors, points, icons, or scaled shapes, helping visualise **spatial distribution of data**.

Use Cases

- Sales distribution by region
- Climate & weather recordings
- Tourism or population density mapping
- Resource allocation and location-based planning

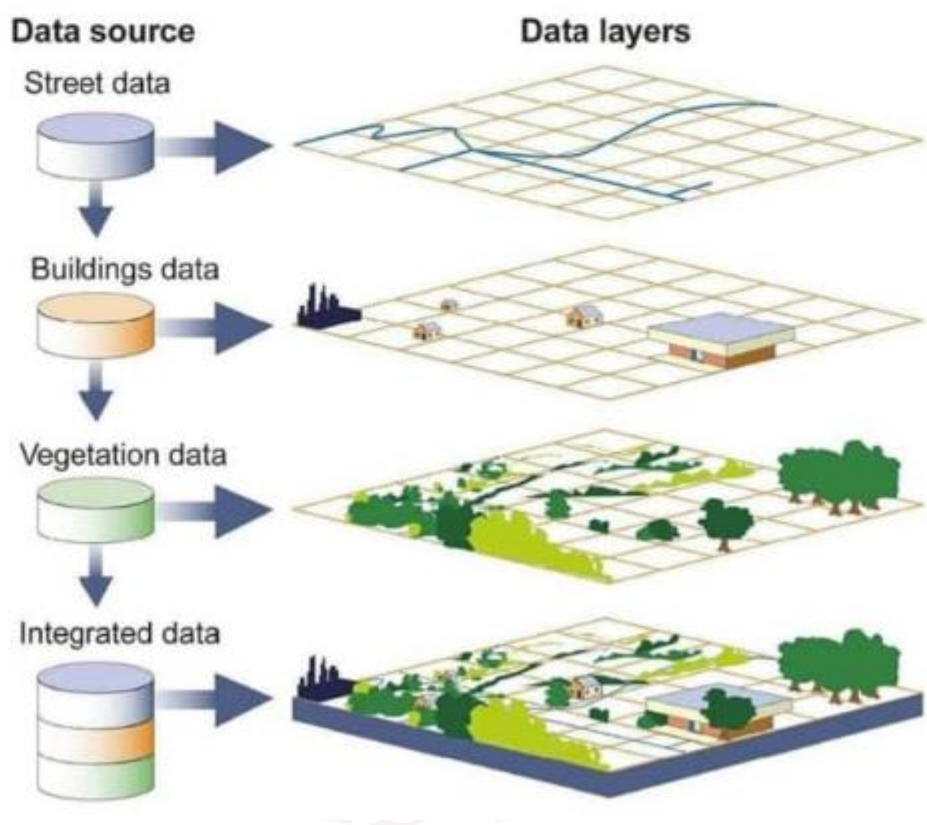

Fig: Geographical plot

The figure shows:

1. **geographic data sources** that are used in mapping.
2. It demonstrates **data layers**, which are commonly used in **geographical plotting tools**.
3. It illustrates how **multiple spatial datasets combine into a single map**.
4. It helps learners understand how **location-based data is visualised and analysed**.

Advantages and Limitations of Geographical Plots

Advantages of Geographical Plots

- **Displays real-world spatial patterns easily**
  Geographical plots visually map data onto actual locations, making patterns like population clusters, rainfall zones, or sales density easy to recognise at a glance.
- **Helps compare data region-wise quickly**
  By viewing values distributed across states, cities, or countries, users can instantly see which areas perform better or weaker without reading long tables or numbers.
- **Useful for decision-making in planning & logistics**
  These plots support real applications such as supply chain routing, resource distribution, market expansion, disaster response, and urban development by showing where attention is most needed.

Limitations of Geographical Plots

- **Requires correct coordinate/region mapping**
  If boundaries or locations are mapped incorrectly, results become misleading. Accuracy of geographic data is essential for meaningful interpretation.
- **Overlapping points may cause clutter**
  When many values lie close together, markers can stack on top of one another, reducing readability and hiding smaller details.
- **Needs consistent color scales to avoid misinterpretation**
  Inaccurate or inconsistent coloring may exaggerate or minimise data differences, so colors must be chosen carefully to represent values correctly

Best Practices for Geographical Plots

- **Use distinct color and size variations**: Colors and marker sizes should be clearly different so users can easily distinguish between high, medium, and low values. This prevents confusion and highlights spatial patterns more effectively.

- **Add tooltips or labels for detailed values**: Since maps often show summarised information, tooltips or value labels help users see exact numbers when hovering over or clicking on a region. This adds depth without cluttering the map.

- **Avoid crowding points — use clustering where needed:** When many data points overlap, grouping them into clusters improves clarity. Cluster markers display summaries instead of overlapping dots, ensuring the map remains readable and visually clean.

Self-Assessment Questions – 5

Multiple Choose Questions	
17	Geographical plots display data on ______ maps.
18	They help visualise regional differences such as weather, sales, or ______ levels.
19	Markers, shapes, or colors are used to represent ______ distribution.
20	Accurate location coordinates are essential to avoid ______ errors.

8. DENSITY MAPS

A **density map** shows how densely data points are distributed within an area. Darker regions indicate higher concentration, while lighter regions show low density. It is ideal for hotspot detection.

Use Cases

- Population density mapping
- Disease outbreak or hotspot tracking
- Crime or traffic congestion analysis
- Wildlife or vegetation density distribution

Fig: Density Maps

- The map displays **locations of shipping ports across the world**.
- Regions such as **Europe, East Asia, and North America show many points**, indicating higher port concentration.
- Areas like **central Africa or polar regions have fewer points**, showing lower density.
- Density maps often convert such point distributions into **heatmaps or intensity-based visualisations** to highlight clusters.

Advantages and Limitations of Density Maps

Advantages of Density Maps

- **Highlights crowded or critical zones visually**
  Areas with higher data concentration appear darker or more intense, helping viewers instantly spot dense clusters such as busy cities, infection zones, or heavy traffic areas.
- **Suitable for large volume data visualisation**
  Density maps can handle thousands or even millions of points by converting them into smooth density gradients, making big data easier to understand in a single view.
- **Helps identify patterns and hotspots quickly**

With clear shading differences, users can detect hotspots, spread zones, or uneven distributions efficiently without manually analysing raw values.

Limitations of Density Maps

- **Overlapping points may hide individual values**
  When multiple data points fall in the same area, individual information may be lost, showing only intensity rather than exact numbers or locations.
- **Requires smoothing techniques for clarity**
  Kernels or smoothing are needed to convert points into gradients. Without proper smoothing, the map may look patchy or unclear.
- **Misleading without proper legend scale**
  Without a clear color legend, users may misinterpret density levels. A well-defined scale ensures interpretation remains accurate and meaningful

Best Practices for Using Density Maps

- **Use gradient color scale (light → dark)**
  A smooth color progression helps users understand variations in density easily. Lighter shades indicate low concentration, while darker shades represent high-density clusters, enabling quick pattern recognition.
- **Avoid too many intensity levels**
  Using too many color steps can make interpretation difficult and visually overwhelming. A limited range of shades improves clarity and helps differentiate hotspots more effectively.
- **Add legends and geographic labels**
  A clearly defined legend explains what each shade value represents, ensuring correct reading of data. Geographic labels such as region names or coordinates help users identify specific areas without confusion.

9. BUBBLE CHART

A Bubble Chart is like a scatter plot with an added dimension—circle size represents a third variable. It allows comparison of three variables at the same time, making it useful for trend analysis or market comparison. However, if many bubbles are similar in size or color, readability decreases.

Use Cases

- Sales revenue (size) vs cost (x-axis) vs profit (y-axis)
- Population vs GDP vs growth rate comparison
- Marketing campaign reach vs cost vs lead generation

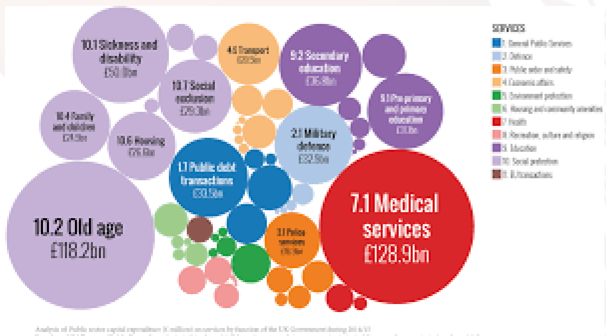

Fig: Bubble CHart

- **bubble charts represent multiple variables visually.**
- In this figure, each circle represents a **medical service category** within the healthcare sector.
- The **size of each bubble indicates the magnitude of spending or value**, where larger bubbles represent higher amounts.
- Colours help distinguish **different service segments**, making the chart easier to interpret.
- For example, the largest bubbles correspond to **major healthcare expenditures**, while smaller bubbles represent less significant categories.
- Bubble charts allow analysts to **compare multiple categories quickly and identify dominant segments.**
- Such visualisations are useful in **market analysis, financial reporting, and business intelligence**, where understanding relative proportions and comparisons is important.

Advantages and Limitations of Bubble Chart

Advantages of Bubble Chart

- **Represents three-dimensional data clearly**

A bubble chart allows visualisation of three variables at once — X-axis, Y-axis, and bubble size — making it easier to observe relationships without multiple charts.

- **Visually appealing for comparisons**

The bubbles create a colorful, engaging display where users can compare groups or categories quickly, especially when different colors are used.

- **Good for business & financial analysis**

Useful for market trend analysis, revenue vs cost comparisons, sales forecasting, customer segmentation, and other business-related insights that involve multiple metrics.

Limitations of Bubble Chart

- **Too many bubbles reduce visibility**
  When datasets are large, bubbles overlap and clutter the chart, making it difficult to see or interpret individual values.
- **Similar sizes are hard to differentiate**
  Small differences in size may not be noticeable visually, leading to misinterpretation if bubbles represent values too close in range.
- **Over-coloring distracts the viewer**
  Excessive or irrelevant colors may shift focus away from actual data, reducing clarity and making the chart more decorative than informative.

Best Practices for Bubble Charts

- **Use contrasting colors and varied bubble sizes**: Color and size differences help users distinguish categories and value magnitudes quickly. Contrasting shades ensure that each bubble stands out clearly rather than blending into the background.

- **Add labels or hover-info for clarity:** Bubble size can sometimes be hard to judge visually, so adding labels or tooltips showing exact values supports better interpretation. This allows users to check details without confusion.

- **Limit number of bubbles for readability:** A bubble chart is most effective when displayed with a manageable number of data points. Too many bubbles overlap and clutter the view, so filtering or grouping ensures the chart remains clear and easy to read.

10. TREE MAPS

A Tree map displays hierarchical data using nested rectangles where the size and color represent proportion and category. It is ideal for showing breakdowns like product sales or budget distribution in compact space. But if too many segments are present, smaller rectangles become difficult to read.

Use Cases

- Market share comparison
- Budget and expenditure breakdown
- File storage usage visualisation
- Sales by product category & subcategory

Fig: Tree Map

The figure shows **nested rectangles of different sizes and colors**, which is the key structure of a **Treemap visualization**.

- **Each rectangle represents a category or item** (such as different projects).
- **The size of the rectangle represents the relative value or proportion**, such as effort, cost, or workload.
- **Colors indicate priority levels**, for example:
    - Red → High priority
    - Yellow → Medium priority
    - Green → Low priority
- The chart allows users to **compare proportions visually in a compact space**.

Advantages and Limitations of Tree maps

Advantages of Tree map

- **Displays large hierarchical data compactly**

  Treemaps allow visualisation of multi-level data within a limited area, making them ideal for summarising large datasets with multiple categories and subcategories.

- **Easy to compare proportions visually**

  The size of each rectangle represents value, allowing viewers to quickly judge which categories contribute more or less to the whole without needing numerical comparison.

- **Fits many categories in a small space**

  Treemaps make efficient use of space, allowing hundreds of values to be displayed at once — useful for dashboards and reports where screen space is limited.

Limitations of Tree map

- **Very small blocks become unreadable**

  When categories are too many or values vary widely, smaller rectangles shrink to a size where labels cannot be read, reducing interpretability.

- **Too many colors can confuse interpretation**

  Color-coding is helpful, but excessive use of different shades makes the chart visually noisy and difficult for users to understand patterns.

- **Not suitable for showing precise values**

  Treemaps are mainly useful for relative comparison, not exact figures. When users need numerical accuracy, tables or bar charts may be more appropriate.

Best Practices for Treemaps

- **Use limited color shades for category grouping:** Colors should be chosen to represent groups or hierarchies clearly. Using a limited palette helps users visually distinguish categories without overwhelming them with too many shades.

- **Add labels or hover-info for small sections:** When smaller blocks are difficult to read, tooltips or interactive labels help provide exact values or category names. This ensures that even tiny elements convey meaningful information.

- **Avoid overcrowding with too many segments:** Treemaps work best when they display a reasonable number of categories. If too many values are shown, rectangles become tiny and unreadable — grouping or filtering improves clarity.

11. DASHBOARD DEVELOPMENT PROCESS

Dashboard development refers to the structured workflow used to design, build, and deploy an interactive data dashboard. It ensures that the dashboard is purposeful, visually meaningful, and supports decision-making with clarity.

Step-by-Step Development Process

1. Requirement Understanding & Goal Identification

The first step is to determine *why* the dashboard is needed and what questions it must answer. Example objectives:

✓ Sales trend tracking

✓ Employee performance monitoring

✓ Project progress visualisation

This step ensures the dashboard solves a real analytical need.

2. Data Collection and Source Identification

Data may come from multiple locations such as databases, Excel sheets, cloud servers, APIs, or ERP systems.
At this stage, developers identify:

- Where data is stored
- How frequently it updates
- What format it is available in

3. Data Cleaning, Processing & Integration

Raw data often contains errors, duplicates, missing values, or inconsistent formats. Data preprocessing ensures accuracy by:

- Removing invalid entries
- Standardising formats
- Merging multiple databases
- Creating calculated fields/KPIs

4. Selection of KPIs and Metrics

A dashboard must show only the most meaningful indicators.

Examples of KPIs include:

- Total revenue
- Conversion rate
- Production output
- Customer satisfaction score

Too many KPIs cause clutter, so only essential ones are chosen.

5. Visualisation Selection

Every dataset requires a suitable chart:

Data Purpose	Best Visual
Trend over time	Line / Area Chart
Category comparison	Bar / Column
Relationship between variables	Scatter Plot
Hierarchical breakdown	Treemap / Sunburst
Target vs Actual	Bullet Graph

Right visualisation = faster and clearer understanding.

6. Dashboard Layout & UI Design

The dashboard layout defines arrangement of charts, spacing, colour scheme, icons, fonts, and navigation.

Good dashboards follow the **eye-flow pattern** — most important visuals appear at the top or center.

7. Development & Implementation

Charts, filters, slicers, drilldowns, and KPI cards are added using BI tools such as Power BI, Tableau, Excel, Google Data Studio, etc.

8. Testing & Refining

Dashboard is tested for:

✓ Performance speed

✓ Data accuracy

✓ Visual clarity

✓ User experience

Feedback is collected and improvements are made.

9. Deployment & Maintenance

Final dashboard is published for users on web, cloud, or application. Scheduled refresh cycles are set to ensure real-time updates.

12. DASHBOARD ARCHITECTURE

Dashboard architecture defines **how data flows** from sources into the visualisation layer. It includes components for storage, processing, visualisation, and user access.

Layers of Dashboard Architecture

1. Data Source Layer

The data source layer is where the raw information originates. It includes a variety of repositories such as databases like SQL, Oracle, and MySQL, files in Excel/CSV format, ERP or CRM systems used in organisations, and cloud-based services like AWS or Azure. This layer acts as the base foundation of dashboard development because it contains the original data that will later be processed, cleaned, and visualised.

2. Data Integration & ETL Layer

In this stage, data is collected from different sources and processed for analysis. ETL stands for *Extract, Transform, and Load* — meaning the system extracts raw data, transforms it by cleaning, formatting, and organising it, and then loads it into a structured environment. Tools like Talend, SSIS, Informatica, or Python scripts are commonly used. This layer ensures that data becomes usable and consistent before entering storage.

3. Data Storage / Data Warehouse Layer

Once cleaned, data moves into a centraliseded storage system known as a data warehouse or data mart. Platforms like BigQuery, Snowflake, Hadoop, or SQL Server store this processed information for future analysis. This layer plays a major role in performance because stored data can be queried quickly, supporting faster report generation and analytics without slowing down operational systems.

4. Data Modelling & Business Logic Layer

At this stage, relationships between datasets are built using joins, measures, and calculated fields. Key performance indicators (KPIs) are derived here — for example, **Sales Growth % = (Current – Previous) / Previous × 100.** This layer transforms raw values into meaningful business insights by applying formulas, logical rules, and aggregation methods. It converts raw data into structured, decision-ready information.

5. Visualisation Layer (Dashboard UI)

This is the layer where visual elements are designed. Charts, graphs, KPI cards, and filters are created using tools like Power BI, Tableau, QlikView, or Google Looker Studio. It is the dashboard's presentation layer and acts as the interface between the processed data and the end user. Here data is converted into attractive visuals that communicate trends, comparisons, and patterns with clarity.

6. User Interaction & Access Layer

In the final layer, users interact with the dashboard through devices such as desktop applications, web browsers, or mobile apps. Features like filters, drill-downs, sorting, and export options help users explore data according to requirements. This layer ensures accessibility and convenience, enabling decision-makers to read, analyse, and act upon insights efficiently in real time.

```
┌─────────────────────────────┐
│    Data Sources      │
│ Databases | Excel | APIs │
└──────────────┬──────────────┘
      │
      ▼
┌─────────────────────────────┐
│  ETL / Data Cleaning  │
│ Transform • Merge • Prep│
└──────────────┬──────────────┘
      │
      ▼
┌─────────────────────────────┐
│ Data Warehouse / Storage │
└──────────────┬──────────────┘
      │
      ▼
┌─────────────────────────────┐
```

| Data Modelling & KPIs |

▼

| Visualisation (Dashboard) |

| Charts • KPIs • Filters |

▼

| User Access Layer |

| Web • Mobile • Desktop |

4. Sample Dashboard Layout (Real-Model Structure)

What This Dashboard Demonstrates

- A sample dashboard layout consists of multiple components, each serving a specific analytical purpose. At the top of the dashboard, **Key Performance Indicators (KPIs)** are placed to provide a quick snapshot of important metrics such as total sales, revenue growth, or conversion rate.

Filters are included to help users explore data by category, time period, or region, allowing flexible and customisable analysis.

- A **trend chart** displays how performance changes over time, making it easy to track growth or decline across months or years.
- A **bar or column chart** is useful for comparing products, departments, or categories side-by-side.
- Meanwhile, a **pie chart** helps visualise distribution or market share percentage among groups.
- For region-based performance tracking, a **map chart** highlights which geographical areas are performing well or underperforming.
- A **highlight table** presents detailed values with color emphasis for quick spotting of highs, lows, and patterns.
- Finally, an **insight or conclusion section** is added to interpret the visuals and summarise results, helping users understand what the data means and what actions may be needed.

Example Dashboard Usage Scenarios

- **Monthly Sales Monitoring Dashboard**: Used by business teams to track total sales, top-selling products, revenue growth, and sales variance month-to-month. It helps identify trends and compare performance across regions or product categories.
- **Student Attendance & Performance Dashboard**: Helpful for schools and colleges to analyse attendance percentage, subject-wise marks, class performance, and identify students who need academic support. It gives teachers a quick academic overview.
- **Hospital Patient Analytics Dashboard**: Used in healthcare to track patient admission rates, bed occupancy, treatment success rate, critical cases, and discharge timelines. It supports hospital management in resource allocation and decision-making.
- **Marketing Campaign KPI Dashboard**: Shows marketing results such as total leads generated, conversion rate, cost per lead, platform performance (Google, Facebook, etc.), and campaign ROI. It helps determine which strategy performs best.
- **HR Attrition & Recruitment Dashboard**: Used by HR teams to monitor employee joining rate, resignation trends, hiring cycle time, department-wise attrition, and workforce demographics. It helps management plan manpower and retention strategies.

13. SUMMARY

- Dashboards are designed, built, and used to convert raw data into meaningful insights. Dashboard development begins with identifying objectives, collecting data from various sources, cleaning and preparing it using ETL processes, and storing it in a data warehouse for quick access.
- Through data modeling, KPIs and calculations are defined, after which visuals—such as charts, graphs, and indicators—are created using BI tools like Power BI, Tableau, or Qlik.
- Finally, dashboards are deployed for users to interact with through filters, drill-downs, and export options.
- The dashboard architecture consists of multiple layers: data source layer, where raw data exists; ETL integration layer, which processes it; data storage layer, where cleaned data is kept; data modeling layer, where logic and KPIs are applied; visualisation layer, where UI and charts are built; and user interaction layer, where viewers analyse and interpret insights. This structure ensures smooth data flow from collection to decision-making.
- Real dashboard applications include sales monitoring, student performance tracking, hospital analytics, marketing performance dashboards, and HR recruitment dashboards.
- A typical dashboard layout contains KPIs at the top, filters for selection, trend charts, bar charts for comparison, pie charts for distribution, maps for geographic insights, highlight tables for value comparison, and an insights section to convert visuals into conclusions.
- Overall, dashboards help organisations analyse data faster, view performance at a glance, and make informed decisions supported by visual evidence.

14. GLOSSARY

Dashboard	–	A visual interface that displays key information using charts, KPIs, and graphs for quick understanding and decision-making.
KPI (Key Performance Indicator)	–	A measurable value that indicates performance such as sales, revenue growth, attendance, or output efficiency.
Data Source	–	Original storage systems from where data is collected (SQL, Excel, Cloud, ERP, CRM, etc.).
ETL (Extract–Transform–Load)	–	A process of extracting raw data, cleaning and formatting it, and loading it into a data warehouse for analysis.
Data Warehouse	–	A centraliseded storage system where processed data is kept for fast queries and reporting.
Data Modelling	–	Creating relationships, hierarchies, KPIs, and calculated measures to convert raw data into meaningful insights.
Business Logic	–	Rules, formulas, and conditions applied to data (e.g., Sales Growth %, profit margin calculation).
Visualisation Layer	–	The dashboard interface where charts, KPIs, and visuals are built using tools like Power BI or Tableau.
User Interaction Layer	–	The section where users interact with the dashboard through filters, drill-downs, sorting, or exporting.
Slicers/Filters	–	Interactive controls used to refine data view based on region, time, product, category, or parameter.
Trend Chart	–	A line/area chart that shows how data changes over time.
Bar/Column Chart	–	Used for comparing values across categories (product sales, departments, branches).
Pie Chart	–	Represents distribution or market share among different categories.

Map Visualisation	–	Displays data geographically to compare performance across locations.
Highlight Table	–	A table using color emphasis to show high/low values clearly.
Drill-Down	–	A dashboard function that lets users view deeper details from a summary level.
Real-time Refresh	–	Automatic updating of dashboards when new data is added to the source.
Dashboard Layout	–	The structured arrangement of KPIs, charts, tables, and filters for readability.
Insights Section	–	A summary panel that explains what the visuals indicate and highlights actionable findings.

15. TERMINAL QUESTIONS

1. What is the main purpose of a Dashboard?
2. What does ETL stand for in Dashboard Development Process?
3. What does a Choropleth Map represent?
4. Give one common use of Choropleth Maps.
5. What does a Word Cloud visualise?
6. What does a Network Diagram show?
7. What is the primary purpose of a Correlation Matrix?
8. What do Geographical Plots help visualise?
9. What does bubble size represent in a Bubble Chart?
10. Why are Treemaps useful?

16. ANSWERS

16.1. Self-Assessment Answers:

1. Answer: colors
2. Answer: higher
3. Answer: legend
4. Answer: normalised
5. Answer: frequency
6. Answer: larger
7. Answer: text
8. Answer: readability
9. Answer: edges/links
10. Answer: edges
11. Answer: communication/computer *(either acceptable)*
12. Answer: cluttered
13. Answer: correlation
14. Answer: strong positive
15. Answer: heatmaps
16. Answer: causation
17. Answer: spatial/geographic *(either accepted)*
18. Answer: population
19. Answer: data
20. Answer: mapping

16.2. Terminal Question Answers

Answer 1: The main purpose of a dashboard is to present important information visually through charts, KPIs, and indicators so that users can quickly understand performance, compare values, and make decisions without analysing raw data.(Refer Section 2 for more information)

Answer 2: ETL stands for Extract, Transform, Load — where data is extracted from sources, cleaned and formatted in the transform stage, and then loaded into a warehouse for visualisation. This step ensures the dashboard uses accurate, structured data.(Ref: Dashboard Development Process / ETL Layer for more information)

Answer 3: A Choropleth Map represents data using color shades to show differences across regions or locations. Darker shades usually indicate higher values, while lighter shades show lower values. (Refer Section 3 for more information)

Answer 4: Choropleth Maps are commonly used to show population density across states or countries, helping compare crowded and less populated regions easily. (Refer Section 3 for more information)

Answer 5: A Word Cloud visualises frequently appearing words in large size and less frequent words in smaller size. This helps quickly identify dominant terms in text data without reading the full content. (Refer Section 4 for more information)

Answer 6: A Network Diagram shows how entities (nodes) are connected using links or edges. It is useful for understanding relationships in social networks, computer networks, workflows, and research connections. (Refer Section 5 for more information)

Answer 7: The primary purpose of a correlation matrix is to show how strongly different variables are related to each other. It helps in feature selection and identifying positive, negative, or weak correlation patterns. (Refer Section 6 for more information)

Answer 8: Geographical plots help visualise data across locations using markers, color intensity, or bubble size. They highlight regional variations and are useful for logistics, sales, and demographic insights. (Refer Section 7 for more information)

Answer 9: In a Bubble Chart, the bubble size represents a third variable, apart from X and Y. This allows three-dimensional comparison in a single chart, making analysis more informative visually. (Refer Section 9 for more information)

Answer 10: Treemaps are useful because they display hierarchical data in nested rectangles, allowing users to compare values quickly even when many categories exist. They make efficient use of available space. (Refer Section 10 for more information)

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